Configuration Summary

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DHW ANALYSIS - CONFIGURATION SUMMARY
Project: oisst_regression
Generated: 2026-10-20 04:32:07
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-- Data Preparation --

  input_dir.............................. /g/data/ia39/aus-ref-clim-data-nci/oisst/data/yearly/
  file_pattern........................... oisst-avhrr-v02r01_{year}.nc
  year_range............................. (1981, 2024)
  variable............................... sst
  detrend_method......................... global_linear
  standardise_method..................... global
  fill_method............................ diffusion
  fill_diffusion_iterations.............. 500
  fill_fallback_value.................... 0
  downscale.............................. True
  target_lat_size........................ 128
  target_lon_size........................ 256
  temporal_average....................... none
  subset_region.......................... False
  lat_bounds............................. None
  lon_bounds............................. None
  diagnostic_year........................ 2020
  diagnostic_pdf......................... diagnostics/data_preparation_diagnostics_128x256.pdf
  output_dir............................. data/processed_128x256/
  save_intermediates..................... True
  downscaled_file........................ sst_downscaled.nc
  temporally_averaged_file............... sst_temporally_averaged.nc
  detrended_file......................... sst_detrended.nc
  standardised_file...................... sst_standardised.nc
  filled_file............................ sst_filled.nc
  final_file............................. sst_final.nc
  subset_file............................ sst_subset.nc
  trend_params_file...................... trend_params.nc
  standardise_params_file................ std_params.nc
  resume_after_stage..................... 0
  chunks................................. {'time': 366, 'lat': 720, 'lon': 720}
  dask_scheduler......................... threads
  dask_n_workers......................... 8
  dask_threads_per_worker................ 2
  dask_memory_limit...................... 12GB
  dask_num_threads....................... 16

-- Model Training --

  model_name............................. sfno
  input_history.......................... 10
  output_channels........................ 1
  embed_dim.............................. 80
  num_layers............................. 5
  num_modes_lat.......................... 32
  num_modes_lon.......................... 64
  mlp_ratio.............................. 2.5
  activation............................. gelu
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norm_type.............................. instance_norm
dropout_rate........................... 0.05
use_residual........................... True
spectral_bias.......................... True
add_grid_features...................... True
add_seasonal_encoding.................. False
forecast_horizon....................... 90
initial_rollout_steps.................. 1
max_rollout_steps...................... 8
rollout_epoch_start.................... 40
rollout_epoch_end...................... 90
rollout_loss_decay..................... 0.92
train_years............................ (1982, 2015)
val_years.............................. (2016, 2019)
test_years............................. (2020, 2024)
learning_rate.......................... 0.0002
min_learning_rate...................... 1e-06
batch_size............................. 16
num_epochs............................. 120
weight_decay........................... 1e-05
optimizer.............................. adamw
scheduler.............................. cosine
scheduler_patience..................... 10
scheduler_factor....................... 0.5
warmup_epochs.......................... 5
gradient_clip_norm..................... 1
loss_function.......................... huber
early_stopping_patience................ 20
early_stopping_metric.................. val_loss
input_data_path........................ data/processed_128x256/sst_final.nc
checkpoint_dir......................... checkpoints_128x256/
best_model_path........................ checkpoints_128x256/best_model.pt
training_log_path...................... results_128x256/training_log.csv
random_seed............................ 42
num_workers............................ 4
distributed............................ True
dist_backend........................... nccl
find_unused_parameters................. False
sync_batchnorm......................... False
master_port............................ 29500

-- Model Inference --

model_path............................. checkpoints_128x256/best_model.pt
input_data_path........................ data/processed_128x256/sst_final.nc
run_validation......................... True
run_test............................... True
forecast_lead_steps.................... 90
chain_stride_steps..................... 90
output_dir............................. results_128x256/
val_predictions_file................... predictions_val.nc
test_predictions_file.................. predictions_test.nc

-- Concat Chains --
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```
  predictions_dir..... /scratch/um09/xl5758/DHW-project/results_128x256
  predictions_file_template..... predictions_{split}.nc
  std_params_path..... data/processed_128x256/std_params.nc
  trend_params_path..... data/processed_128x256/trend_params.nc
  output_dir..... /scratch/um09/xl5758/DHW-project/results_128x256
  actual_file_template..... sst_actual_{split}_continuous.nc
  predicted_file_template..... sst_predicted_{split}_chains.nc
  splits..... ['test', 'val']
  lon_correction_threshold..... 0.6

-- Calc DHW --

  mmm_path..... /home/272/xl5758/ct5km_climatology_v3.1.nc
  results_dir..... /scratch/um09/xl5758/DHW-project/results_128x256
  predicted_file_template..... sst_predicted_{split}_chains.nc
  actual_file_template..... sst_actual_{split}_continuous.nc
  dhw_output_dir..... /scratch/um09/xl5758/DHW-project/dhw_output_128x256
  output_file_template..... dhw_products_{split}.nc
  run_test..... True
  run_validation..... True
  window_days..... 84
  hotspot_threshold..... 1

-- DHW Plots --

  dhw_dir..... /scratch/um09/xl5758/DHW-project/dhw_output_128x256
  output_pdf..... dhw_analysis_combined.pdf
  output_csv..... dhw_metrics_combined.csv
  spatial_lead_steps..... [1, 7, 14, 30, 60, 90]
  n_sample_chains..... 6
  plot_init_dates..... ['2016-05-20', '2016-10-15', '2017-02-10', '2017-07-09', '20
  dhw_vmax..... 20
  dhw_diff_vmax..... 8
  tropic_lat..... 30
  dhw_cmap..... ['#CCe6F0', '#441C94', '#6F47BF', '#815CCC', '#A383E6', '#ED
  error_cmap..... ['#053061', '#2166AC', '#92C5DE', '#F7F7F7', '#F4A582', '#D6
  rmse_cmap..... YlOrRd
  bias_cmap..... RdBu_r
  bleaching_thresholds..... [(0, 4, 'No Stress', '#27AE60'), (4, 8, 'Warning', '#F39C12'
  gbr_lat_bounds..... (-25.0, -10.0)
  gbr_lon_bounds..... (142.0, 157.0)
  reef_boxes..... [{'name': 'Great Barrier Reef', 'lat': (-25.0, -10.0), 'lon'
```

DHW Analysis - Summary

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DEGREE HEATING WEEKS (DHW) - COMBINED ANALYSIS
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METHODOLOGY  (NOAA Coral Reef Watch v3.1)
  SSTA      = SST - DC              (DC = daily climatology)
  HotSpot   = max(SST - MMM, 0)     (MMM = Maximum Monthly Mean)
  DHW_i     = (1/7) x sum(HotSpot_j) for j=i-83..i, HotSpot_j >= 1.0C

SLIDING WINDOW CONSTRUCTION  (always 84 days ending on valid date)
  Lead L:  (84-L) actual + L predicted SST days
  Lead 1:  83 actual +  1 predicted
  Lead 30: 54 actual + 30 predicted
  Lead 84:  0 actual + 84 predicted
  Lead 90: previous 84 predicted days (days 7-90)

VERIFICATION
  Actual DHW  : 84 observed days ending on the forecast valid date
  Persistence : observed DHW on the day before lead 1, held constant
  Sample      : chains defined at every lead step (same chains for
                model and persistence at all leads)

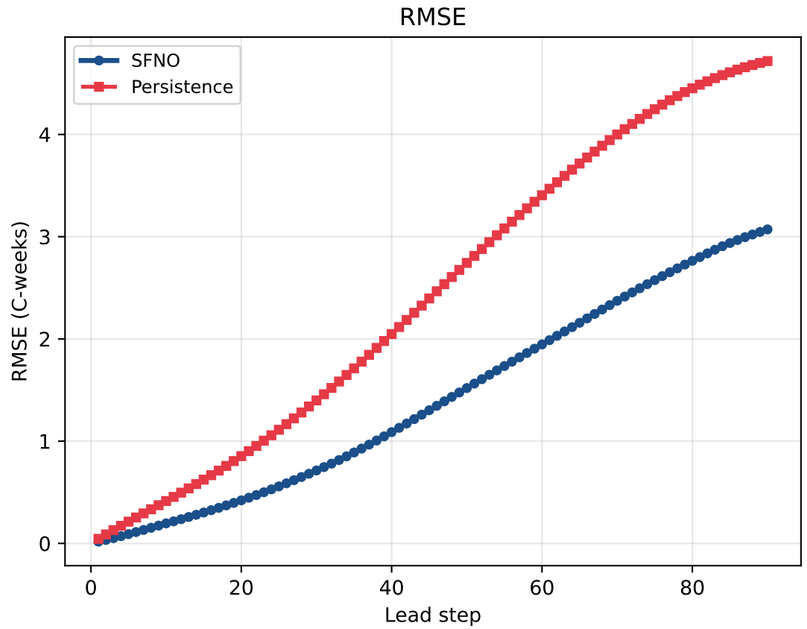
BLEACHING ALERT THRESHOLDS
  DHW  0-4  : No Stress
  DHW  4-8  : Warning
  DHW  8-12 : Alert Level 1
  DHW 12-16 : Alert Level 2
  DHW 16-20 : Alert Level 3
  DHW 20->= : Alert Level 4-5

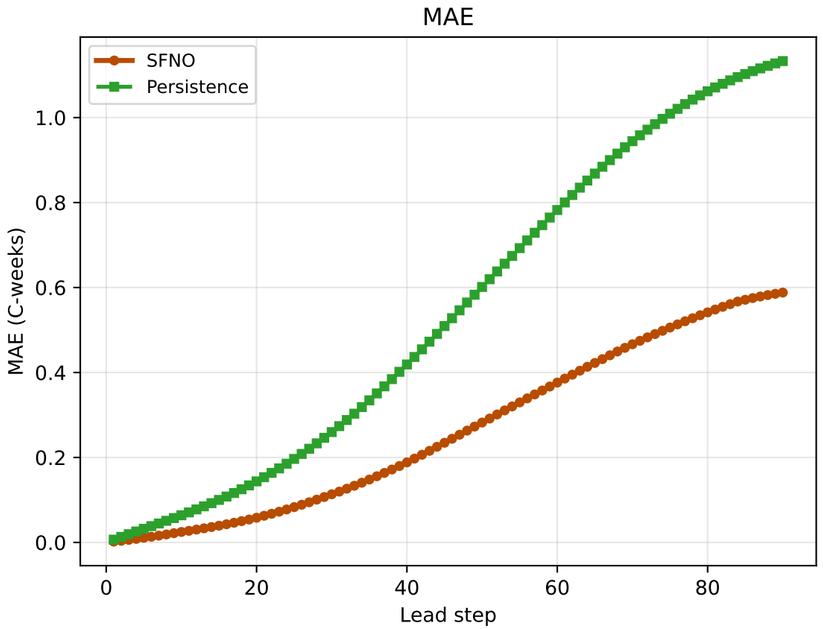
DATASETS
  Test          19 chains x 90 leads x 128 lat x 256 lon
  Val           16 chains x 90 leads x 128 lat x 256 lon
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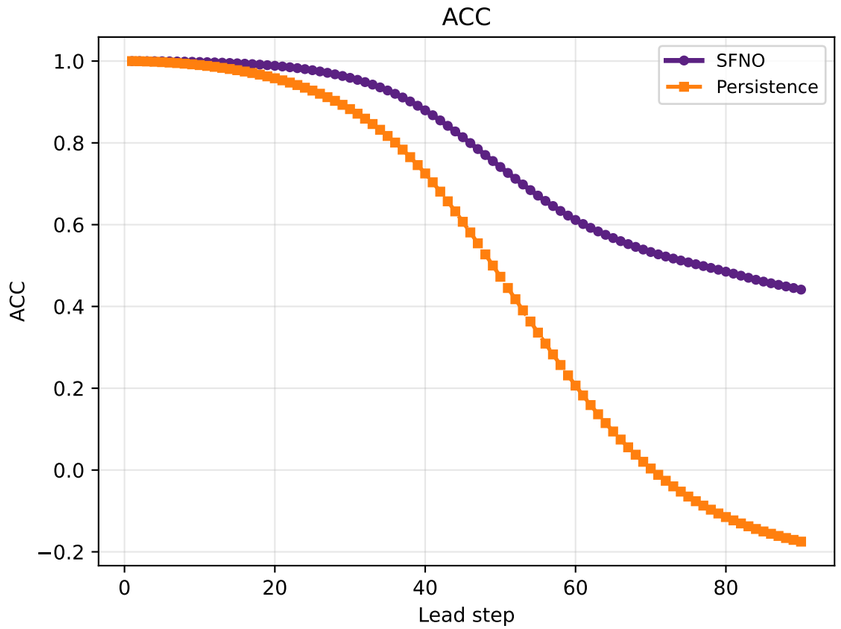
DHW Metrics by Lead Step - Test (18 chains scored at every lead)

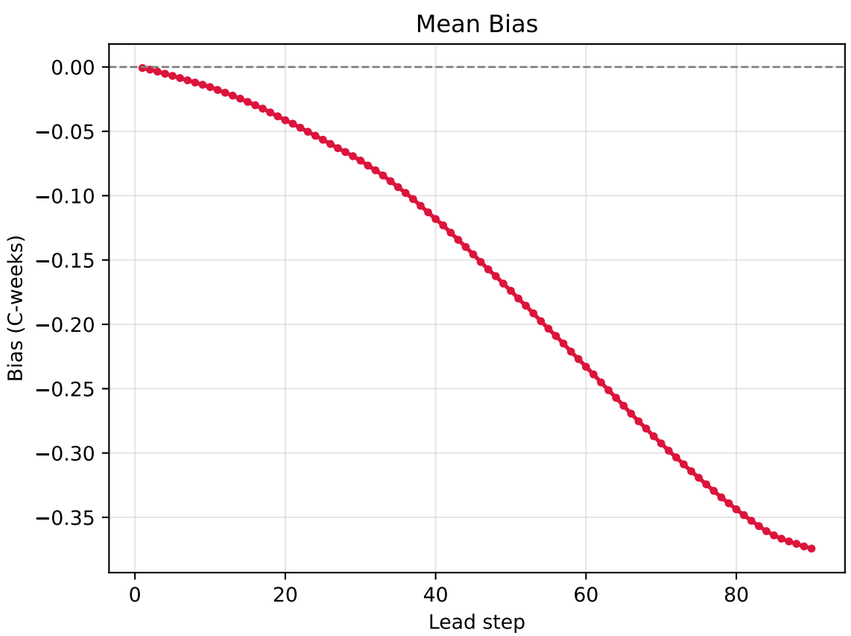
Lead Step	SFNO RMSE	Persist RMSE	Skill %	SFNO MAE	Persist MAE	SFNO Bias	SFNO ACC	Persist ACC	SFNO R²
1	0.0173	0.0446	61.2%	0.0019	0.0066	-0.0008	1.0000	0.9999	1.0000
2	0.0339	0.0878	61.4%	0.0039	0.0130	-0.0021	0.9999	0.9996	0.9999
3	0.0517	0.1301	60.2%	0.0060	0.0194	-0.0036	0.9998	0.9990	0.9997
4	0.0707	0.1717	58.8%	0.0084	0.0258	-0.0052	0.9997	0.9983	0.9995
5	0.0906	0.2128	57.4%	0.0109	0.0321	-0.0069	0.9995	0.9974	0.9992
6	0.1110	0.2534	56.2%	0.0134	0.0383	-0.0086	0.9993	0.9964	0.9988
7	0.1315	0.2936	55.2%	0.0160	0.0446	-0.0103	0.9990	0.9951	0.9984
8	0.1527	0.3337	54.2%	0.0188	0.0509	-0.0121	0.9987	0.9937	0.9978
9	0.1742	0.3737	53.4%	0.0218	0.0573	-0.0138	0.9983	0.9920	0.9971
10	0.1958	0.4140	52.7%	0.0247	0.0639	-0.0156	0.9978	0.9902	0.9963
11	0.2157	0.4549	52.6%	0.0273	0.0707	-0.0178	0.9973	0.9882	0.9955
12	0.2368	0.4961	52.3%	0.0302	0.0778	-0.0200	0.9968	0.9859	0.9945
13	0.2583	0.5380	52.0%	0.0332	0.0850	-0.0222	0.9961	0.9834	0.9934
14	0.2801	0.5805	51.8%	0.0362	0.0924	-0.0246	0.9954	0.9807	0.9922
15	0.3021	0.6234	51.5%	0.0394	0.1001	-0.0271	0.9945	0.9777	0.9908
16	0.3246	0.6671	51.3%	0.0427	0.1080	-0.0297	0.9936	0.9744	0.9893
17	0.3480	0.7119	51.1%	0.0464	0.1164	-0.0324	0.9925	0.9709	0.9875
18	0.3717	0.7579	51.0%	0.0501	0.1251	-0.0353	0.9913	0.9669	0.9856
19	0.3961	0.8052	50.8%	0.0540	0.1343	-0.0384	0.9900	0.9627	0.9834
20	0.4211	0.8536	50.7%	0.0581	0.1437	-0.0414	0.9885	0.9580	0.9810
21	0.4466	0.9027	50.5%	0.0628	0.1535	-0.0441	0.9868	0.9529	0.9783
22	0.4730	0.9530	50.4%	0.0675	0.1637	-0.0472	0.9849	0.9474	0.9753
23	0.5001	1.0048	50.2%	0.0724	0.1743	-0.0503	0.9828	0.9414	0.9719
24	0.5284	1.0576	50.0%	0.0776	0.1853	-0.0534	0.9804	0.9349	0.9682
25	0.5574	1.1114	49.8%	0.0831	0.1966	-0.0565	0.9777	0.9278	0.9640
26	0.5875	1.1666	49.6%	0.0887	0.2083	-0.0598	0.9746	0.9201	0.9593
27	0.6183	1.2230	49.4%	0.0946	0.2205	-0.0631	0.9713	0.9118	0.9542
28	0.6492	1.2805	49.3%	0.1007	0.2332	-0.0661	0.9676	0.9028	0.9486
29	0.6810	1.3394	49.2%	0.1070	0.2463	-0.0693	0.9635	0.8930	0.9425
30	0.7133	1.3987	49.0%	0.1133	0.2598	-0.0727	0.9590	0.8826	0.9358
31	0.7462	1.4591	48.9%	0.1196	0.2737	-0.0766	0.9541	0.8713	0.9285
32	0.7801	1.5206	48.7%	0.1265	0.2881	-0.0803	0.9486	0.8592	0.9205
33	0.8152	1.5834	48.5%	0.1334	0.3031	-0.0843	0.9425	0.8461	0.9116
34	0.8516	1.6473	48.3%	0.1406	0.3185	-0.0887	0.9357	0.8320	0.9020
35	0.8892	1.7121	48.1%	0.1480	0.3345	-0.0933	0.9281	0.8169	0.8914
36	0.9274	1.7778	47.8%	0.1558	0.3508	-0.0979	0.9199	0.8007	0.8800
37	0.9663	1.8444	47.6%	0.1636	0.3674	-0.1026	0.9110	0.7835	0.8677
38	1.0062	1.9117	47.4%	0.1716	0.3844	-0.1079	0.9013	0.7651	0.8545
39	1.0469	1.9795	47.1%	0.1799	0.4015	-0.1129	0.8908	0.7456	0.8404
40	1.0884	2.0477	46.8%	0.1884	0.4189	-0.1181	0.8795	0.7252	0.8255
41	1.1301	2.1165	46.6%	0.1973	0.4365	-0.1231	0.8676	0.7035	0.8098
42	1.1727	2.1861	46.4%	0.2063	0.4545	-0.1287	0.8549	0.6808	0.7935
43	1.2158	2.2558	46.1%	0.2156	0.4725	-0.1343	0.8417	0.6571	0.7766
44	1.2591	2.3259	45.9%	0.2251	0.4908	-0.1398	0.8280	0.6325	0.7593
45	1.3023	2.3960	45.6%	0.2346	0.5093	-0.1456	0.8139	0.6070	0.7417
46	1.3455	2.4661	45.4%	0.2440	0.5277	-0.1515	0.7995	0.5810	0.7240
47	1.3892	2.5360	45.2%	0.2535	0.5463	-0.1573	0.7849	0.5544	0.7062
48	1.4326	2.6057	45.0%	0.2632	0.5647	-0.1626	0.7703	0.5273	0.6884
49	1.4761	2.6749	44.8%	0.2727	0.5830	-0.1683	0.7556	0.5000	0.6706
50	1.5194	2.7437	44.6%	0.2822	0.6013	-0.1739	0.7410	0.4726	0.6531
51	1.5629	2.8119	44.4%	0.2916	0.6195	-0.1799	0.7265	0.4452	0.6357
52	1.6062	2.8796	44.2%	0.3012	0.6378	-0.1855	0.7122	0.4177	0.6186
53	1.6492	2.9468	44.0%	0.3107	0.6561	-0.1914	0.6982	0.3904	0.6020
54	1.6921	3.0138	43.9%	0.3202	0.6744	-0.1976	0.6844	0.3631	0.5856
55	1.7350	3.0805	43.7%	0.3297	0.6926	-0.2033	0.6710	0.3360	0.5696
56	1.7775	3.1466	43.5%	0.3391	0.7108	-0.2090	0.6581	0.3092	0.5541
57	1.8199	3.2122	43.3%	0.3484	0.7289	-0.2149	0.6456	0.2827	0.5389
58	1.8622	3.2770	43.2%	0.3576	0.7469	-0.2212	0.6335	0.2568	0.5243
59	1.9042	3.3414	43.0%	0.3669	0.7647	-0.2270	0.6222	0.2314	0.5104
60	1.9461	3.4054	42.9%	0.3762	0.7824	-0.2330	0.6115	0.2065	0.4971
61	1.9882	3.4692	42.7%	0.3856	0.8002	-0.2389	0.6016	0.1823	0.4846
62	2.0303	3.5325	42.5%	0.3949	0.8178	-0.2451	0.5922	0.1589	0.4726
63	2.0727	3.5947	42.3%	0.4042	0.8350	-0.2512	0.5834	0.1364	0.4610
64	2.1152	3.6556	42.1%	0.4135	0.8518	-0.2571	0.5751	0.1149	0.4499
65	2.1578	3.7155	41.9%	0.4225	0.8682	-0.2633	0.5673	0.0943	0.4391
66	2.2009	3.7745	41.7%	0.4315	0.8842	-0.2694	0.5597	0.0746	0.4286
67	2.2442	3.8325	41.4%	0.4405	0.8999	-0.2754	0.5525	0.0556	0.4182
68	2.2875	3.8894	41.2%	0.4494	0.9152	-0.2810	0.5457	0.0375	0.4081
69	2.3304	3.9451	40.9%	0.4581	0.9300	-0.2870	0.5391	0.0202	0.3984
70	2.3727	3.9993	40.7%	0.4665	0.9444	-0.2925	0.5331	0.0037	0.3891
71	2.4142	4.0518	40.4%	0.4746	0.9582	-0.2982	0.5274	-0.0117	0.3804
72	2.4551	4.1026	40.2%	0.4827	0.9716	-0.3034	0.5221	-0.0262	0.3720
73	2.4955	4.1521	39.9%	0.4905	0.9845	-0.3088	0.5172	-0.0399	0.3641
74	2.5353	4.2000	39.6%	0.4981	0.9969	-0.3142	0.5123	-0.0527	0.3565
75	2.5750	4.2466	39.4%	0.5058	1.0090	-0.3192	0.5078	-0.0648	0.3490
76	2.6141	4.2915	39.1%	0.5133	1.0206	-0.3243	0.5032	-0.0762	0.3416
77	2.6525	4.3346	38.8%	0.5206	1.0318	-0.3294	0.4988	-0.0869	0.3343
78	2.6901	4.3756	38.5%	0.5277	1.0423	-0.3345	0.4943	-0.0969	0.3270
79	2.7273	4.4146	38.2%	0.5347	1.0523	-0.3391	0.4898	-0.1062	0.3195
80	2.7640	4.4516	37.9%	0.5415	1.0618	-0.3438	0.4851	-0.1149	0.3118
81	2.8006	4.4868	37.6%	0.5482	1.0709	-0.3482	0.4802	-0.1231	0.3038
82	2.8368	4.5199	37.2%	0.5546	1.0795	-0.3526	0.4752	-0.1306	0.2956
83	2.8725	4.5510	36.9%	0.5609	1.0876	-0.3568	0.4701	-0.1375	0.2871
84	2.9067	4.5803	36.5%	0.5669	1.0953	-0.3607	0.4652	-0.1441	0.2790
85	2.9380	4.6080	36.2%	0.5712	1.1025	-0.3640	0.4608	-0.1502	0.2719
86	2.9674	4.6338	36.0%	0.5752	1.1093	-0.3666	0.4567	-0.1558	0.2652
87	2.9951	4.6576	35.7%	0.5788	1.1158	-0.3687	0.4528	-0.1612	0.2587
88	3.0214	4.6794	35.4%	0.5822	1.1218	-0.3706	0.4490	-0.1662	0.2521
89	3.0465	4.6993	35.2%	0.5852	1.1275	-0.3726	0.4451	-0.1708	0.2456
90	3.0710	4.7176	34.9%	0.5881	1.1330	-0.3743	0.4412	-0.1750	0.2389

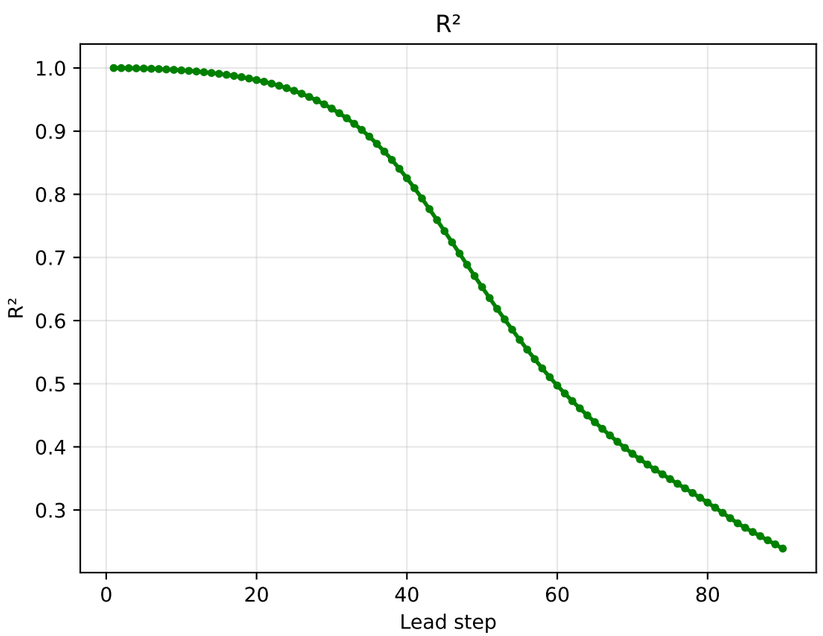
DHW Forecast Skill vs Lead Step - Test

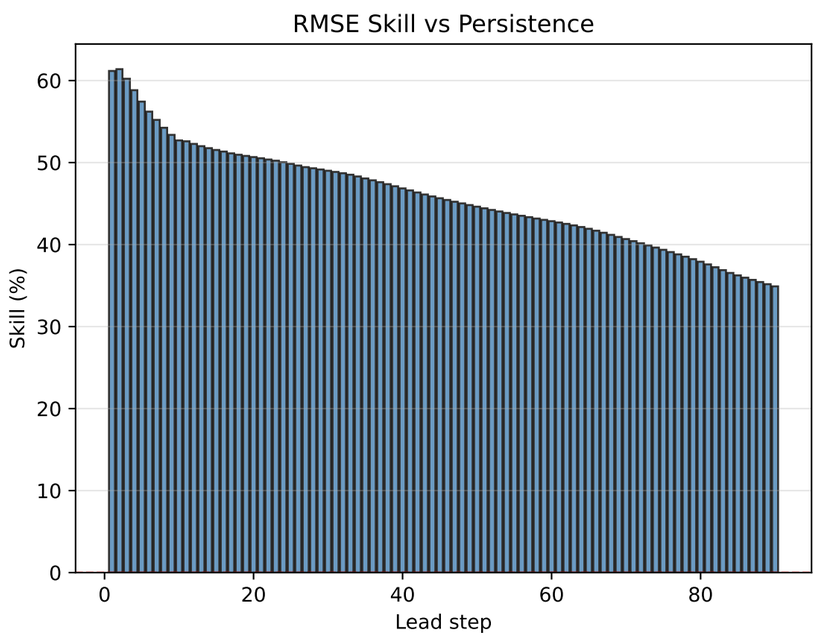


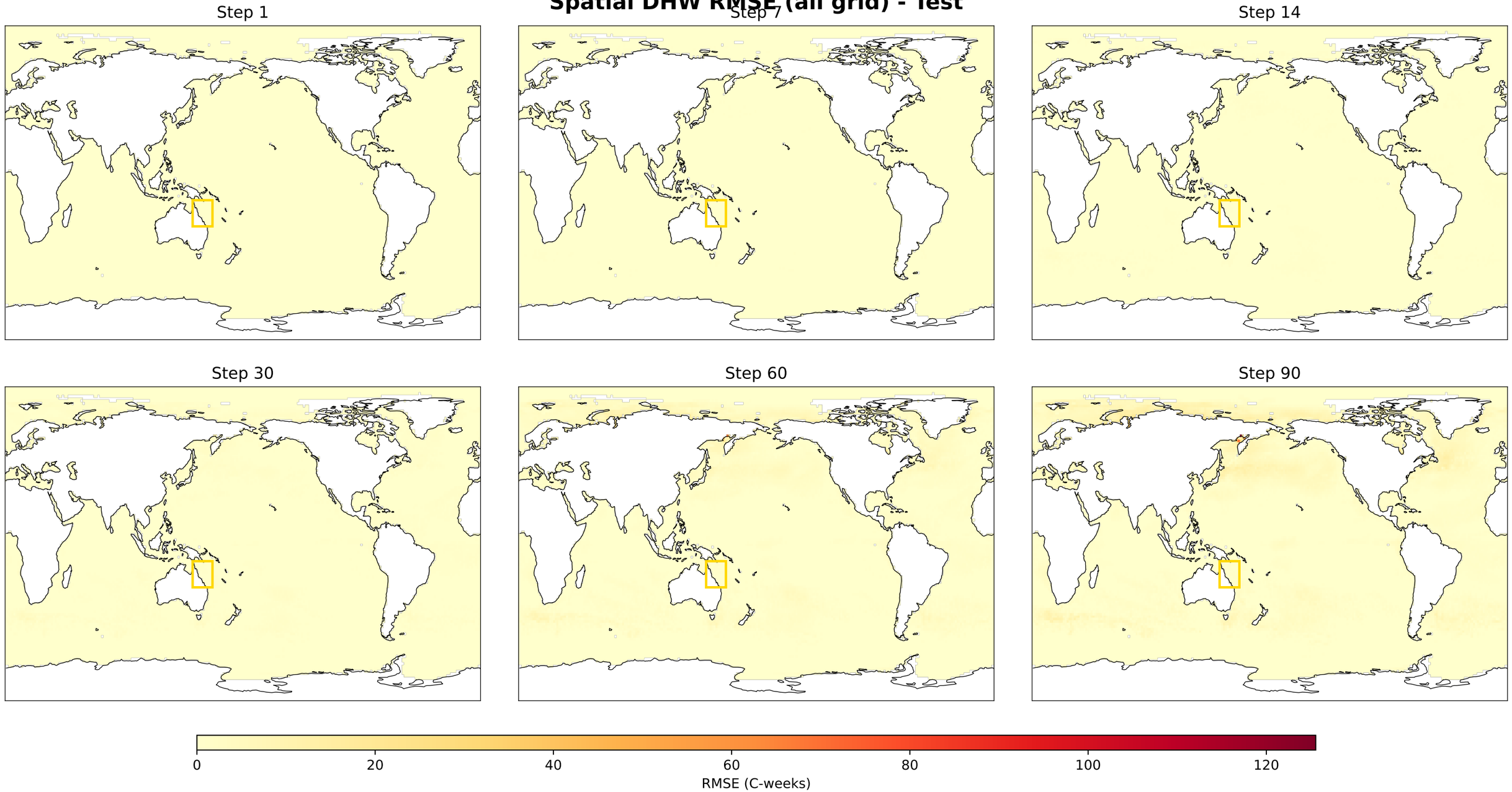


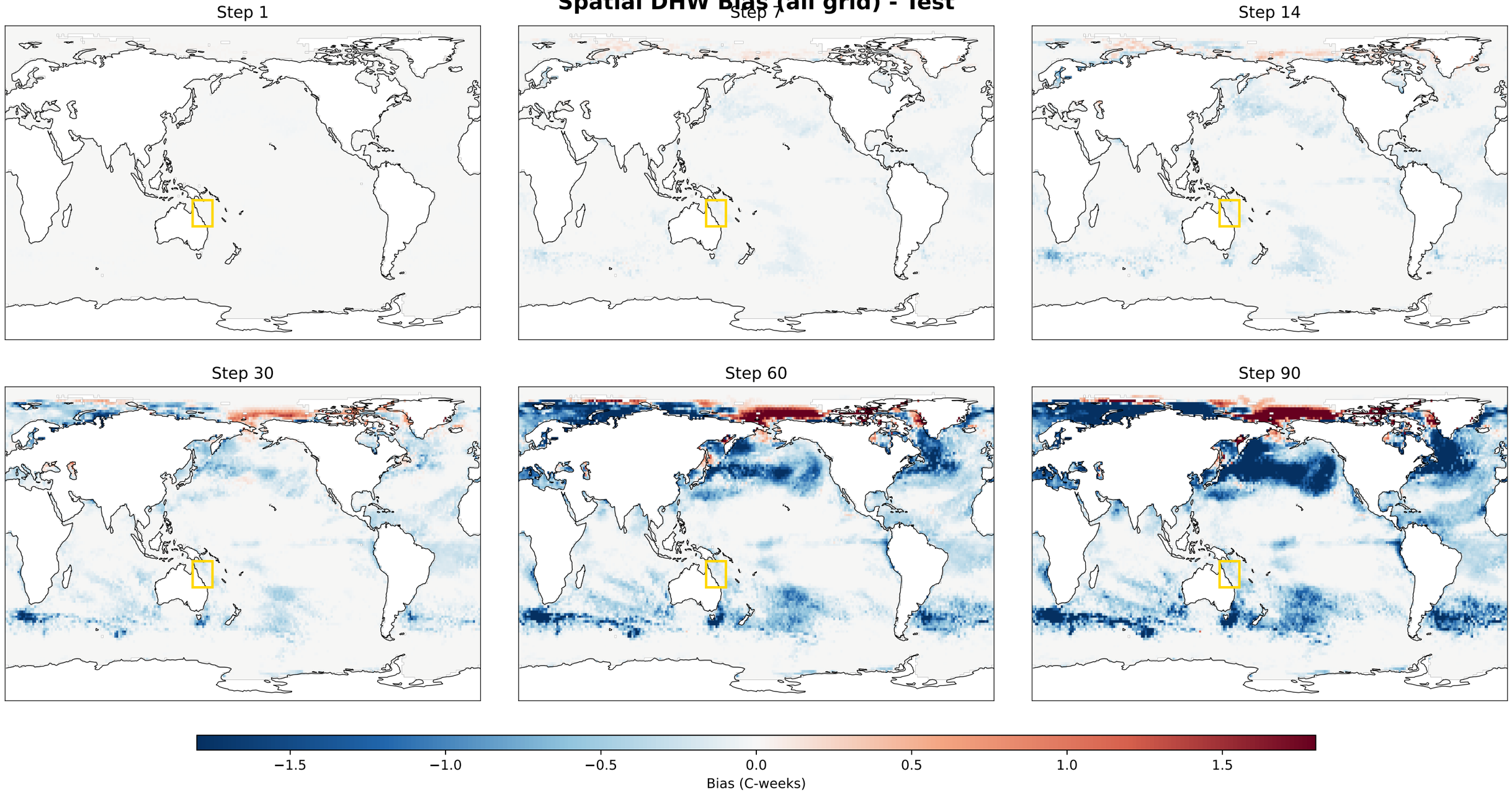


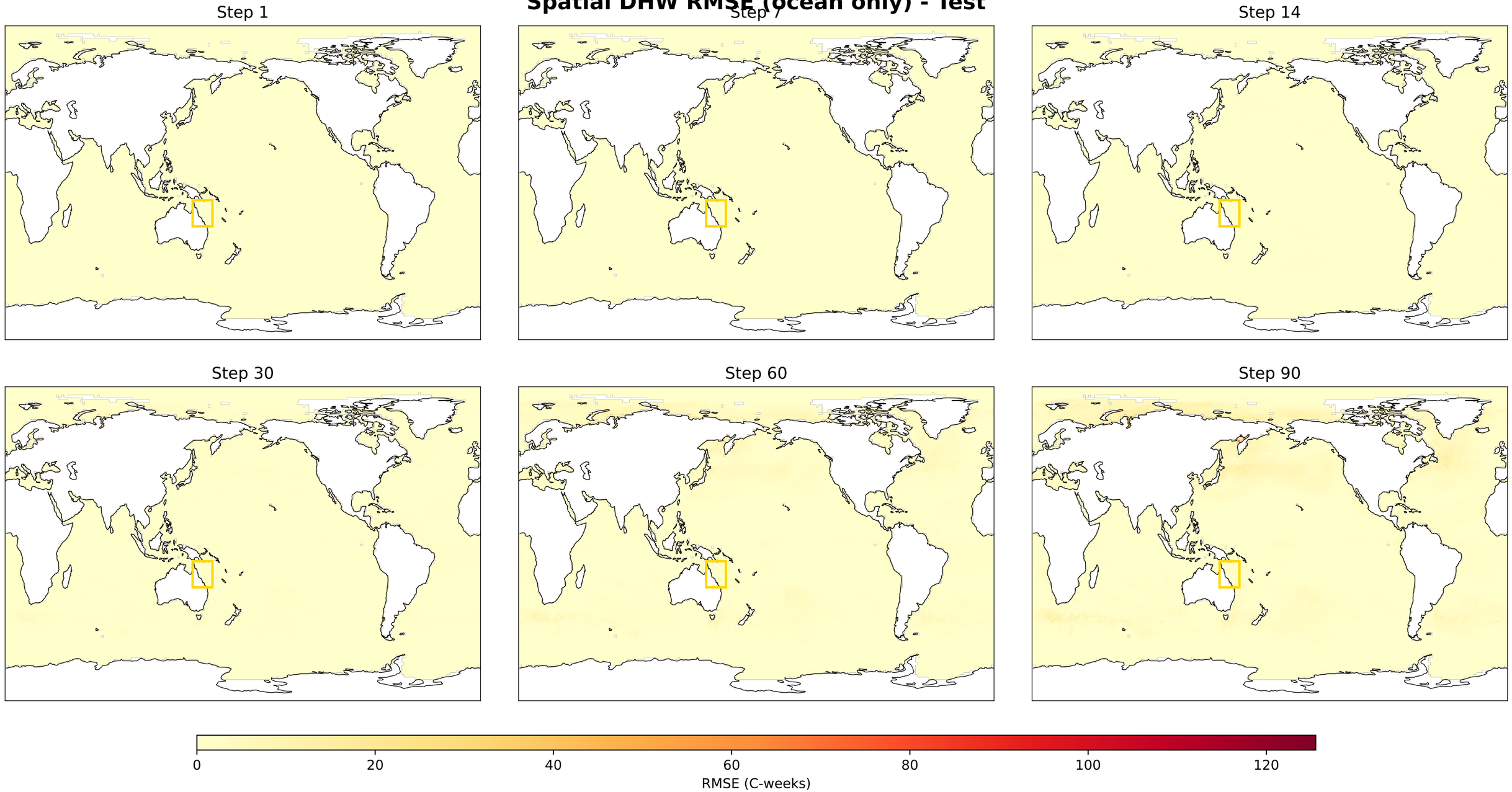


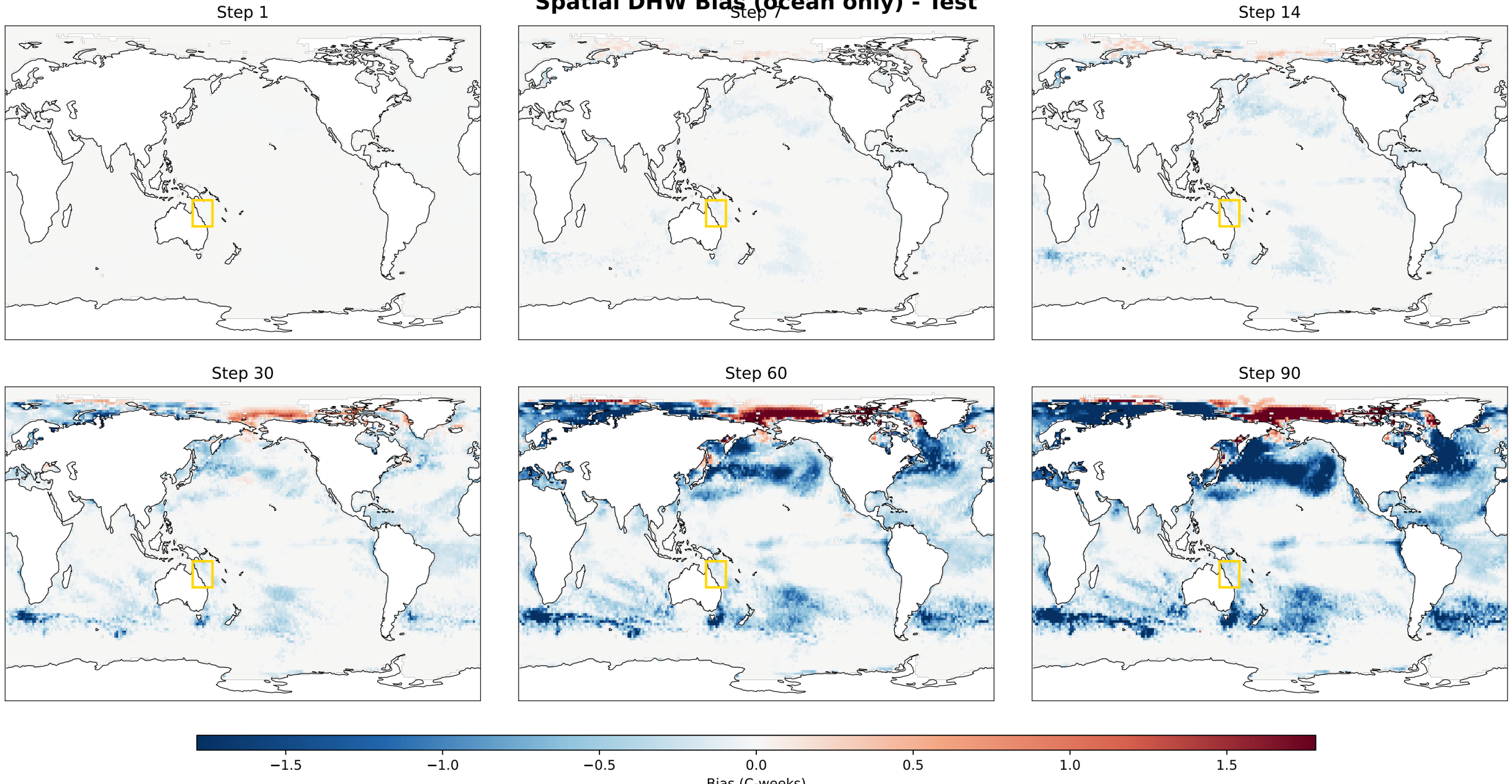





Spatial DHW RMSE (all grid) - Test
Step 1
Step 7
Step 14
Step 30
Step 60
Step 90
0
20
40
60
80
100
120
RMSE (C-weeks)

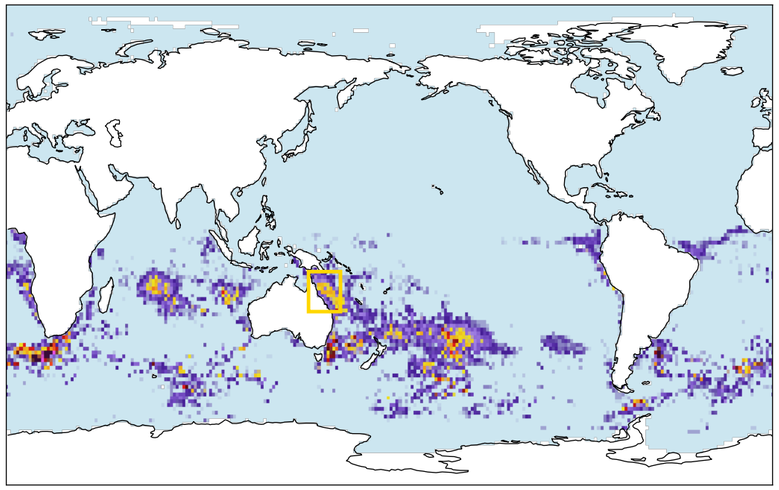

Spatial DHW Bias (all grid) - Test
Step 1
Step 7
Step 14
Step 30
Step 60
Step 90
−1.5
−1.0
−0.5
0.0
0.5
1.0
1.5
Bias (C-weeks)


Spatial DHW RMSE (ocean only) - Test
Step 1
Step 7
Step 14
Step 30
Step 60
Step 90
0
20
40
60
80
100
120
RMSE (C-weeks)

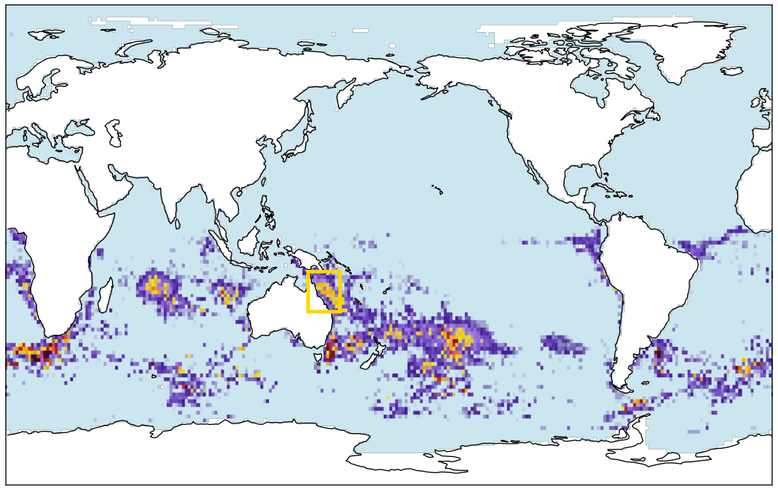

Spatial DHW Bias (ocean only) - Test
Step 1
Step 7
Step 14
Step 30
Step 60
Step 90
−1.5
−1.0
−0.5
0.0
0.5
1.0
1.5
Bias (C-weeks)

DHW Chain init: 2020-04-10 - Test (C-weeks)

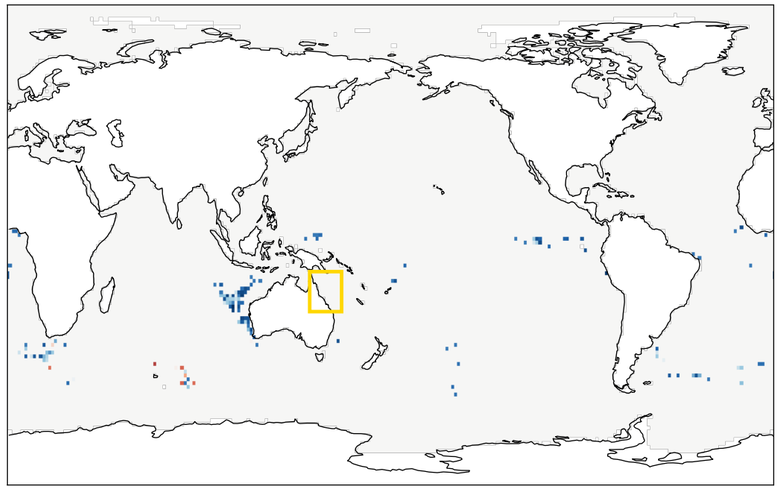
Actual DHW - Step 1



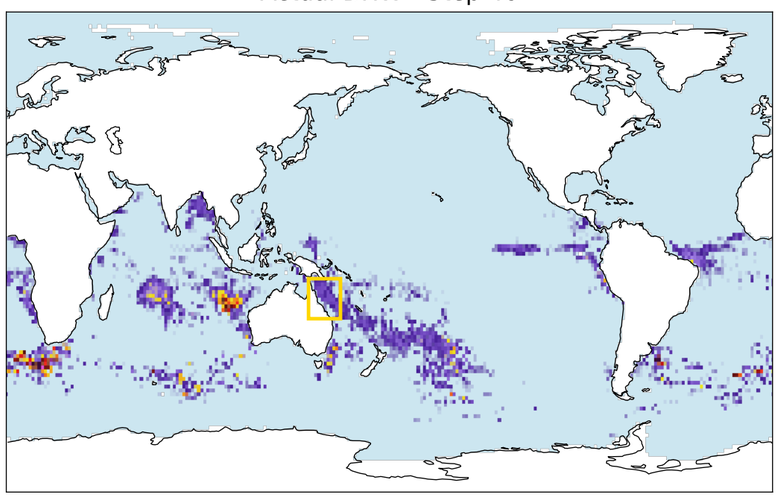
Predicted DHW - Step 1



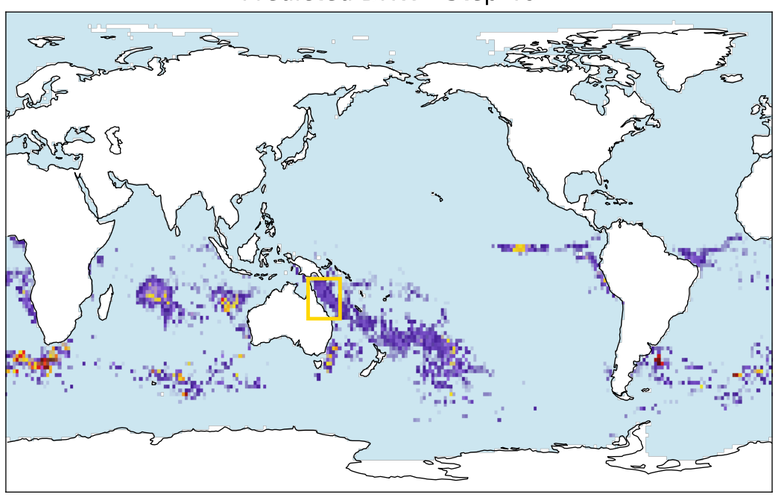
Error (Pred - Actual) - Step 1 [scale ±0.25]



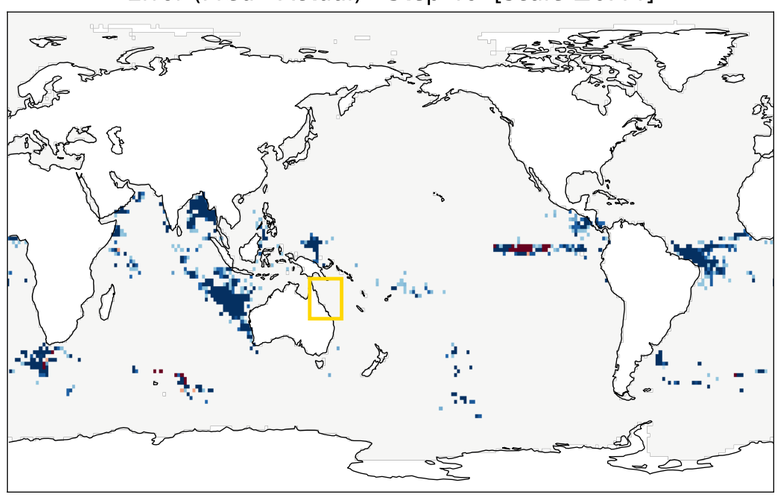
Actual DHW - Step 46



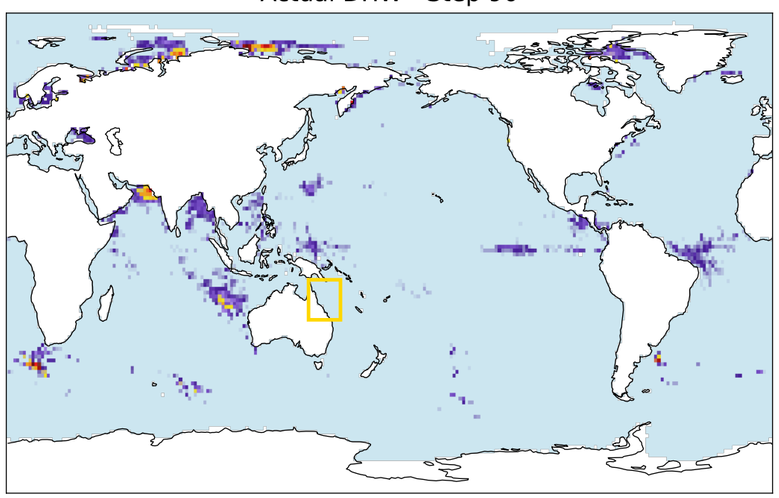
Predicted DHW - Step 46



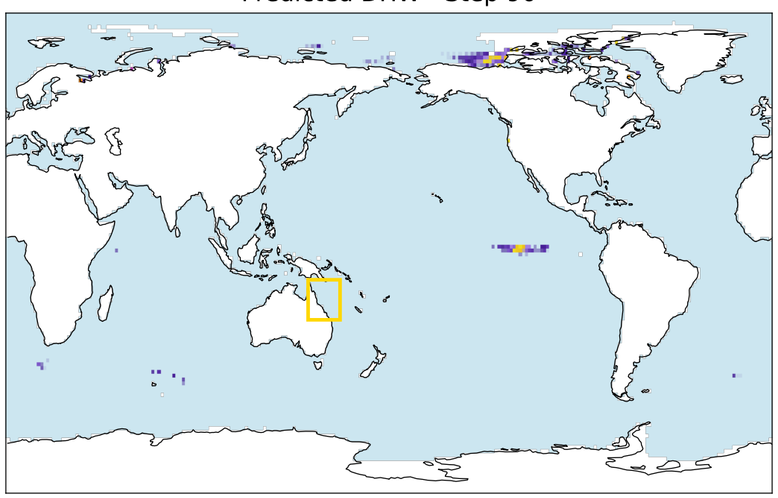
Error (Pred - Actual) - Step 46 [scale ±0.44]



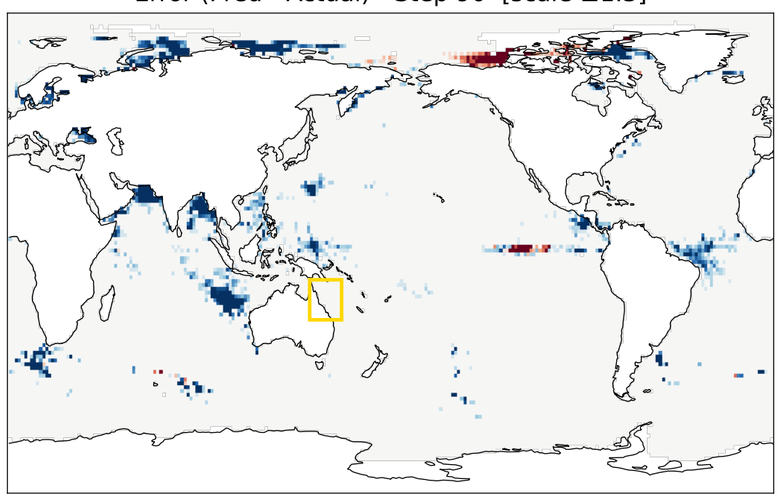
Actual DHW - Step 90

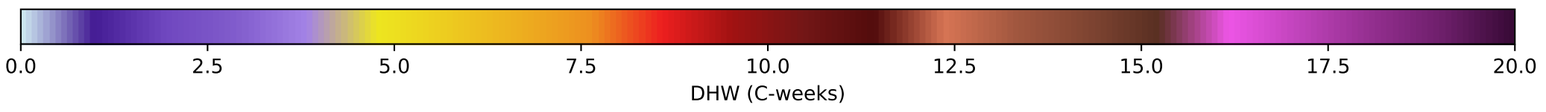


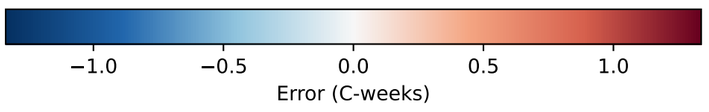
Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±1.3]

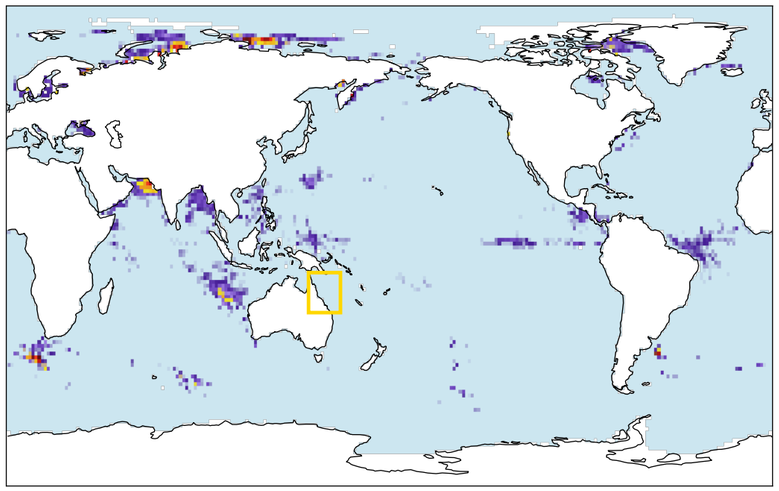




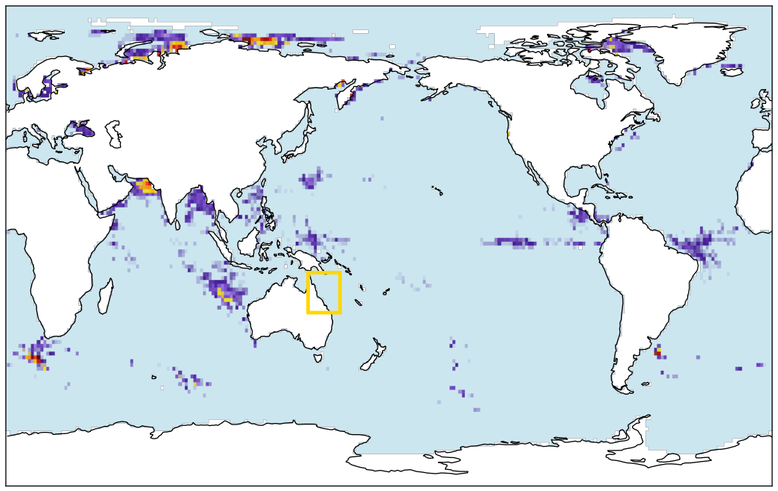


DHW Chain init: 2020-07-09 - Test (C-weeks)

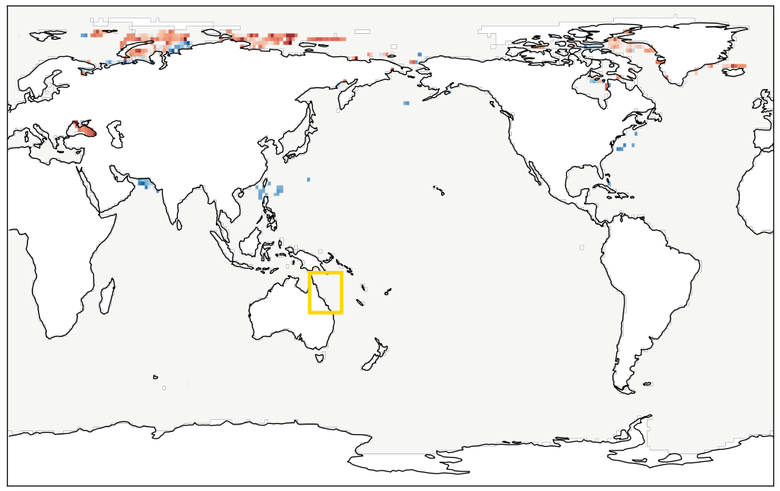
Actual DHW - Step 1



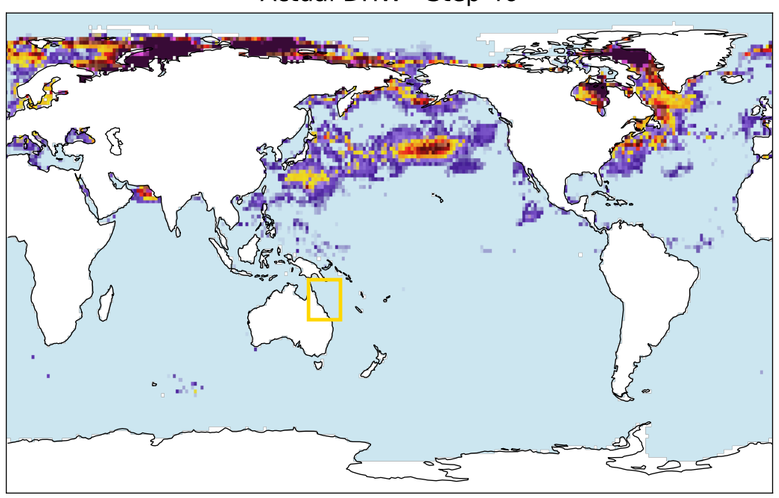
Predicted DHW - Step 1



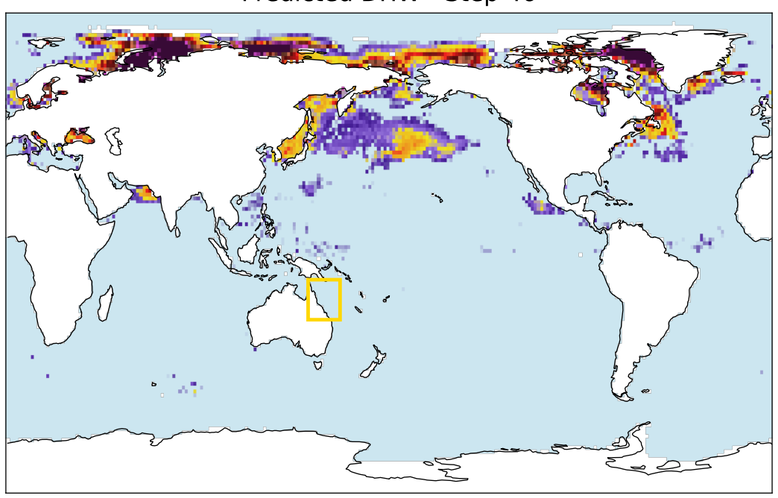
Error (Pred - Actual) - Step 1 [scale ±0.36]



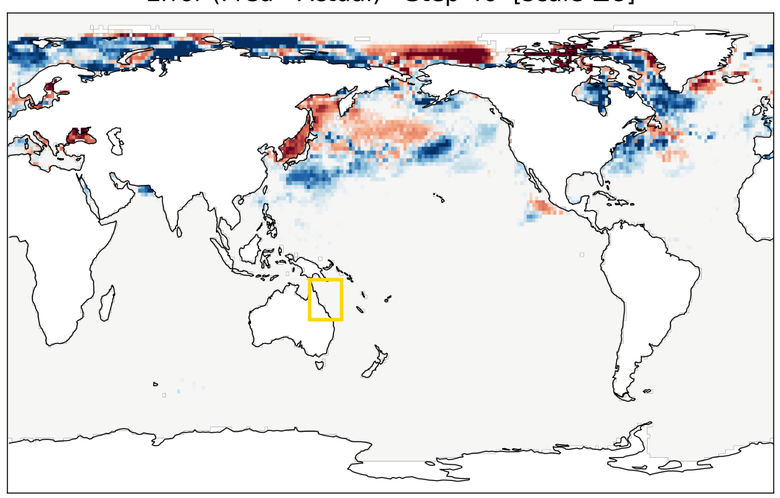
Actual DHW - Step 46



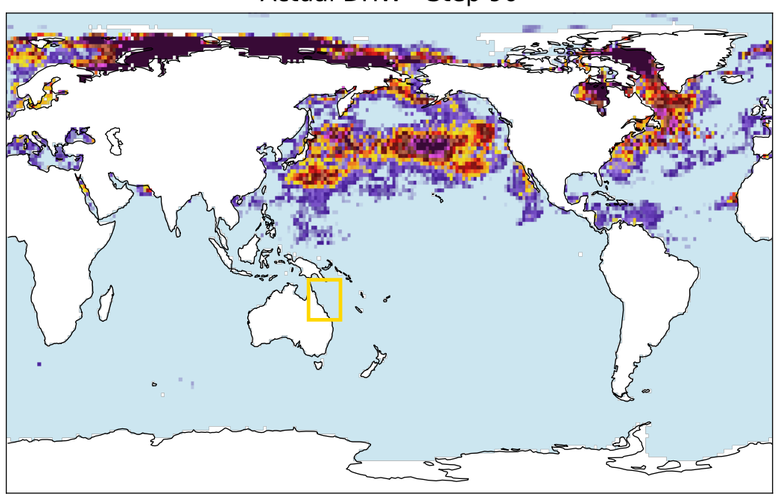
Predicted DHW - Step 46



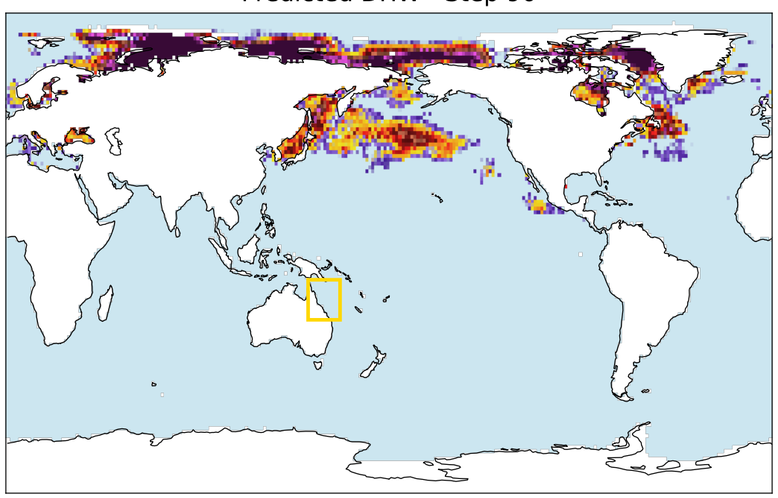
Error (Pred - Actual) - Step 46 [scale ±8]



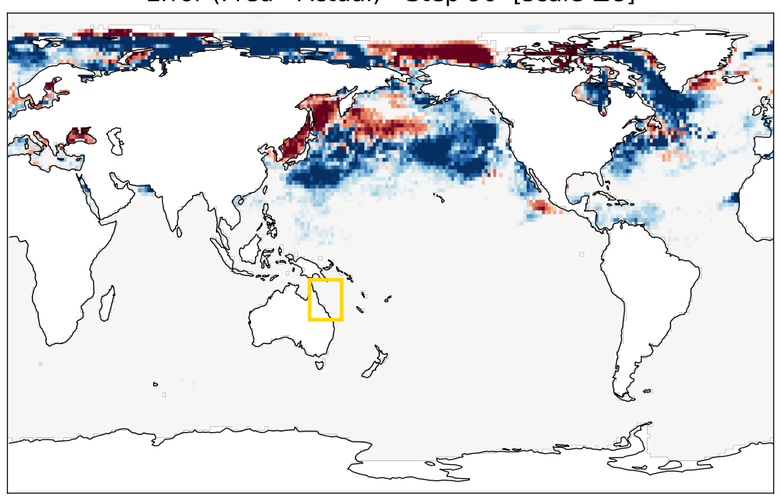
Actual DHW - Step 90

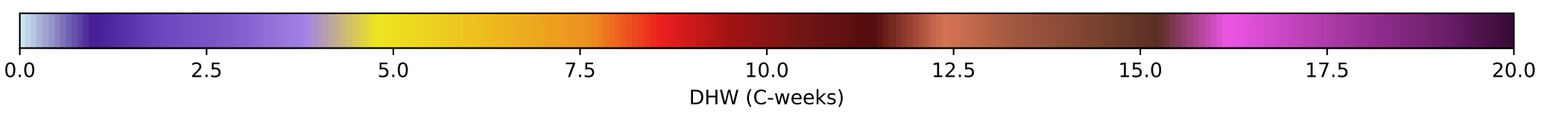


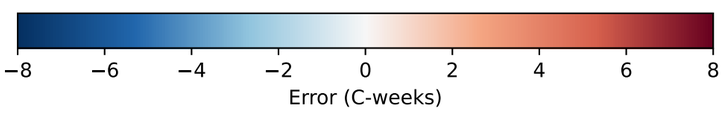
Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±8]

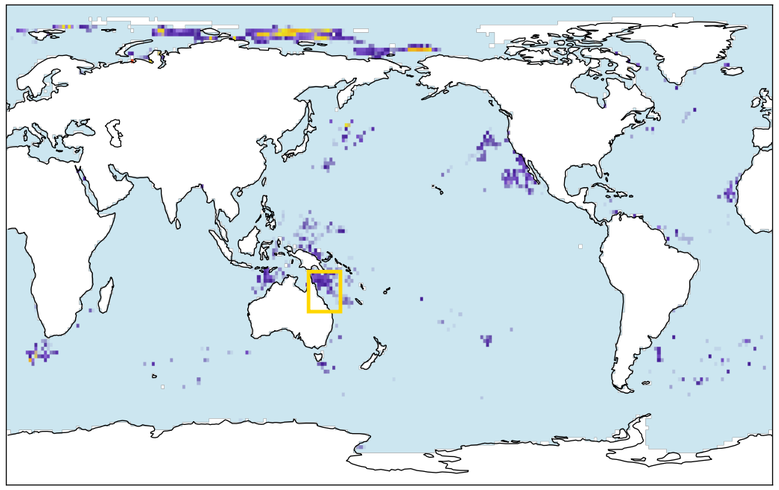




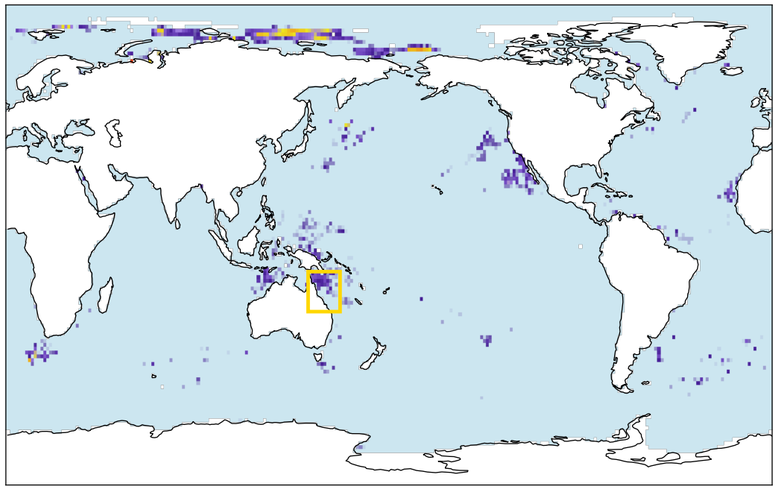


DHW Chain init: 2021-01-06 - Test (C-weeks)

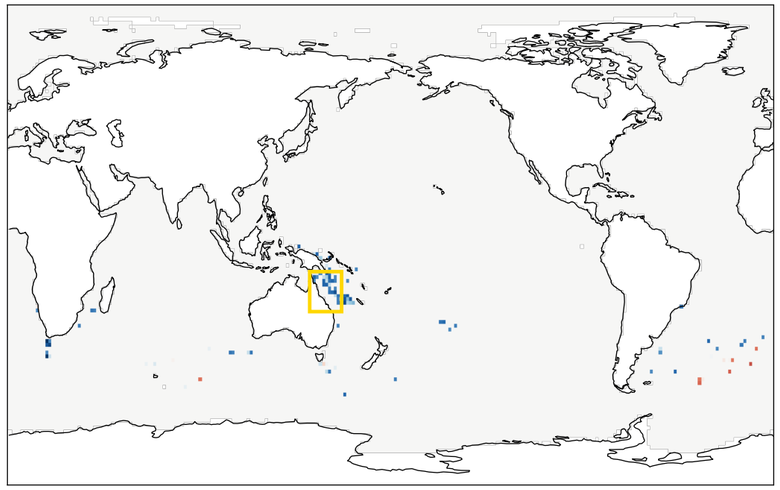
Actual DHW - Step 1



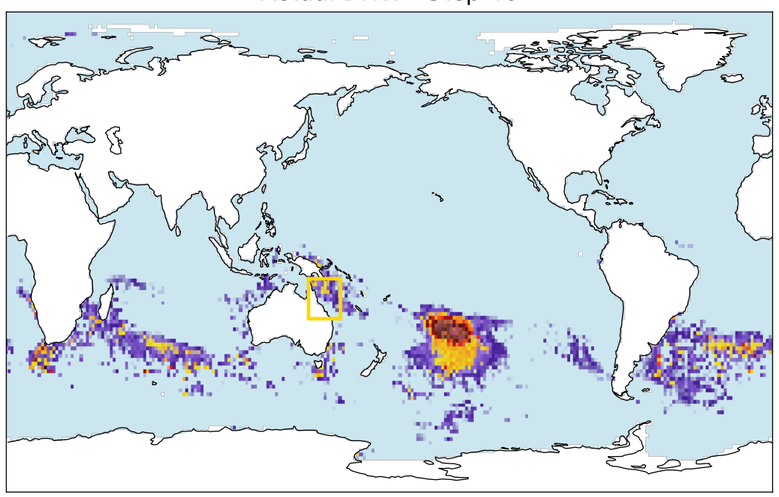
Predicted DHW - Step 1



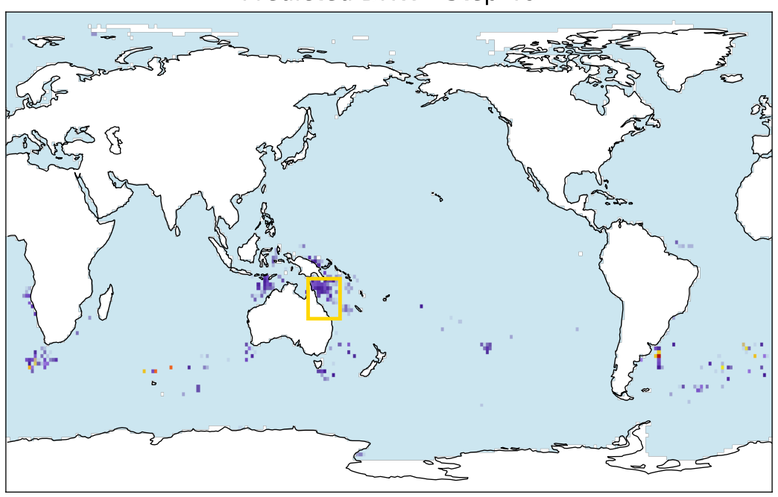
Error (Pred - Actual) - Step 1 [scale ±0.27]



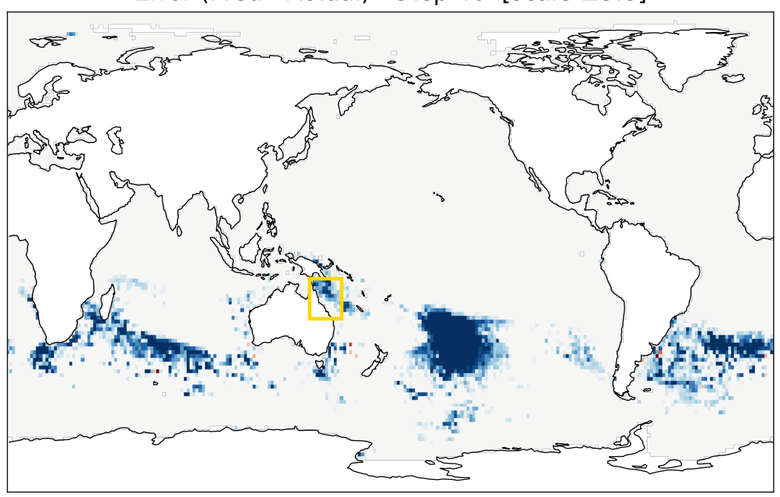
Actual DHW - Step 46



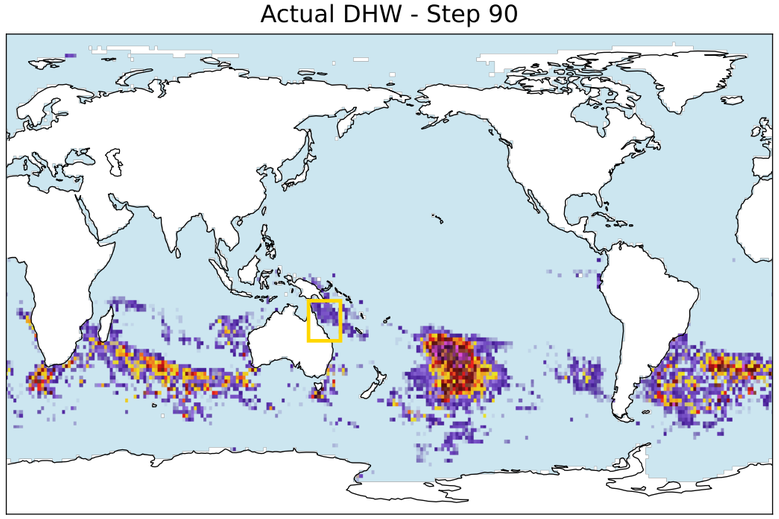
Predicted DHW - Step 46



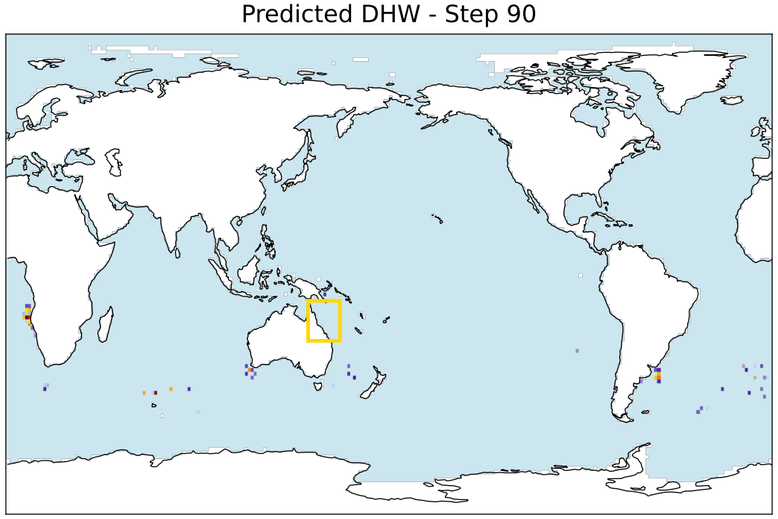
Error (Pred - Actual) - Step 46 [scale ±3.8]



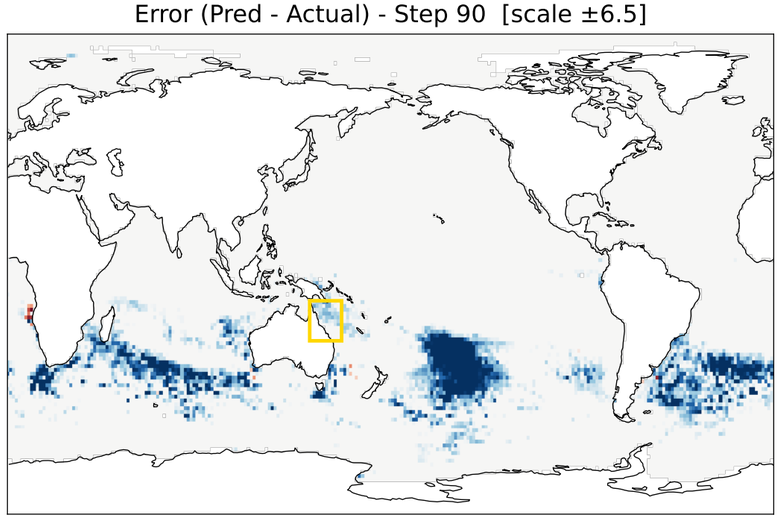
Actual DHW - Step 90

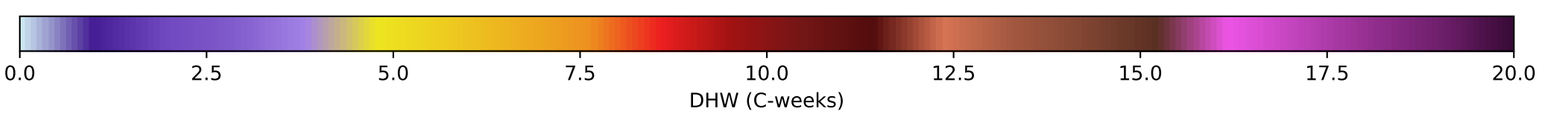


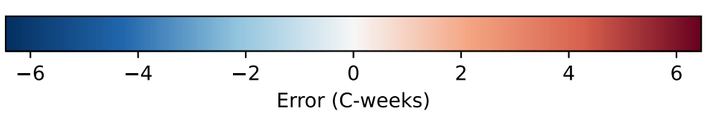
Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±6.5]

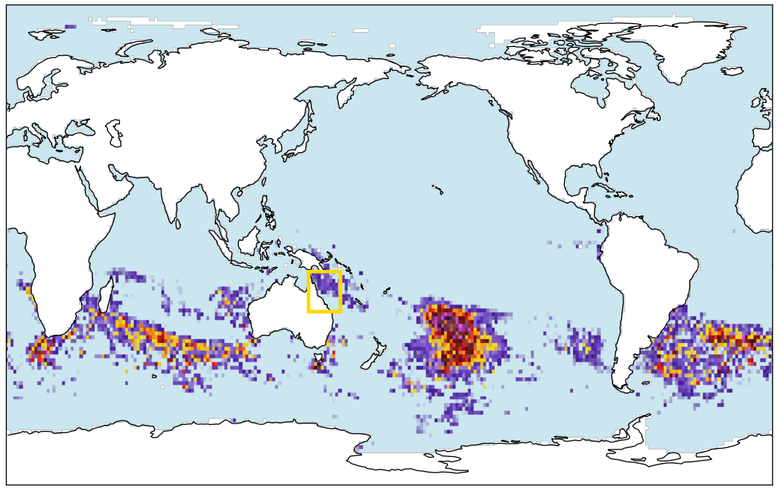




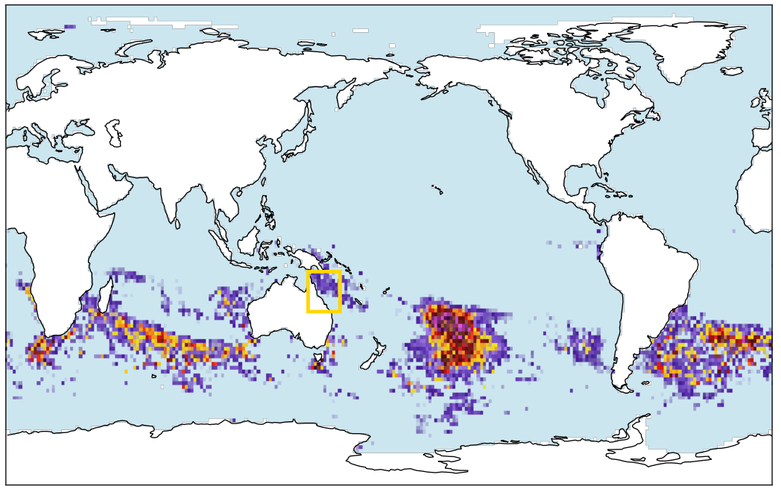


DHW Chain init: 2021-04-06 - Test (C-weeks)

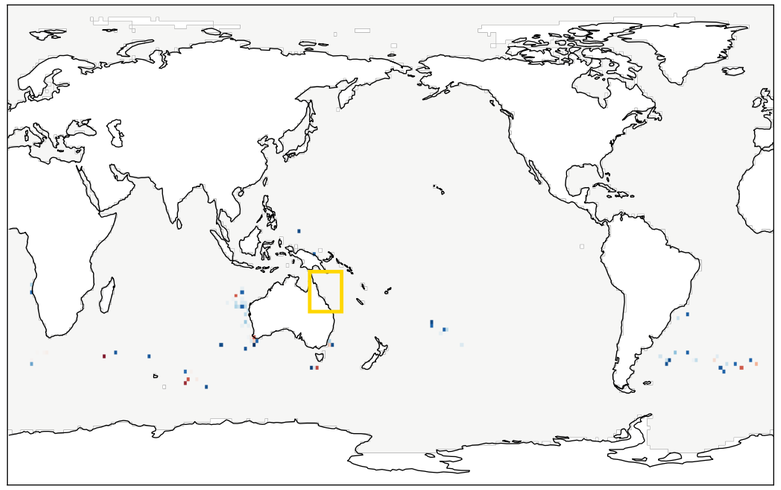
Actual DHW - Step 1



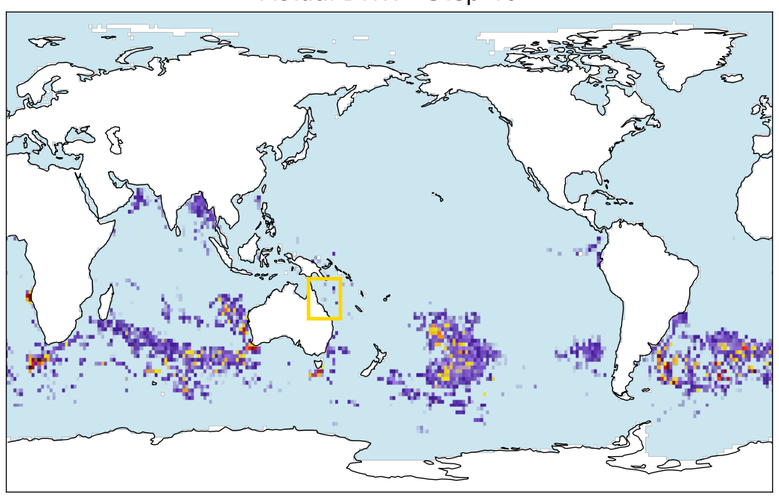
Predicted DHW - Step 1



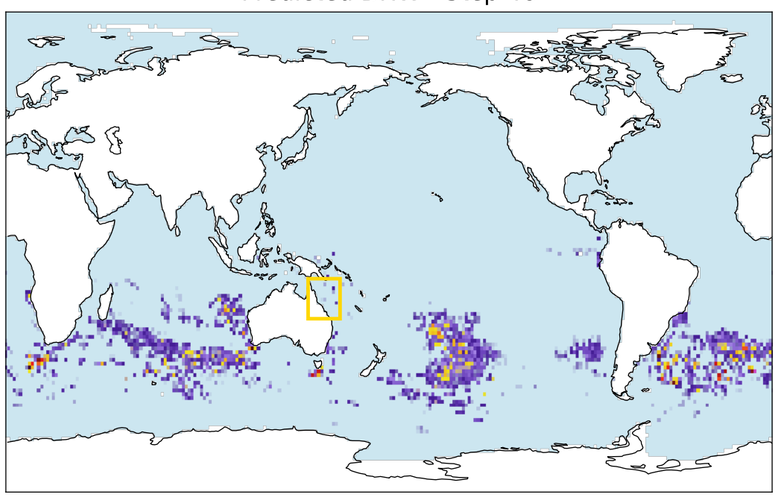
Error (Pred - Actual) - Step 1 [scale ±0.22]



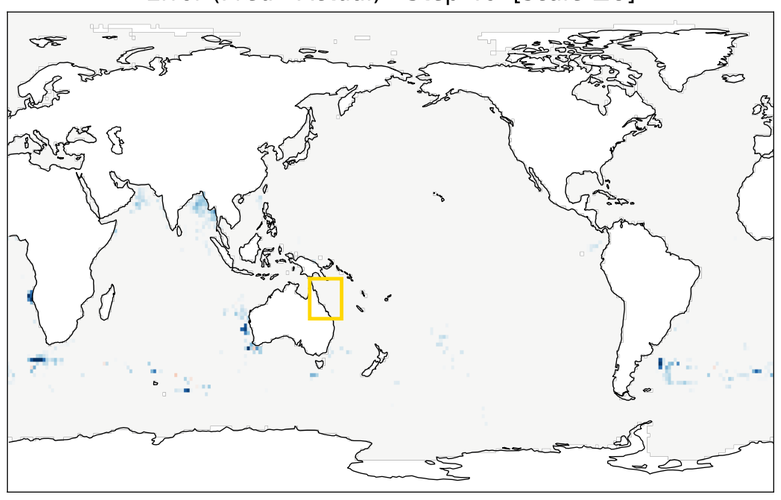
Actual DHW - Step 46



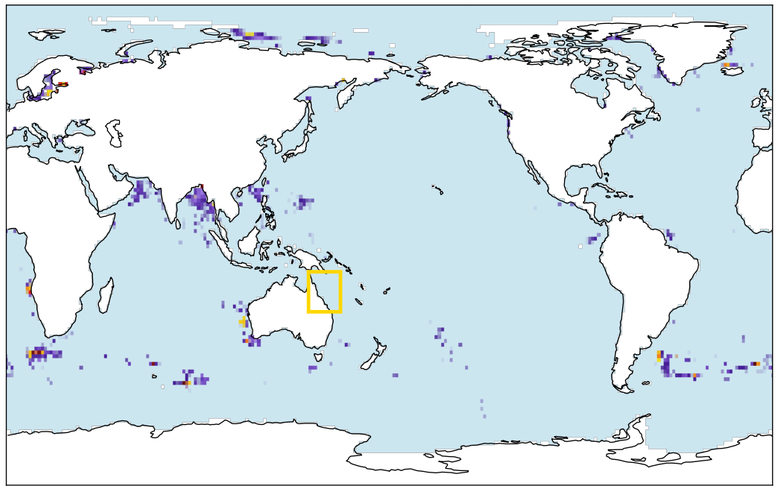
Predicted DHW - Step 46



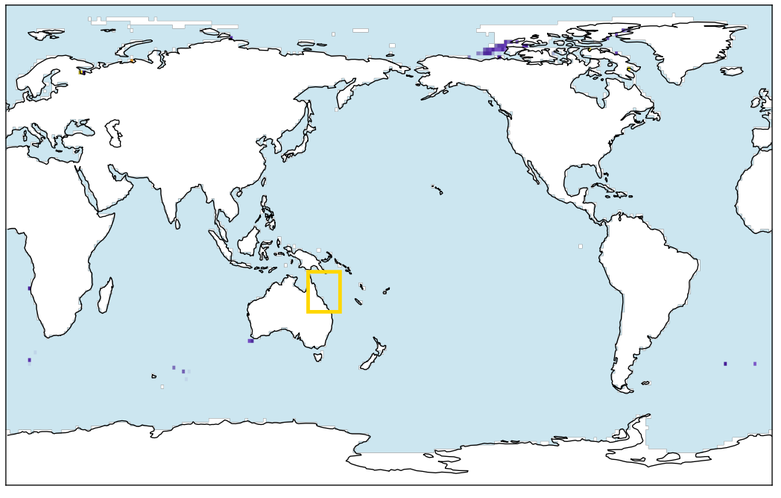
Error (Pred - Actual) - Step 46 [scale ±8]



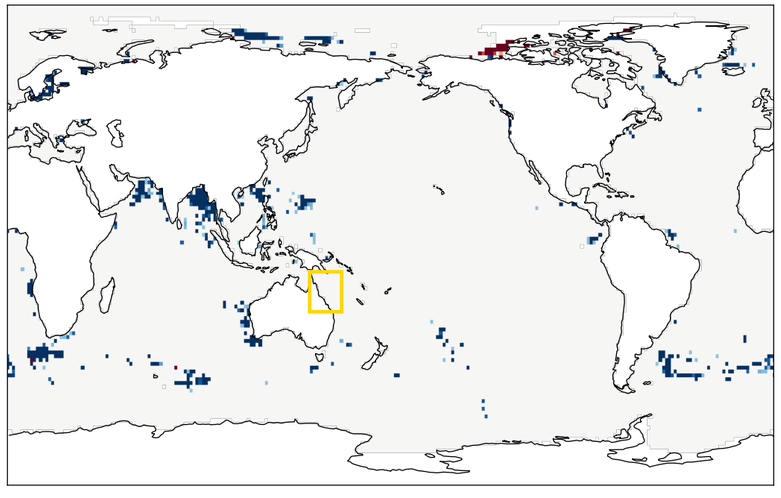
Actual DHW - Step 90

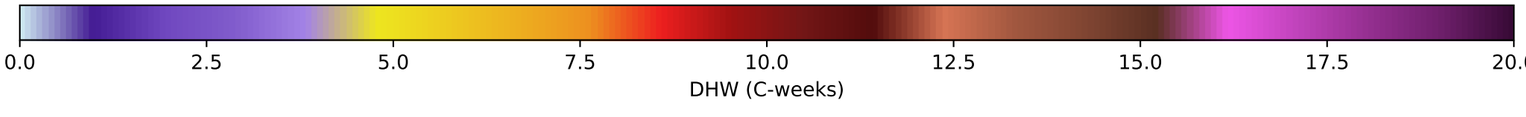


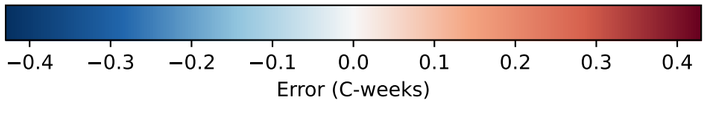
Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±0.43]

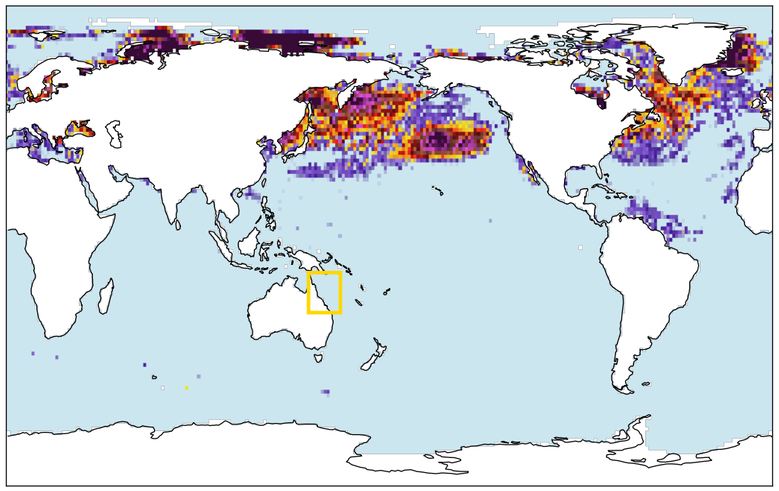




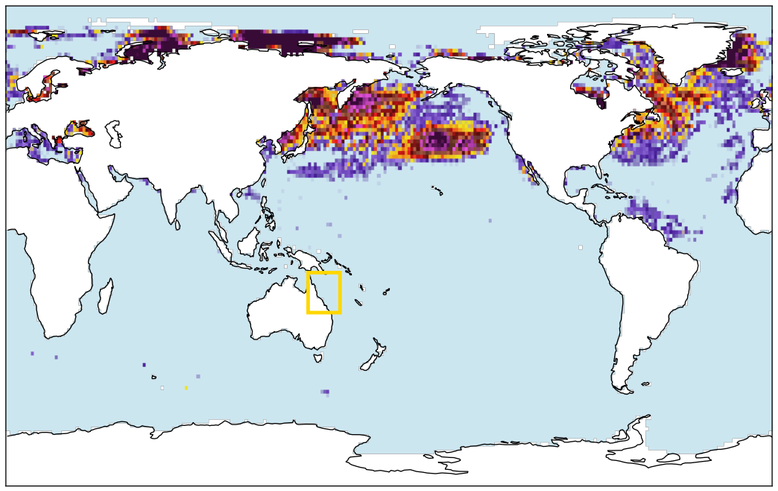


DHW Chain init: 2021-10-03 - Test (C-weeks)

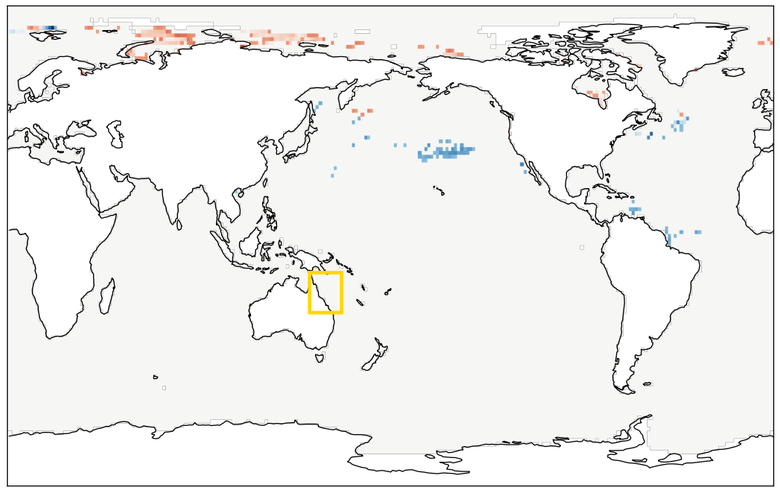
Actual DHW - Step 1



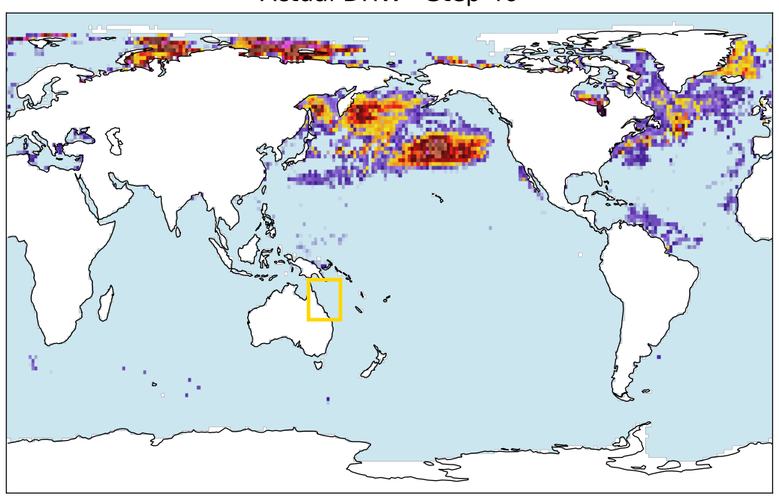
Predicted DHW - Step 1



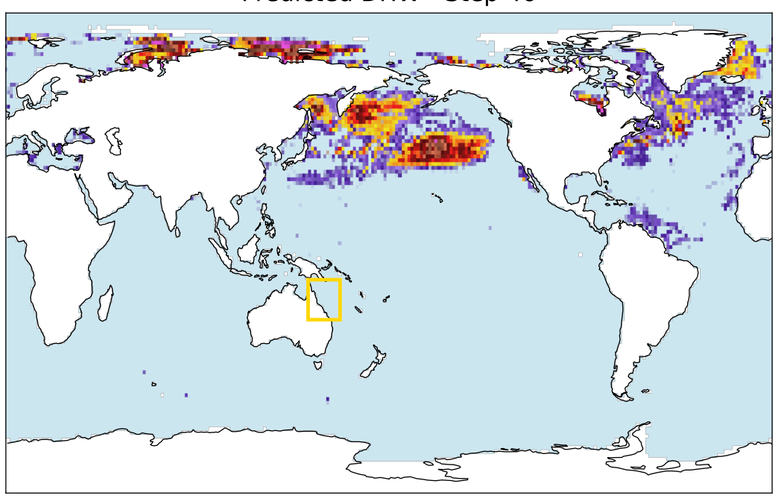
Error (Pred - Actual) - Step 1 [scale ±0.4]



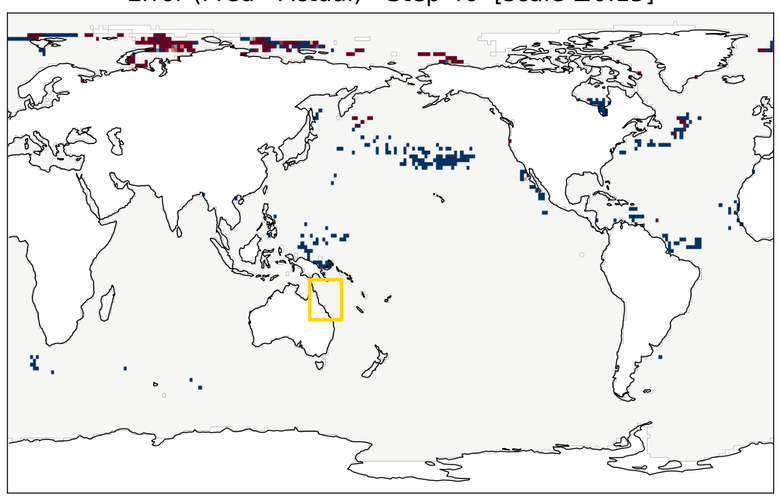
Actual DHW - Step 46



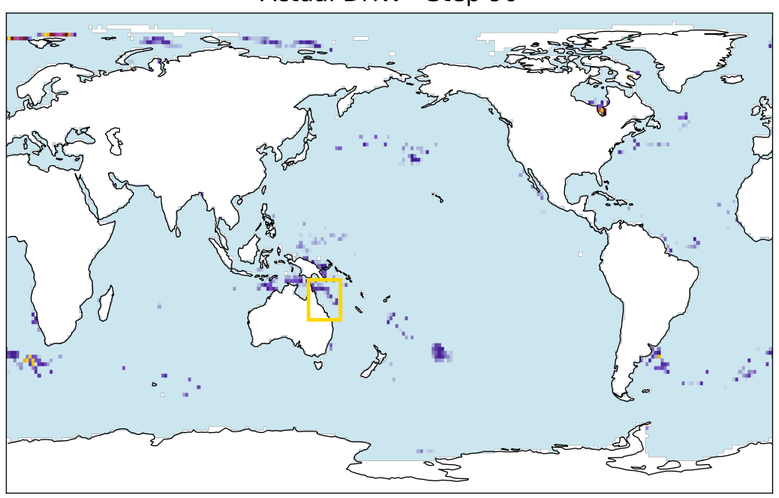
Predicted DHW - Step 46



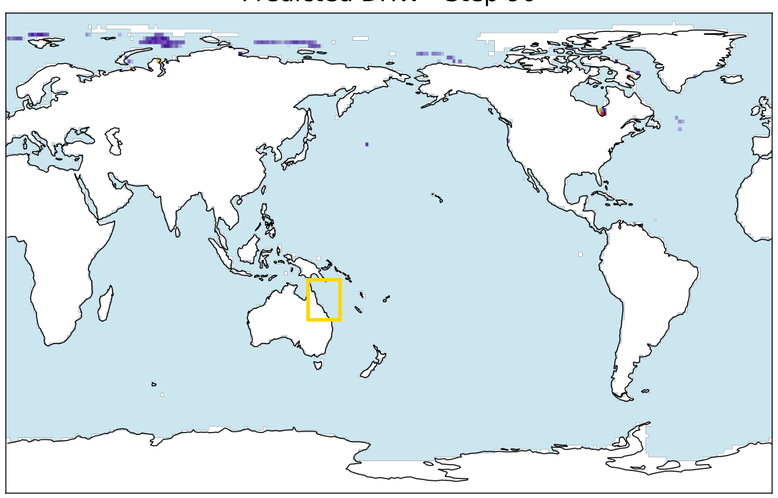
Error (Pred - Actual) - Step 46 [scale ±0.15]



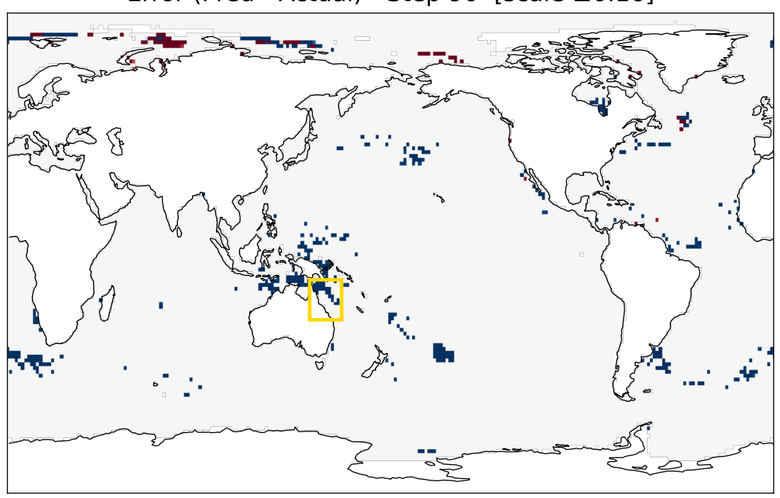
Actual DHW - Step 90

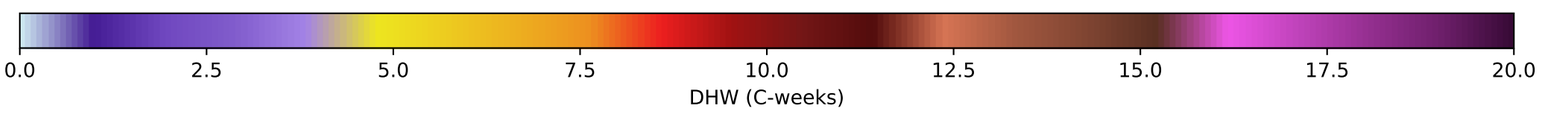


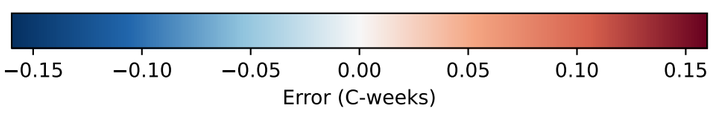
Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±0.16]

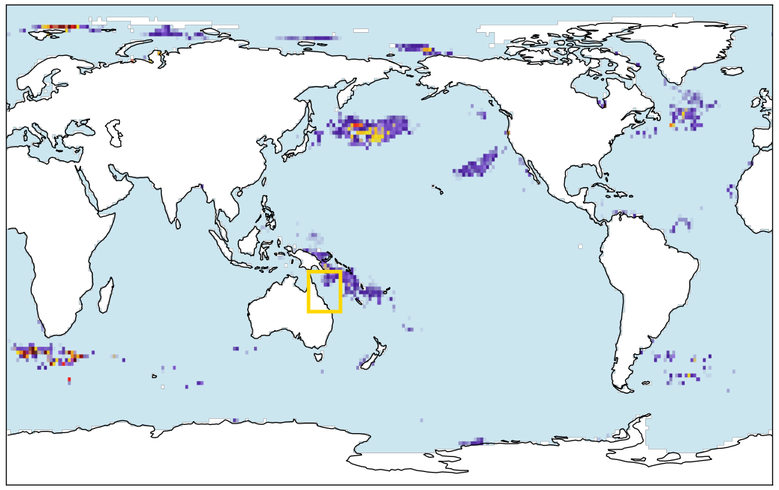




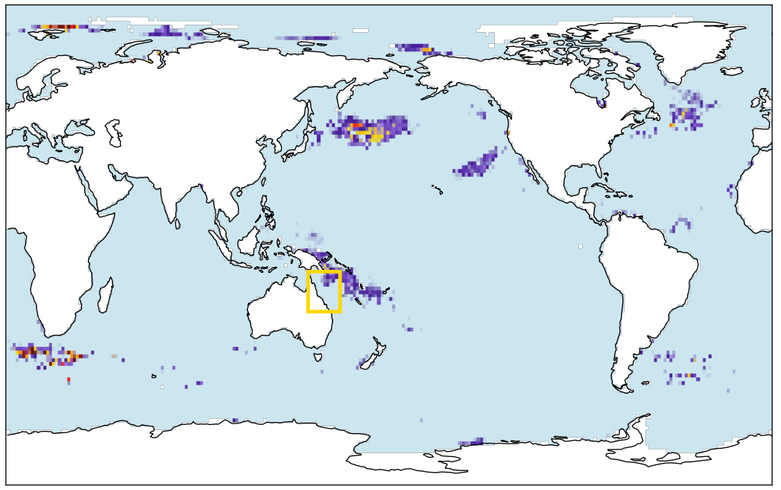


DHW Chain init: 2022-12-27 - Test (C-weeks)

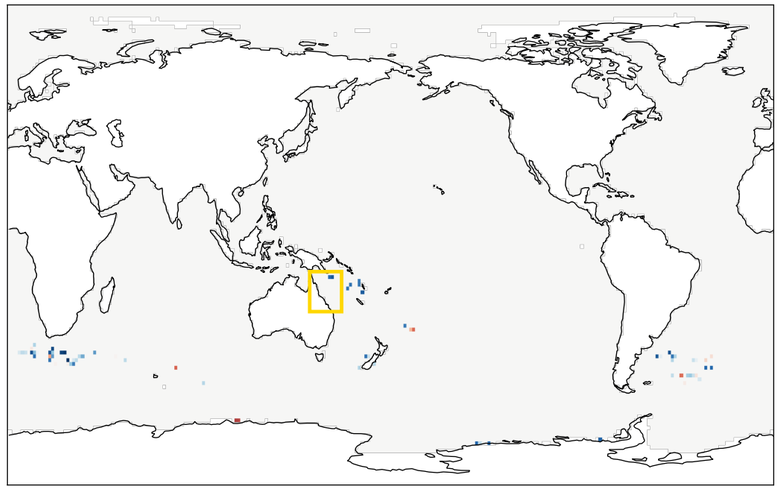
Actual DHW - Step 1



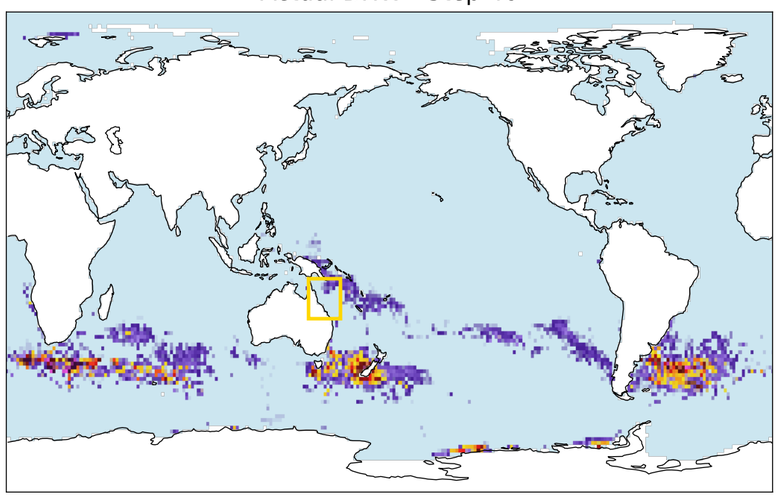
Predicted DHW - Step 1



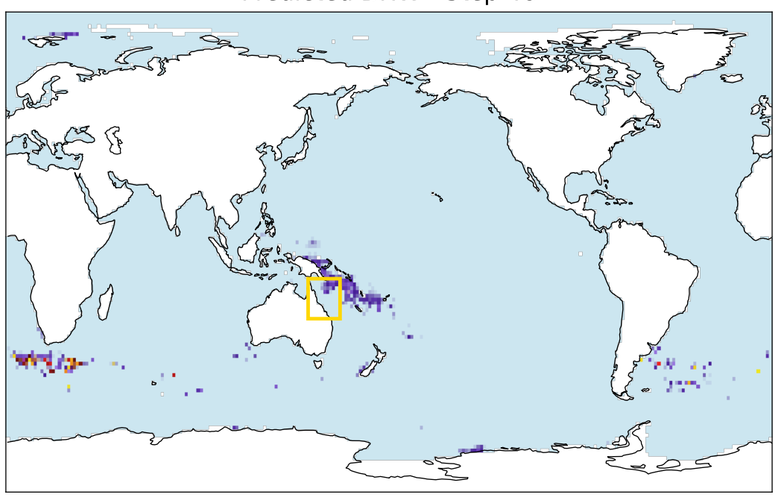
Error (Pred - Actual) - Step 1 [scale ±0.24]



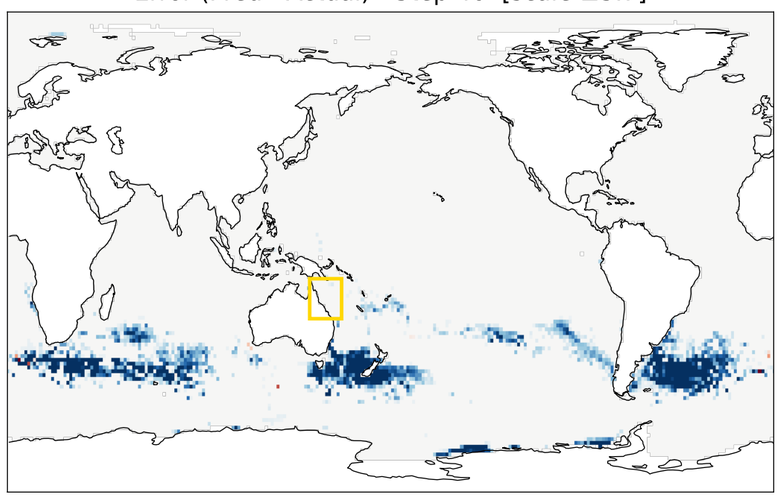
Actual DHW - Step 46



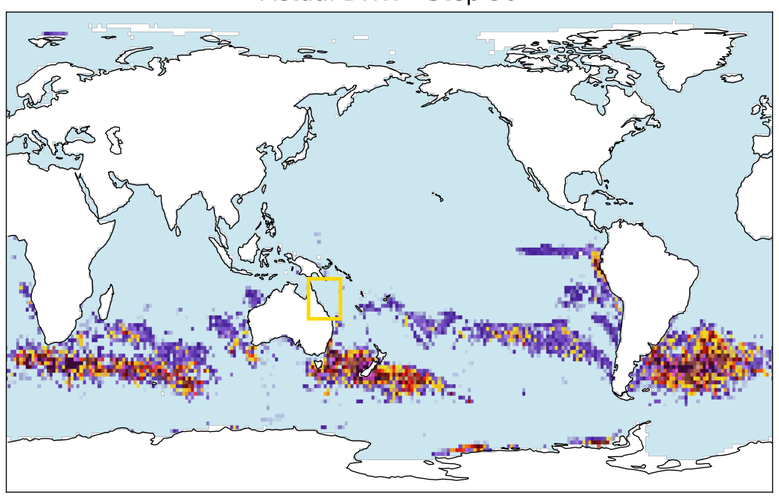
Predicted DHW - Step 46



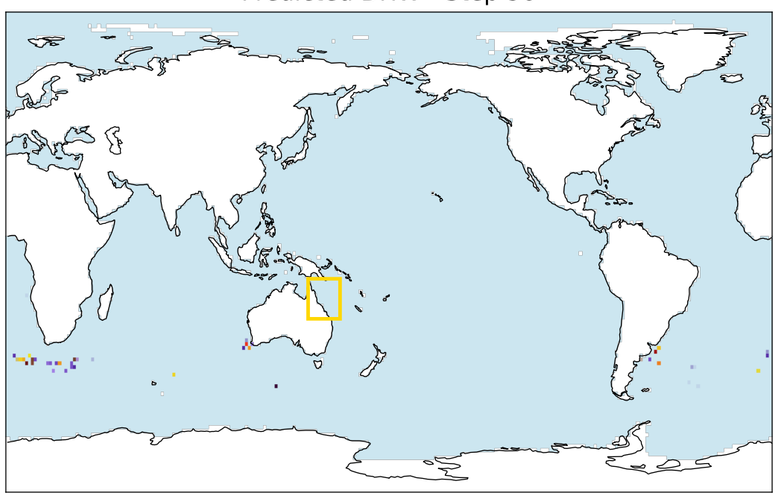
Error (Pred - Actual) - Step 46 [scale ±3.7]



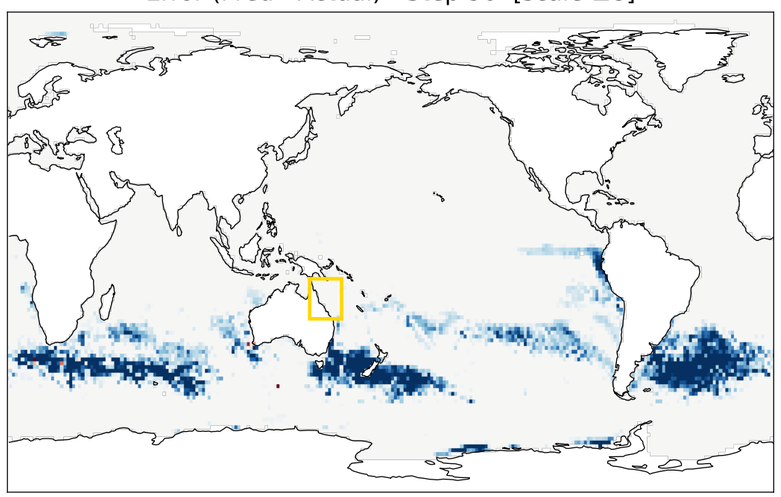
Actual DHW - Step 90

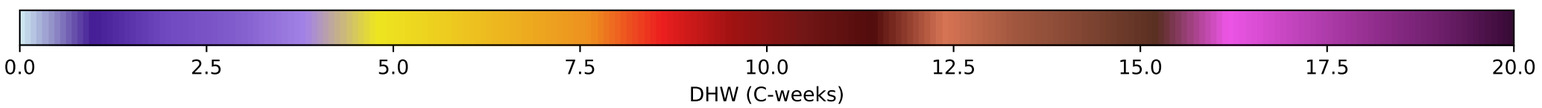


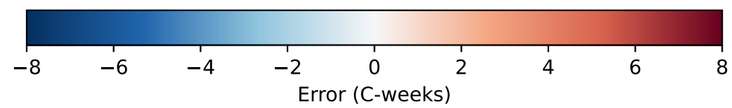
Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±8]

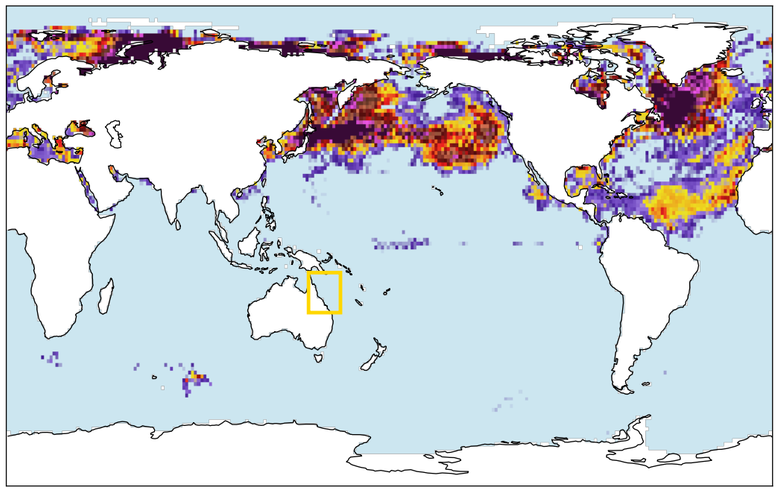




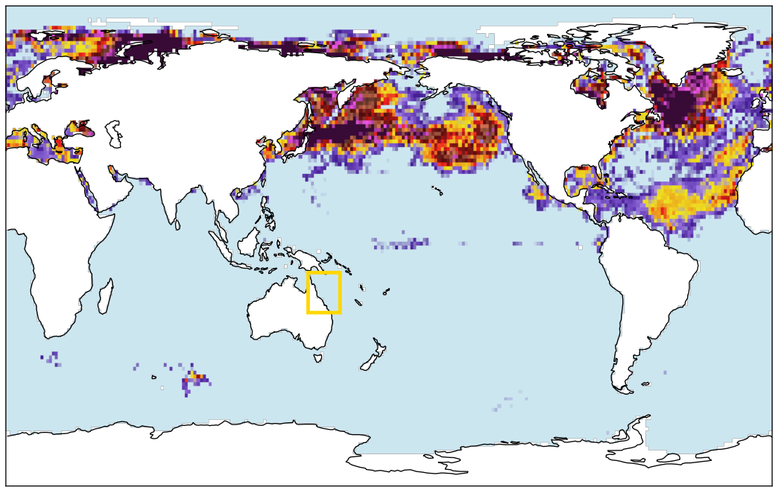


DHW Chain init: 2023-09-23 - Test (C-weeks)

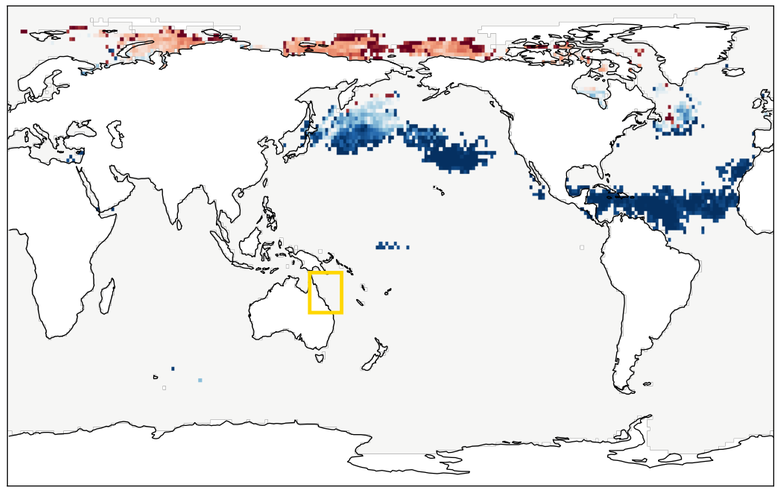
Actual DHW - Step 1



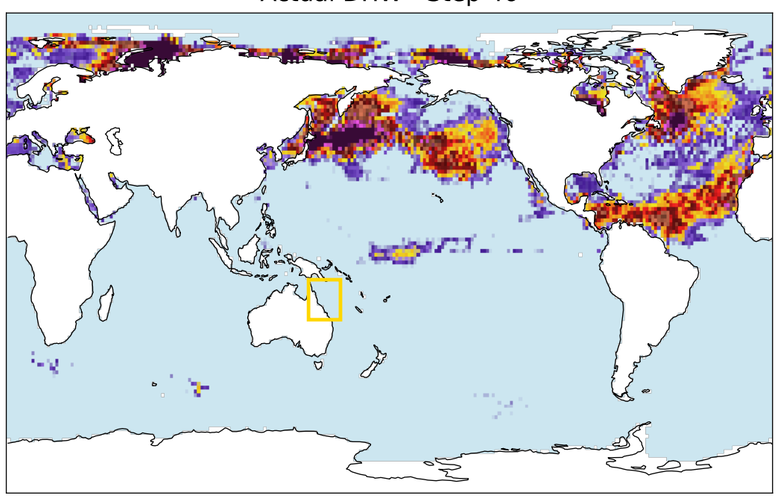
Predicted DHW - Step 1



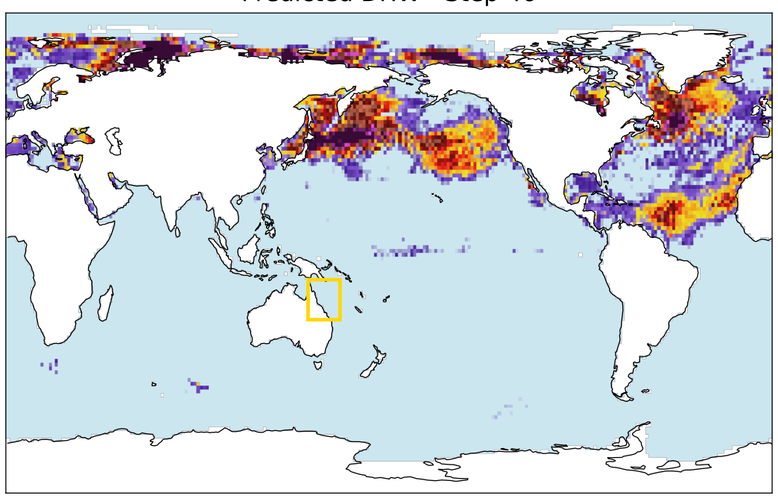
Error (Pred - Actual) - Step 1 [scale ±0.18]



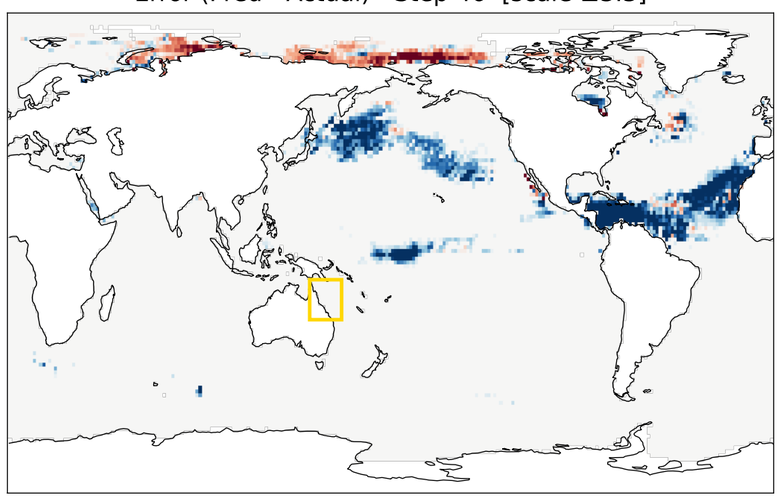
Actual DHW - Step 46



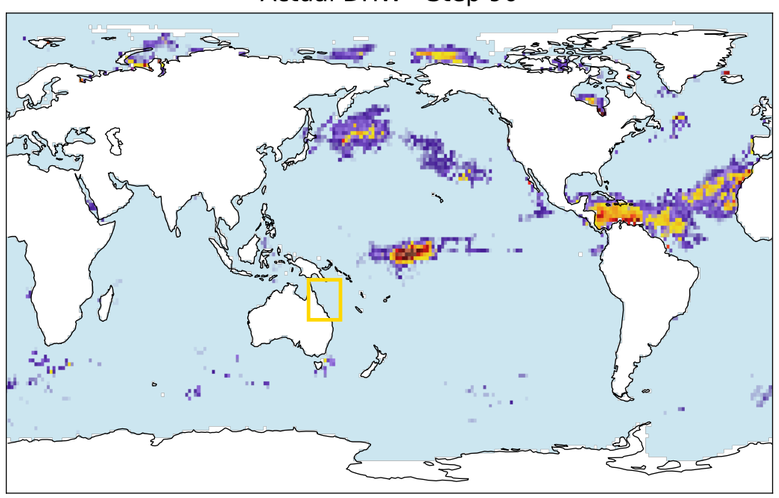
Predicted DHW - Step 46



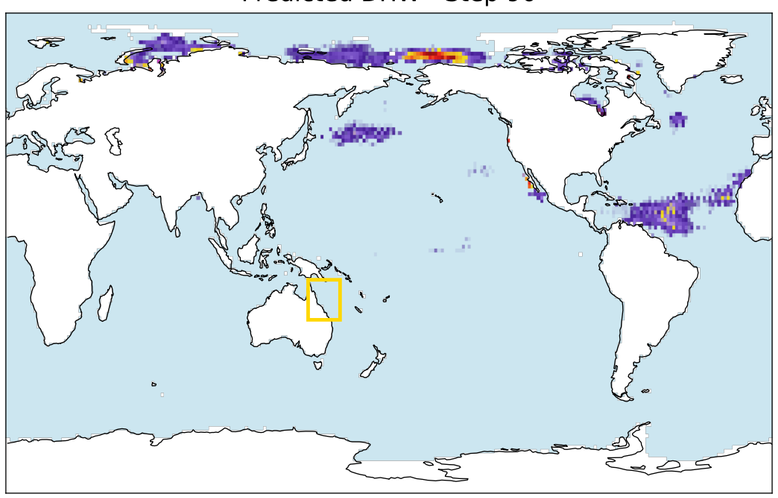
Error (Pred - Actual) - Step 46 [scale ±3.3]



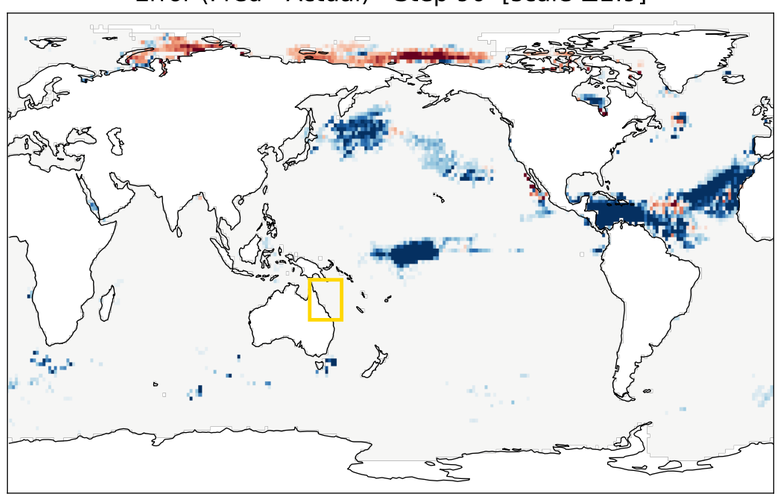
Actual DHW - Step 90

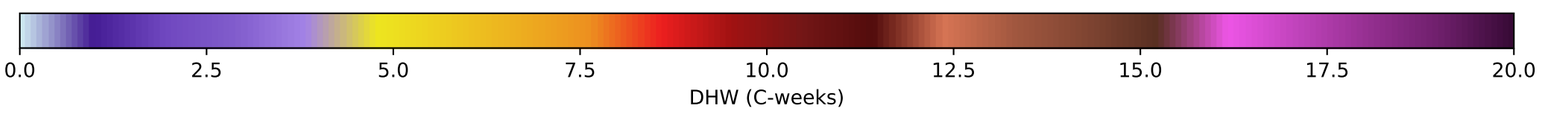


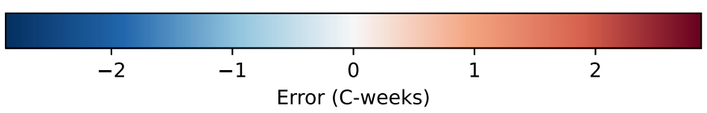
Predicted DHW - Step 90



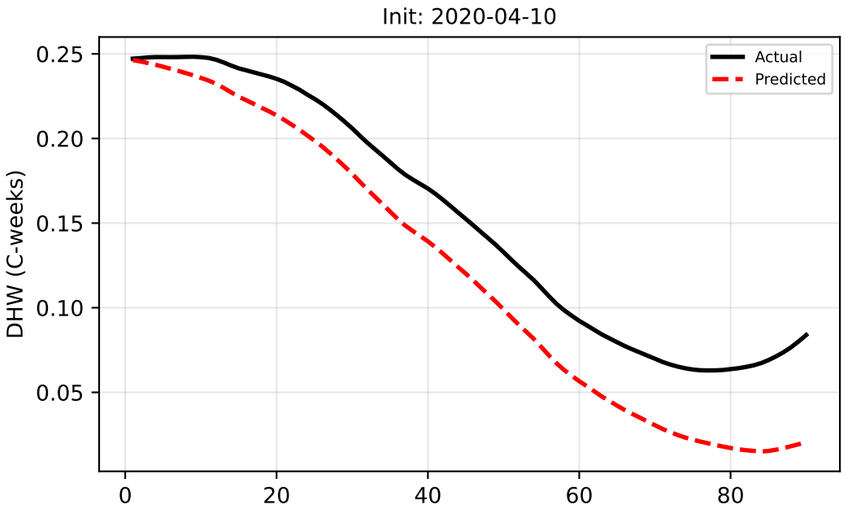
Error (Pred - Actual) - Step 90 [scale ±2.9]

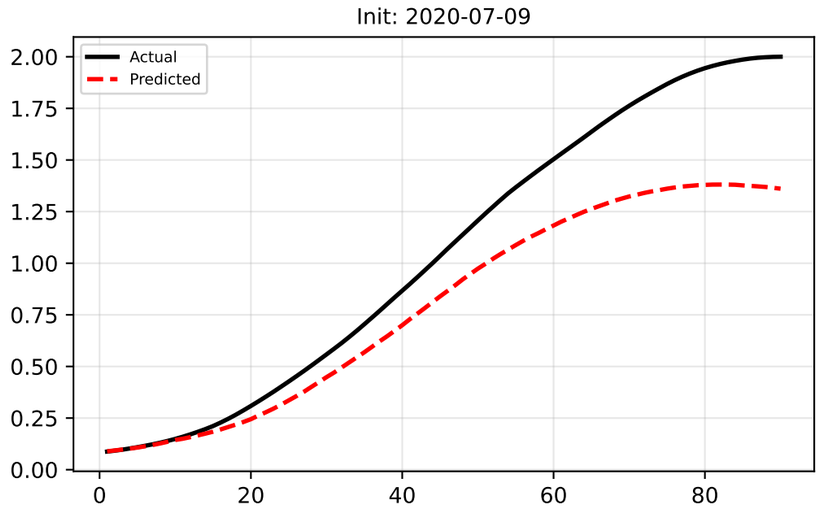


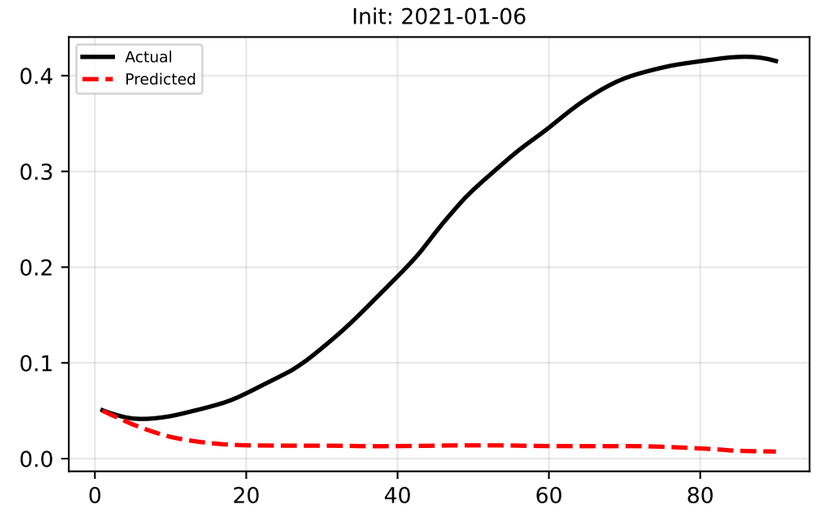


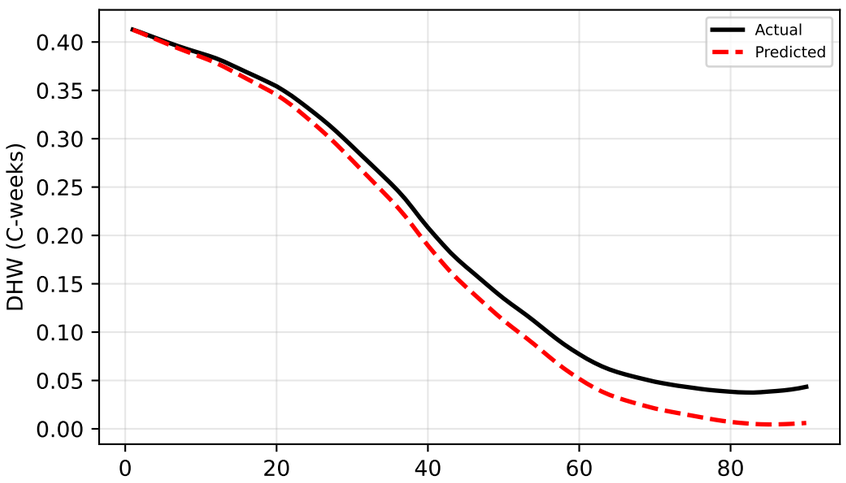


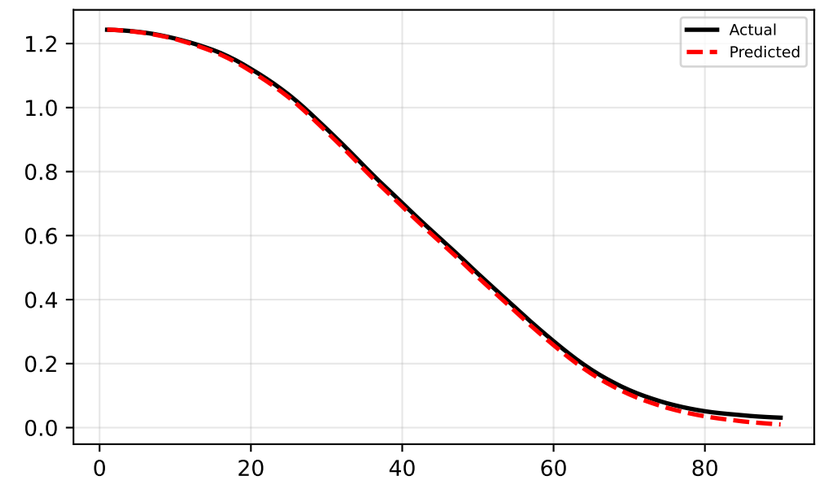
Global-Mean DHW - Test (C-weeks)

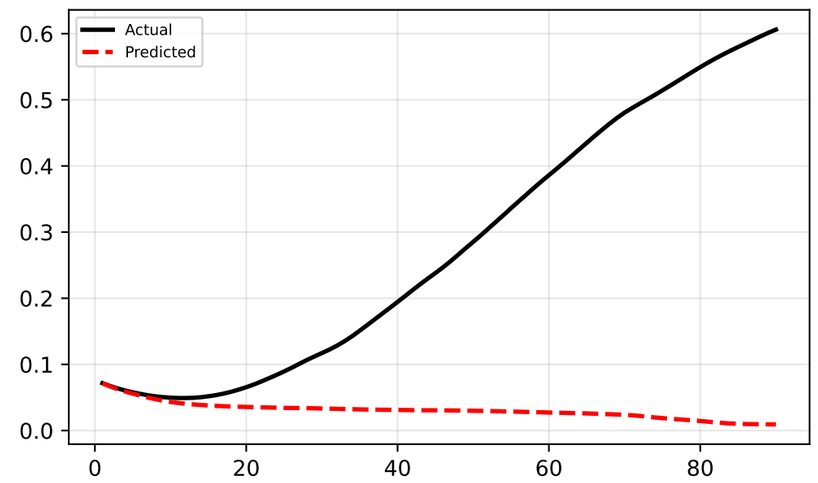


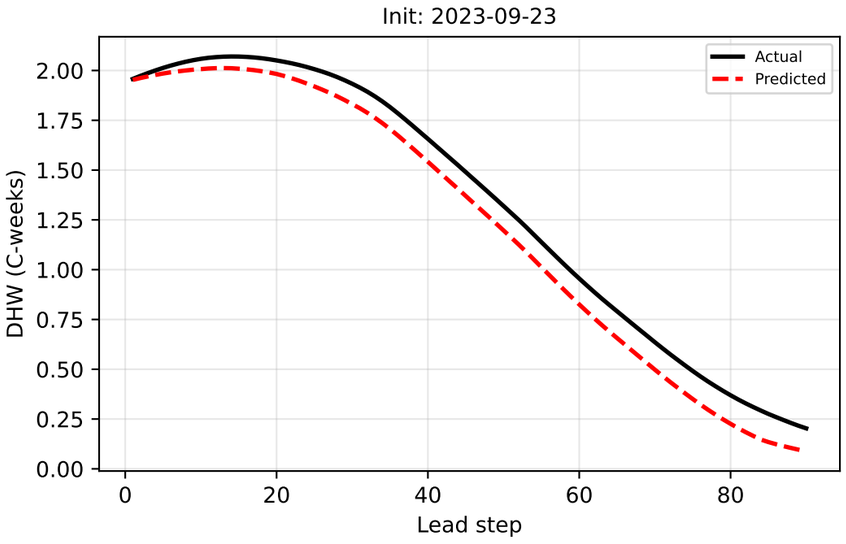




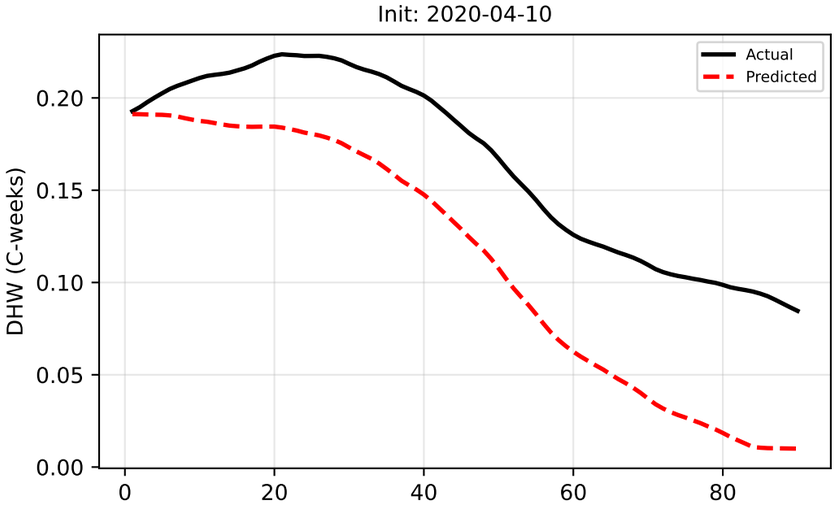


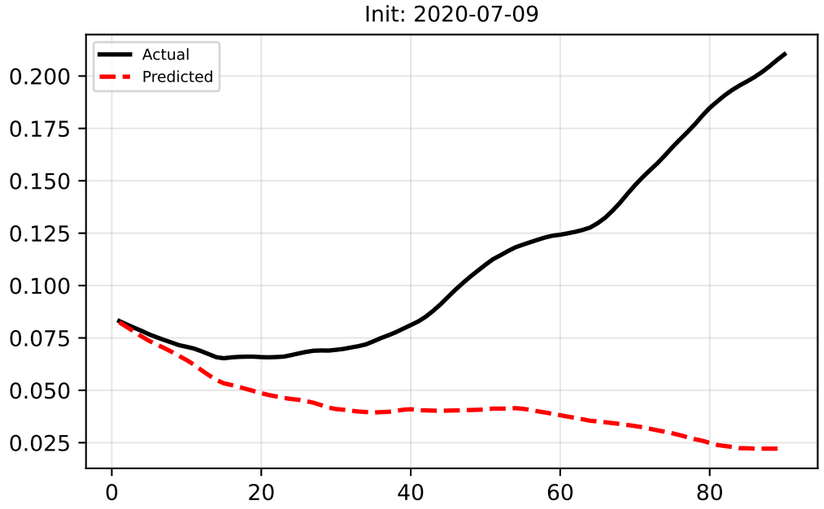


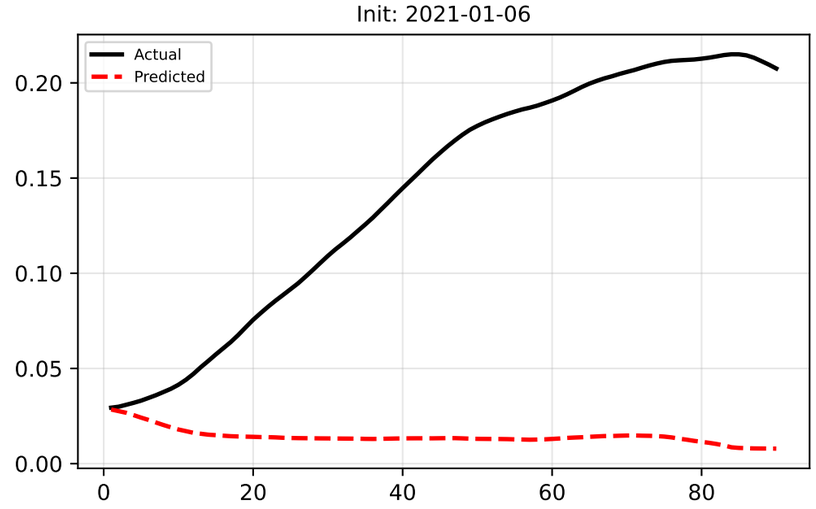


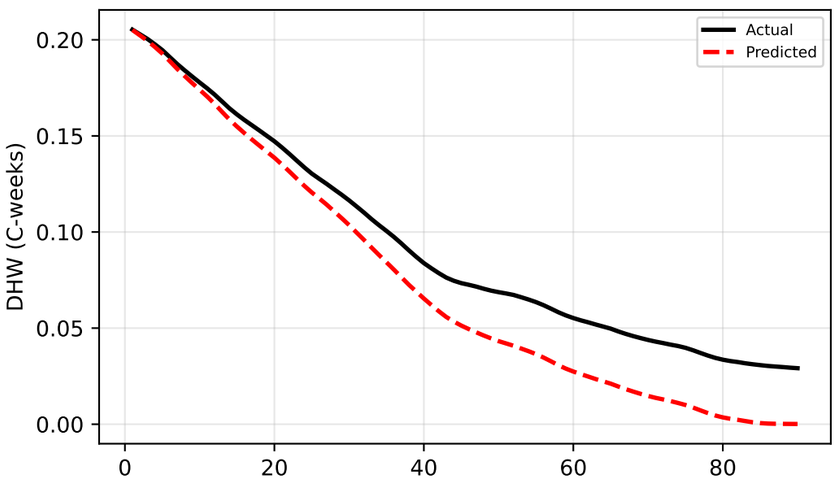


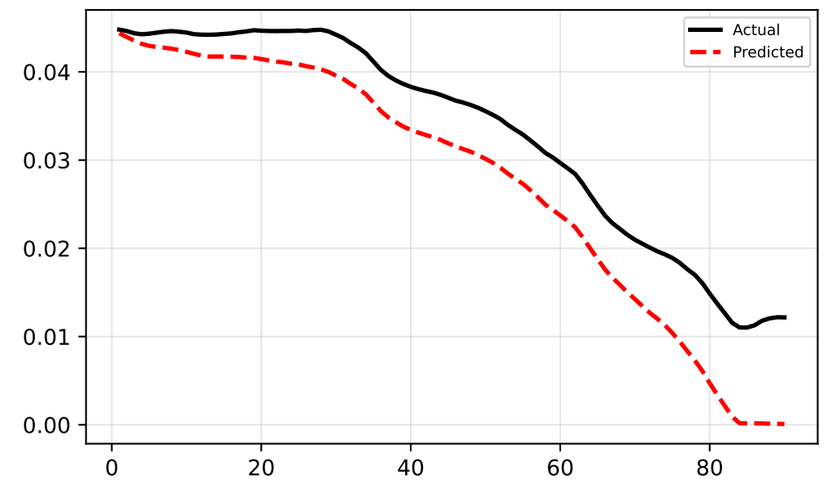
Tropical-Mean DHW (|lat|<=30) - Test (C-weeks)

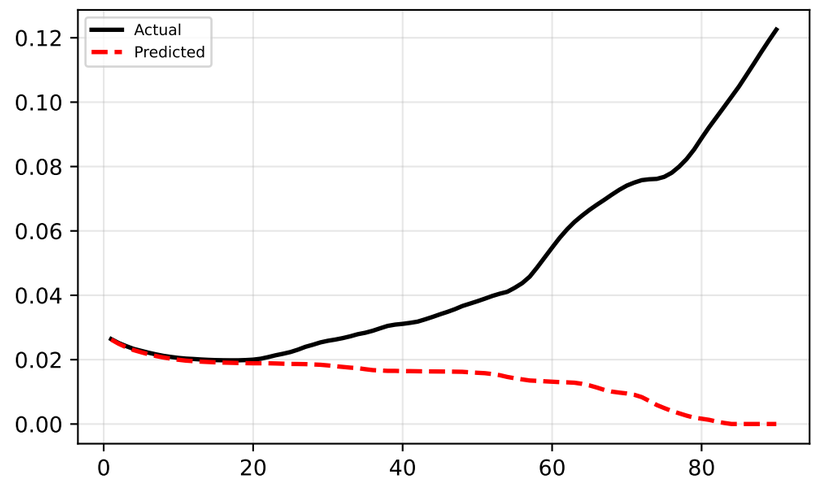


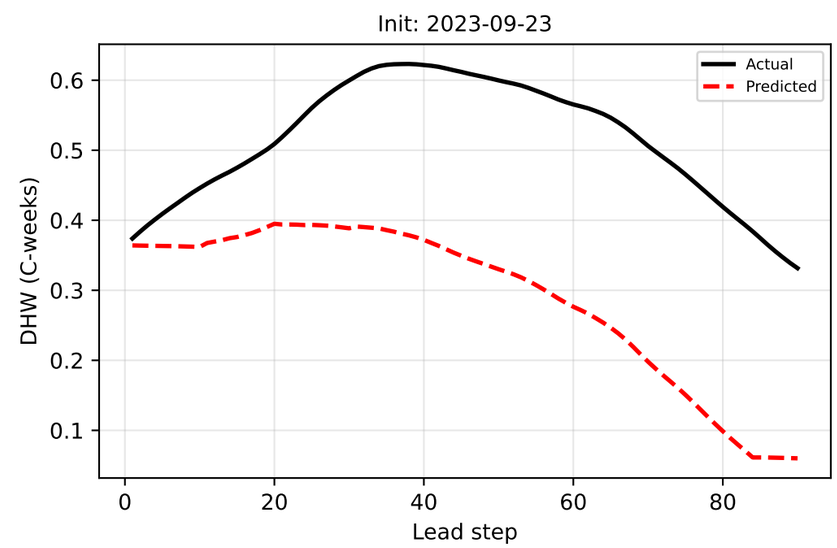




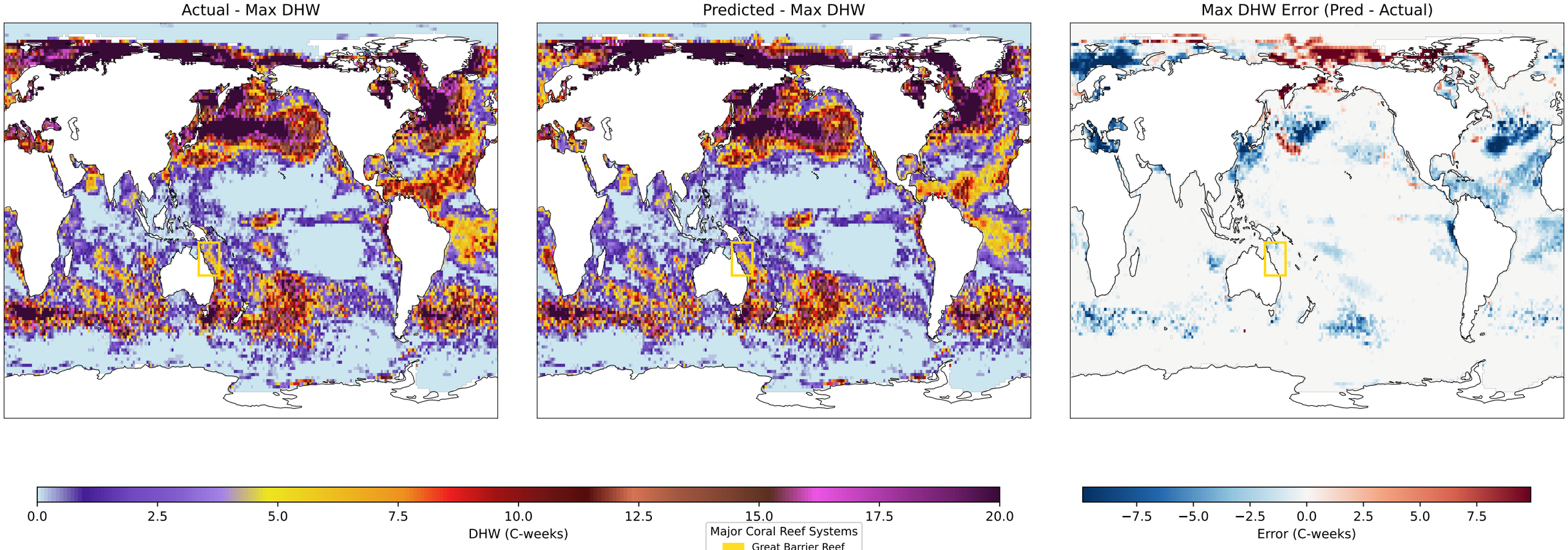


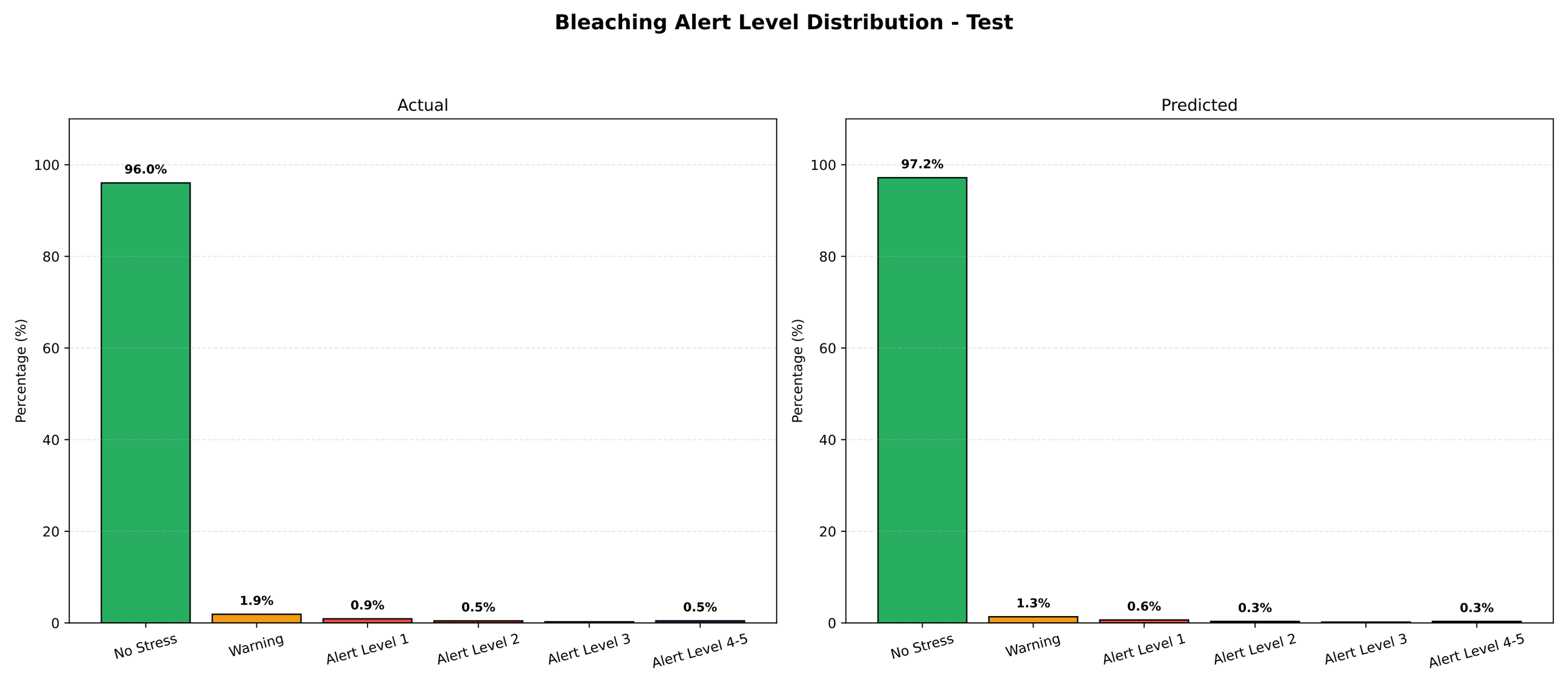




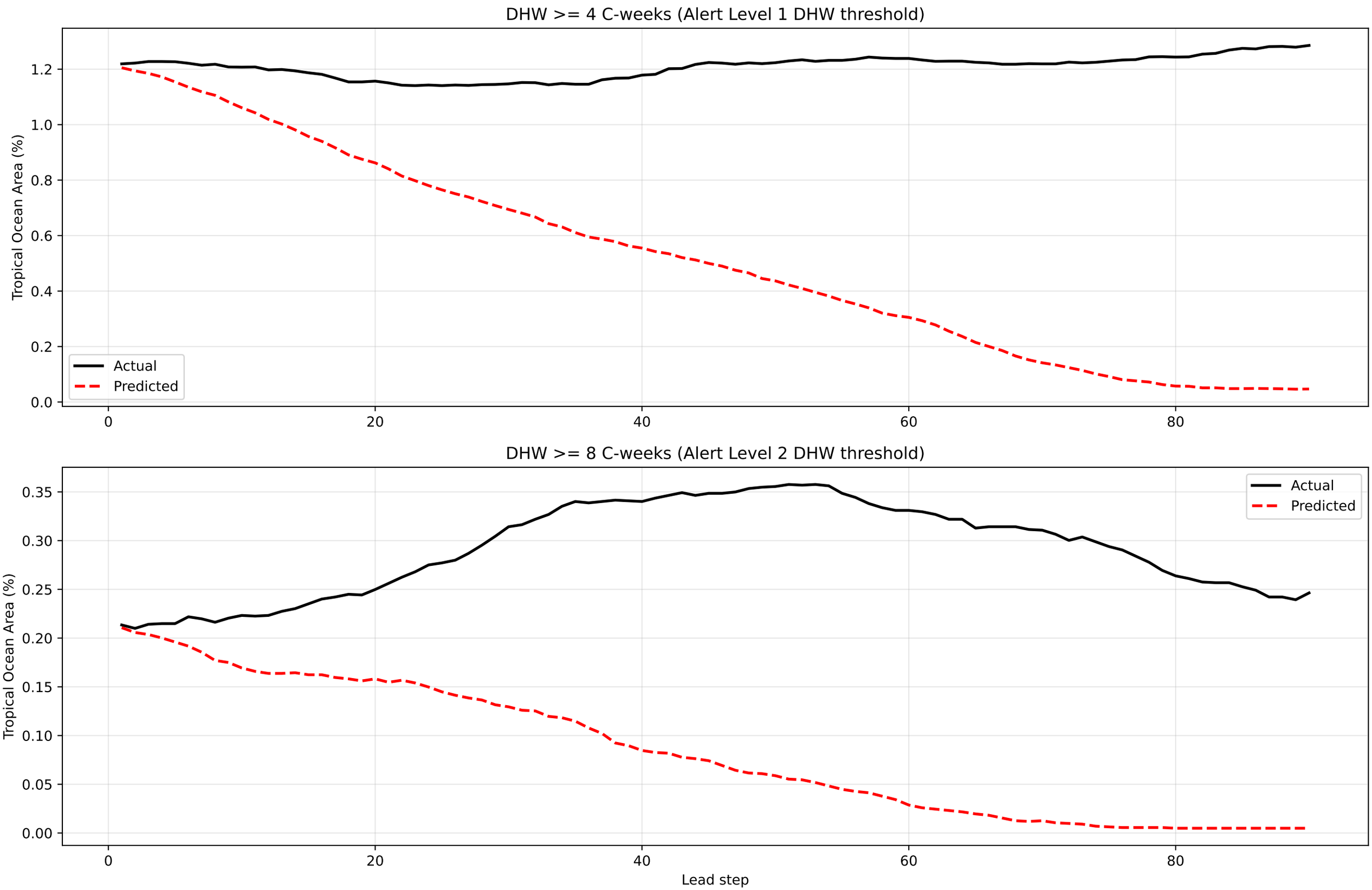


Peak DHW (max over all chains & leads) - Test (C-weeks)




Bleaching Alert Level Distribution - Test
Actual
Predicted
Percentage (%)
0
20
40
60
80
100
96.0%
1.9%
0.9%
0.5%
0.5%
97.2%
1.3%
0.6%
0.3%
0.3%
No Stress
Warning
Alert Level 1
Alert Level 2
Alert Level 3
Alert Level 4-5

Tropical Ocean Area Under Bleaching Stress - Test



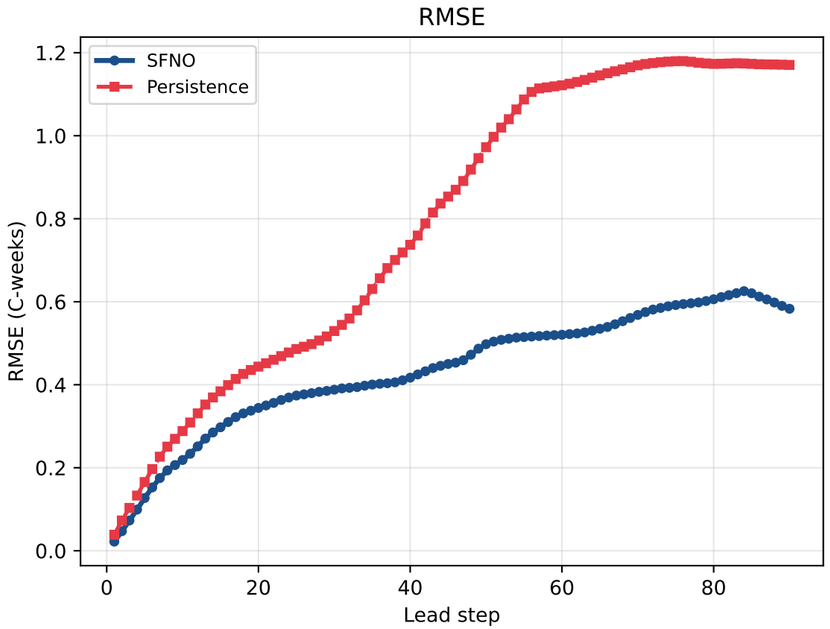
Great Barrier Reef Analysis - Test

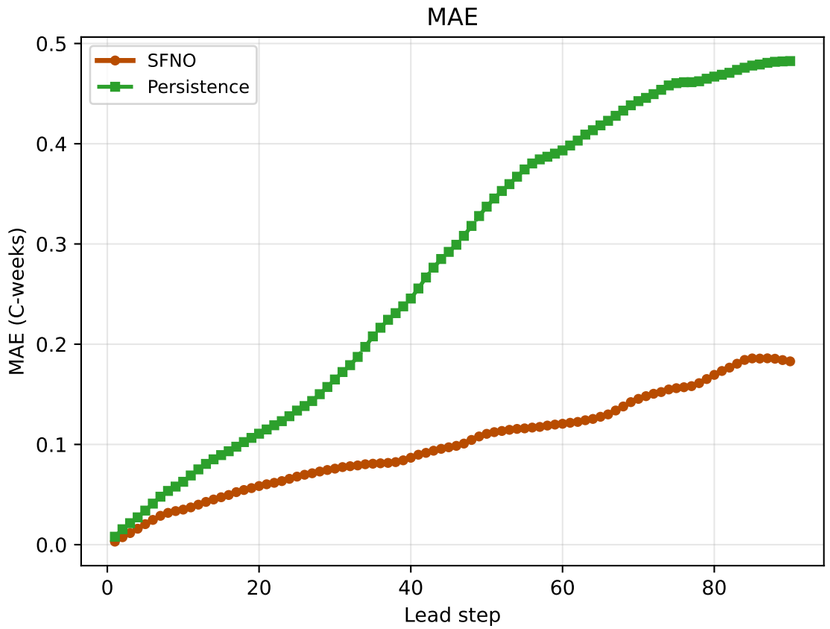
```
                    GREAT BARRIER REEF - TEST
Lat -25.0 deg to -10.0 deg  |  Lon 142.0 deg to 157.0 deg  (0-360)
                    Grid: 11 lat x 11 lon points
```

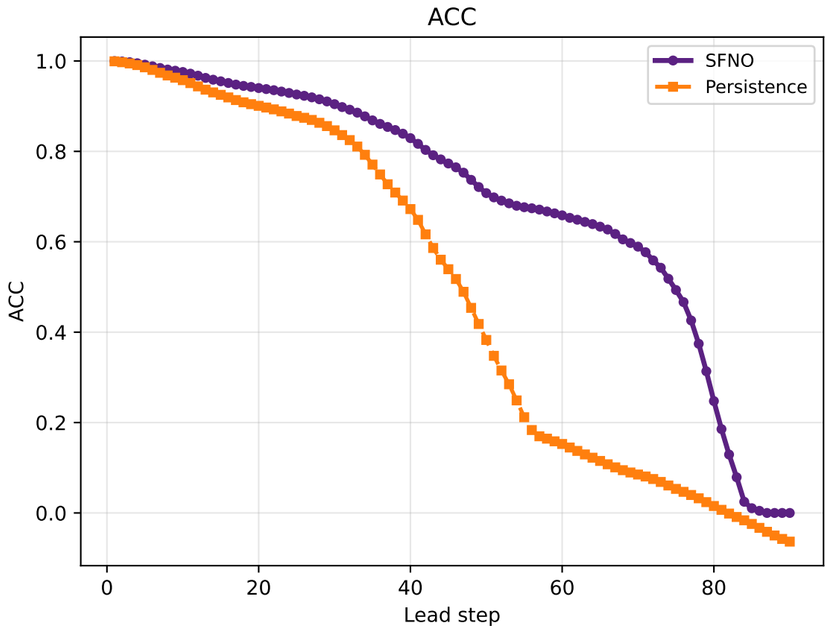
DHW Metrics by Lead Step - Test - GBR (18 chains scored at every lead)

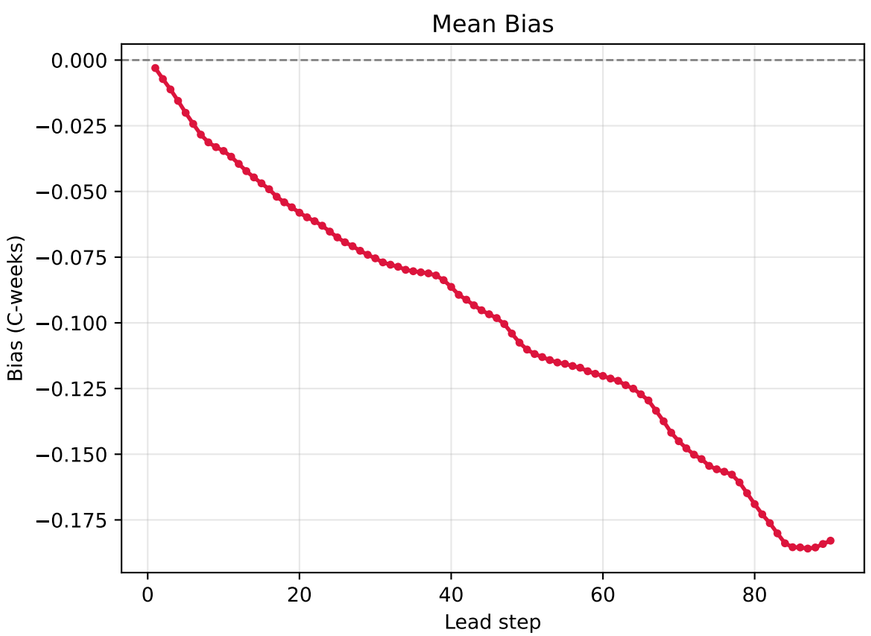
Lead Step	SFNO RMSE	Persist RMSE	Skill %	SFNO MAE	Persist MAE	SFNO Bias	SFNO ACC	Persist ACC	SFNO R²
1	0.0220	0.0386	43.0%	0.0030	0.0079	-0.0030	0.9998	0.9992	0.9995
2	0.0471	0.0729	35.4%	0.0074	0.0154	-0.0072	0.9989	0.9973	0.9978
3	0.0729	0.1032	29.3%	0.0116	0.0214	-0.0112	0.9973	0.9945	0.9947
4	0.0992	0.1329	25.4%	0.0159	0.0272	-0.0155	0.9950	0.9909	0.9903
5	0.1270	0.1655	23.2%	0.0205	0.0340	-0.0201	0.9918	0.9860	0.9842
6	0.1525	0.1968	22.5%	0.0247	0.0409	-0.0243	0.9882	0.9802	0.9773
7	0.1751	0.2266	22.7%	0.0288	0.0479	-0.0284	0.9845	0.9738	0.9702
8	0.1936	0.2509	22.8%	0.0317	0.0536	-0.0313	0.9810	0.9679	0.9637
9	0.2065	0.2696	23.4%	0.0335	0.0580	-0.0331	0.9784	0.9630	0.9587
10	0.2186	0.2883	24.2%	0.0350	0.0627	-0.0346	0.9756	0.9575	0.9534
11	0.2337	0.3090	24.4%	0.0372	0.0690	-0.0368	0.9719	0.9510	0.9465
12	0.2515	0.3310	24.0%	0.0399	0.0751	-0.0395	0.9675	0.9438	0.9380
13	0.2702	0.3523	23.3%	0.0427	0.0806	-0.0423	0.9625	0.9365	0.9288
14	0.2850	0.3694	22.8%	0.0451	0.0851	-0.0447	0.9585	0.9305	0.9213
15	0.2975	0.3842	22.5%	0.0473	0.0894	-0.0469	0.9550	0.9250	0.9148
16	0.3102	0.3991	22.3%	0.0495	0.0932	-0.0491	0.9513	0.9194	0.9079
17	0.3219	0.4140	22.2%	0.0524	0.0977	-0.0520	0.9477	0.9135	0.9013
18	0.3309	0.4260	22.3%	0.0545	0.1023	-0.0541	0.9448	0.9086	0.8959
19	0.3375	0.4355	22.5%	0.0564	0.1065	-0.0560	0.9425	0.9044	0.8917
20	0.3441	0.4437	22.4%	0.0585	0.1107	-0.0581	0.9401	0.9007	0.8872
21	0.3501	0.4518	22.5%	0.0602	0.1149	-0.0598	0.9378	0.8970	0.8831
22	0.3562	0.4602	22.6%	0.0617	0.1191	-0.0613	0.9354	0.8929	0.8786
23	0.3632	0.4690	22.5%	0.0634	0.1232	-0.0630	0.9323	0.8885	0.8732
24	0.3691	0.4777	22.7%	0.0657	0.1281	-0.0653	0.9292	0.8836	0.8677
25	0.3739	0.4861	23.1%	0.0679	0.1338	-0.0675	0.9258	0.8783	0.8617
26	0.3768	0.4916	23.3%	0.0697	0.1383	-0.0693	0.9230	0.8743	0.8568
27	0.3797	0.4979	23.7%	0.0712	0.1433	-0.0708	0.9195	0.8695	0.8508
28	0.3828	0.5066	24.4%	0.0730	0.1501	-0.0726	0.9153	0.8632	0.8434
29	0.3851	0.5163	25.4%	0.0745	0.1573	-0.0741	0.9105	0.8559	0.8352
30	0.3881	0.5294	26.7%	0.0759	0.1648	-0.0755	0.9045	0.8465	0.8249
31	0.3912	0.5442	28.1%	0.0774	0.1722	-0.0770	0.8980	0.8360	0.8140
32	0.3926	0.5596	29.8%	0.0783	0.1791	-0.0779	0.8922	0.8249	0.8041
33	0.3944	0.5792	31.9%	0.0791	0.1873	-0.0787	0.8857	0.8106	0.7932
34	0.3977	0.6035	34.1%	0.0802	0.1973	-0.0798	0.8775	0.7924	0.7796
35	0.4005	0.6307	36.5%	0.0808	0.2078	-0.0804	0.8686	0.7708	0.7651
36	0.4024	0.6566	38.7%	0.0812	0.2164	-0.0808	0.8608	0.7489	0.7525
37	0.4036	0.6811	40.7%	0.0816	0.2244	-0.0812	0.8540	0.7272	0.7417
38	0.4058	0.7006	42.1%	0.0824	0.2310	-0.0820	0.8472	0.7089	0.7310
39	0.4107	0.7187	42.9%	0.0841	0.2378	-0.0837	0.8391	0.6912	0.7183
40	0.4170	0.7372	43.4%	0.0867	0.2456	-0.0863	0.8293	0.6723	0.7028
41	0.4248	0.7596	44.1%	0.0897	0.2556	-0.0893	0.8167	0.6484	0.6832
42	0.4324	0.7886	45.2%	0.0916	0.2666	-0.0912	0.8032	0.6164	0.6621
43	0.4402	0.8150	46.0%	0.0937	0.2764	-0.0933	0.7915	0.5862	0.6434
44	0.4455	0.8367	46.7%	0.0956	0.2850	-0.0952	0.7822	0.5603	0.6284
45	0.4500	0.8533	47.3%	0.0971	0.2921	-0.0967	0.7732	0.5393	0.6140
46	0.4532	0.8698	47.9%	0.0986	0.2992	-0.0982	0.7647	0.5176	0.6003
47	0.4591	0.8909	48.5%	0.1009	0.3081	-0.1005	0.7529	0.4893	0.5818
48	0.4723	0.9184	48.6%	0.1045	0.3180	-0.1041	0.7370	0.4538	0.5573
49	0.4869	0.9460	48.5%	0.1079	0.3279	-0.1075	0.7211	0.4181	0.5330
50	0.4977	0.9723	48.8%	0.1106	0.3372	-0.1102	0.7076	0.3828	0.5125
51	0.5040	0.9973	49.5%	0.1123	0.3453	-0.1119	0.6982	0.3476	0.4977
52	0.5082	1.0196	50.2%	0.1134	0.3529	-0.1130	0.6911	0.3152	0.4860
53	0.5108	1.0399	50.9%	0.1146	0.3597	-0.1142	0.6851	0.2847	0.4761
54	0.5136	1.0634	51.7%	0.1155	0.3671	-0.1151	0.6796	0.2490	0.4669
55	0.5148	1.0873	52.7%	0.1160	0.3743	-0.1156	0.6764	0.2118	0.4606
56	0.5161	1.1055	53.3%	0.1168	0.3803	-0.1164	0.6741	0.1835	0.4562
57	0.5173	1.1139	53.6%	0.1175	0.3844	-0.1171	0.6712	0.1696	0.4514
58	0.5186	1.1164	53.6%	0.1188	0.3870	-0.1184	0.6670	0.1644	0.4460
59	0.5195	1.1190	53.6%	0.1198	0.3902	-0.1194	0.6629	0.1585	0.4400
60	0.5205	1.1215	53.6%	0.1206	0.3933	-0.1202	0.6585	0.1526	0.4334
61	0.5220	1.1253	53.6%	0.1216	0.3982	-0.1212	0.6529	0.1448	0.4247
62	0.5236	1.1295	53.6%	0.1225	0.4031	-0.1221	0.6487	0.1373	0.4178
63	0.5261	1.1345	53.6%	0.1241	0.4091	-0.1237	0.6439	0.1295	0.4112
64	0.5302	1.1400	53.5%	0.1254	0.4135	-0.1250	0.6392	0.1223	0.4050
65	0.5348	1.1455	53.3%	0.1276	0.4183	-0.1272	0.6334	0.1151	0.3980
66	0.5392	1.1506	53.1%	0.1299	0.4229	-0.1295	0.6271	0.1077	0.3900
67	0.5457	1.1553	52.8%	0.1338	0.4278	-0.1334	0.6173	0.1009	0.3791
68	0.5531	1.1599	52.3%	0.1379	0.4330	-0.1375	0.6052	0.0946	0.3671
69	0.5608	1.1649	51.9%	0.1422	0.4384	-0.1418	0.5972	0.0895	0.3590
70	0.5682	1.1696	51.4%	0.1454	0.4425	-0.1450	0.5892	0.0850	0.3508
71	0.5750	1.1728	51.0%	0.1482	0.4457	-0.1478	0.5769	0.0807	0.3393
72	0.5812	1.1751	50.5%	0.1506	0.4495	-0.1502	0.5588	0.0751	0.3220
73	0.5850	1.1771	50.3%	0.1523	0.4538	-0.1519	0.5425	0.0686	0.3057
74	0.5891	1.1786	50.0%	0.1548	0.4582	-0.1544	0.5182	0.0608	0.2834
75	0.5920	1.1796	49.8%	0.1561	0.4604	-0.1557	0.4932	0.0533	0.2615
76	0.5944	1.1798	49.6%	0.1571	0.4614	-0.1567	0.4665	0.0467	0.2400
77	0.5964	1.1783	49.4%	0.1582	0.4614	-0.1578	0.4259	0.0398	0.2089
78	0.5983	1.1759	49.1%	0.1611	0.4623	-0.1607	0.3743	0.0324	0.1716
79	0.6017	1.1743	48.8%	0.1652	0.4649	-0.1648	0.3135	0.0239	0.1295
80	0.6059	1.1731	48.3%	0.1694	0.4670	-0.1690	0.2475	0.0154	0.0849
81	0.6112	1.1734	47.9%	0.1733	0.4688	-0.1729	0.1855	0.0069	0.0430
82	0.6159	1.1740	47.5%	0.1766	0.4708	-0.1762	0.1292	-0.0013	0.0036
83	0.6206	1.1746	47.2%	0.1805	0.4737	-0.1801	0.0790	-0.0089	-0.0330
84	0.6255	1.1741	46.7%	0.1843	0.4758	-0.1839	0.0247	-0.0162	-0.0754
85	0.6205	1.1732	47.1%	0.1858	0.4780	-0.1854	0.0105	-0.0243	-0.0895
86	0.6123	1.1722	47.8%	0.1857	0.4791	-0.1855	0.0047	-0.0330	-0.0969
87	0.6056	1.1718	48.3%	0.1859	0.4807	-0.1859	-0.0000	-0.0417	-0.1040
88	0.5982	1.1717	48.9%	0.1855	0.4817	-0.1855	-0.0000	-0.0498	-0.1064
89	0.5900	1.1712	49.6%	0.1842	0.4821	-0.1842	-0.0000	-0.0575	-0.1079
90	0.5830	1.1706	50.2%	0.1829	0.4825	-0.1829	0.0000	-0.0634	-0.1091

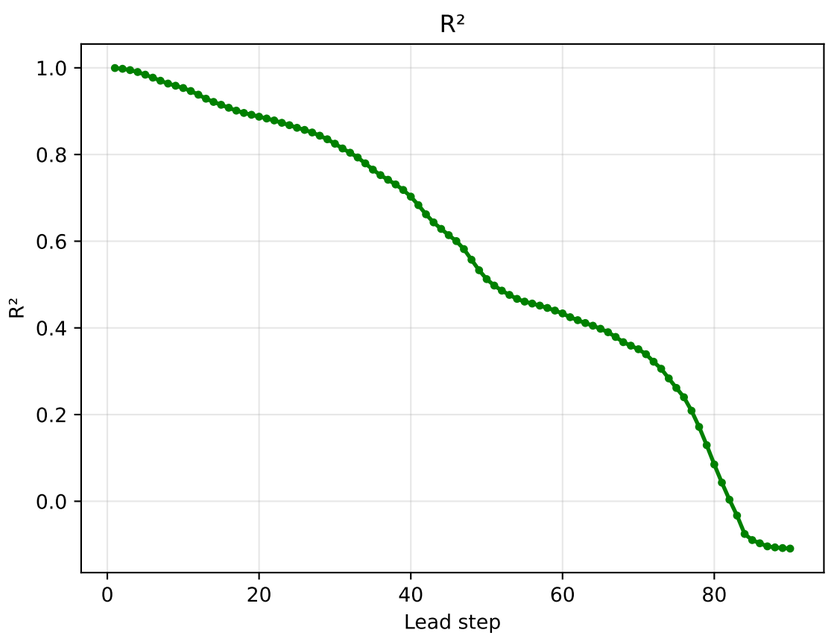
DHW Forecast Skill vs Lead Step - Test - GBR

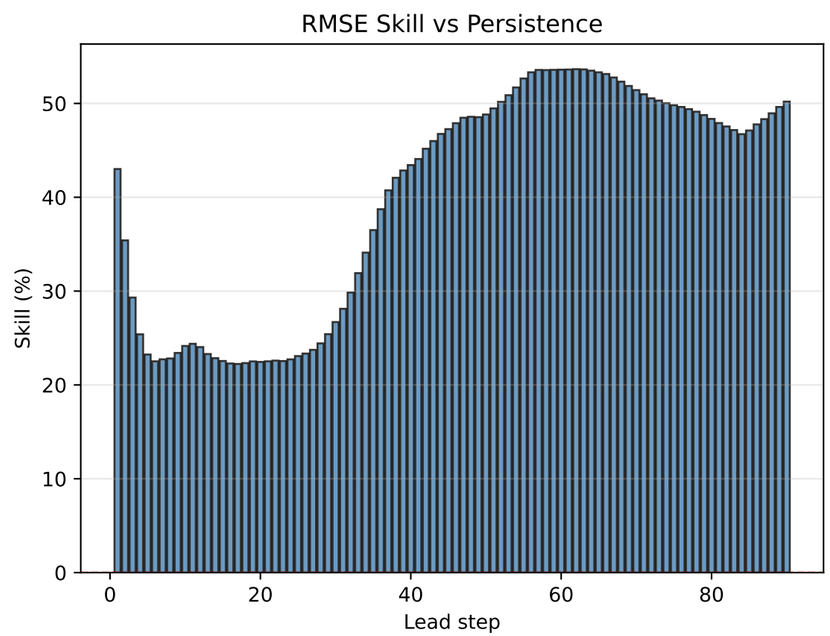


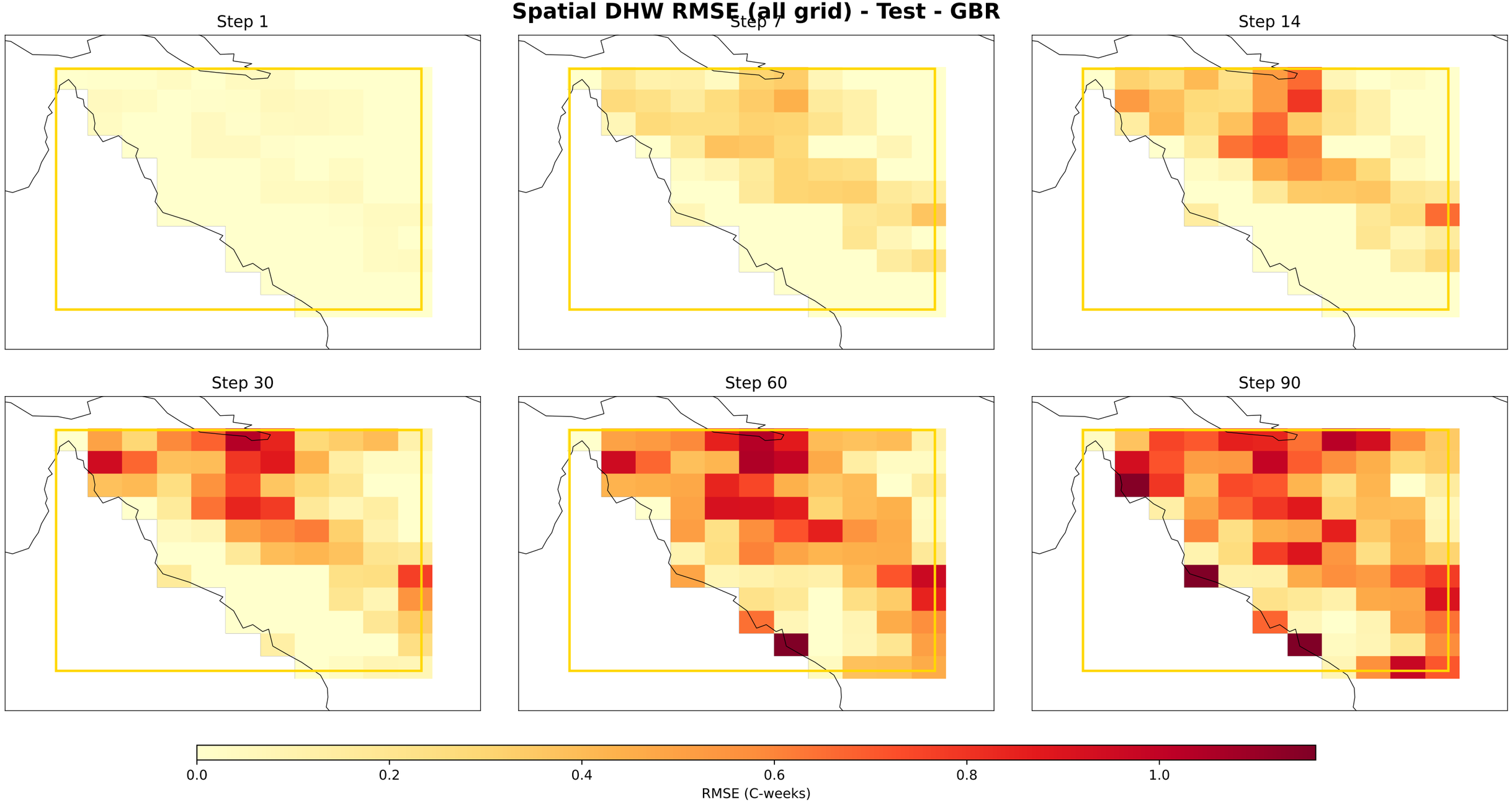


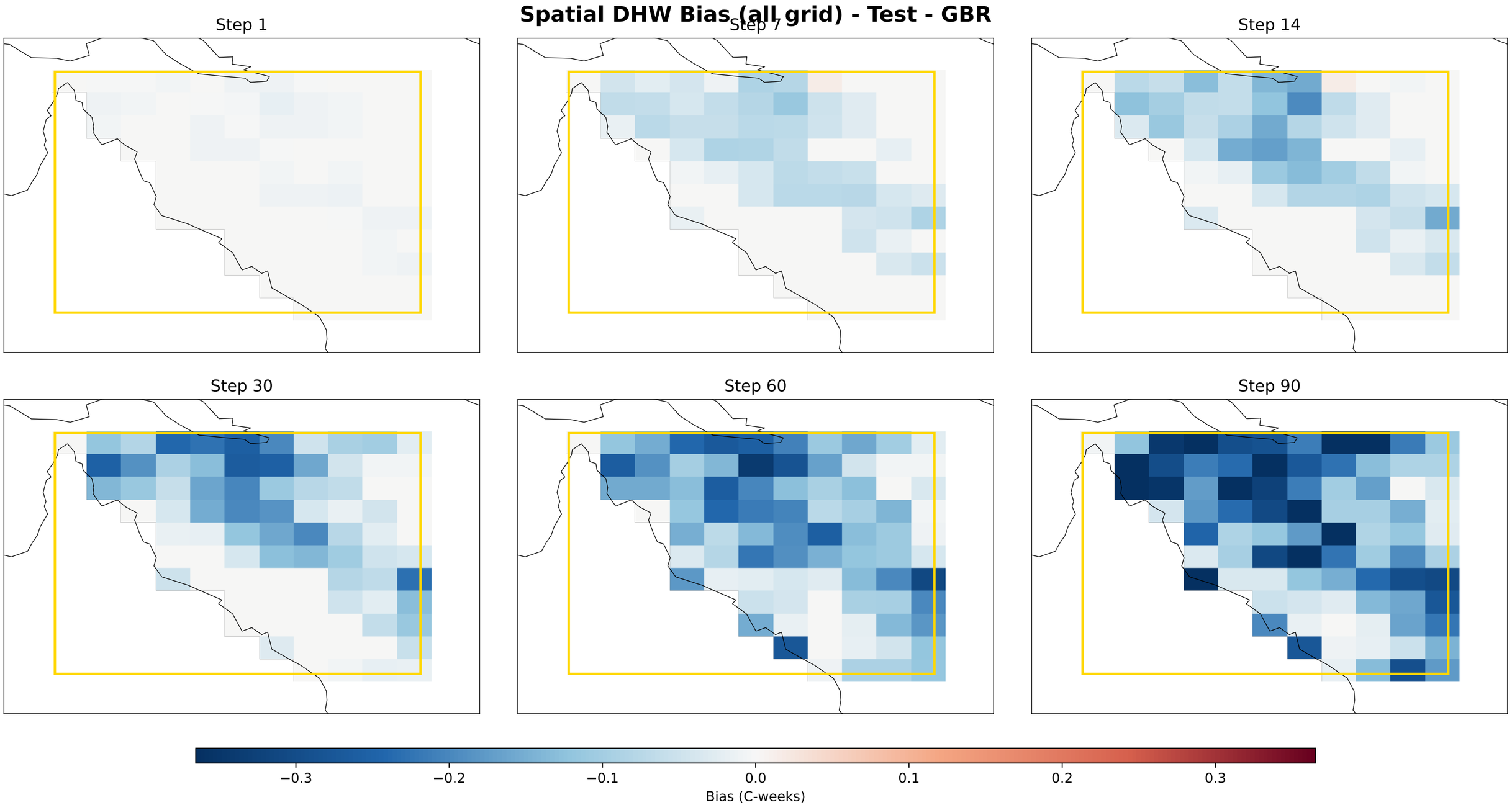


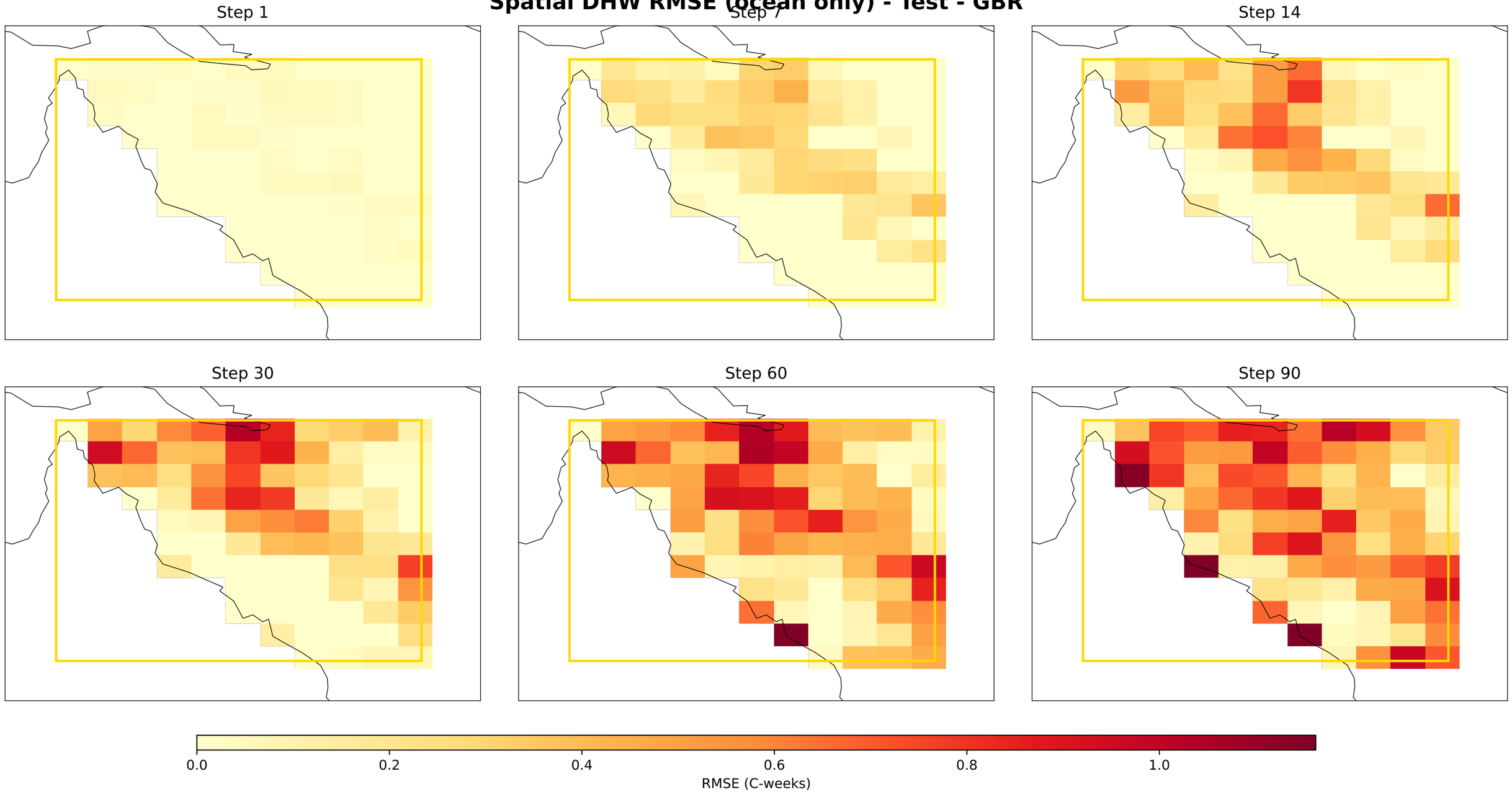


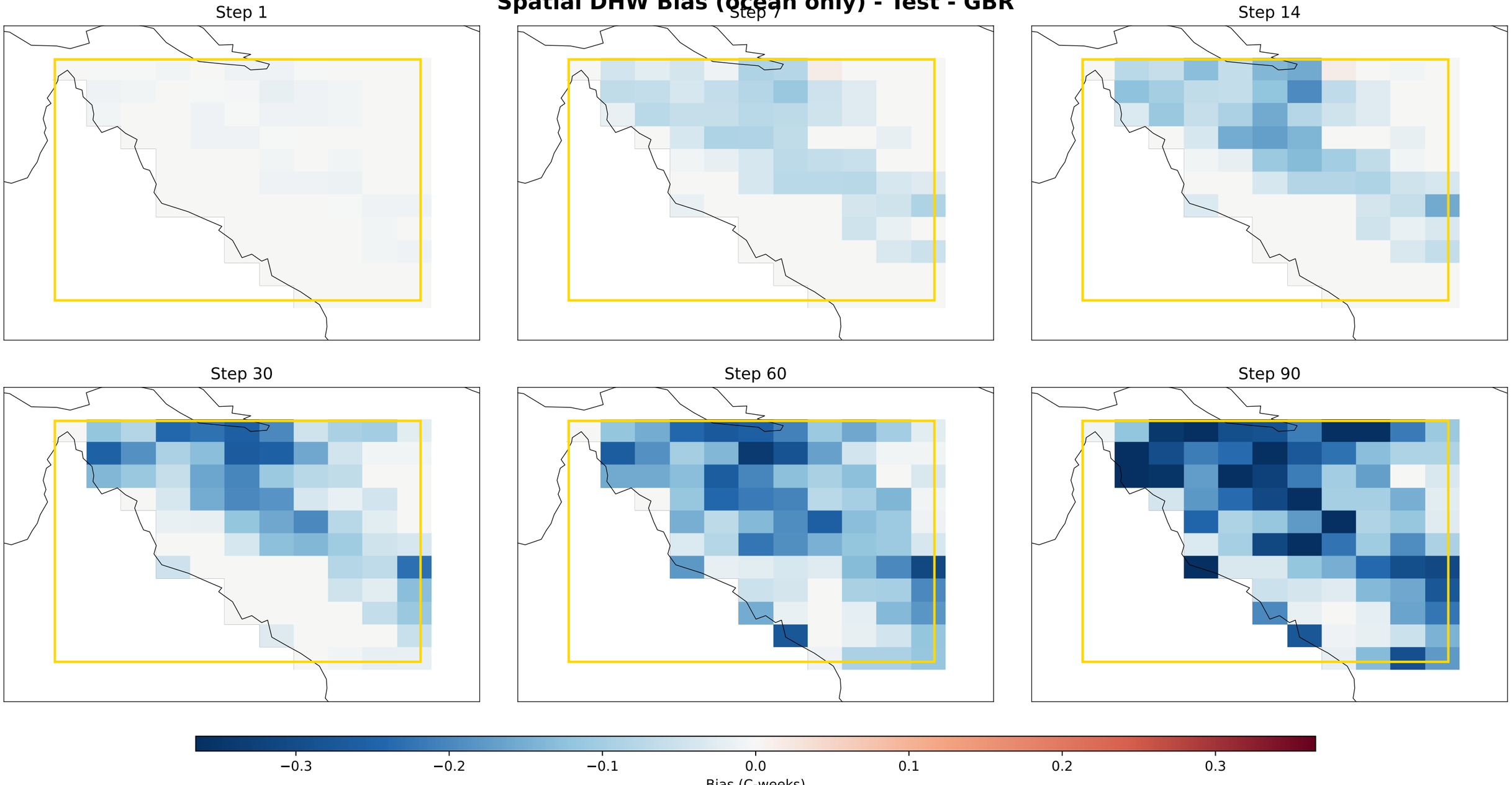





Spatial DHW RMSE (all grid) - Test - GBR
Step 1
Step 7
Step 14
Step 30
Step 60
Step 90
0.0
0.2
0.4
0.6
0.8
1.0
RMSE (C-weeks)

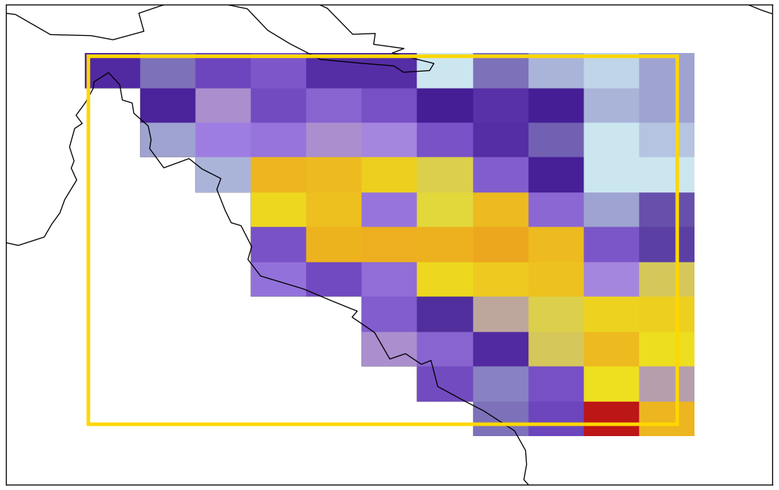

Spatial DHW Bias (all grid) - Test - GBR
Step 1
Step 7
Step 14
Step 30
Step 60
Step 90
−0.3
−0.2
−0.1
0.0
0.1
0.2
0.3
Bias (C-weeks)


Spatial DHW RMSE (ocean only) - Test - GBR
Step 1
Step 7
Step 14
Step 30
Step 60
Step 90
0.0
0.2
0.4
0.6
0.8
1.0
RMSE (C-weeks)

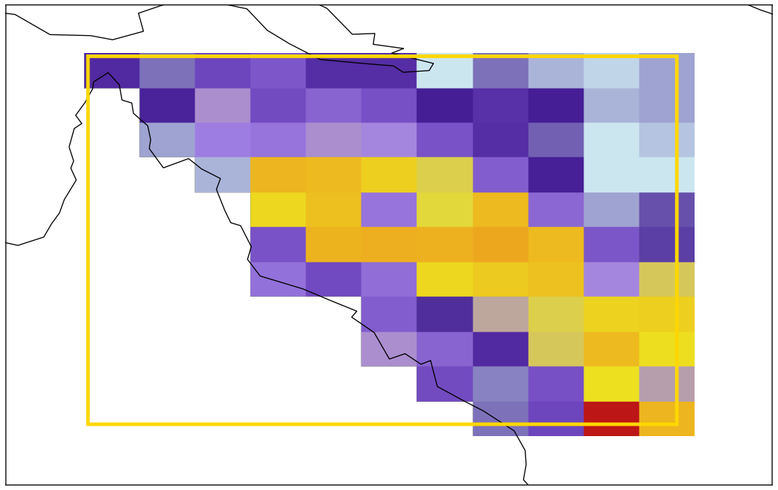

Spatial DHW Bias (ocean only) - Test - GBR
Step 1
Step 7
Step 14
Step 30
Step 60
Step 90
−0.3
−0.2
−0.1
0.0
0.1
0.2
0.3
Bias (C-weeks)

DHW Chain init: 2020-04-10 - Test - GBR (C-weeks)

Actual DHW - Step 1



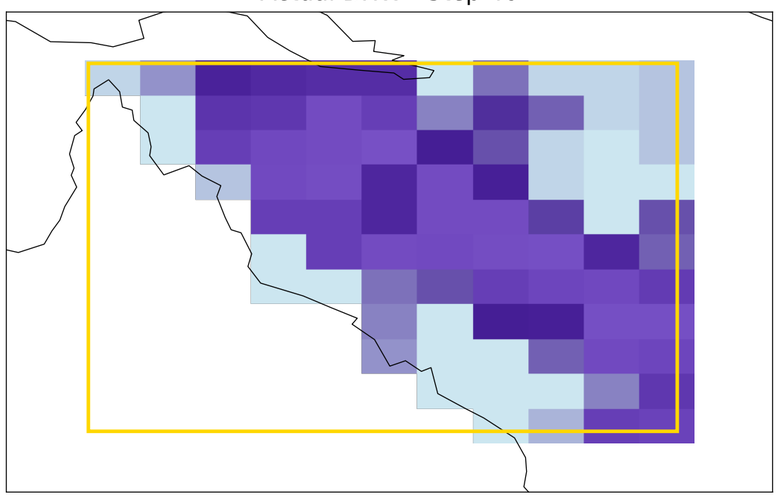
Predicted DHW - Step 1



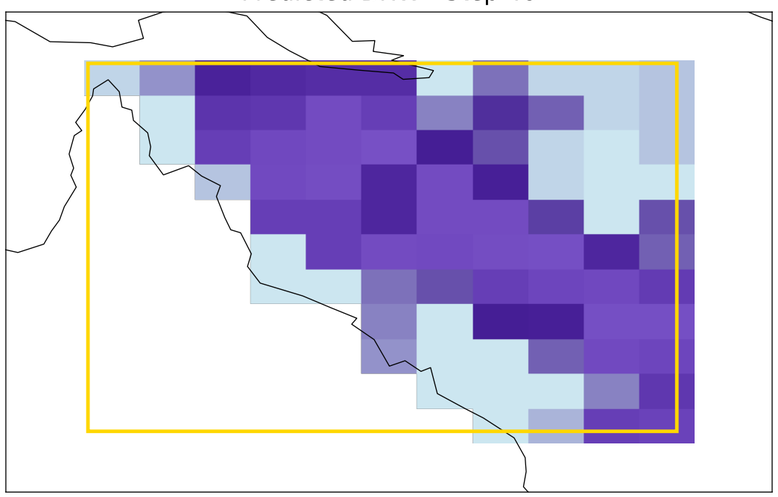
Error (Pred - Actual) - Step 1 [scale ±8]



Actual DHW - Step 46



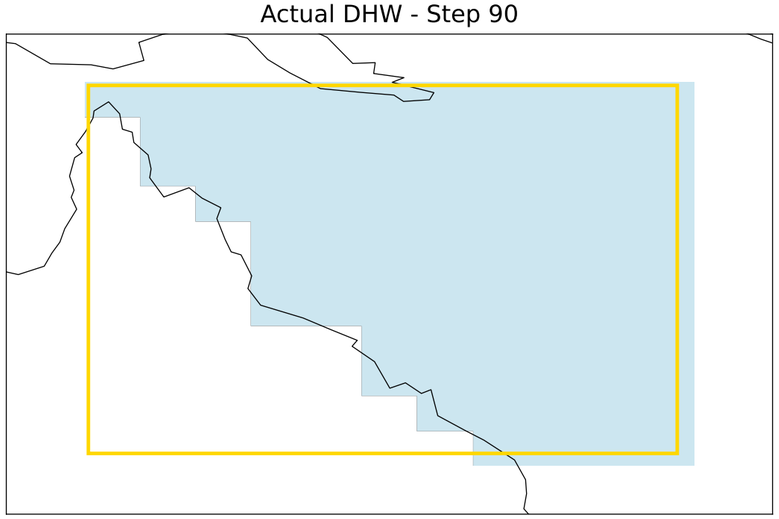
Predicted DHW - Step 46



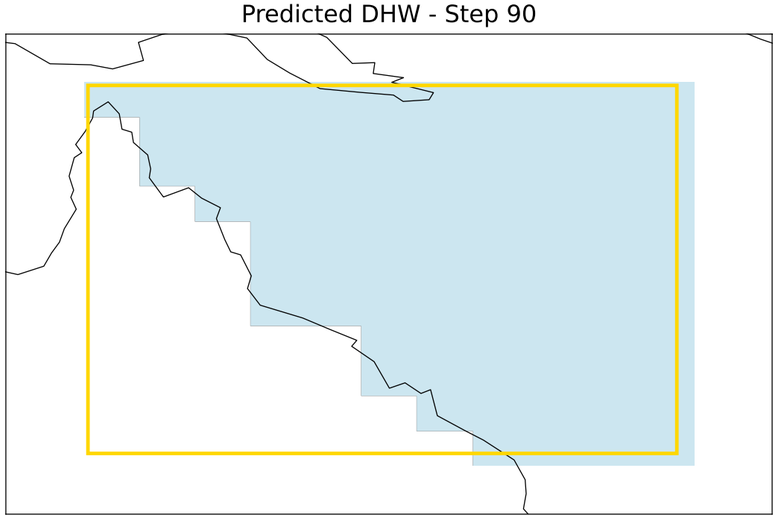
Error (Pred - Actual) - Step 46 [scale ±8]



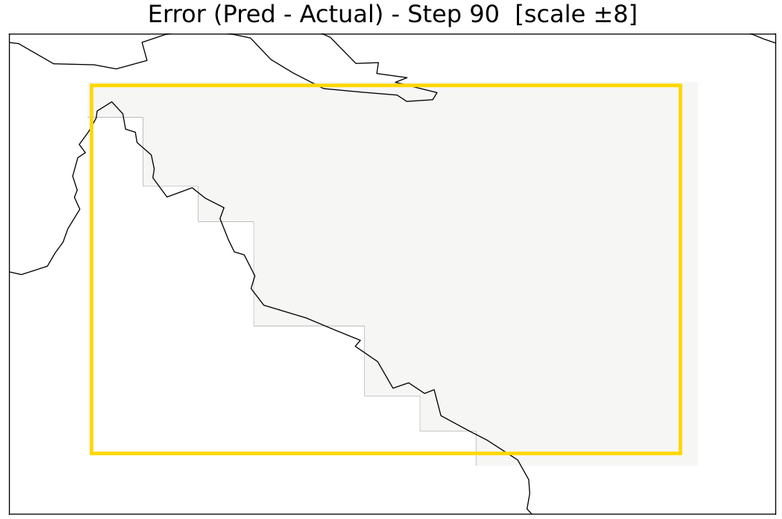
Actual DHW - Step 90

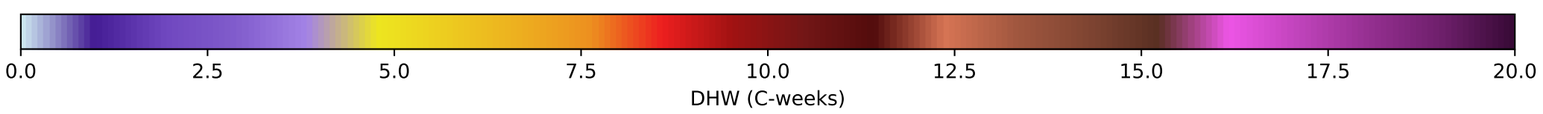


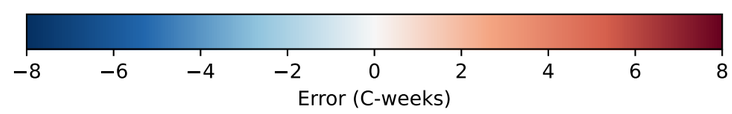
Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±8]

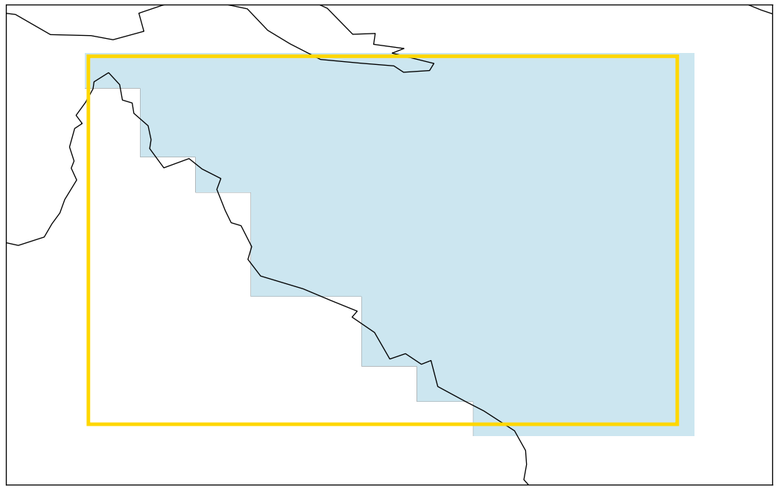




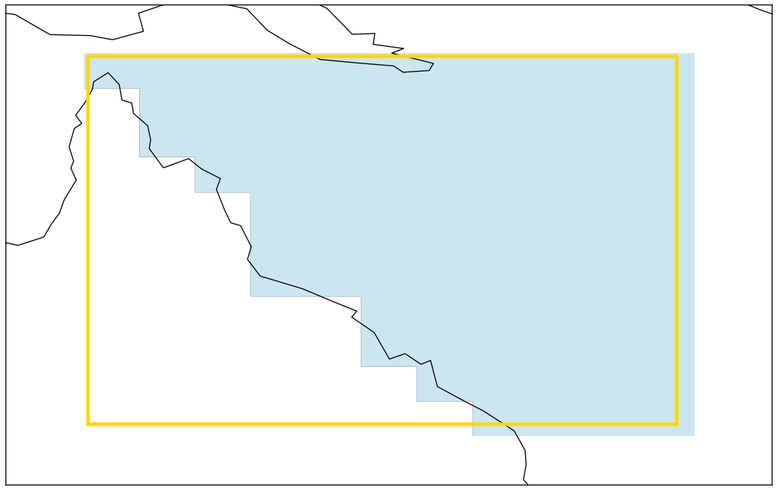


DHW Chain init: 2020-07-09 - Test - GBR (C-weeks)

Actual DHW - Step 1



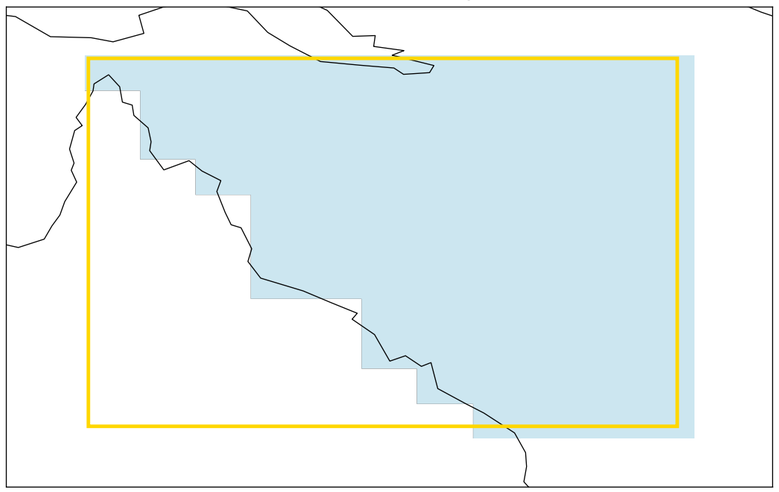
Predicted DHW - Step 1



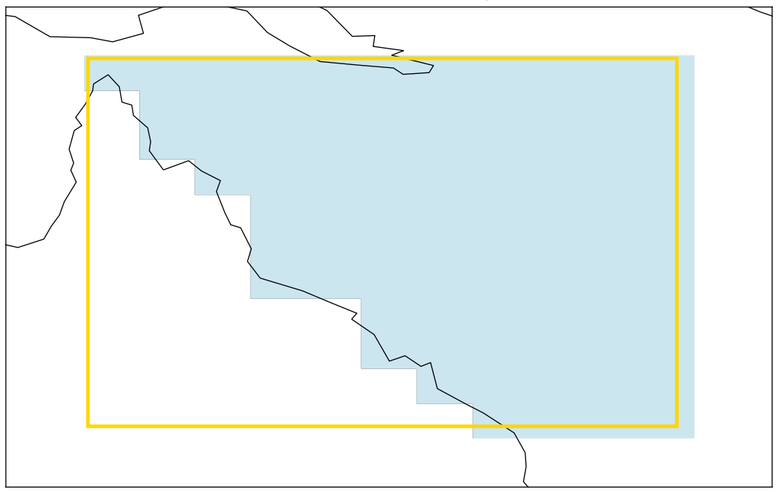
Error (Pred - Actual) - Step 1 [scale ±8]



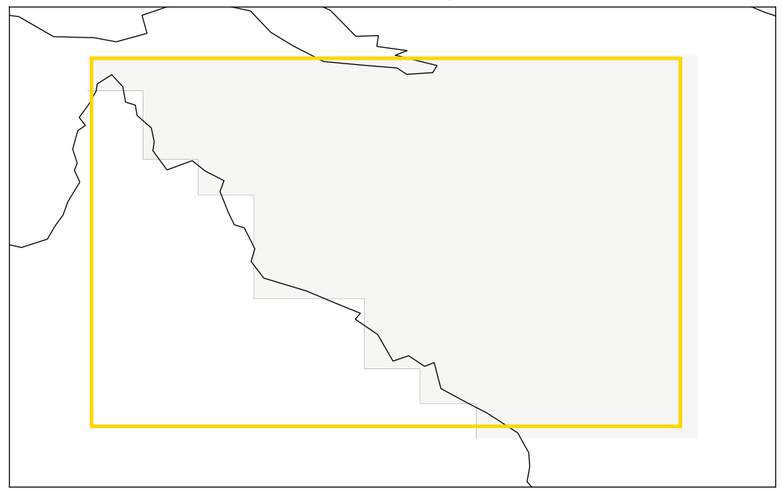
Actual DHW - Step 46



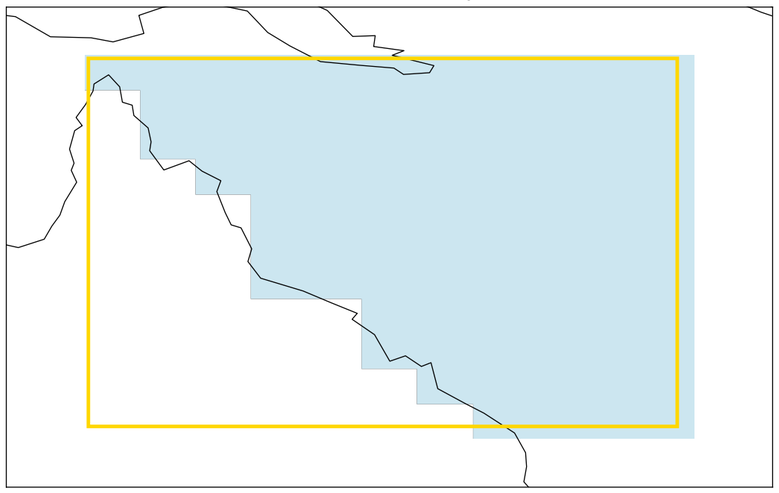
Predicted DHW - Step 46



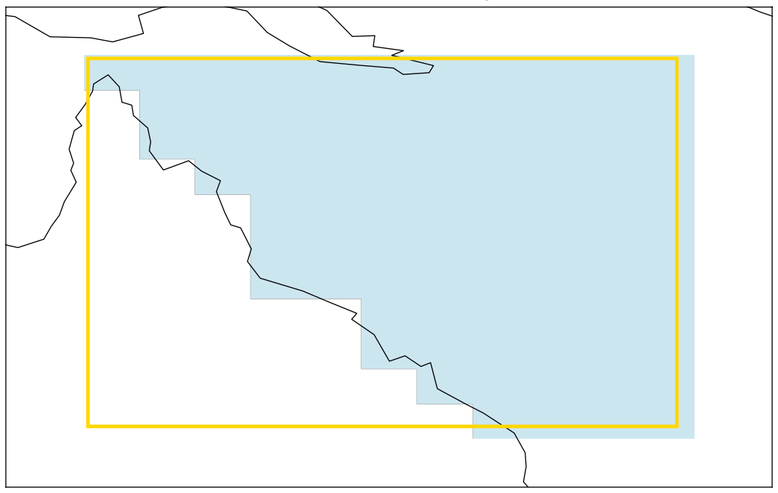
Error (Pred - Actual) - Step 46 [scale ±8]



Actual DHW - Step 90

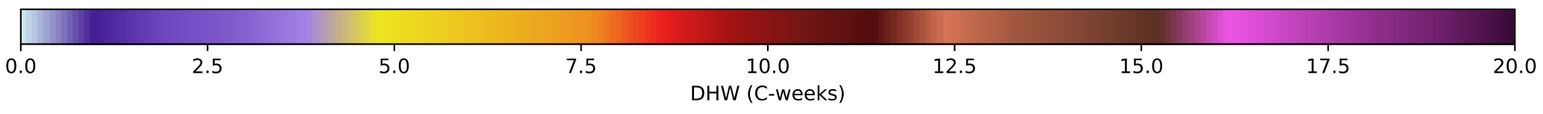


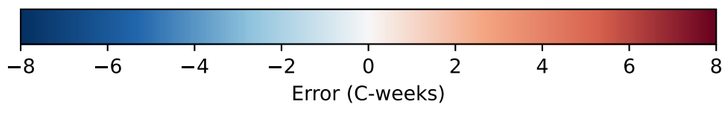
Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±8]

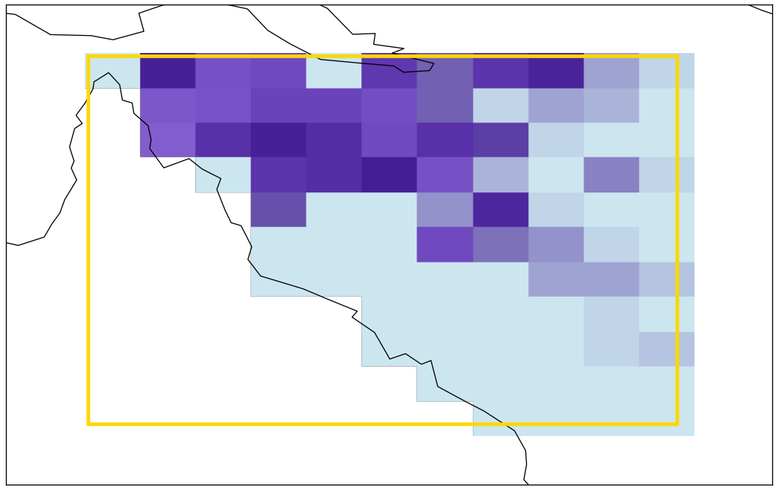




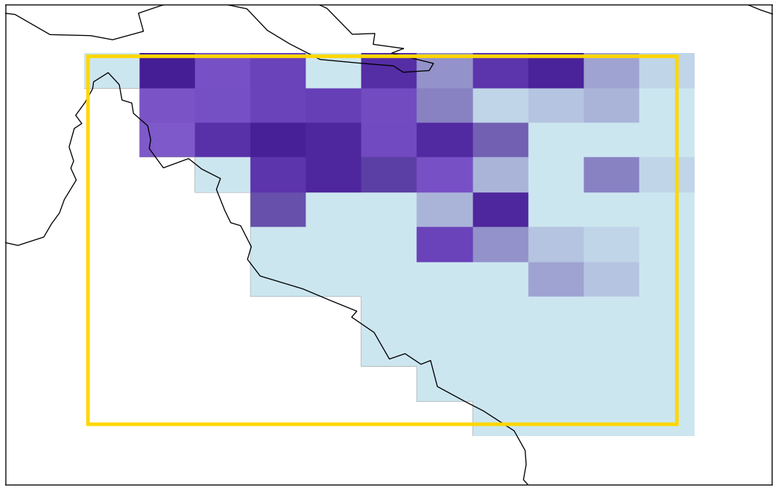


DHW Chain init: 2021-01-06 - Test - GBR (C-weeks)

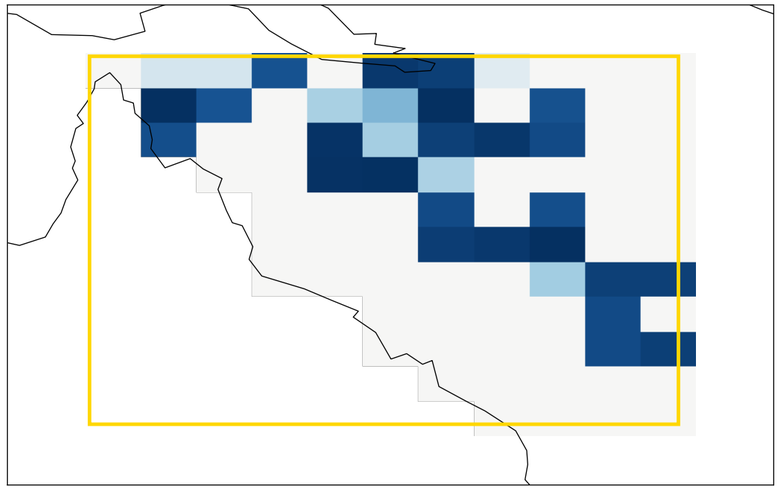
Actual DHW - Step 1



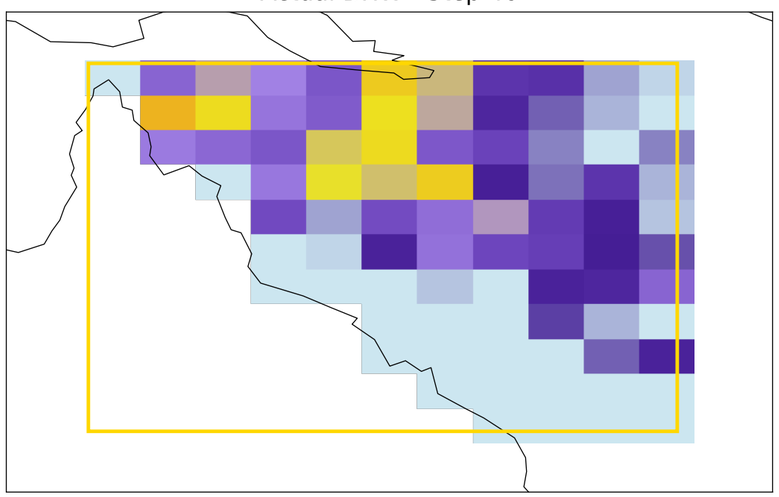
Predicted DHW - Step 1



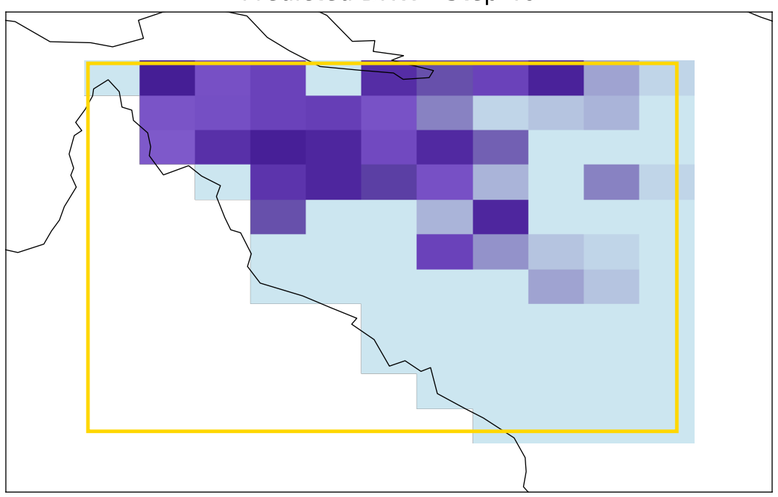
Error (Pred - Actual) - Step 1 [scale ±0.18]



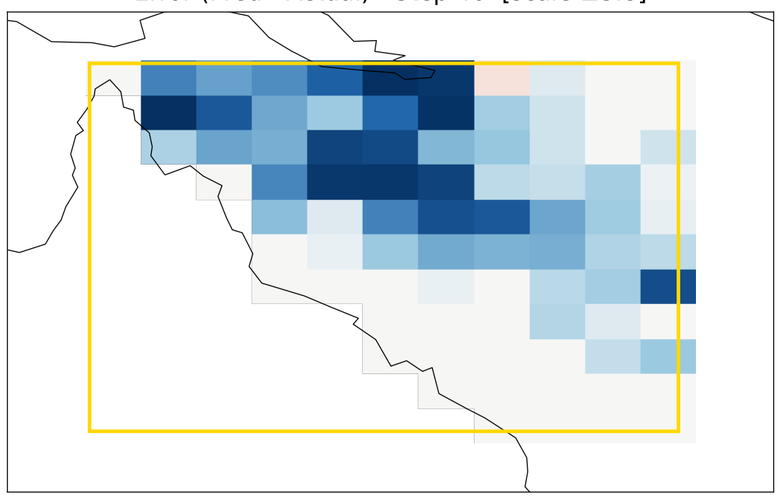
Actual DHW - Step 46



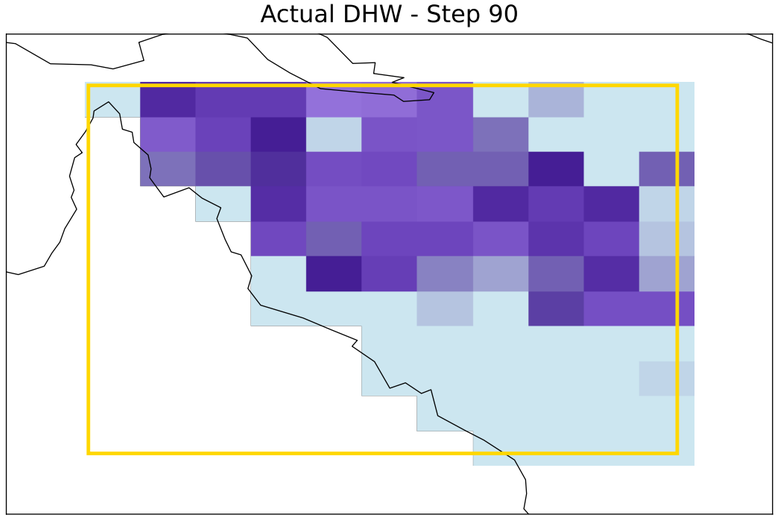
Predicted DHW - Step 46



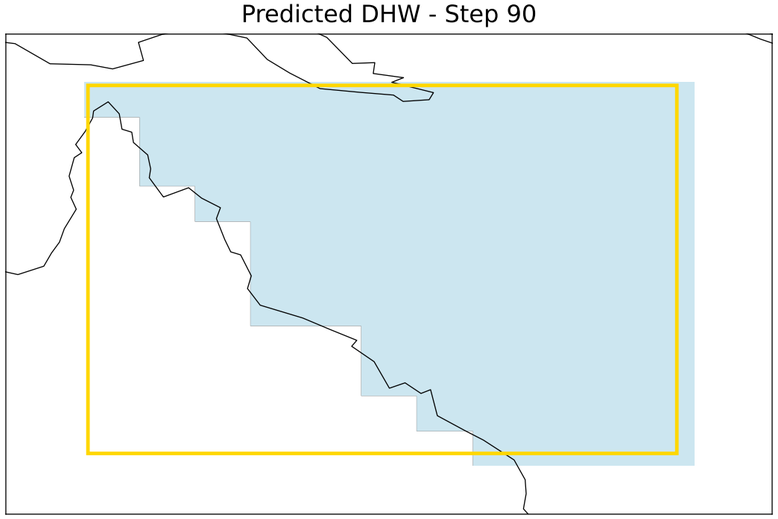
Error (Pred - Actual) - Step 46 [scale ±3.8]



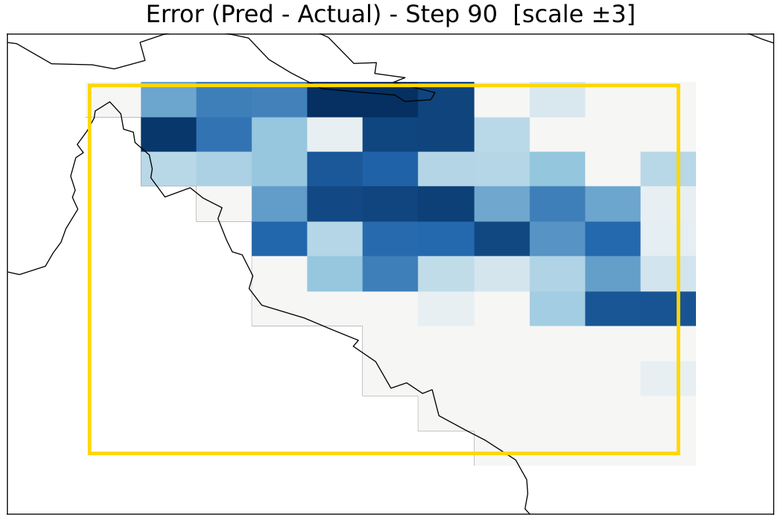
Actual DHW - Step 90

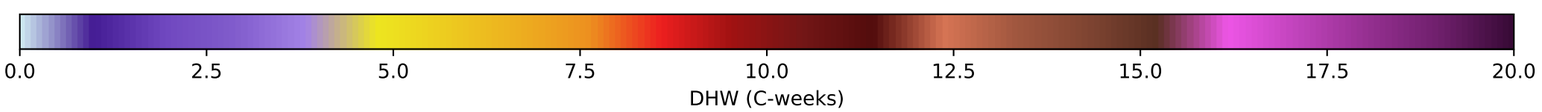


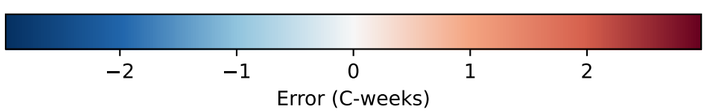
Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±3]

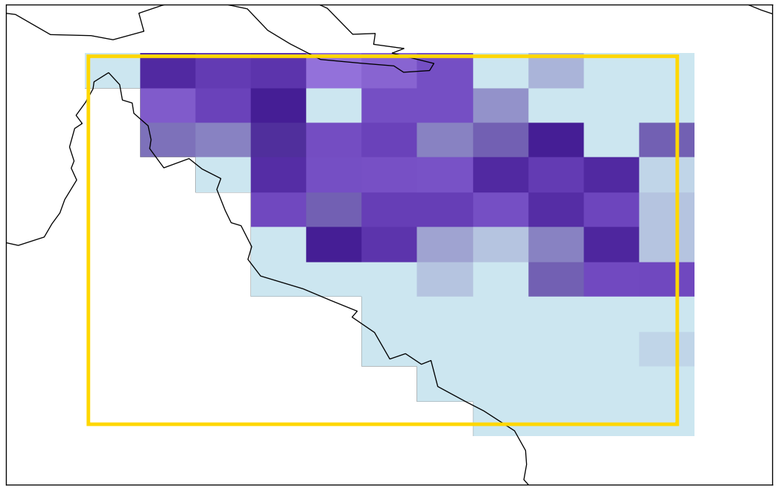




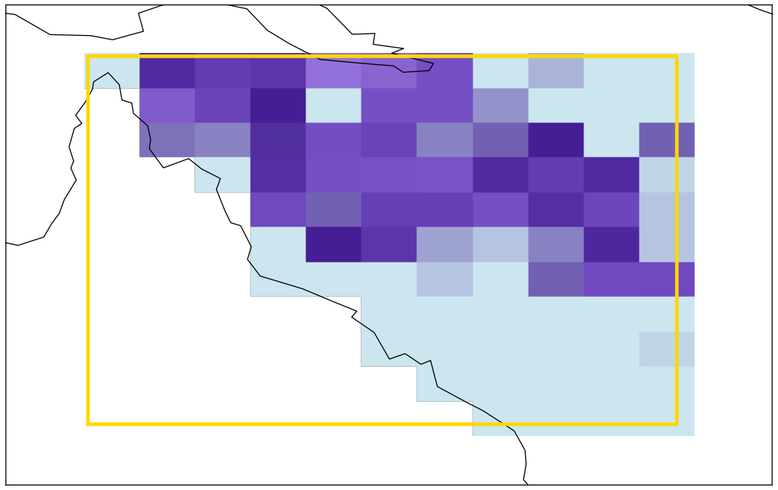


DHW Chain init: 2021-04-06 - Test - GBR (C-weeks)

Actual DHW - Step 1



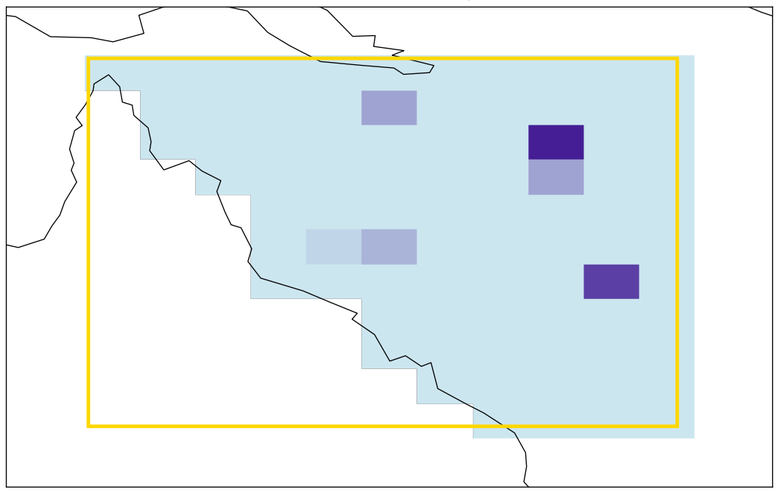
Predicted DHW - Step 1



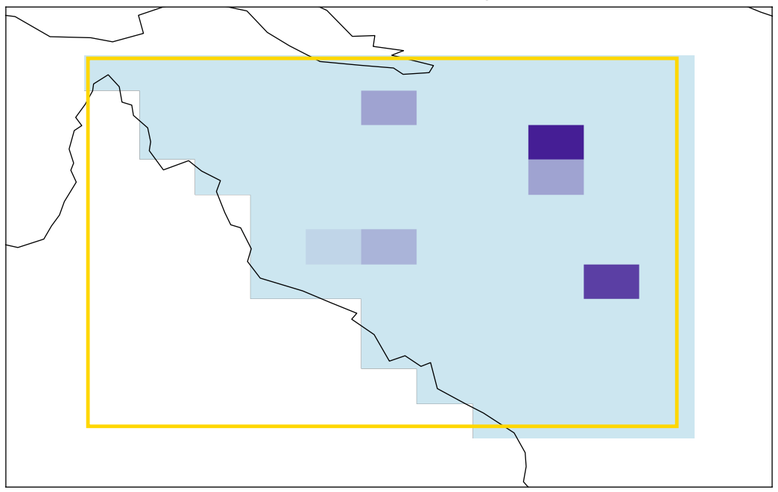
Error (Pred - Actual) - Step 1 [scale ±8]



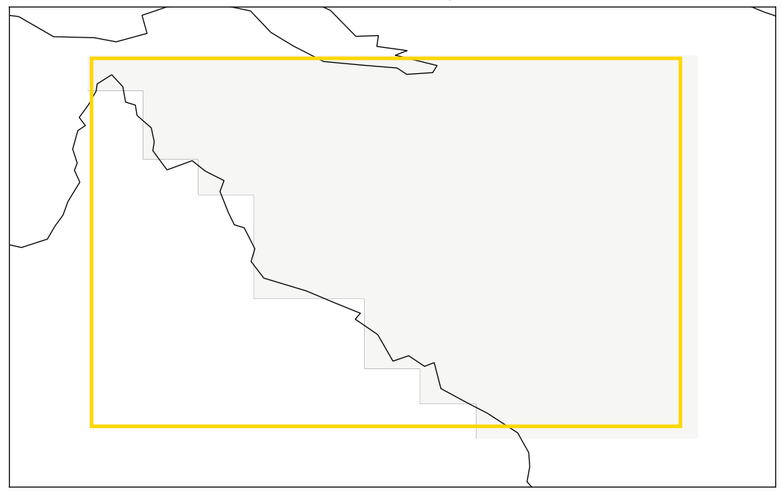
Actual DHW - Step 46



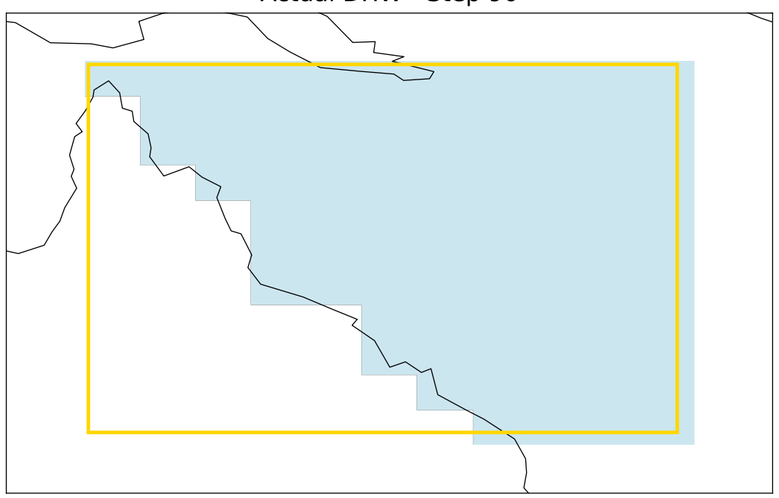
Predicted DHW - Step 46



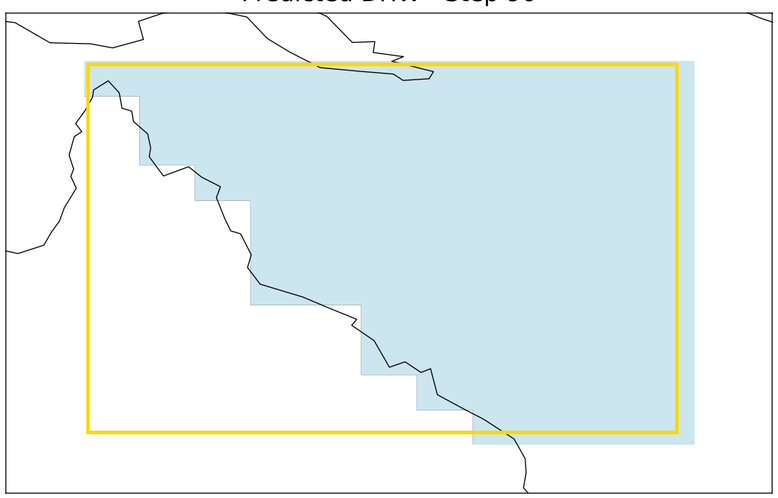
Error (Pred - Actual) - Step 46 [scale ±8]



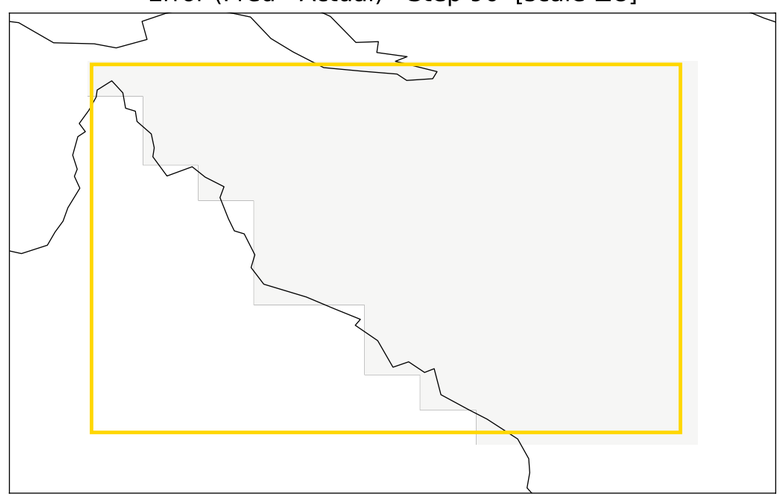
Actual DHW - Step 90

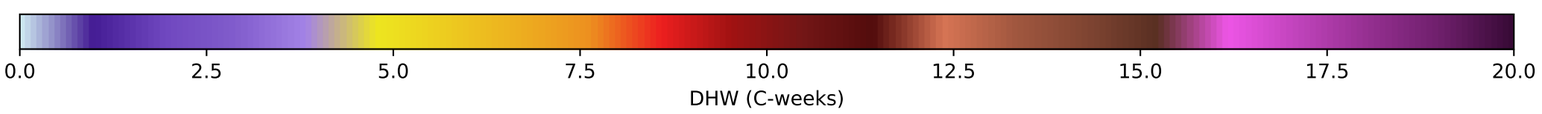


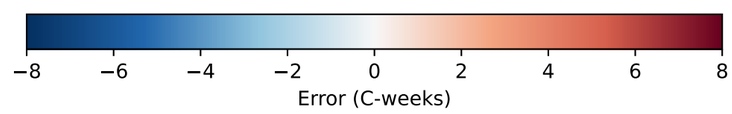
Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±8]

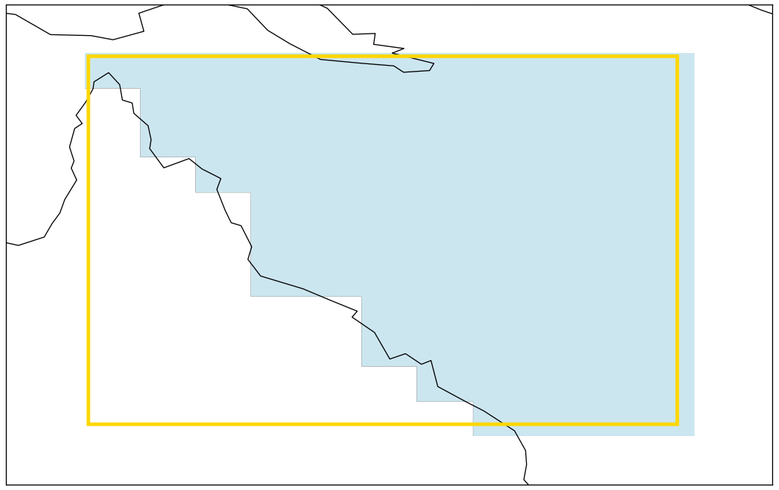




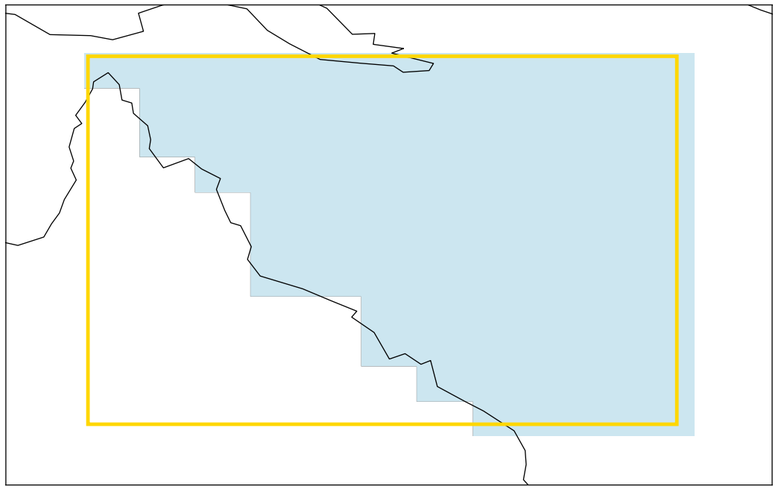


DHW Chain init: 2021-10-03 - Test - GBR (C-weeks)

Actual DHW - Step 1



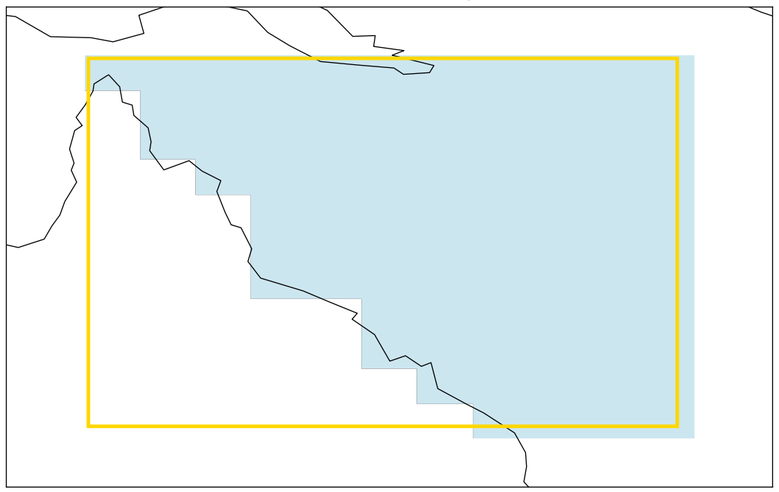
Predicted DHW - Step 1



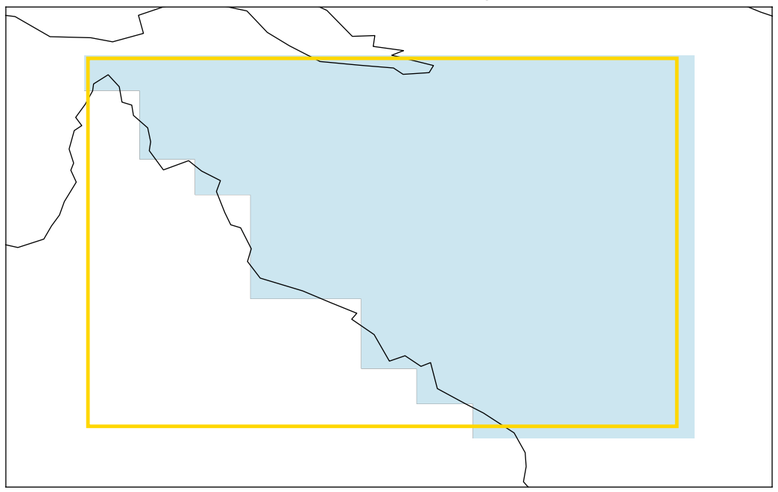
Error (Pred - Actual) - Step 1 [scale ±8]



Actual DHW - Step 46



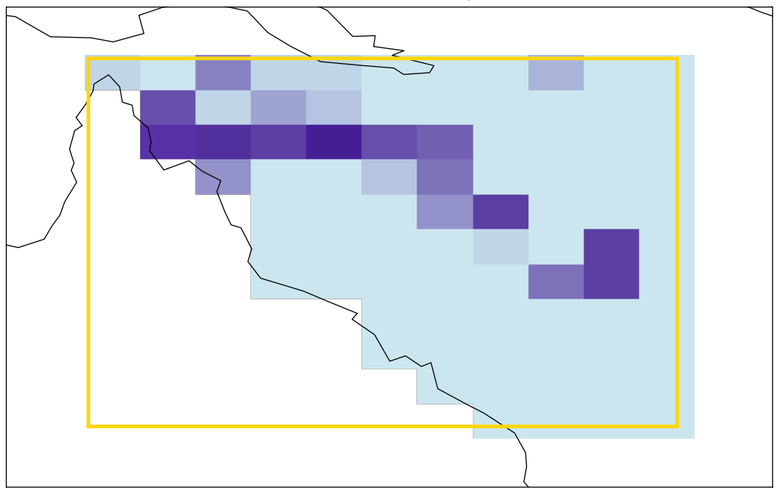
Predicted DHW - Step 46



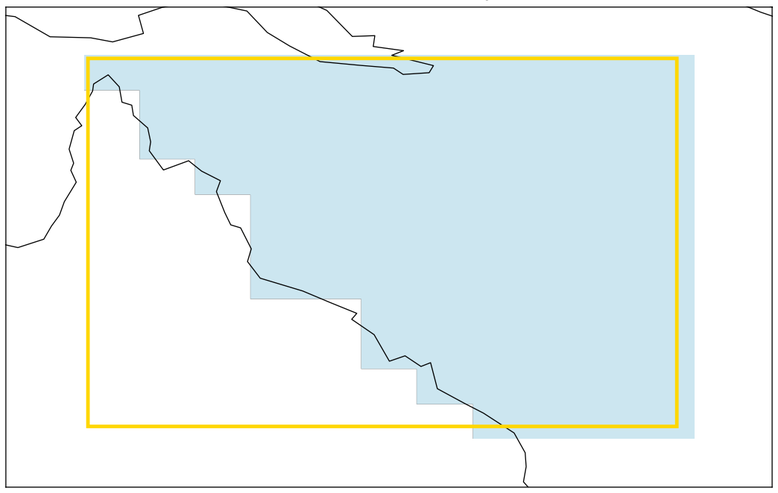
Error (Pred - Actual) - Step 46 [scale ±8]



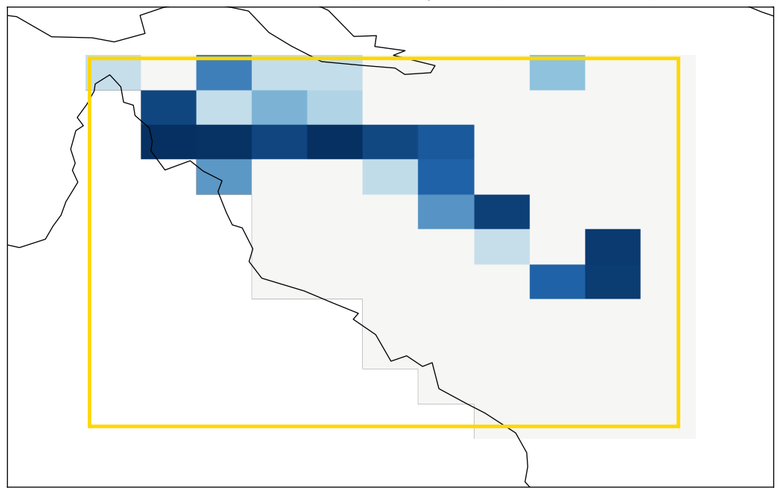
Actual DHW - Step 90

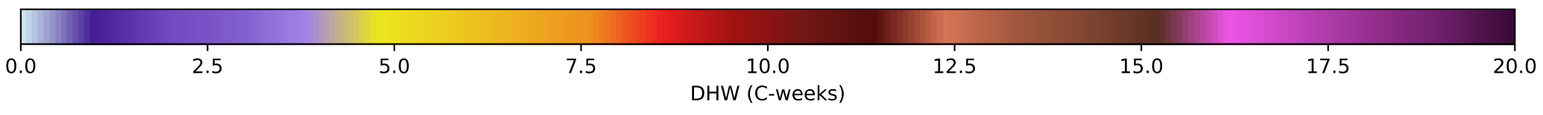


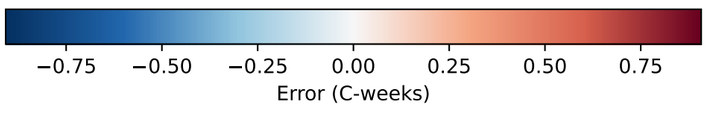
Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±0.91]

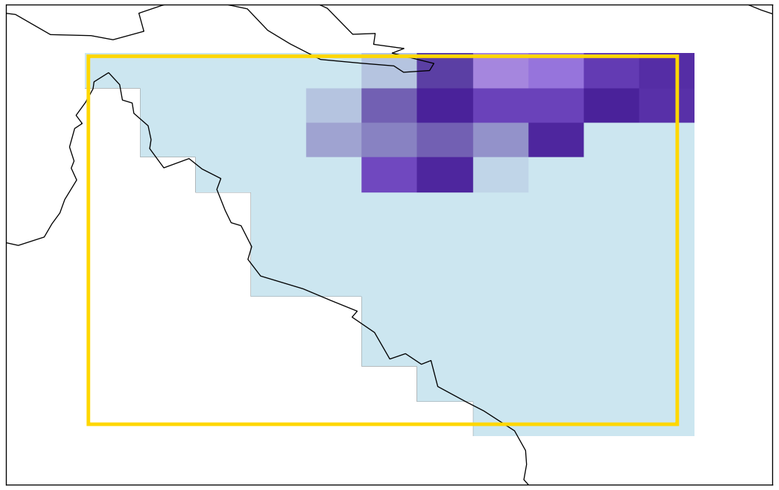




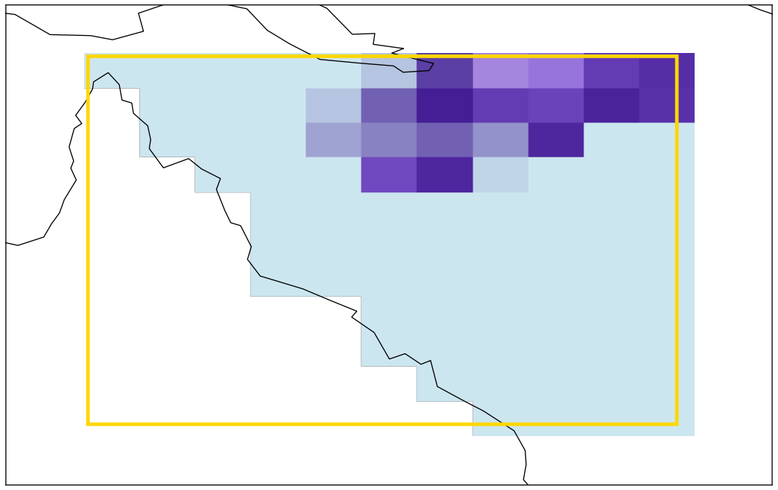


DHW Chain init: 2022-12-27 - Test - GBR (C-weeks)

Actual DHW - Step 1



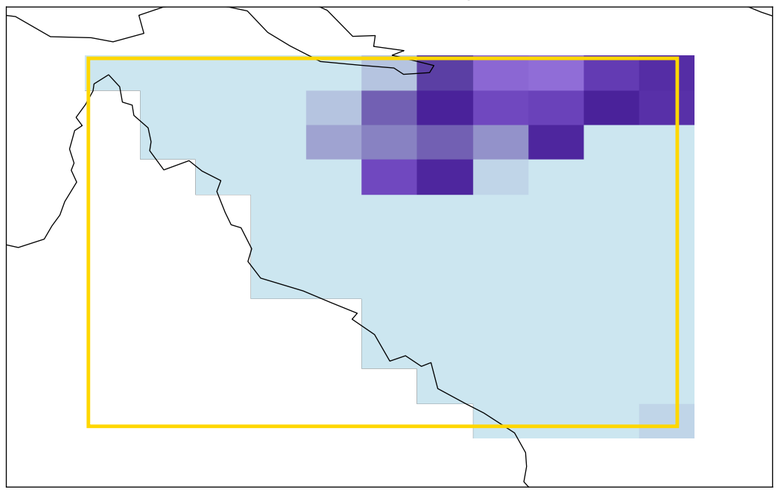
Predicted DHW - Step 1



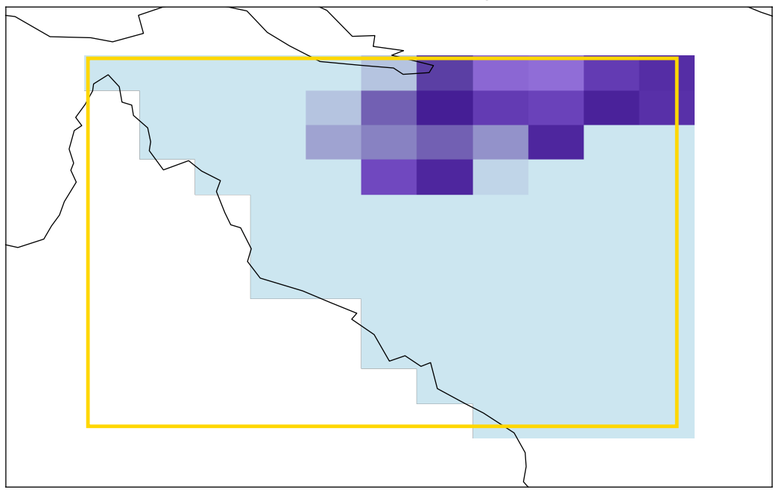
Error (Pred - Actual) - Step 1 [scale ±0.051]



Actual DHW - Step 46



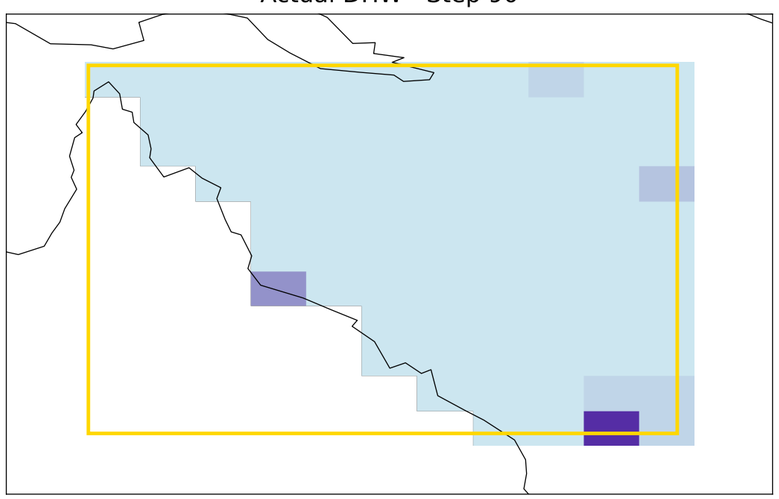
Predicted DHW - Step 46



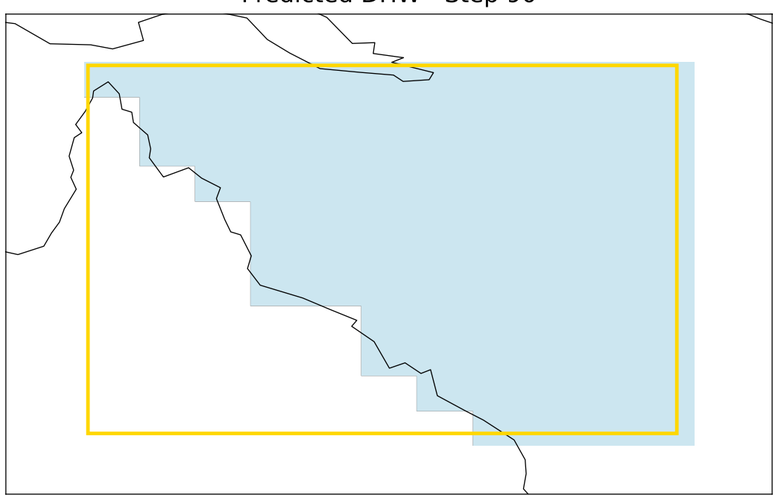
Error (Pred - Actual) - Step 46 [scale ±0.15]



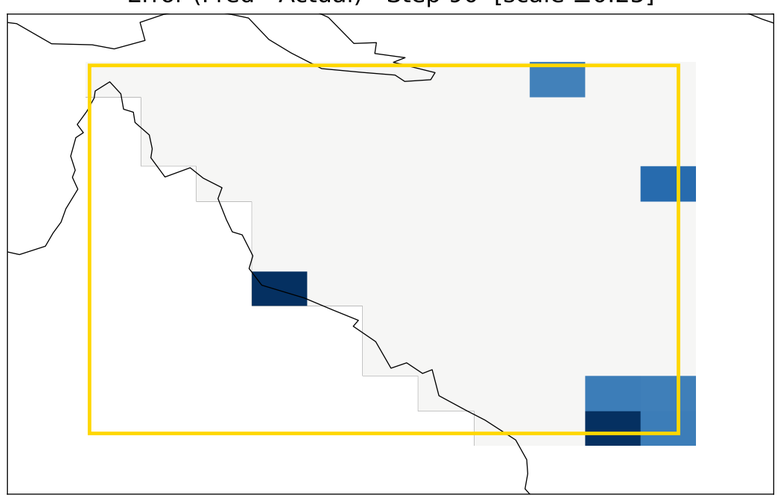
Actual DHW - Step 90

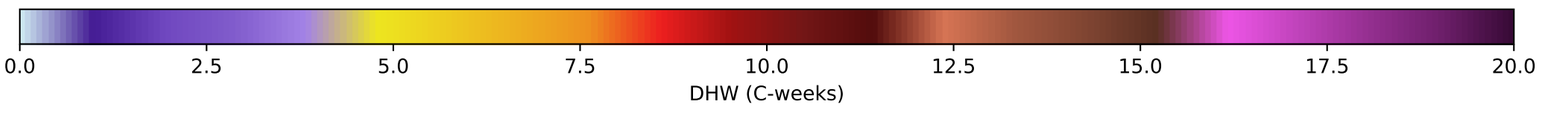


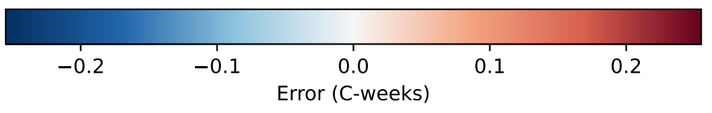
Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±0.25]

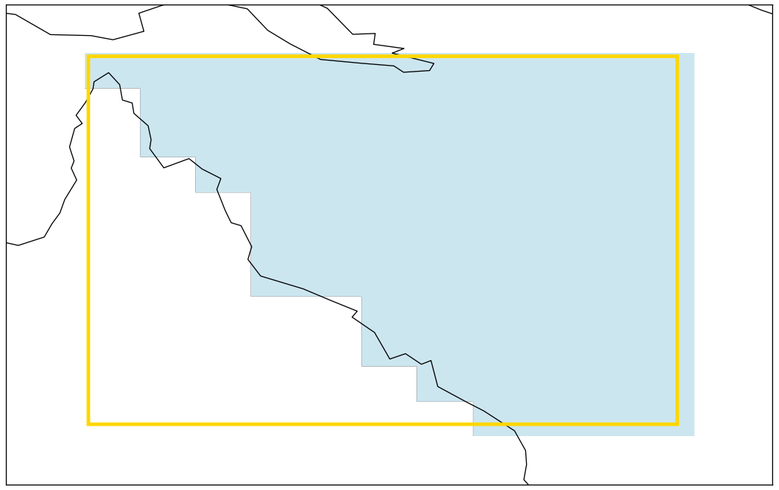




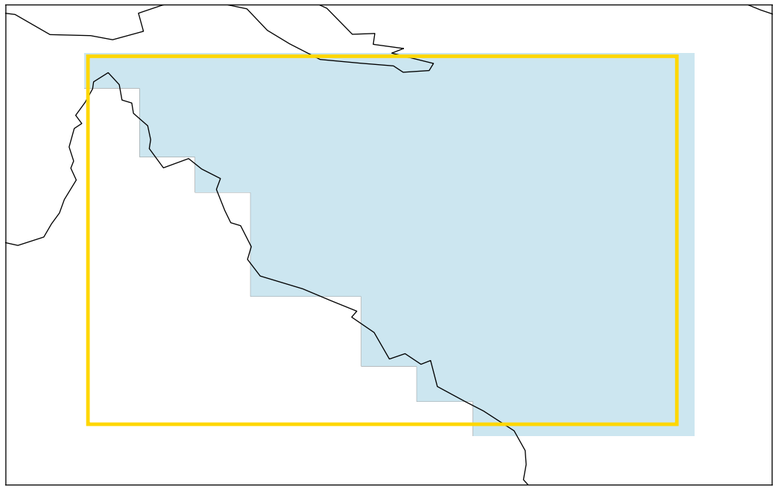


DHW Chain init: 2023-09-23 - Test - GBR (C-weeks)

Actual DHW - Step 1



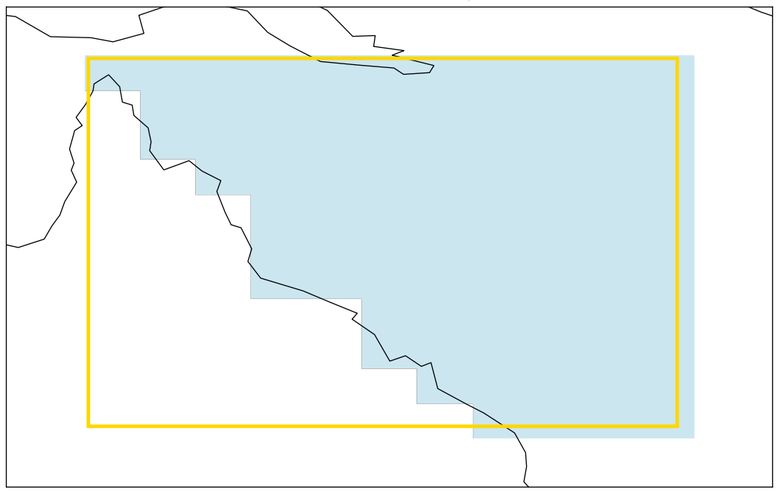
Predicted DHW - Step 1



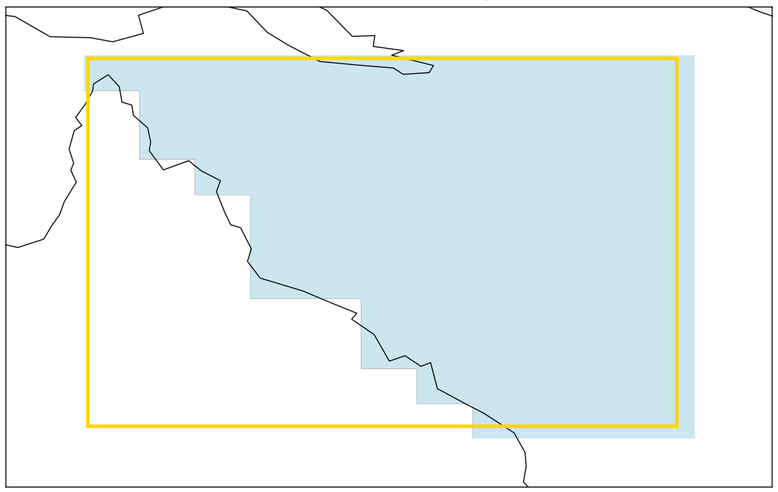
Error (Pred - Actual) - Step 1 [scale ±8]



Actual DHW - Step 46



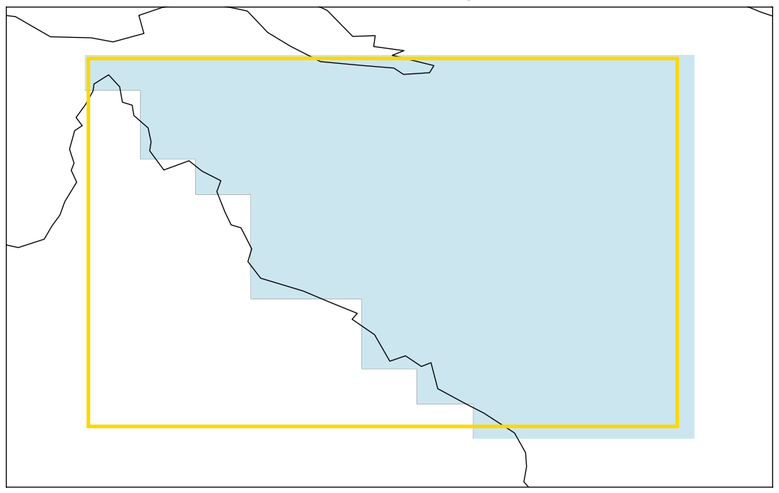
Predicted DHW - Step 46



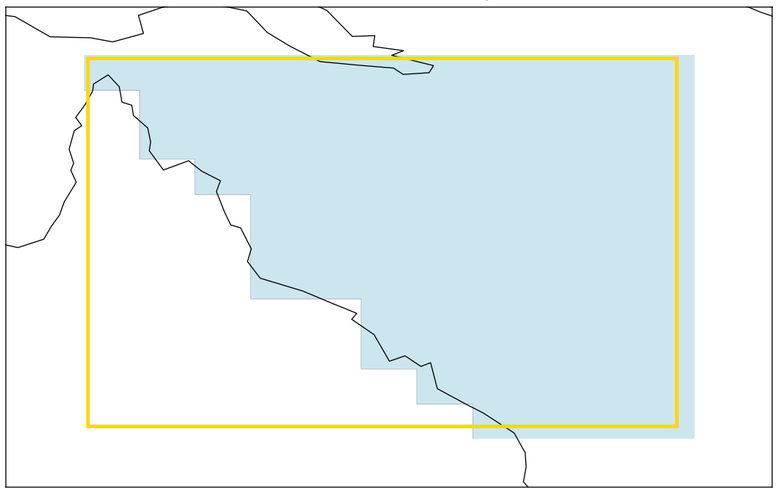
Error (Pred - Actual) - Step 46 [scale ±8]



Actual DHW - Step 90

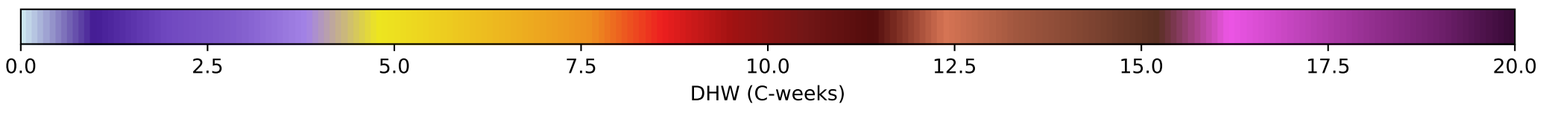


Predicted DHW - Step 90



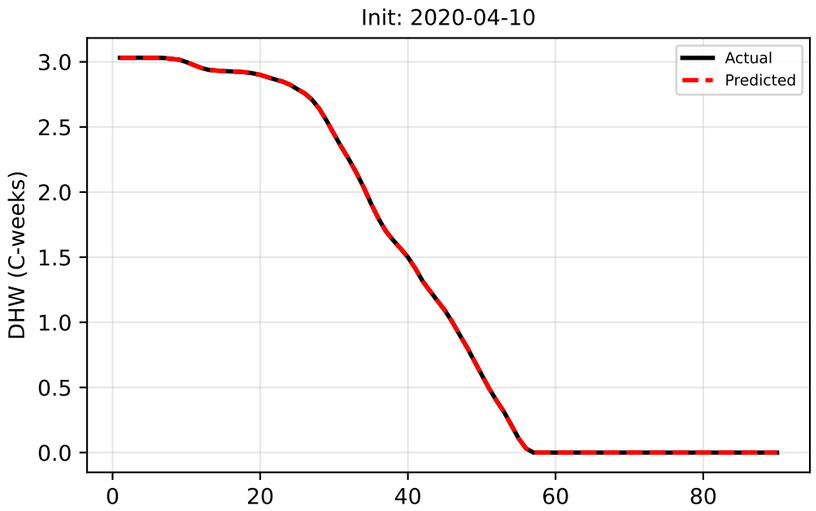
Error (Pred - Actual) - Step 90 [scale ±8]

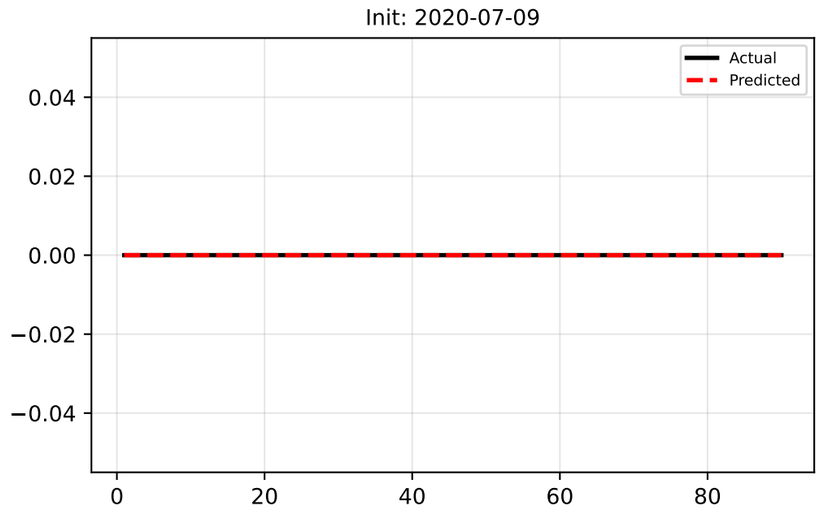


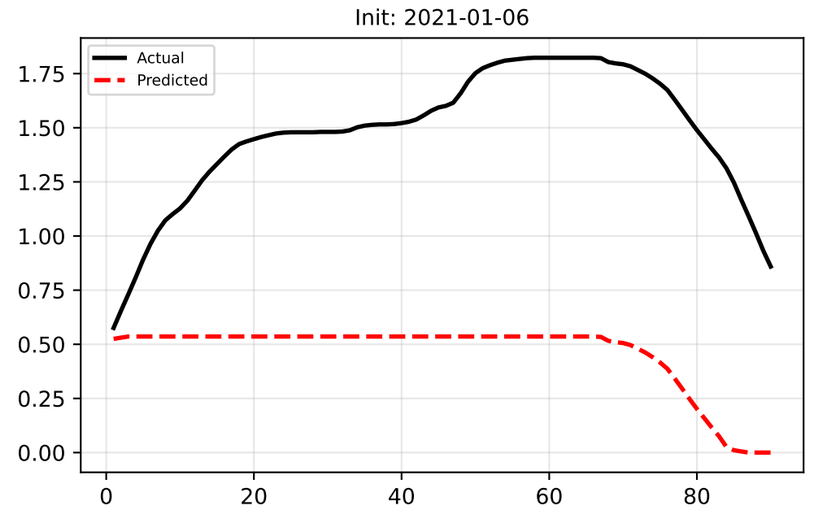


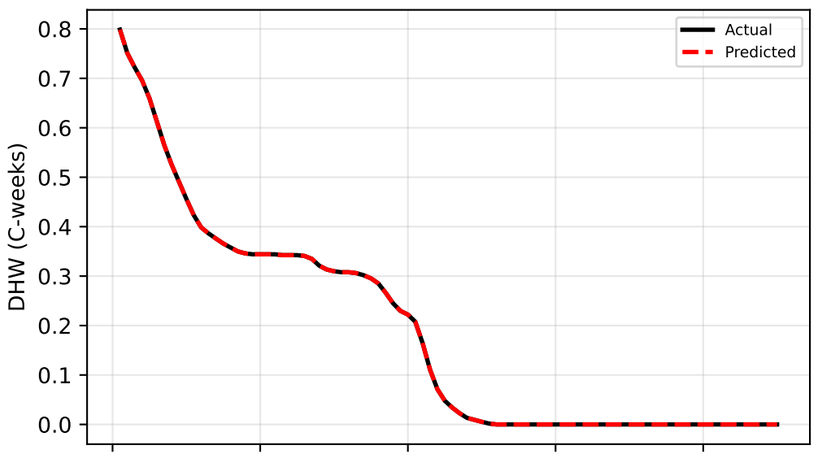


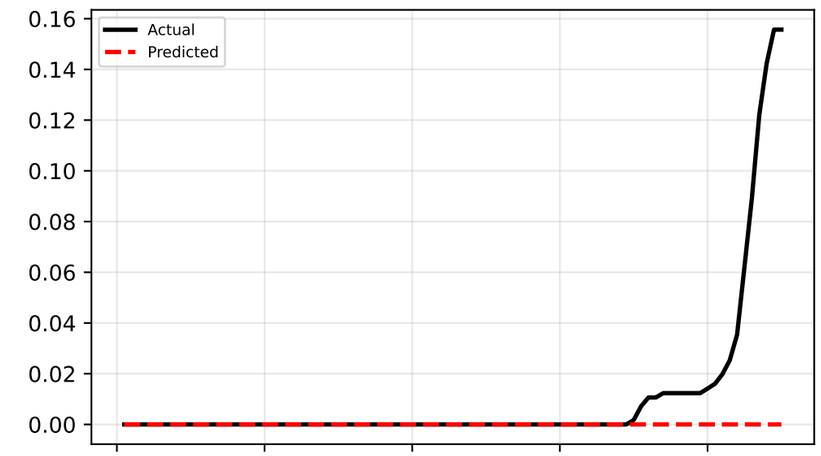
Global-Mean DHW - Test - GBR (C-weeks)

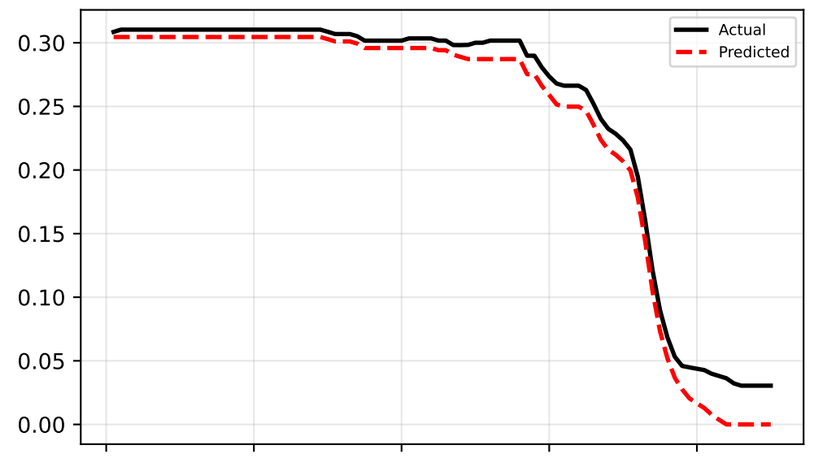


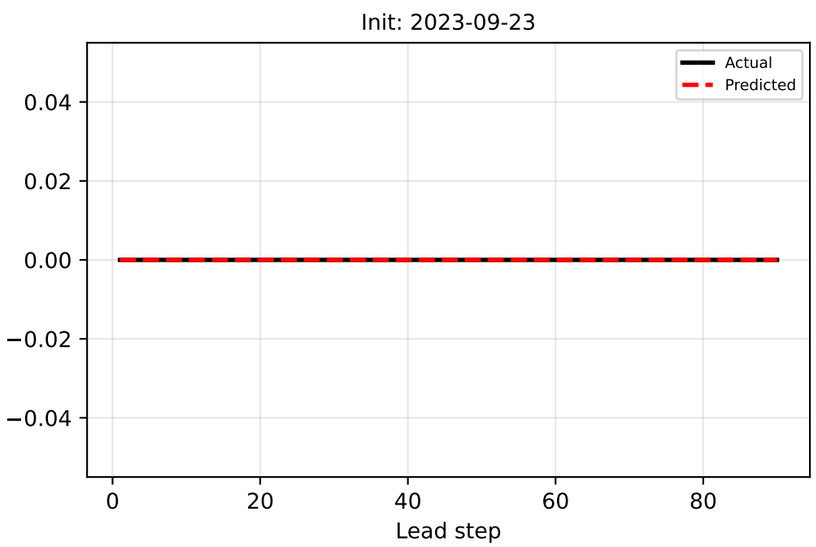


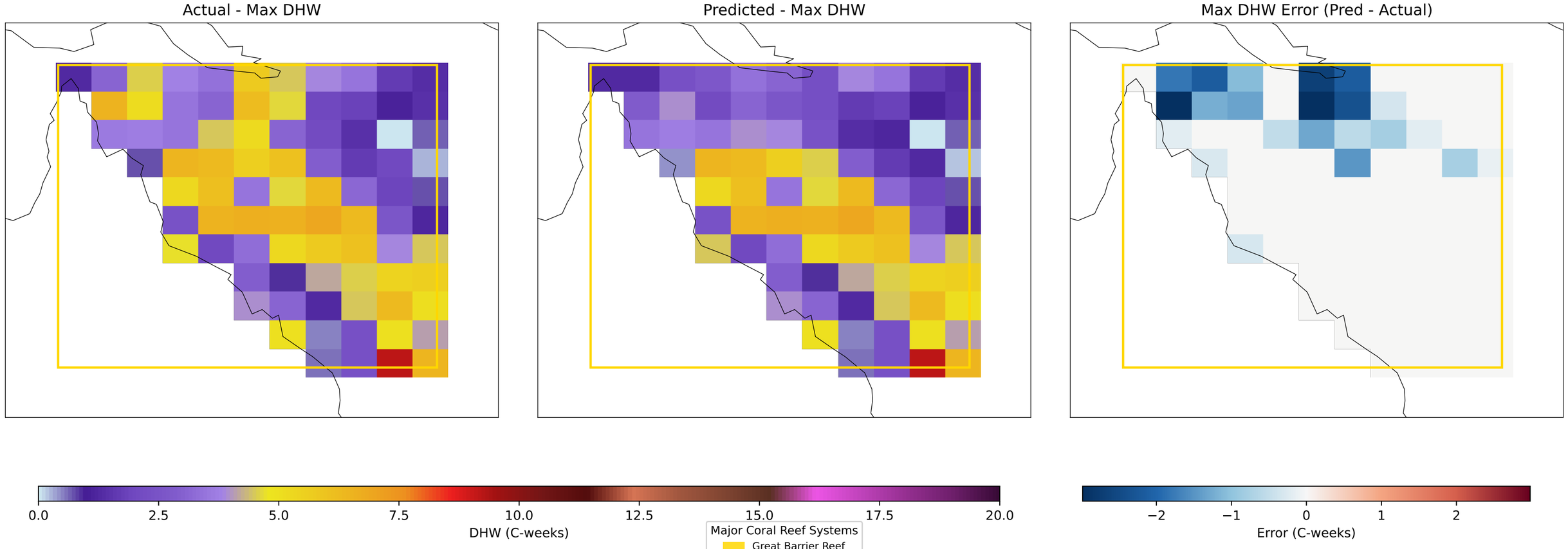


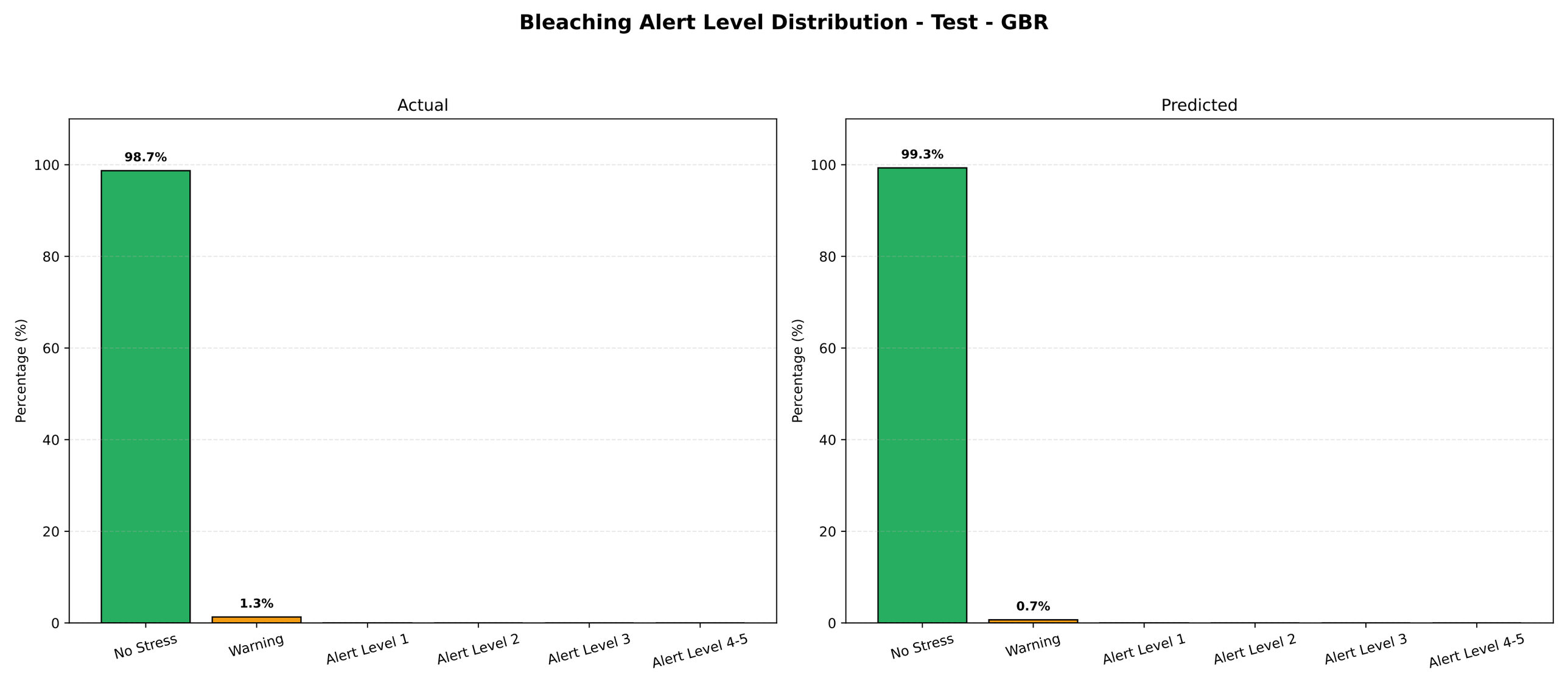






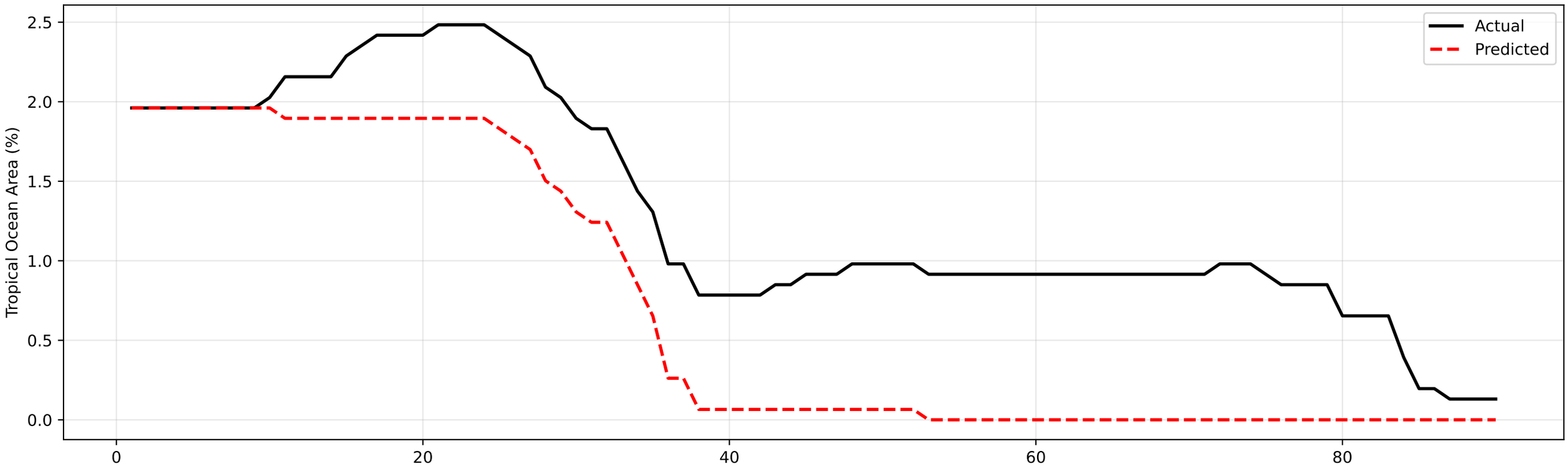



Peak DHW (max over all chains & leads) - Test - GBR (C-weeks)
Actual - Max DHW
Predicted - Max DHW
Max DHW Error (Pred - Actual)
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
DHW (C-weeks)
Major Coral Reef Systems
Great Barrier Reef
−2
−1
0
1
2
Error (C-weeks)

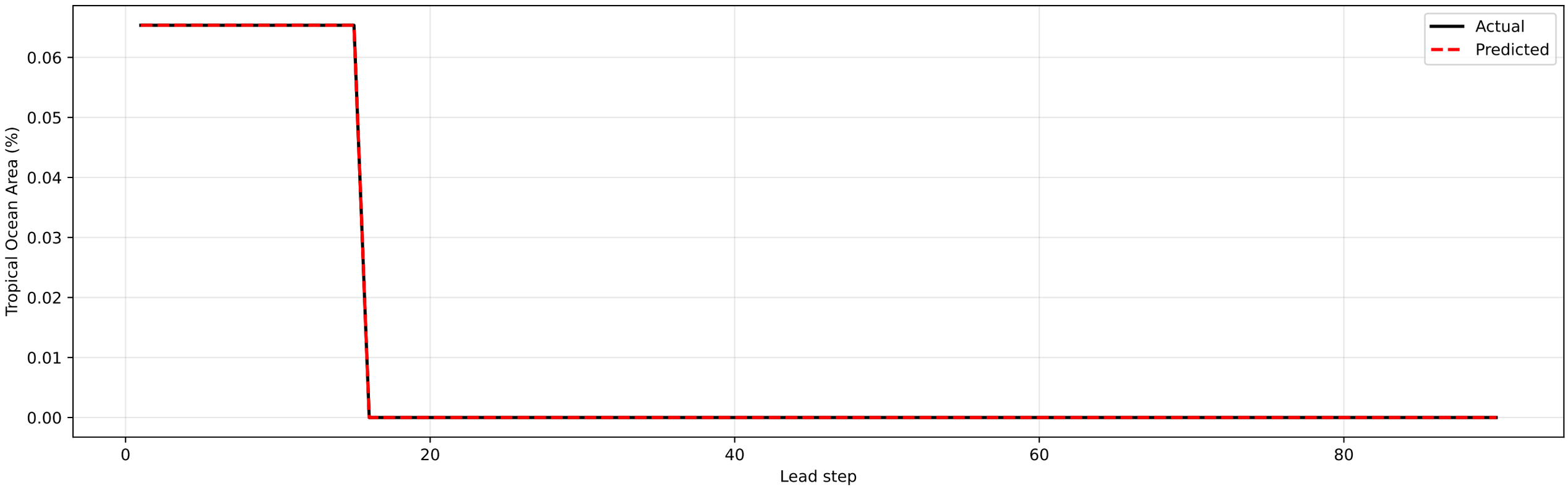

Bleaching Alert Level Distribution - Test - GBR
Actual
Predicted
Percentage (%)
0
20
40
60
80
100
98.7%
1.3%
99.3%
0.7%
No Stress
Warning
Alert Level 1
Alert Level 2
Alert Level 3
Alert Level 4-5

Tropical Ocean Area Under Bleaching Stress - Test - GBR

DHW >= 4 C-weeks (Alert Level 1 DHW threshold)



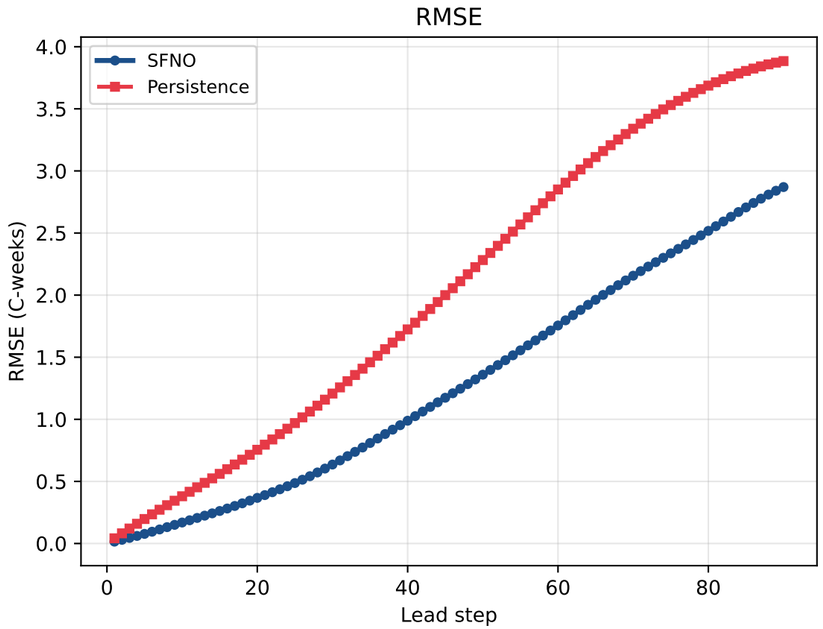
DHW >= 8 C-weeks (Alert Level 2 DHW threshold)

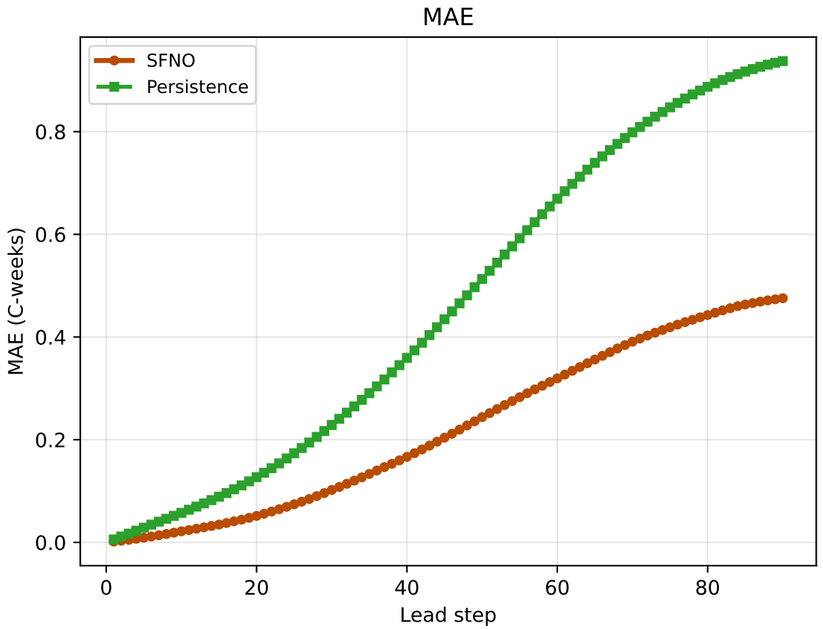


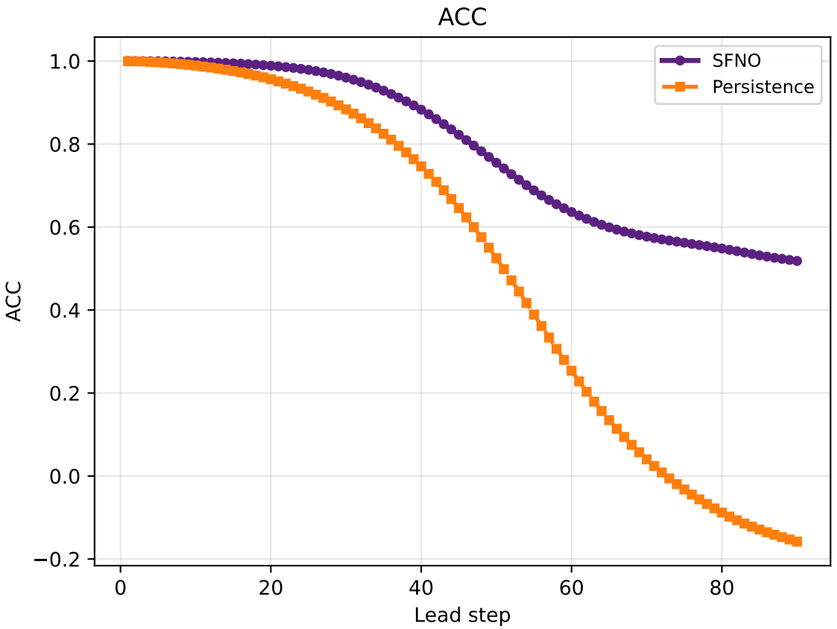
DHW Metrics by Lead Step - Val (15 chains scored at every lead)

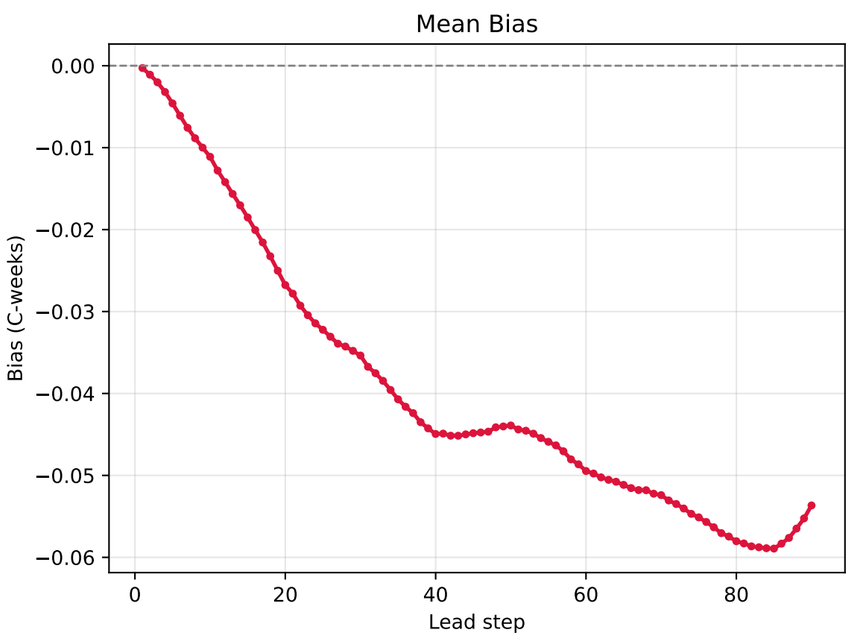
Lead Step	SFNO RMSE	Persist RMSE	Skill %	SFNO MAE	Persist MAE	SFNO Bias	SFNO ACC	Persist ACC	SFNO R²
1	0.0151	0.0418	63.9%	0.0015	0.0057	-0.0003	1.0000	0.9999	1.0000
2	0.0298	0.0818	63.6%	0.0032	0.0114	-0.0011	0.9999	0.9995	0.9999
3	0.0451	0.1205	62.6%	0.0051	0.0171	-0.0020	0.9998	0.9989	0.9997
4	0.0609	0.1589	61.7%	0.0071	0.0228	-0.0032	0.9997	0.9981	0.9995
5	0.0775	0.1972	60.7%	0.0092	0.0287	-0.0046	0.9996	0.9971	0.9993
6	0.0950	0.2350	59.6%	0.0115	0.0346	-0.0061	0.9993	0.9959	0.9989
7	0.1131	0.2720	58.4%	0.0139	0.0403	-0.0076	0.9990	0.9944	0.9984
8	0.1317	0.3084	57.3%	0.0163	0.0460	-0.0089	0.9987	0.9928	0.9979
9	0.1502	0.3444	56.4%	0.0189	0.0517	-0.0100	0.9983	0.9911	0.9972
10	0.1690	0.3804	55.6%	0.0216	0.0576	-0.0111	0.9978	0.9891	0.9965
11	0.1871	0.4166	55.1%	0.0241	0.0636	-0.0128	0.9973	0.9869	0.9956
12	0.2057	0.4526	54.5%	0.0267	0.0697	-0.0142	0.9968	0.9845	0.9947
13	0.2243	0.4885	54.1%	0.0293	0.0760	-0.0156	0.9961	0.9819	0.9937
14	0.2427	0.5245	53.7%	0.0319	0.0824	-0.0170	0.9954	0.9791	0.9926
15	0.2618	0.5608	53.3%	0.0347	0.0890	-0.0185	0.9946	0.9760	0.9913
16	0.2816	0.5981	52.9%	0.0377	0.0961	-0.0201	0.9937	0.9727	0.9898
17	0.3023	0.6362	52.5%	0.0410	0.1034	-0.0216	0.9927	0.9691	0.9882
18	0.3235	0.6749	52.1%	0.0443	0.1111	-0.0233	0.9915	0.9651	0.9864
19	0.3452	0.7145	51.7%	0.0479	0.1189	-0.0250	0.9903	0.9608	0.9843
20	0.3673	0.7547	51.3%	0.0516	0.1271	-0.0268	0.9888	0.9562	0.9821
21	0.3897	0.7957	51.0%	0.0559	0.1356	-0.0278	0.9872	0.9512	0.9796
22	0.4129	0.8377	50.7%	0.0601	0.1446	-0.0293	0.9855	0.9457	0.9768
23	0.4365	0.8806	50.4%	0.0646	0.1539	-0.0304	0.9835	0.9399	0.9738
24	0.4611	0.9245	50.1%	0.0693	0.1636	-0.0314	0.9813	0.9335	0.9704
25	0.4866	0.9697	49.8%	0.0742	0.1736	-0.0322	0.9788	0.9266	0.9665
26	0.5135	1.0157	49.4%	0.0793	0.1839	-0.0331	0.9759	0.9191	0.9622
27	0.5421	1.0625	49.0%	0.0845	0.1945	-0.0339	0.9727	0.9112	0.9573
28	0.5720	1.1099	48.5%	0.0902	0.2055	-0.0343	0.9690	0.9026	0.9518
29	0.6033	1.1584	47.9%	0.0960	0.2170	-0.0348	0.9648	0.8934	0.9457
30	0.6358	1.2074	47.3%	0.1020	0.2288	-0.0354	0.9602	0.8836	0.9388
31	0.6691	1.2569	46.8%	0.1079	0.2407	-0.0367	0.9550	0.8732	0.9313
32	0.7034	1.3067	46.2%	0.1141	0.2527	-0.0375	0.9493	0.8621	0.9230
33	0.7380	1.3569	45.6%	0.1204	0.2650	-0.0385	0.9431	0.8504	0.9140
34	0.7731	1.4076	45.1%	0.1267	0.2776	-0.0396	0.9362	0.8380	0.9042
35	0.8093	1.4593	44.5%	0.1333	0.2906	-0.0407	0.9287	0.8247	0.8935
36	0.8456	1.5117	44.1%	0.1400	0.3038	-0.0416	0.9206	0.8106	0.8820
37	0.8814	1.5646	43.7%	0.1466	0.3173	-0.0424	0.9120	0.7957	0.8699
38	0.9167	1.6177	43.3%	0.1531	0.3312	-0.0435	0.9029	0.7800	0.8573
39	0.9525	1.6708	43.0%	0.1598	0.3452	-0.0443	0.8932	0.7636	0.8438
40	0.9887	1.7241	42.7%	0.1666	0.3594	-0.0449	0.8829	0.7464	0.8295
41	1.0255	1.7780	42.3%	0.1738	0.3740	-0.0449	0.8718	0.7282	0.8142
42	1.0625	1.8327	42.0%	0.1810	0.3888	-0.0452	0.8602	0.7090	0.7982
43	1.0997	1.8883	41.8%	0.1885	0.4038	-0.0452	0.8480	0.6888	0.7814
44	1.1369	1.9443	41.5%	0.1962	0.4191	-0.0450	0.8354	0.6676	0.7639
45	1.1736	1.9999	41.3%	0.2039	0.4345	-0.0448	0.8225	0.6458	0.7460
46	1.2098	2.0555	41.1%	0.2117	0.4499	-0.0448	0.8096	0.6233	0.7278
47	1.2463	2.1115	41.0%	0.2196	0.4656	-0.0447	0.7963	0.5999	0.7090
48	1.2836	2.1682	40.8%	0.2278	0.4813	-0.0441	0.7826	0.5756	0.6894
49	1.3211	2.2251	40.6%	0.2358	0.4971	-0.0440	0.7689	0.5505	0.6694
50	1.3594	2.2823	40.4%	0.2440	0.5130	-0.0439	0.7551	0.5248	0.6487
51	1.3977	2.3396	40.3%	0.2518	0.5290	-0.0444	0.7412	0.4984	0.6276
52	1.4367	2.3967	40.1%	0.2597	0.5449	-0.0445	0.7275	0.4717	0.6060
53	1.4758	2.4537	39.9%	0.2675	0.5607	-0.0449	0.7140	0.4446	0.5843
54	1.5154	2.5113	39.7%	0.2752	0.5766	-0.0454	0.7009	0.4170	0.5625
55	1.5555	2.5689	39.4%	0.2829	0.5923	-0.0459	0.6883	0.3893	0.5406
56	1.5956	2.6262	39.2%	0.2905	0.6080	-0.0463	0.6763	0.3615	0.5191
57	1.6353	2.6833	39.1%	0.2979	0.6237	-0.0471	0.6653	0.3339	0.4984
58	1.6745	2.7398	38.9%	0.3051	0.6392	-0.0480	0.6551	0.3066	0.4785
59	1.7148	2.7954	38.7%	0.3123	0.6543	-0.0486	0.6453	0.2799	0.4585
60	1.7553	2.8507	38.4%	0.3195	0.6692	-0.0495	0.6362	0.2536	0.4389
61	1.7969	2.9054	38.2%	0.3268	0.6839	-0.0498	0.6277	0.2280	0.4194
62	1.8385	2.9593	37.9%	0.3341	0.6983	-0.0502	0.6197	0.2031	0.4005
63	1.8804	3.0117	37.6%	0.3415	0.7123	-0.0505	0.6123	0.1793	0.3818
64	1.9221	3.0626	37.2%	0.3489	0.7259	-0.0508	0.6055	0.1565	0.3636
65	1.9625	3.1120	36.9%	0.3562	0.7391	-0.0512	0.5993	0.1348	0.3464
66	2.0017	3.1601	36.7%	0.3633	0.7519	-0.0515	0.5939	0.1140	0.3304
67	2.0407	3.2071	36.4%	0.3704	0.7641	-0.0518	0.5891	0.0941	0.3150
68	2.0800	3.2530	36.1%	0.3776	0.7761	-0.0518	0.5847	0.0752	0.2997
69	2.1182	3.2971	35.8%	0.3843	0.7876	-0.0522	0.5806	0.0572	0.2850
70	2.1563	3.3397	35.4%	0.3909	0.7986	-0.0524	0.5769	0.0402	0.2705
71	2.1931	3.3806	35.1%	0.3970	0.8092	-0.0531	0.5734	0.0240	0.2567
72	2.2296	3.4203	34.8%	0.4029	0.8194	-0.0535	0.5704	0.0088	0.2433
73	2.2651	3.4585	34.5%	0.4084	0.8291	-0.0540	0.5676	-0.0057	0.2306
74	2.3003	3.4954	34.2%	0.4137	0.8384	-0.0547	0.5649	-0.0195	0.2179
75	2.3362	3.5305	33.8%	0.4190	0.8474	-0.0551	0.5622	-0.0325	0.2044
76	2.3723	3.5641	33.4%	0.4241	0.8560	-0.0557	0.5593	-0.0446	0.1906
77	2.4083	3.5969	33.0%	0.4290	0.8644	-0.0563	0.5567	-0.0562	0.1771
78	2.4439	3.6284	32.6%	0.4336	0.8724	-0.0570	0.5540	-0.0673	0.1634
79	2.4806	3.6583	32.2%	0.4383	0.8801	-0.0575	0.5512	-0.0779	0.1482
80	2.5172	3.6871	31.7%	0.4428	0.8873	-0.0580	0.5483	-0.0880	0.1328
81	2.5548	3.7142	31.2%	0.4474	0.8942	-0.0583	0.5452	-0.0976	0.1160
82	2.5926	3.7392	30.7%	0.4518	0.9006	-0.0587	0.5419	-0.1065	0.0983
83	2.6306	3.7626	30.1%	0.4560	0.9064	-0.0588	0.5386	-0.1145	0.0801
84	2.6689	3.7845	29.5%	0.4602	0.9119	-0.0589	0.5351	-0.1220	0.0614
85	2.7061	3.8050	28.9%	0.4634	0.9171	-0.0589	0.5318	-0.1289	0.0430
86	2.7421	3.8241	28.3%	0.4662	0.9218	-0.0583	0.5288	-0.1354	0.0249
87	2.7766	3.8419	27.7%	0.4689	0.9263	-0.0576	0.5260	-0.1415	0.0071
88	2.8096	3.8579	27.2%	0.4714	0.9304	-0.0565	0.5234	-0.1473	-0.0109
89	2.8404	3.8719	26.6%	0.4735	0.9341	-0.0552	0.5208	-0.1528	-0.0291
90	2.8701	3.8845	26.1%	0.4755	0.9374	-0.0537	0.5183	-0.1581	-0.0474

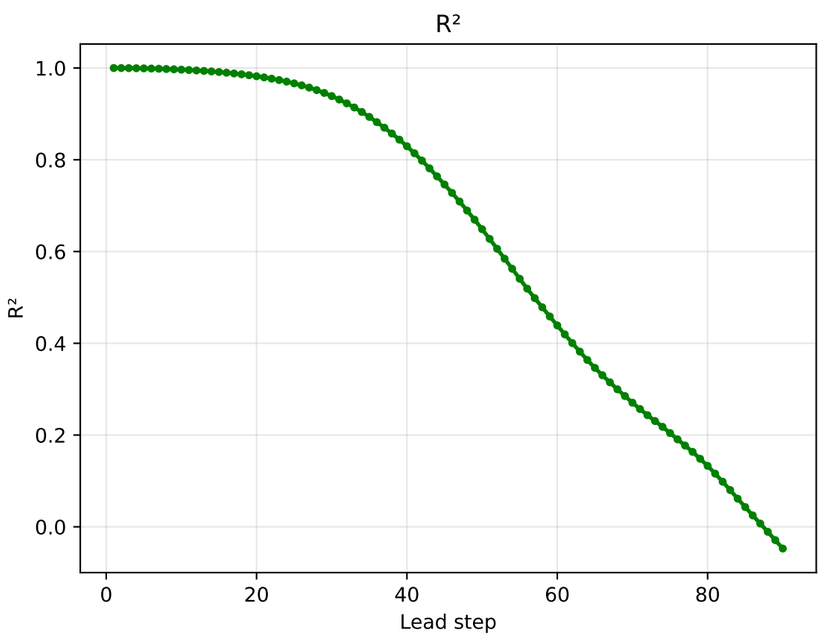
DHW Forecast Skill vs Lead Step - Val

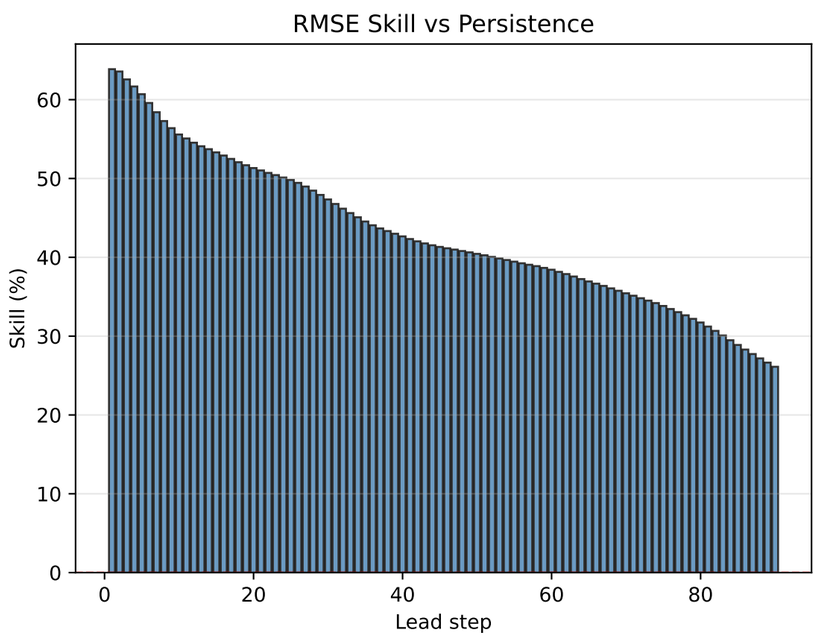


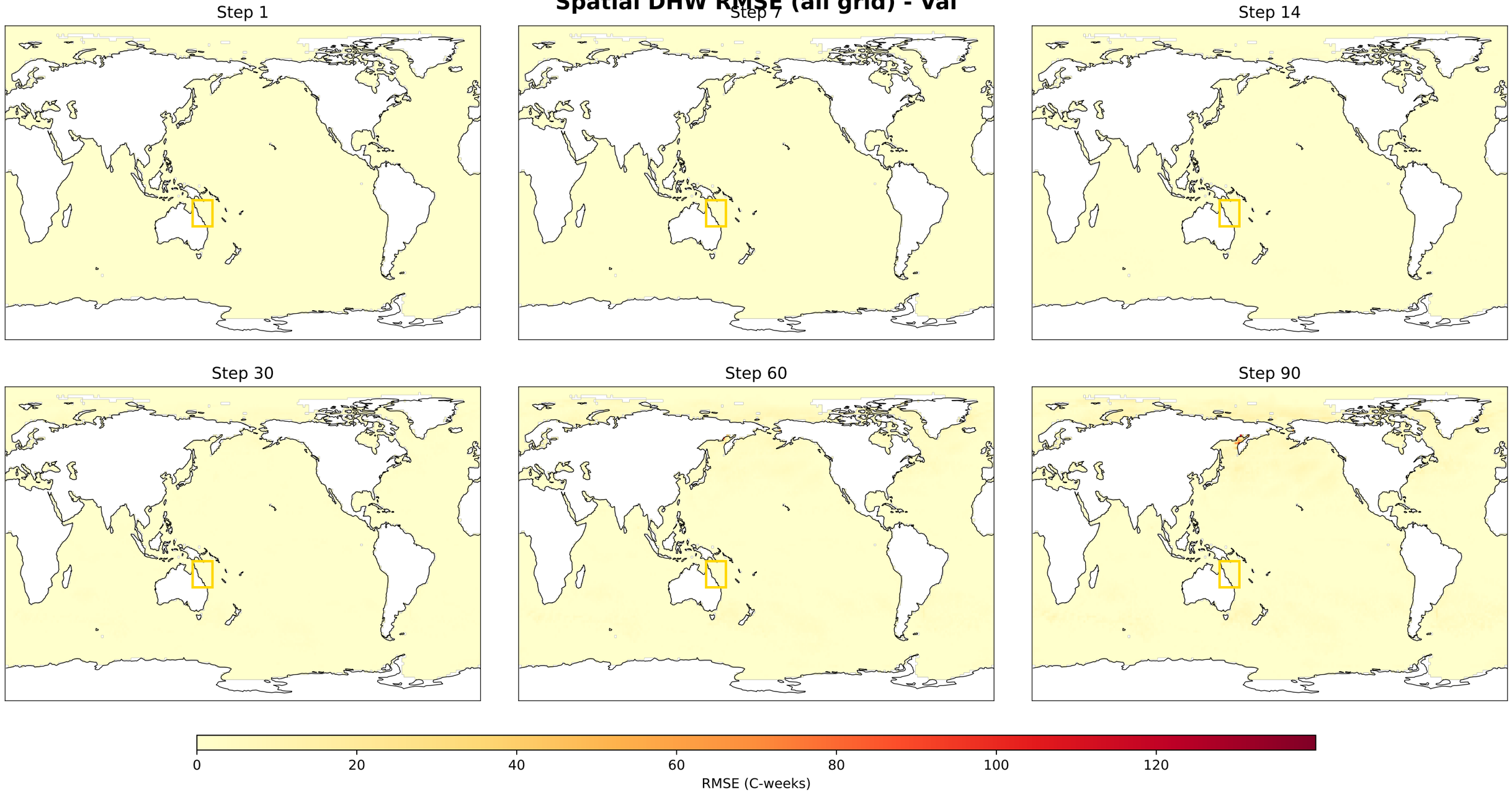


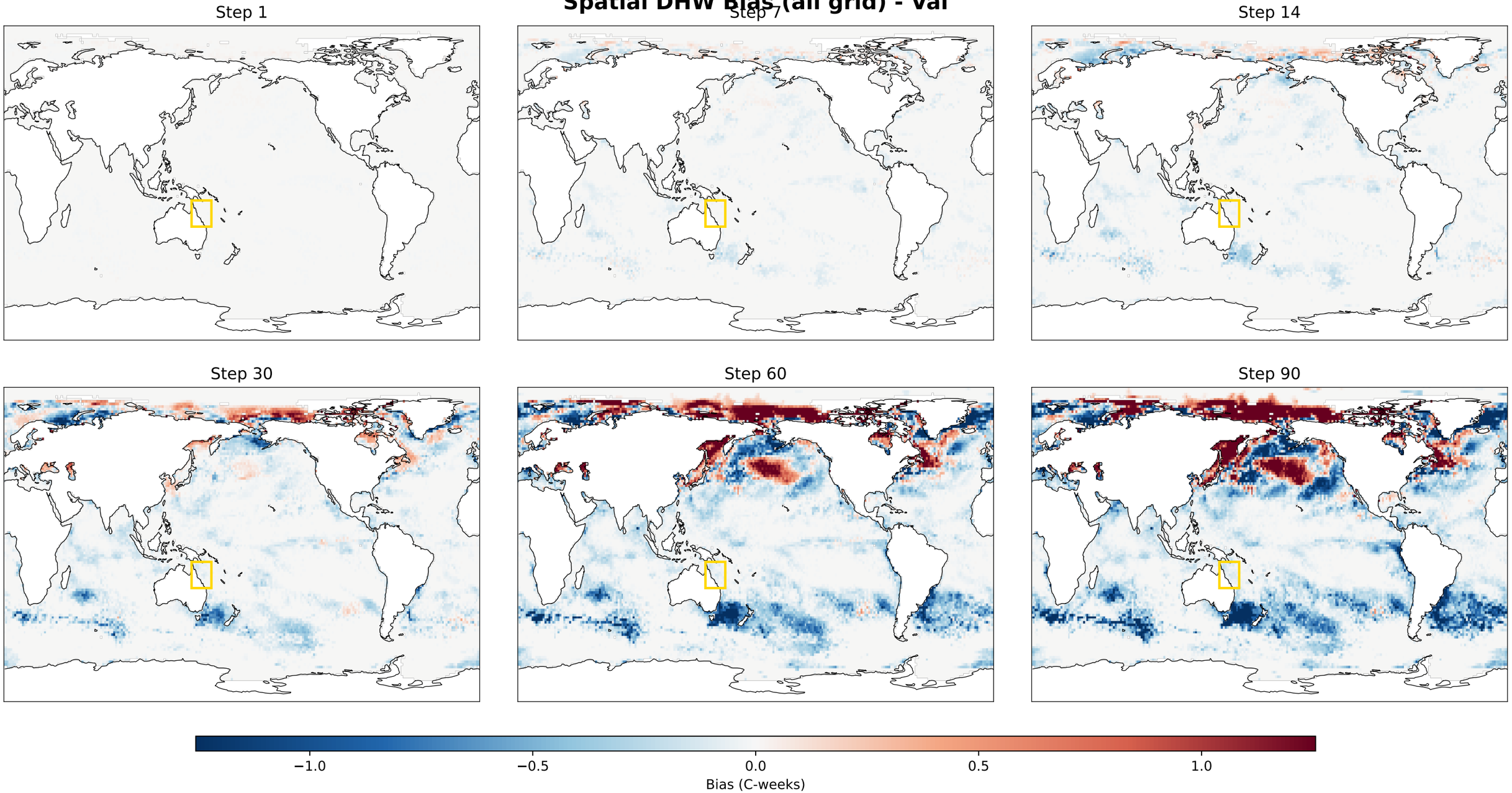


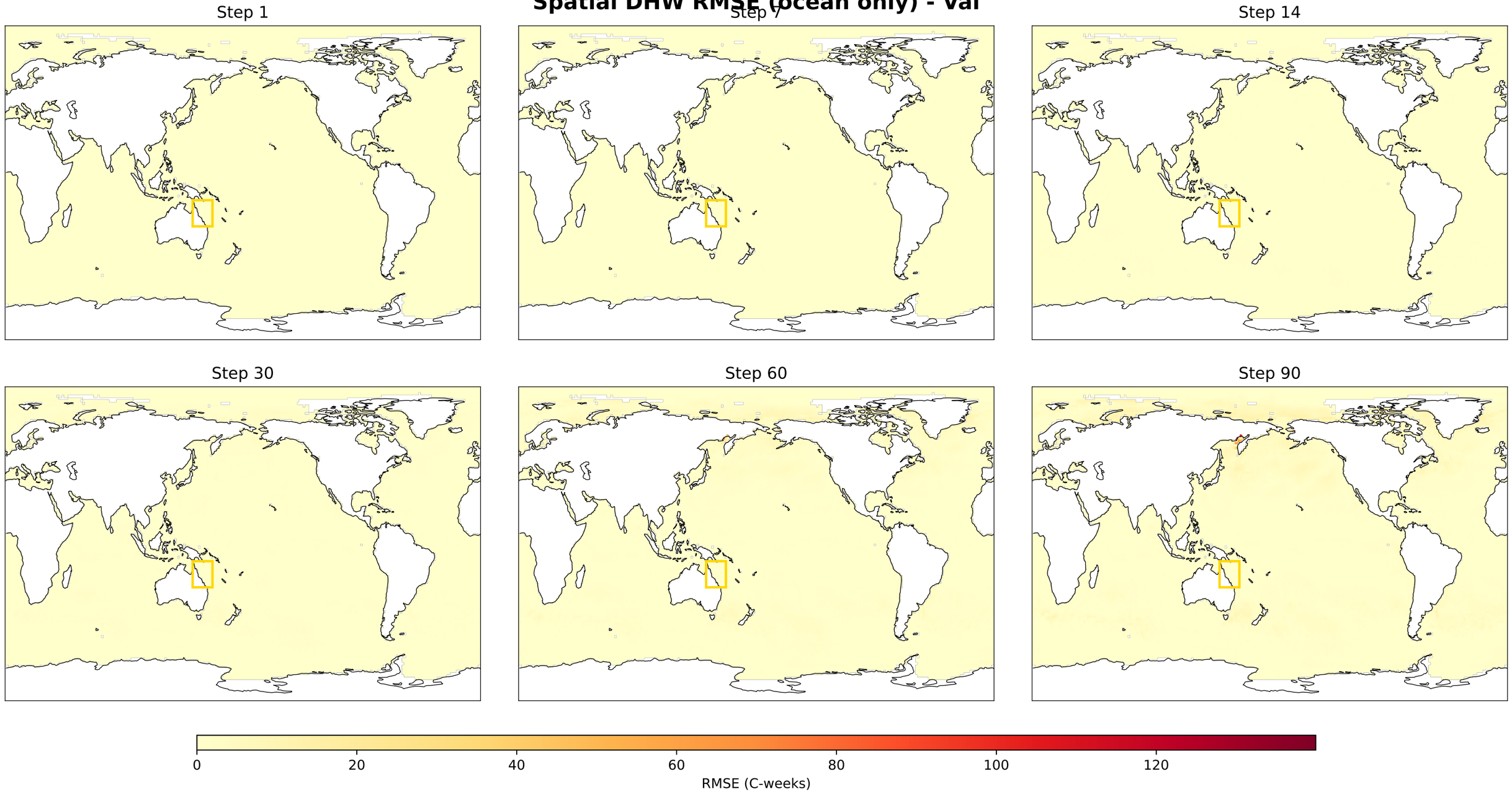


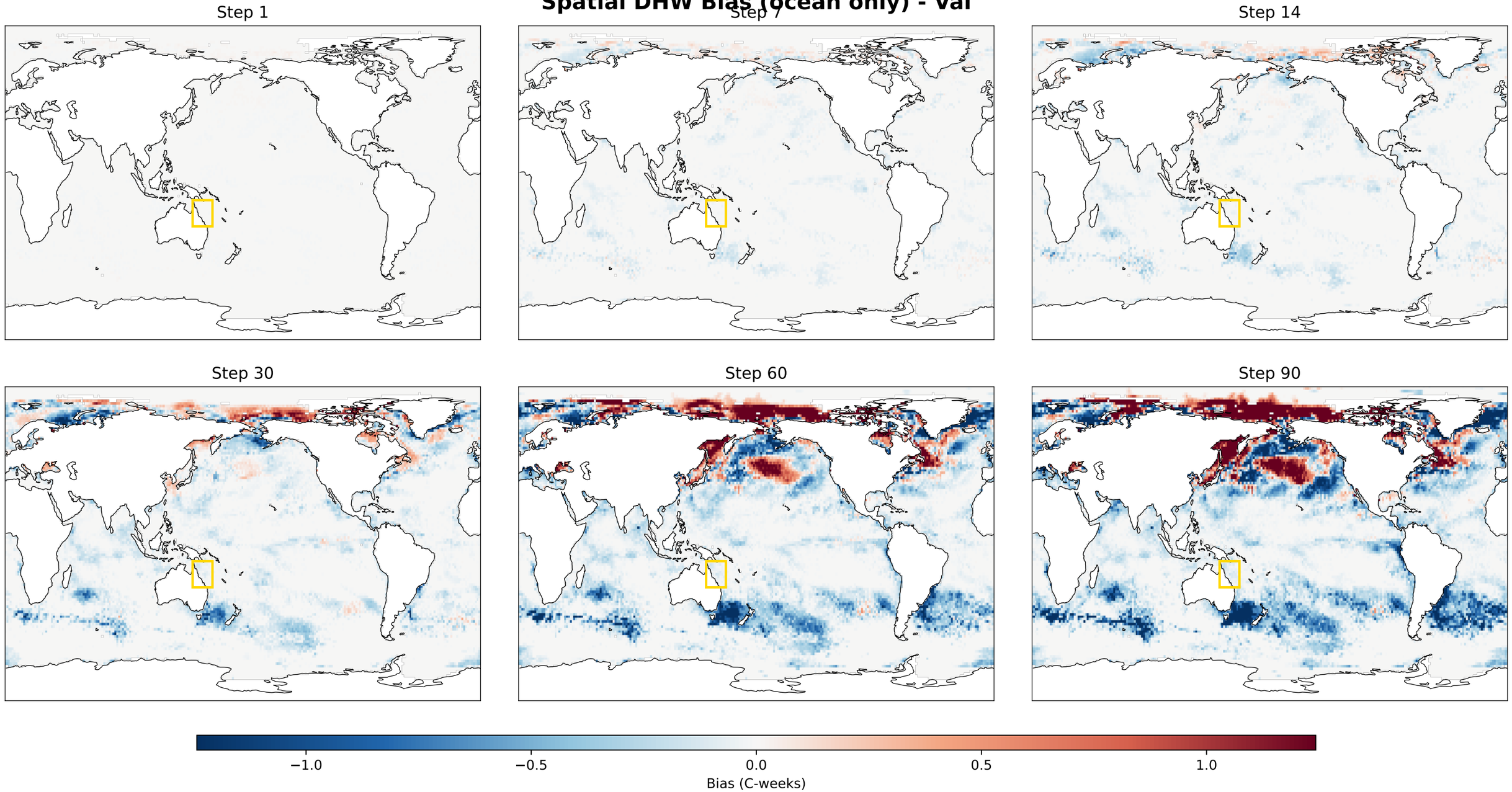





Spatial DHW RMSE (all grid) - Val
Step 1
Step 7
Step 14
Step 30
Step 60
Step 90
0
20
40
60
80
100
120
RMSE (C-weeks)

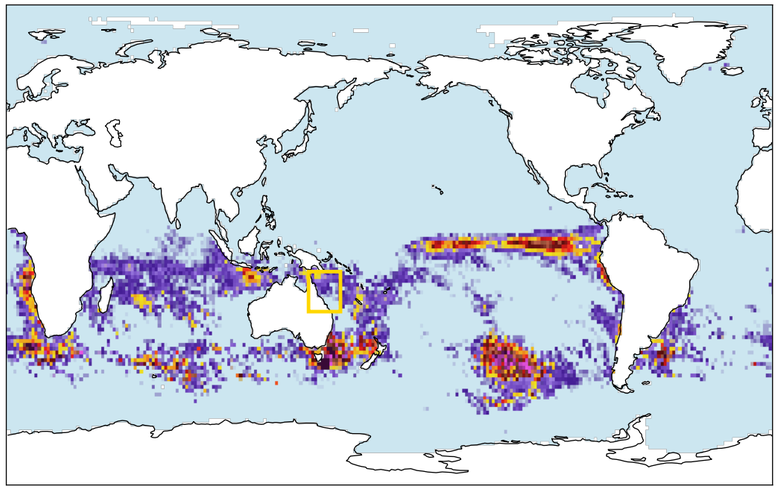

Spatial DHW Bias (all grid) - Val
Step 1
Step 7
Step 14
Step 30
Step 60
Step 90
−1.0
−0.5
0.0
0.5
1.0
Bias (C-weeks)


Spatial DHW RMSE (ocean only) - Val
Step 1
Step 7
Step 14
Step 30
Step 60
Step 90
0
20
40
60
80
100
120
RMSE (C-weeks)

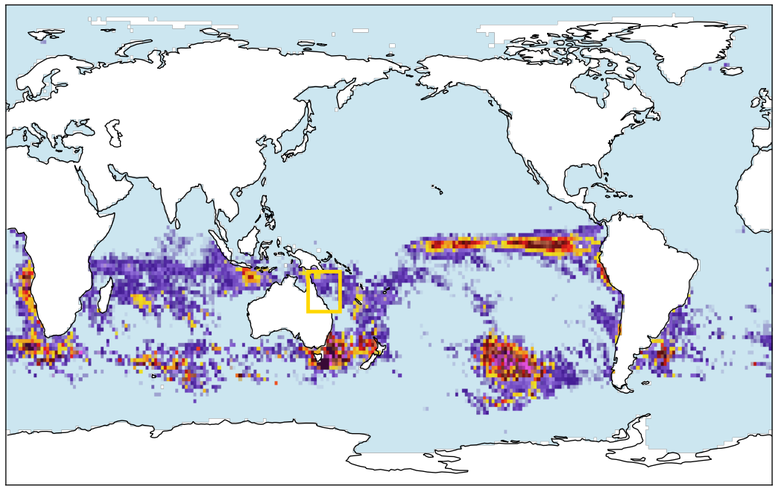

Spatial DHW Bias (ocean only) - Val
Step 1
Step 7
Step 14
Step 30
Step 60
Step 90
−1.0
−0.5
0.0
0.5
1.0
Bias (C-weeks)

DHW Chain init: 2016-04-10 - Val (C-weeks)

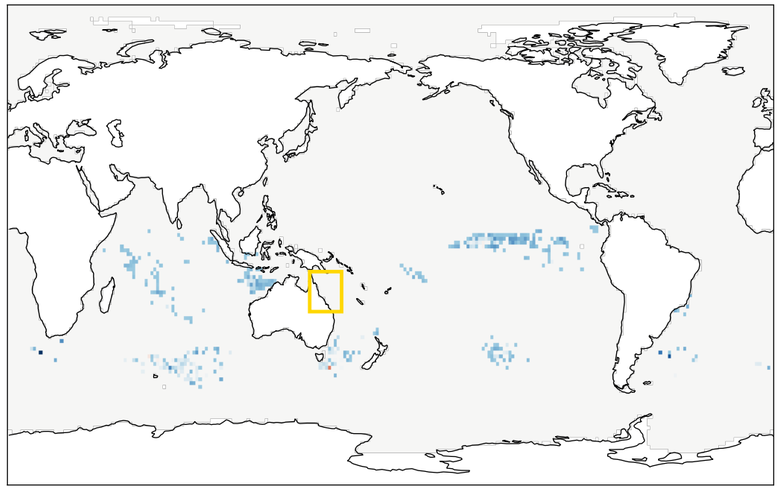
Actual DHW - Step 1



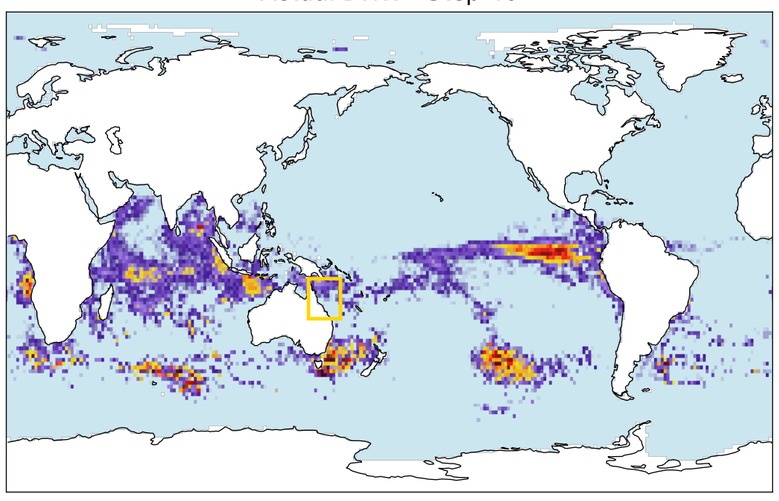
Predicted DHW - Step 1



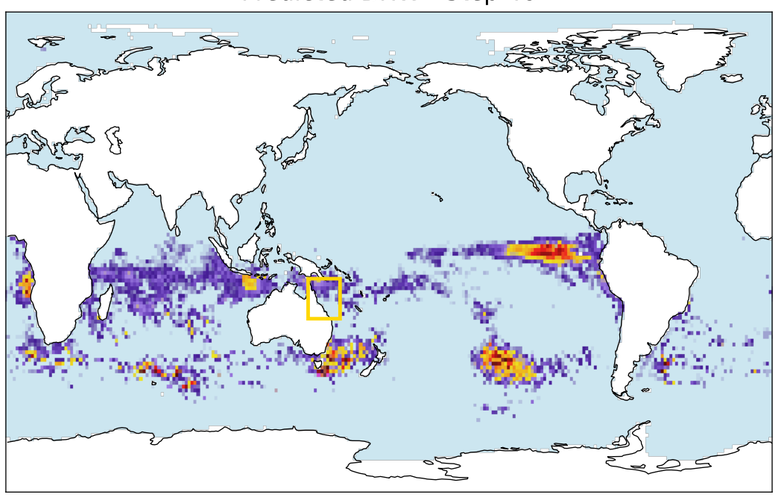
Error (Pred - Actual) - Step 1 [scale ±0.48]



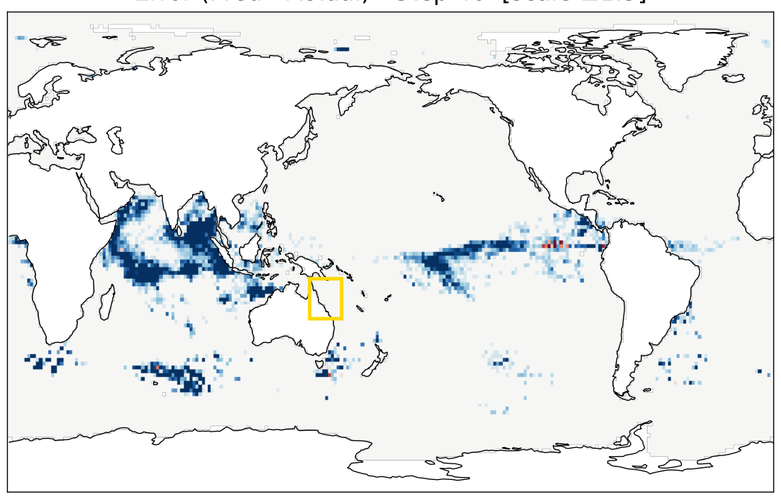
Actual DHW - Step 46



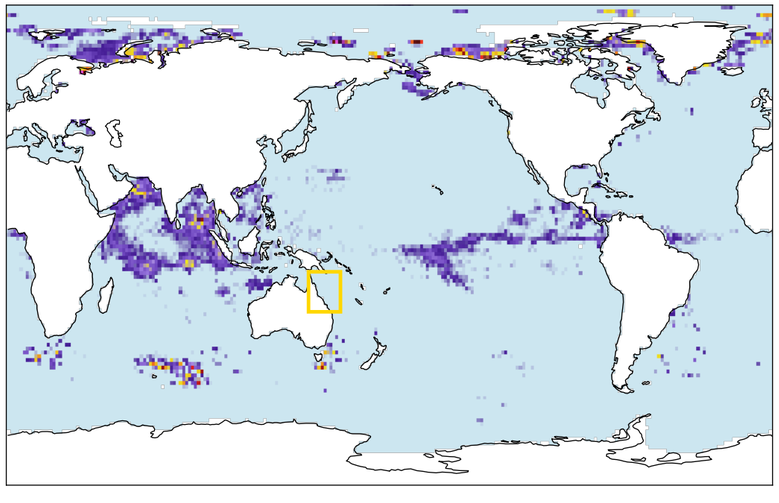
Predicted DHW - Step 46



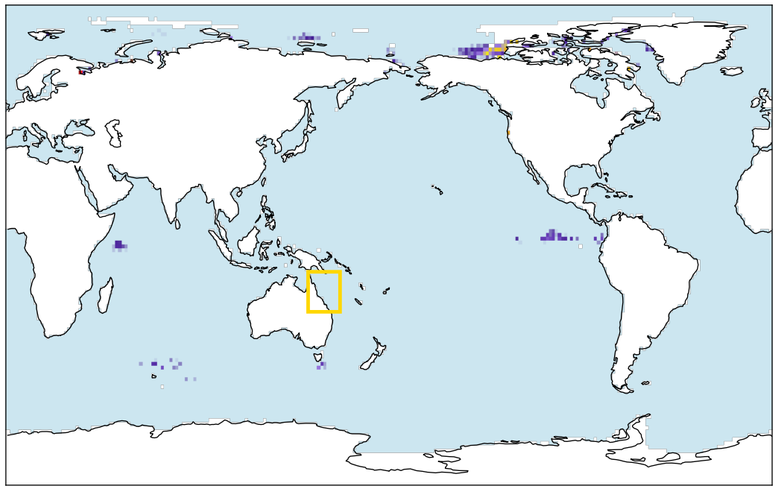
Error (Pred - Actual) - Step 46 [scale ±1.5]



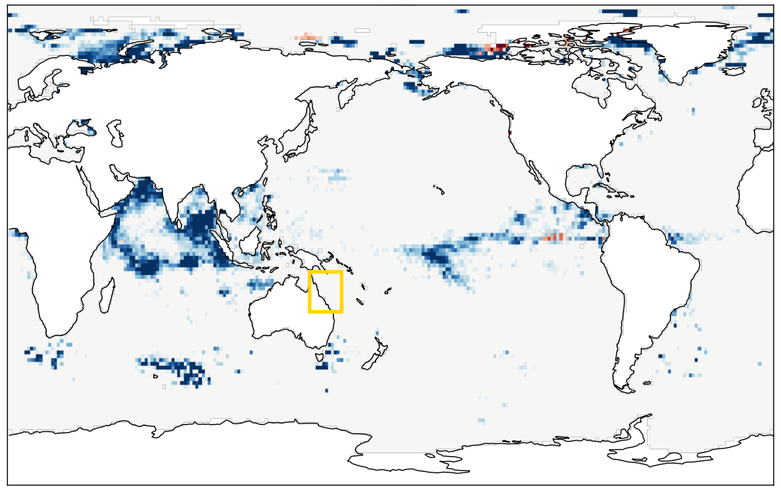
Actual DHW - Step 90

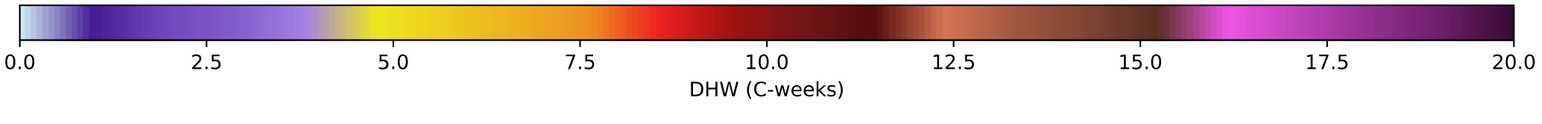


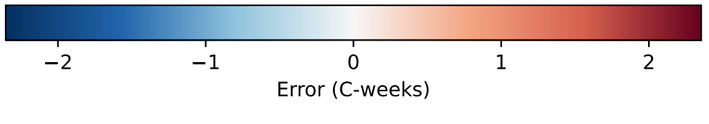
Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±2.4]

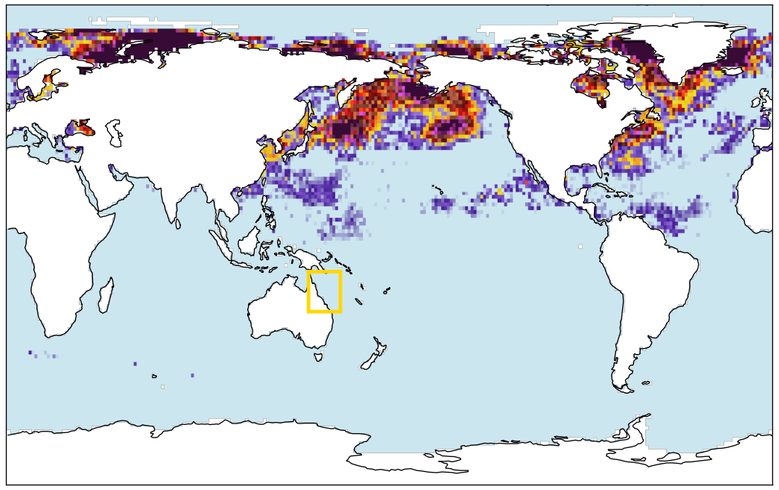




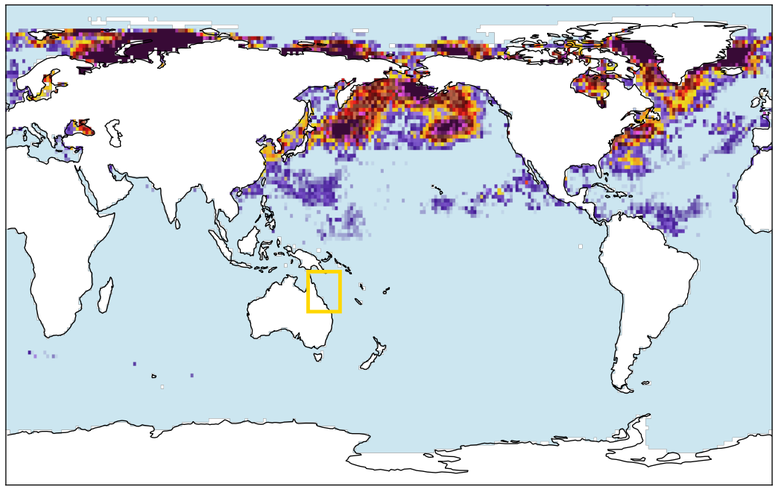


DHW Chain init: 2016-10-07 - Val (C-weeks)

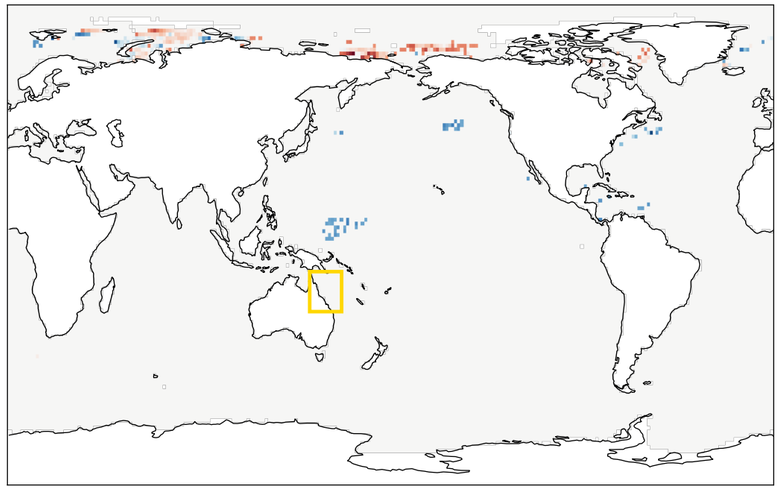
Actual DHW - Step 1



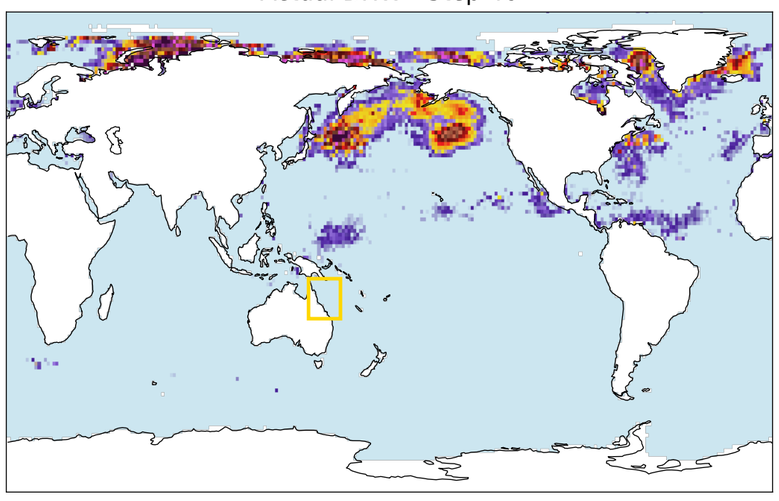
Predicted DHW - Step 1



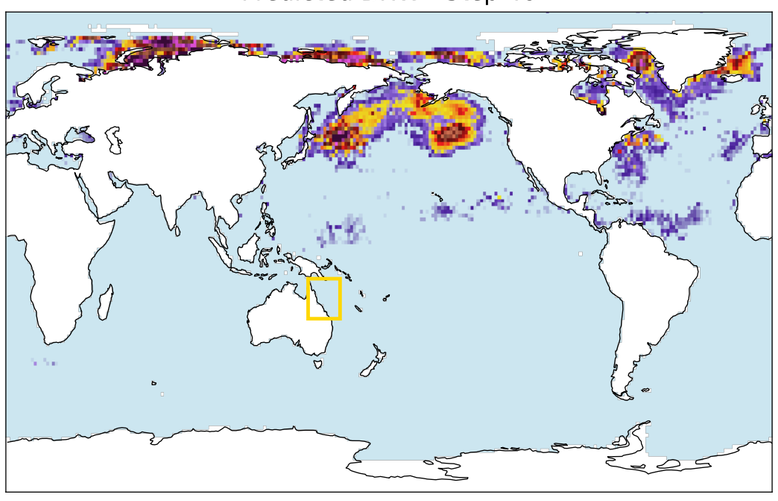
Error (Pred - Actual) - Step 1 [scale ±0.34]



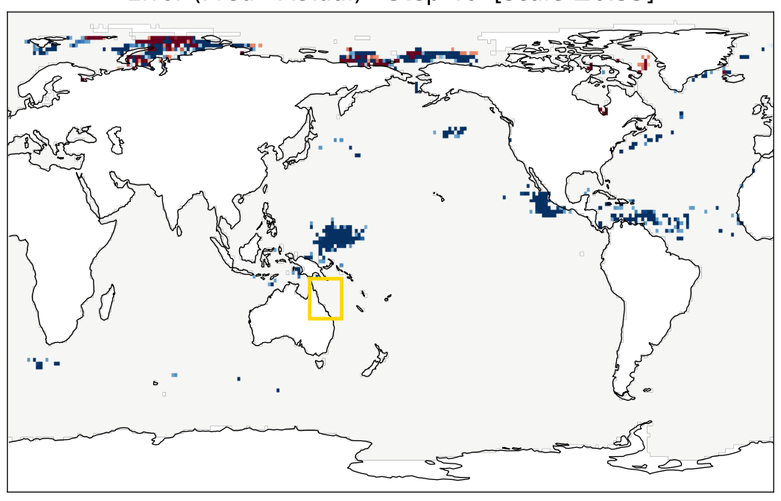
Actual DHW - Step 46



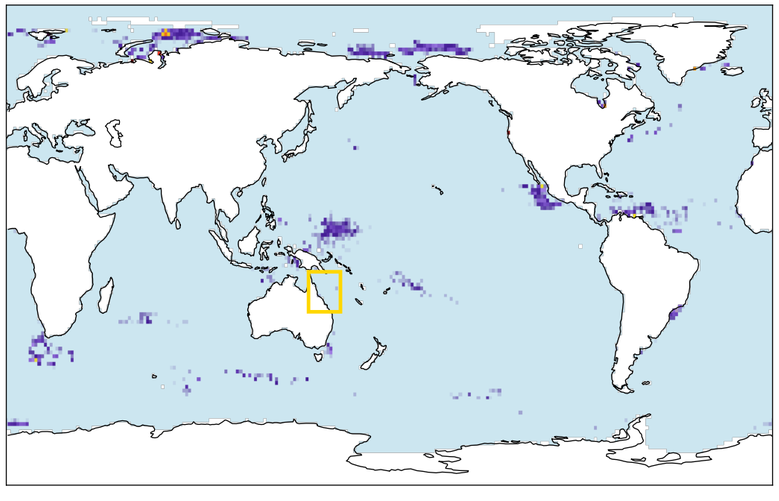
Predicted DHW - Step 46



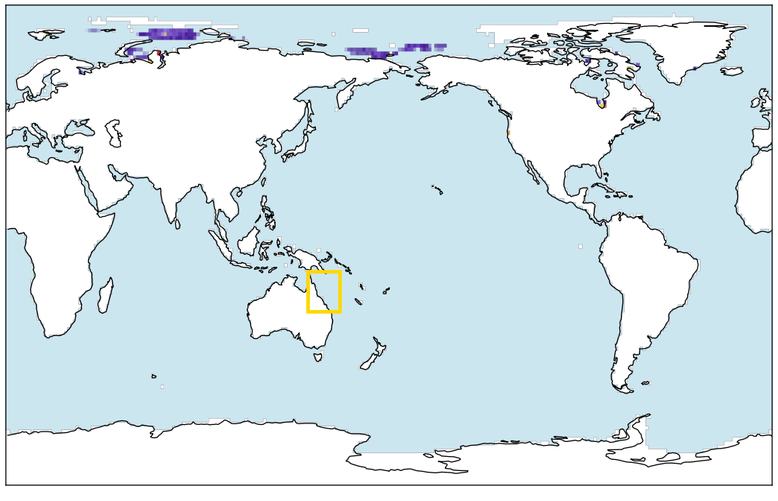
Error (Pred - Actual) - Step 46 [scale ±0.33]



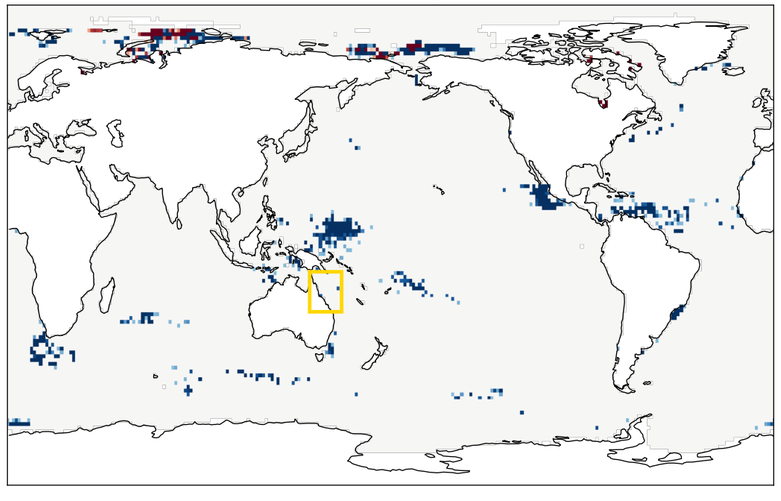
Actual DHW - Step 90

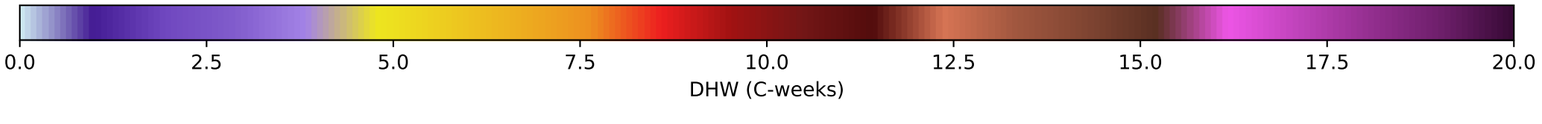


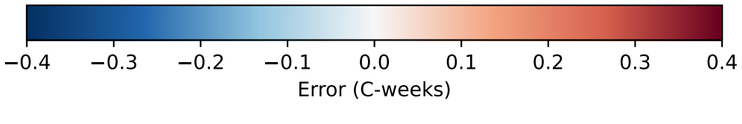
Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±0.4]

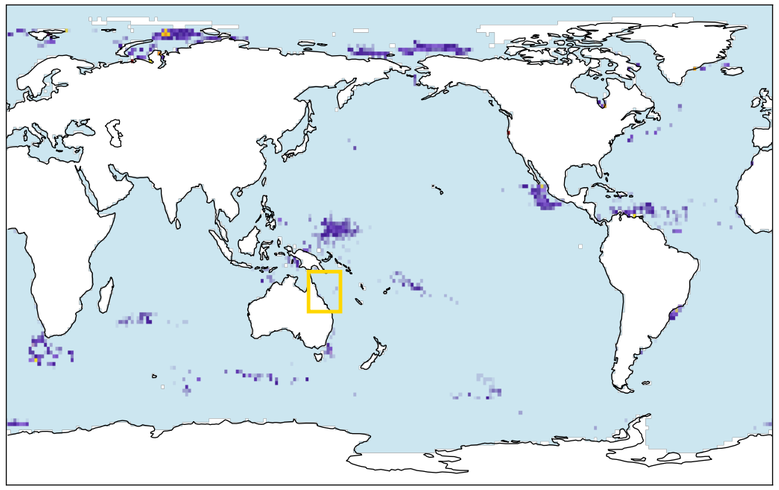




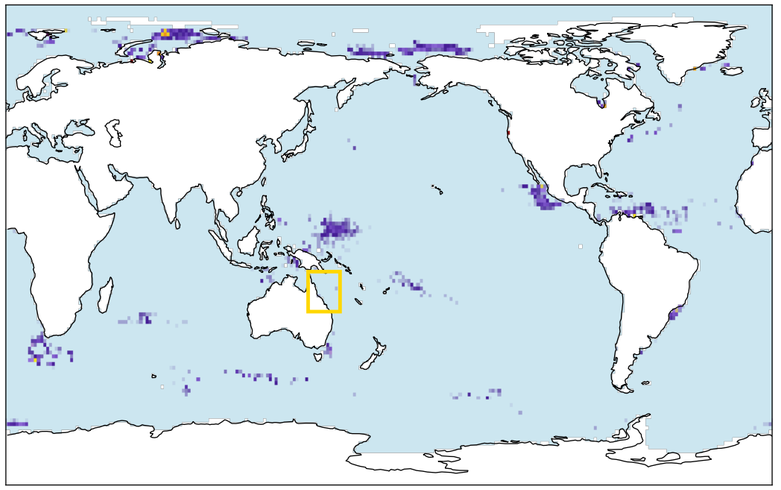


DHW Chain init: 2017-01-05 - Val (C-weeks)

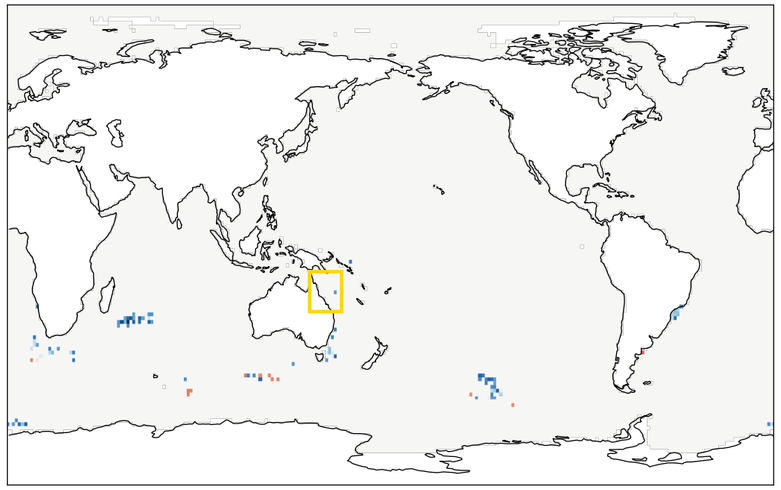
Actual DHW - Step 1



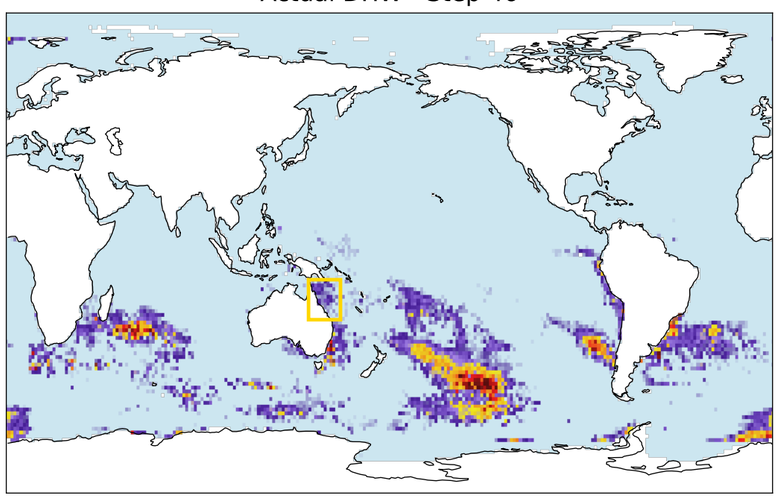
Predicted DHW - Step 1



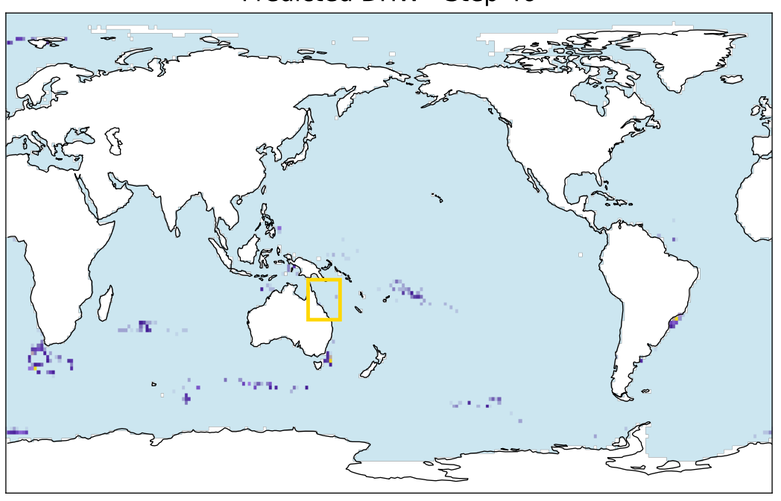
Error (Pred - Actual) - Step 1 [scale ±0.31]



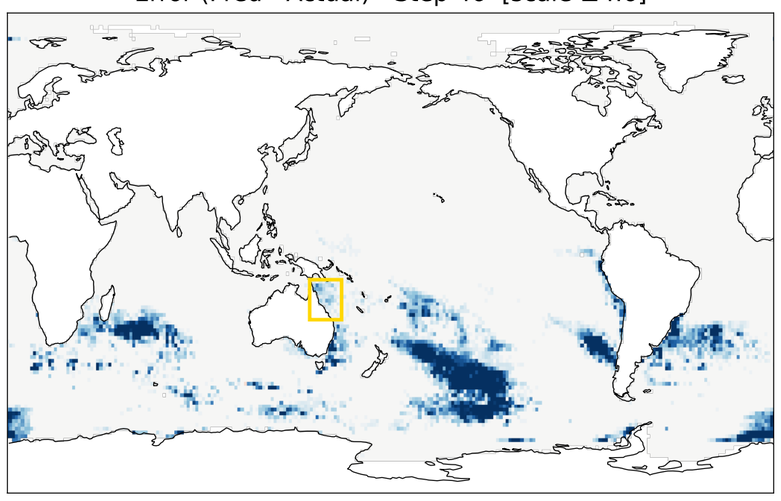
Actual DHW - Step 46



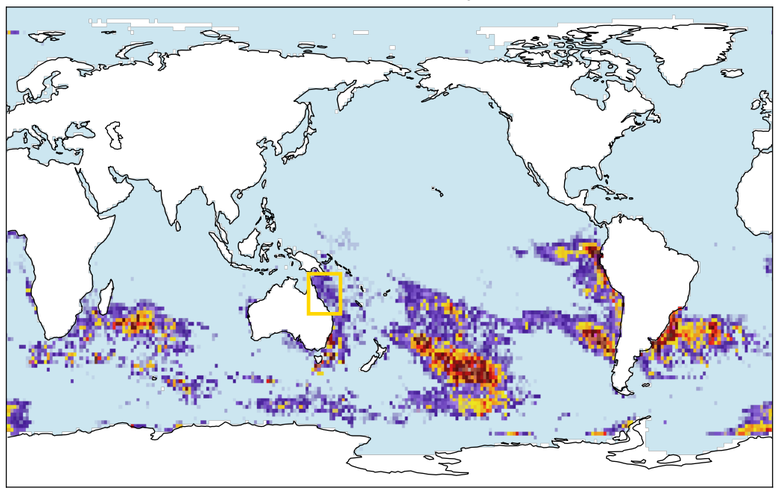
Predicted DHW - Step 46



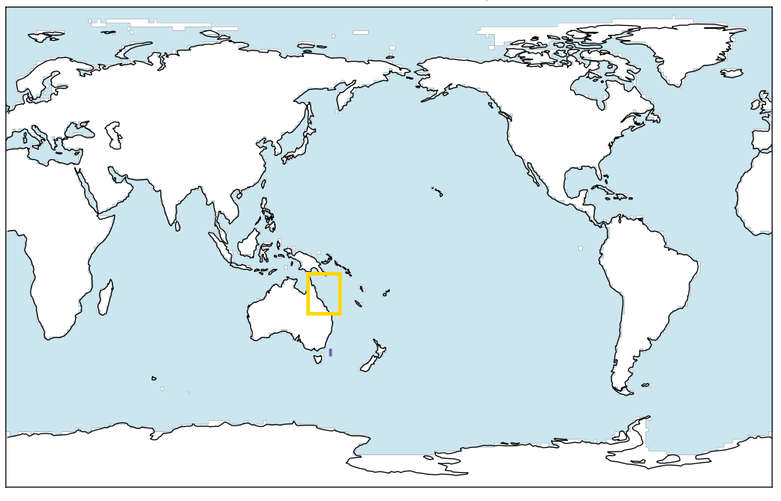
Error (Pred - Actual) - Step 46 [scale ±4.6]



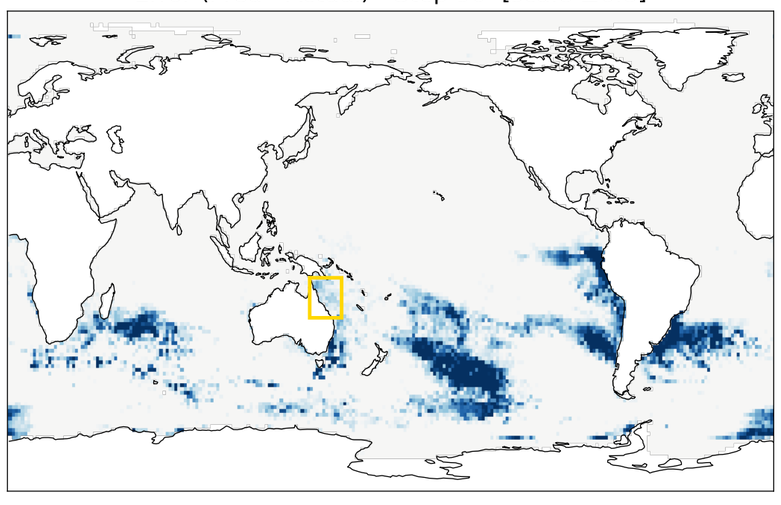
Actual DHW - Step 90

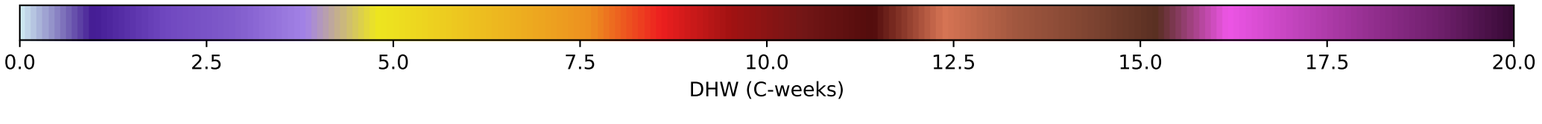


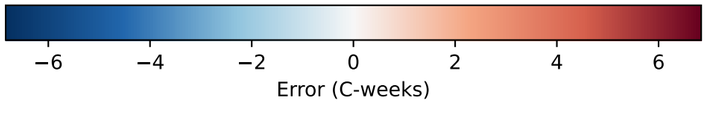
Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±6.8]

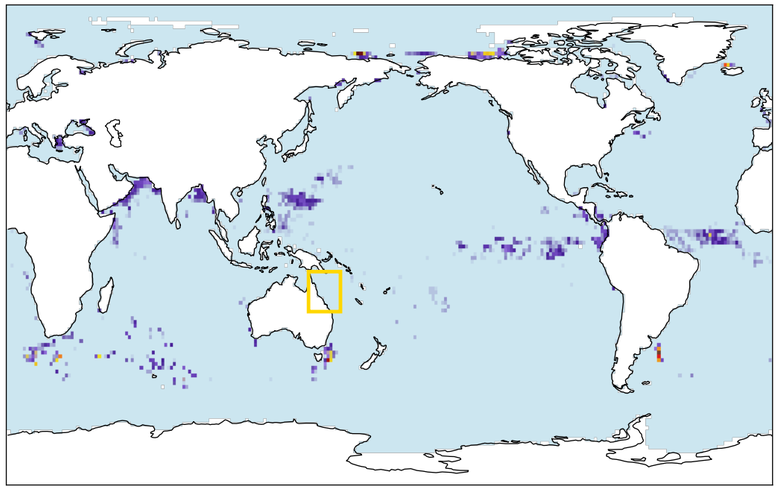




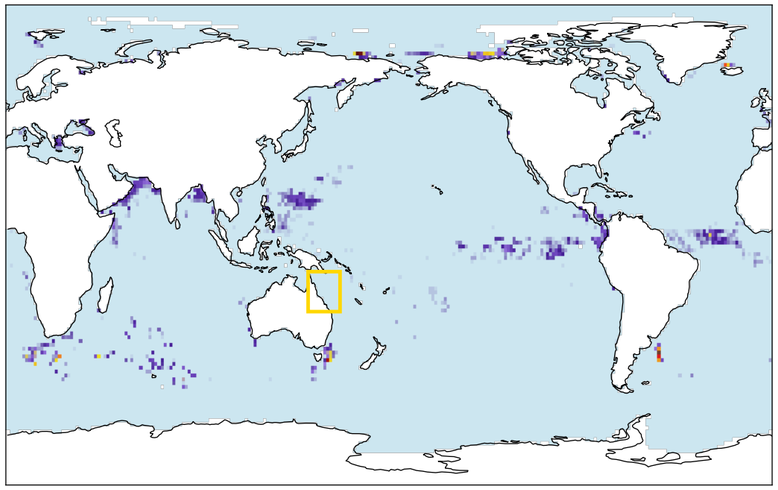


DHW Chain init: 2017-07-04 - Val (C-weeks)

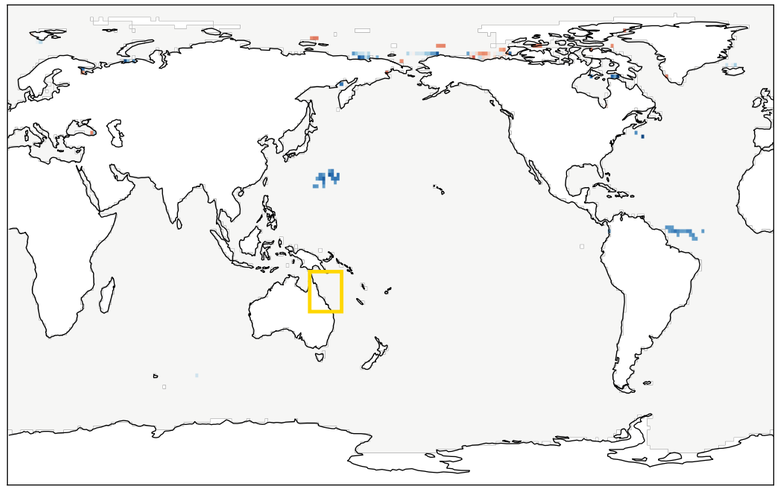
Actual DHW - Step 1



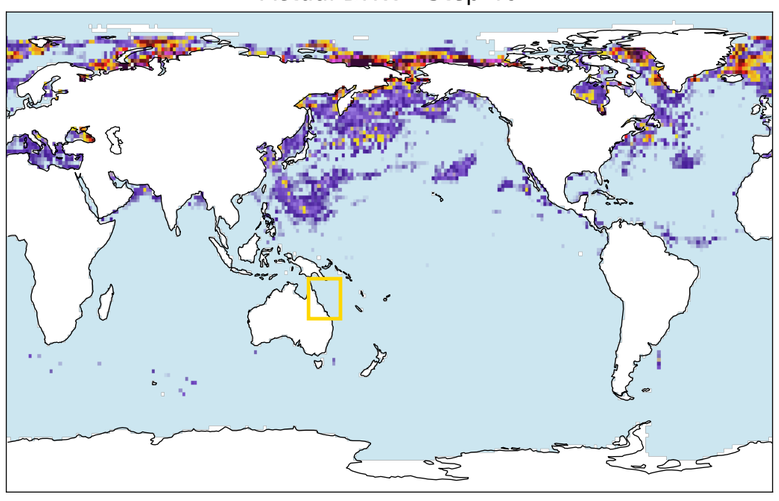
Predicted DHW - Step 1



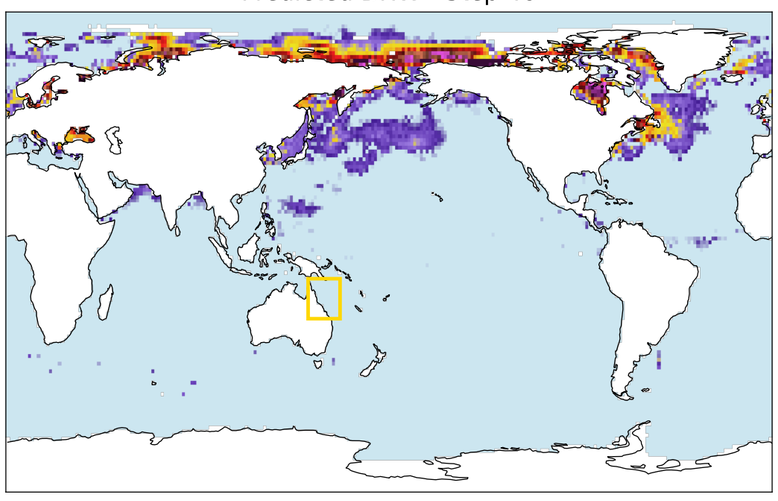
Error (Pred - Actual) - Step 1 [scale ±0.32]



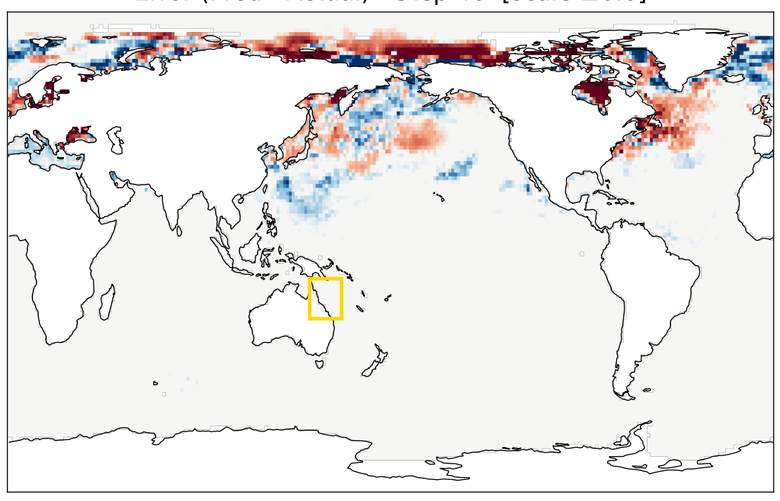
Actual DHW - Step 46



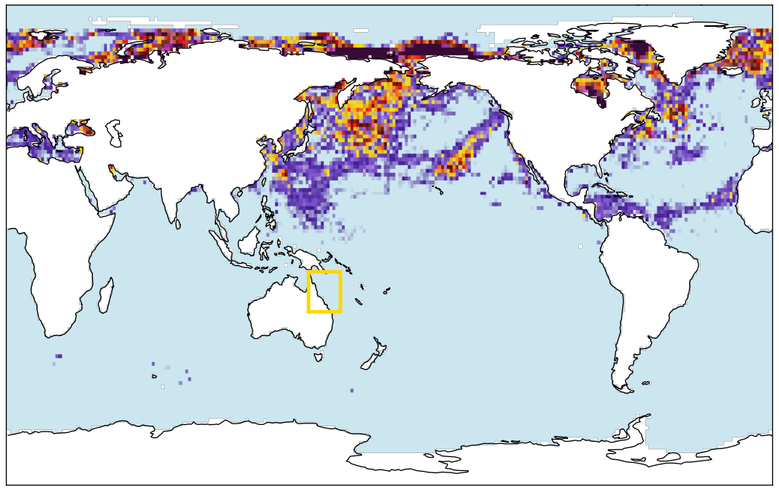
Predicted DHW - Step 46



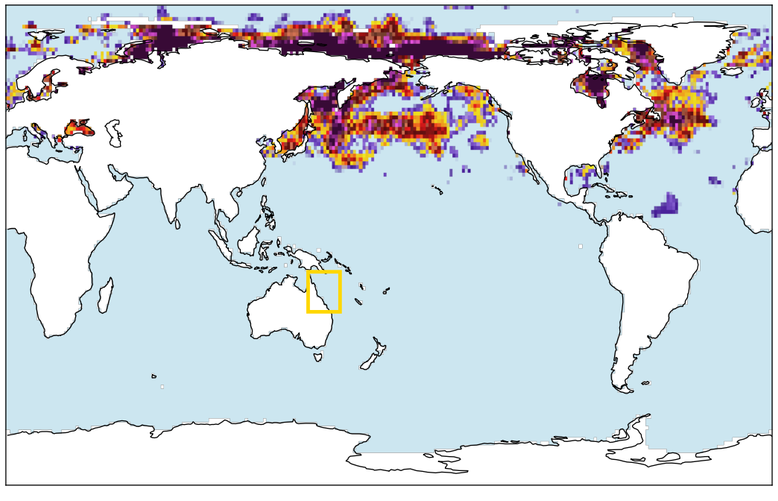
Error (Pred - Actual) - Step 46 [scale ±6.6]



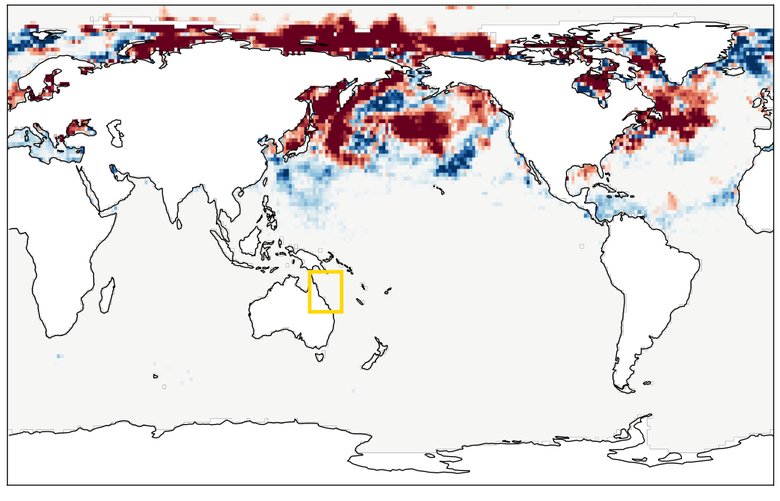
Actual DHW - Step 90

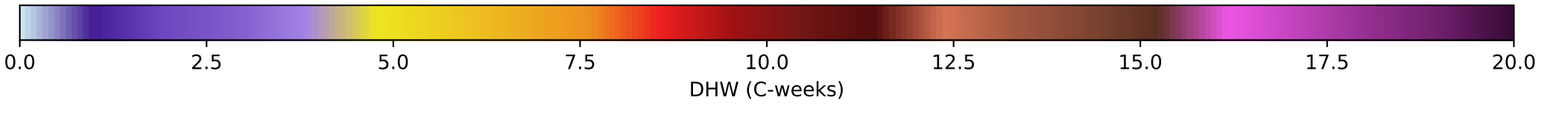


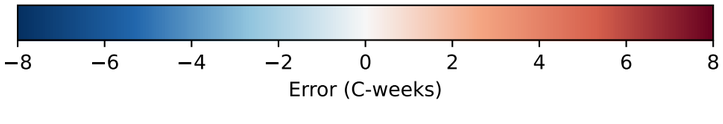
Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±8]

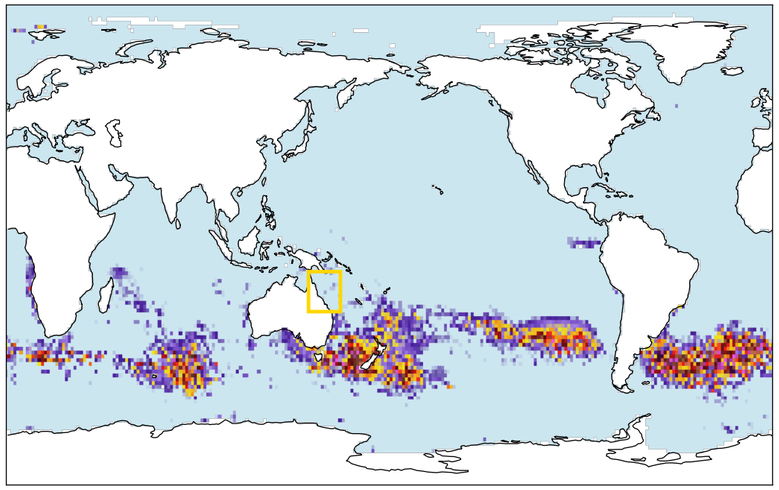




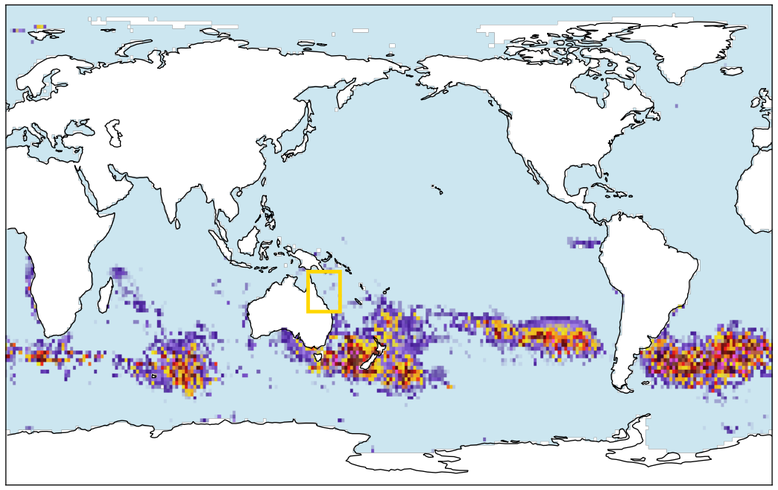


DHW Chain init: 2018-03-31 - Val (C-weeks)

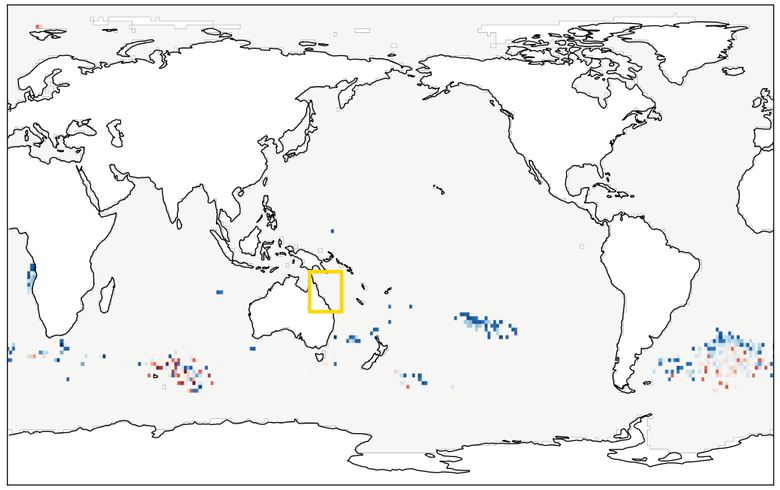
Actual DHW - Step 1



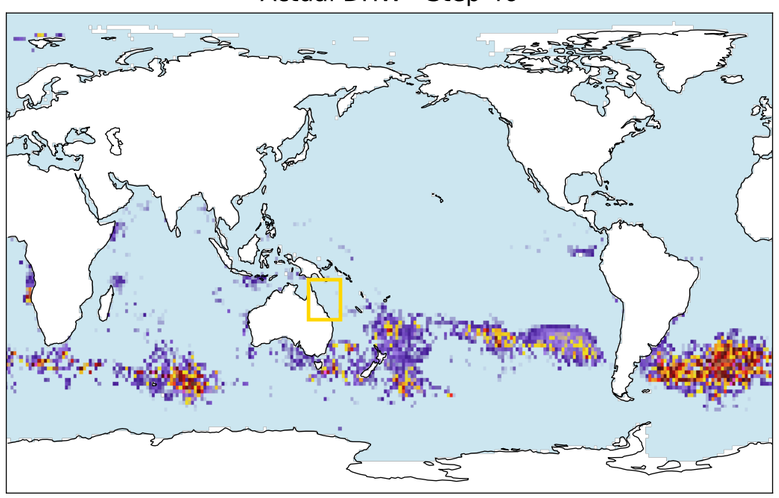
Predicted DHW - Step 1



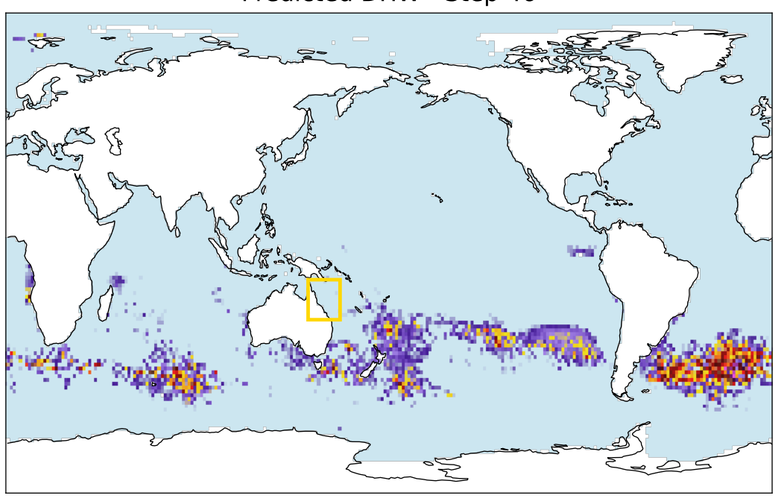
Error (Pred - Actual) - Step 1 [scale ±0.24]



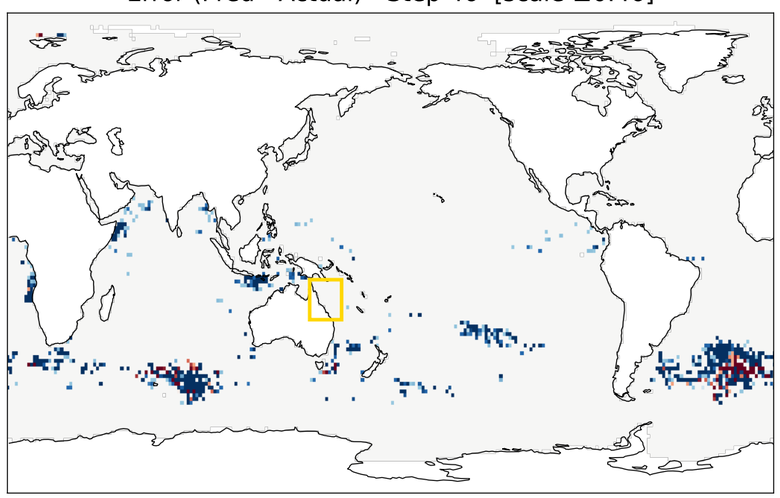
Actual DHW - Step 46



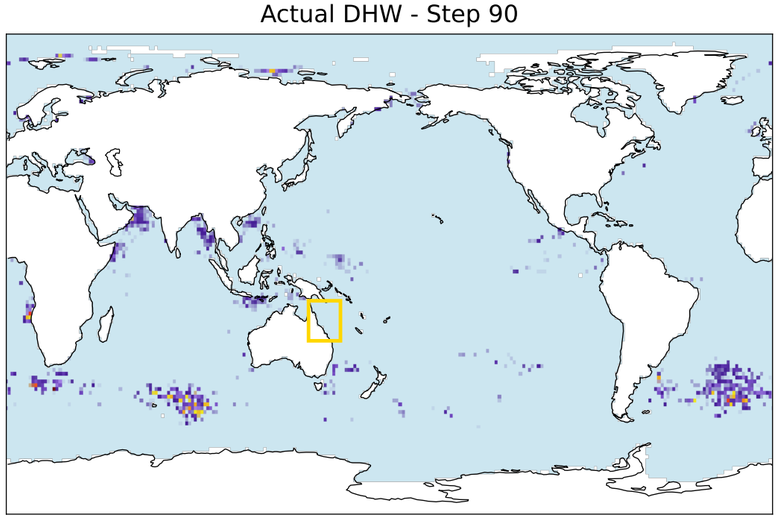
Predicted DHW - Step 46



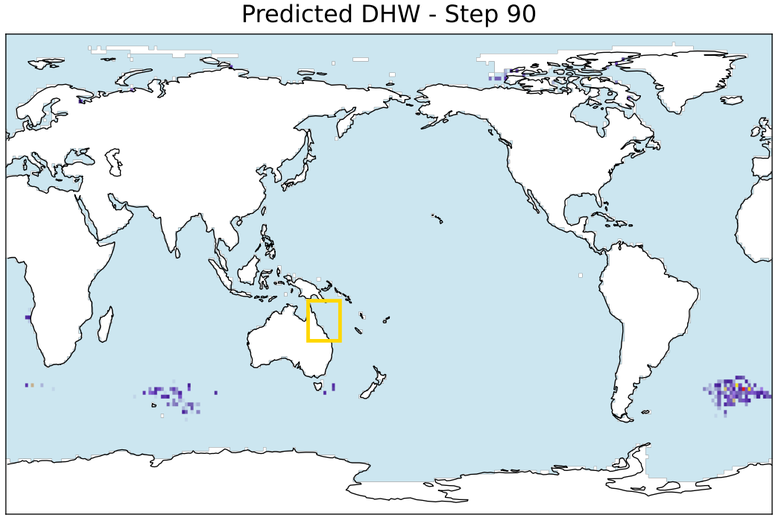
Error (Pred - Actual) - Step 46 [scale ±0.46]



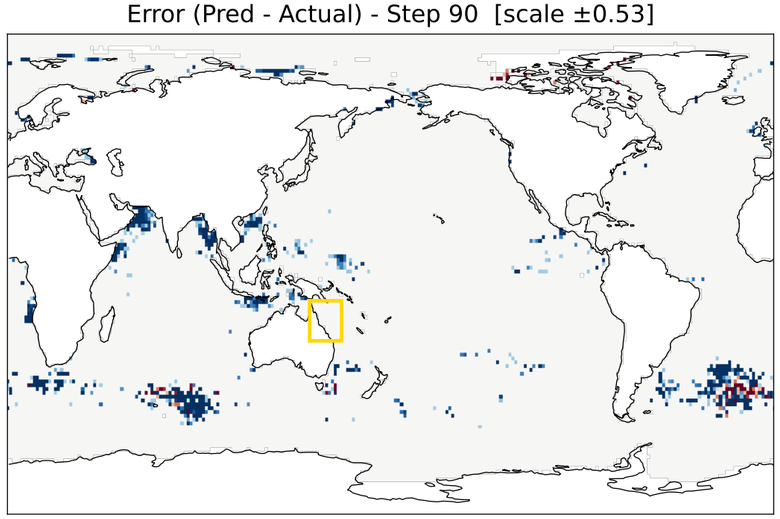
Actual DHW - Step 90

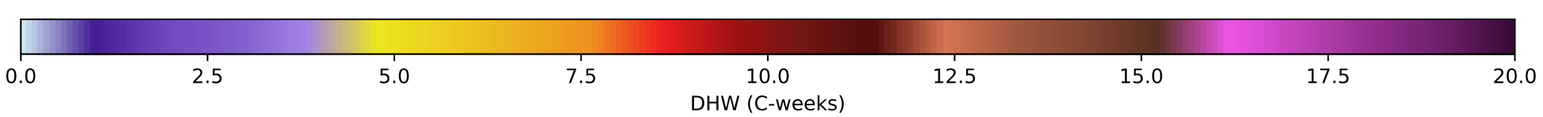


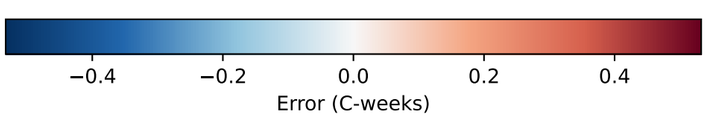
Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±0.53]

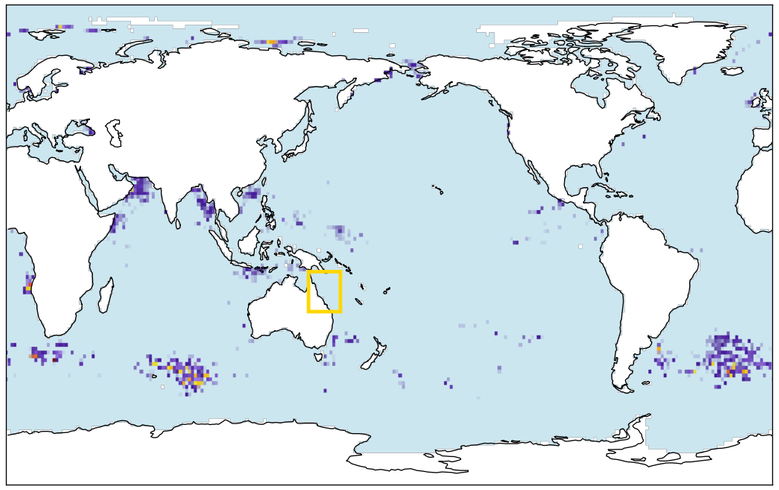




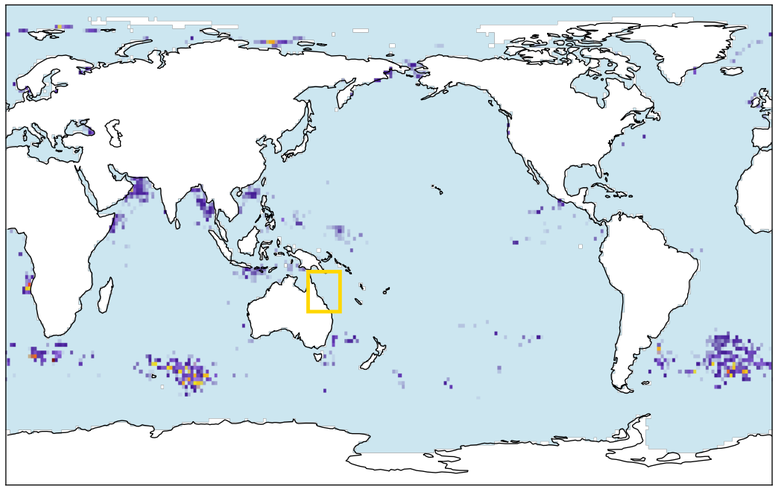


DHW Chain init: 2018-06-29 - Val (C-weeks)

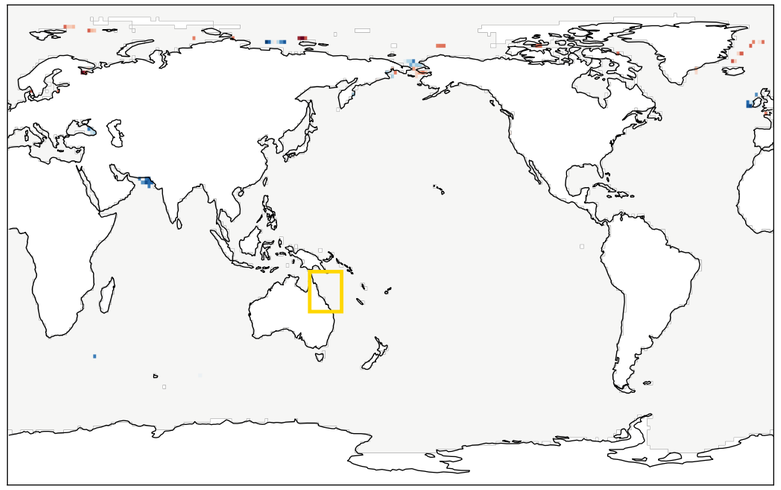
Actual DHW - Step 1



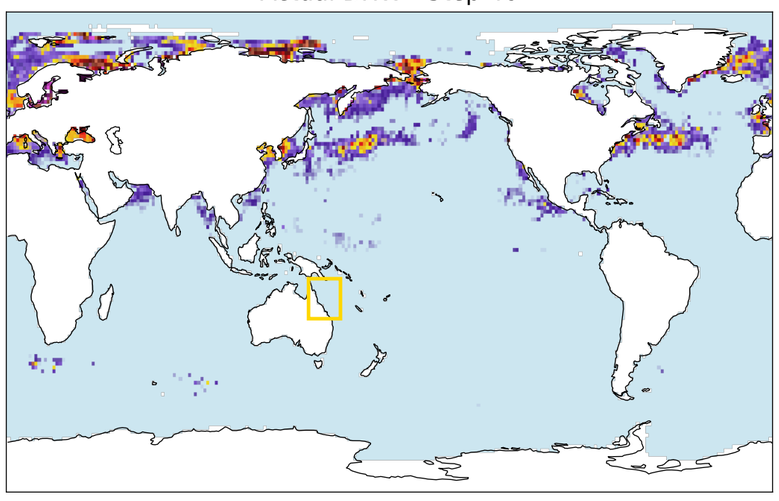
Predicted DHW - Step 1



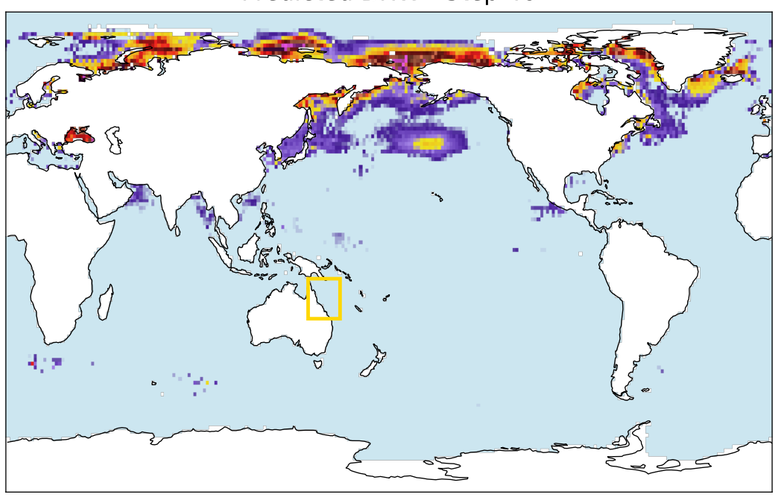
Error (Pred - Actual) - Step 1 [scale ±0.29]



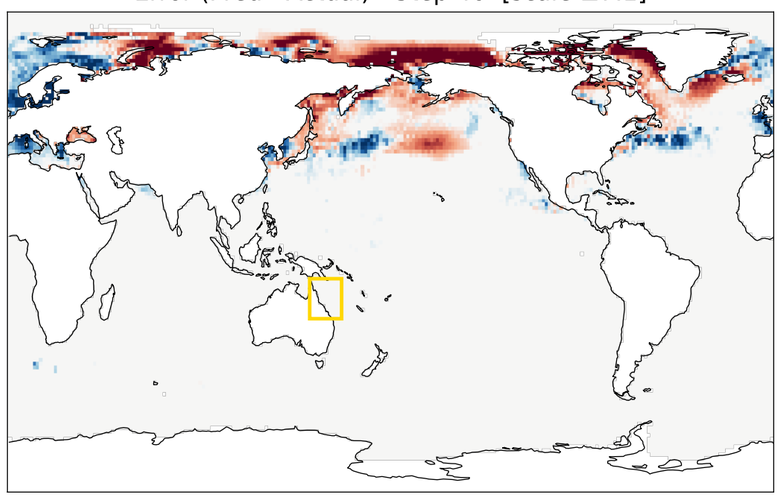
Actual DHW - Step 46



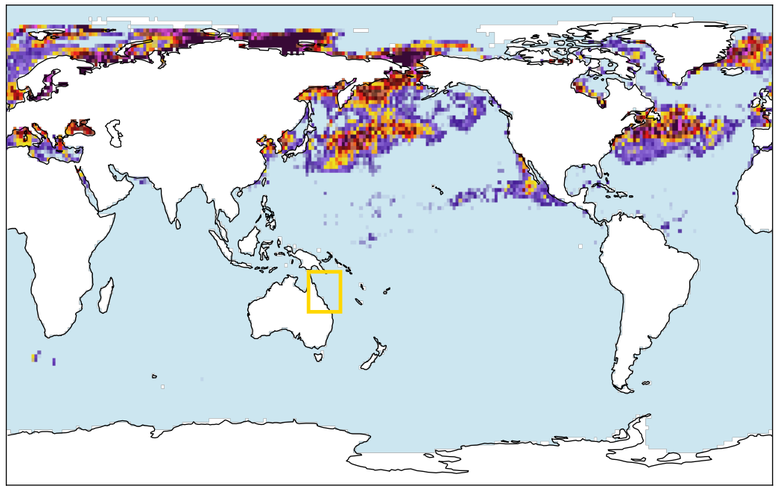
Predicted DHW - Step 46



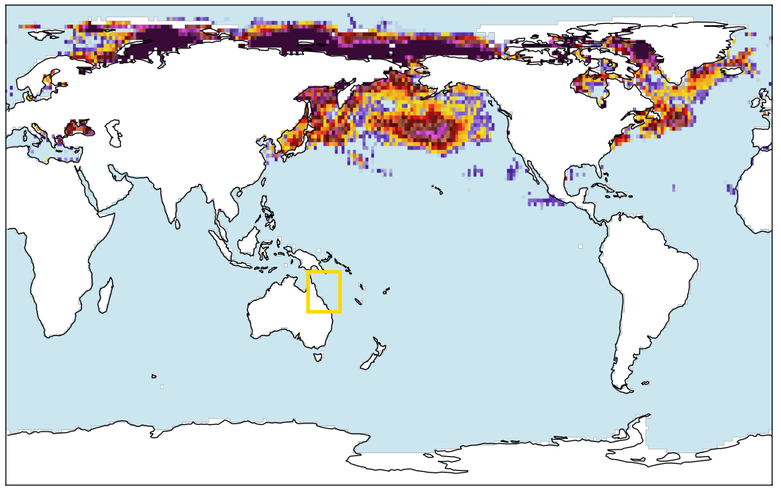
Error (Pred - Actual) - Step 46 [scale ±7.1]



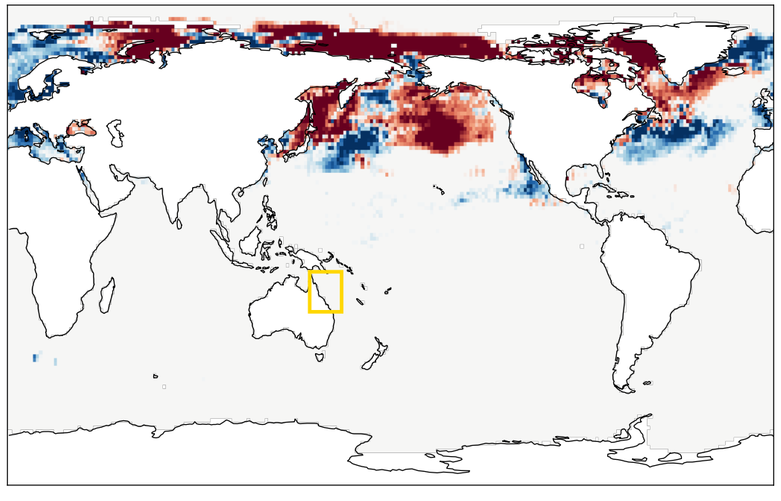
Actual DHW - Step 90

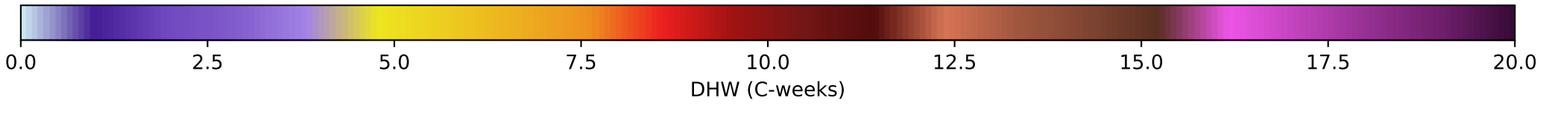


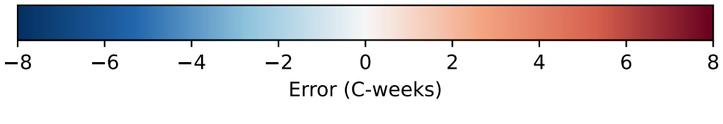
Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±8]

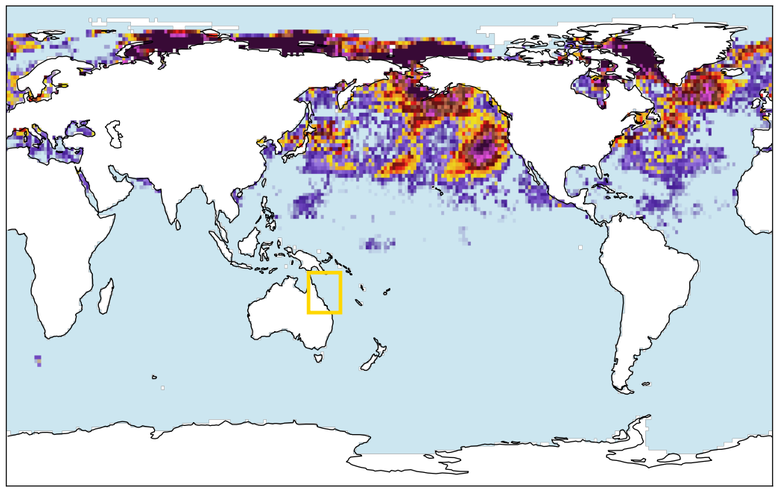




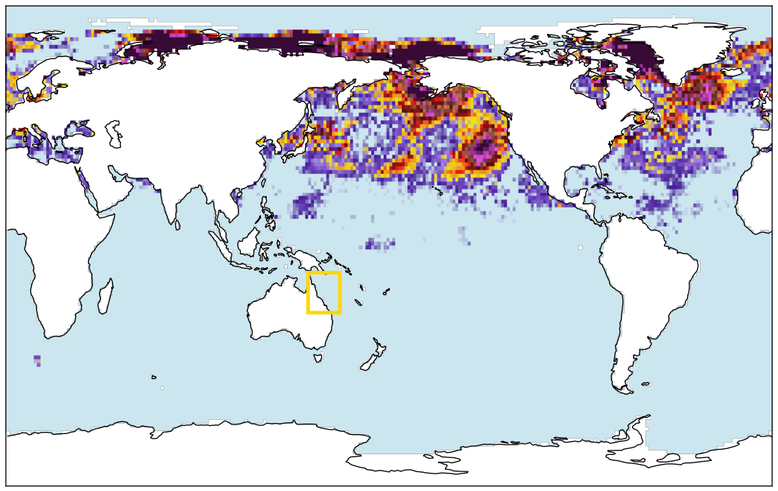


DHW Chain init: 2019-09-22 - Val (C-weeks)

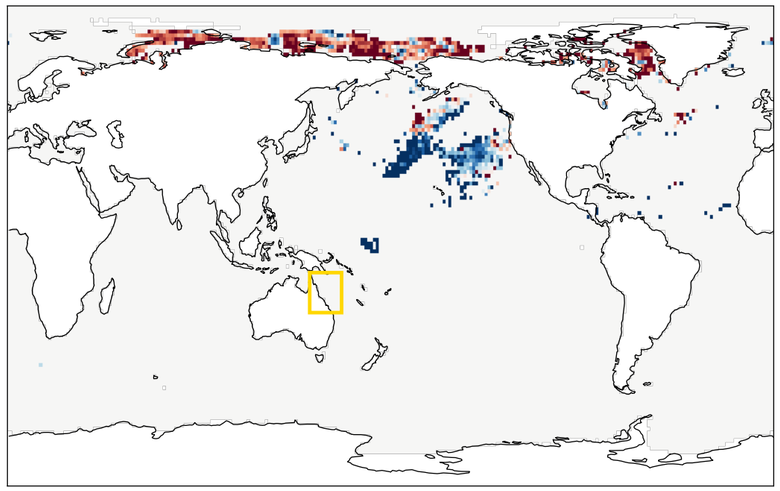
Actual DHW - Step 1



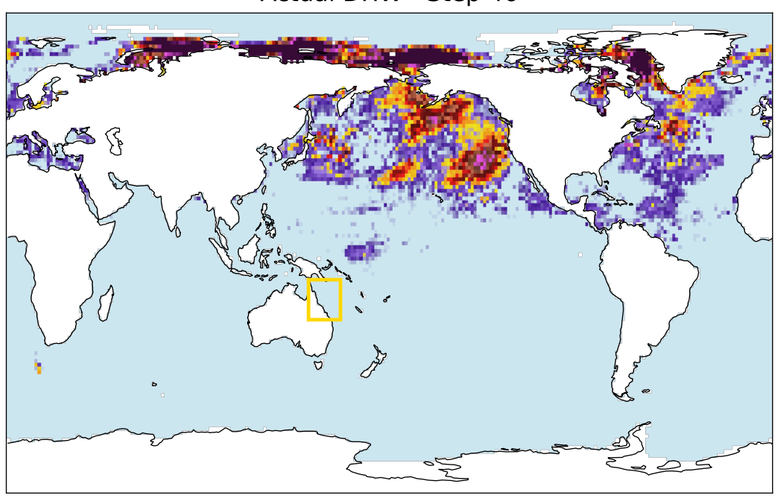
Predicted DHW - Step 1



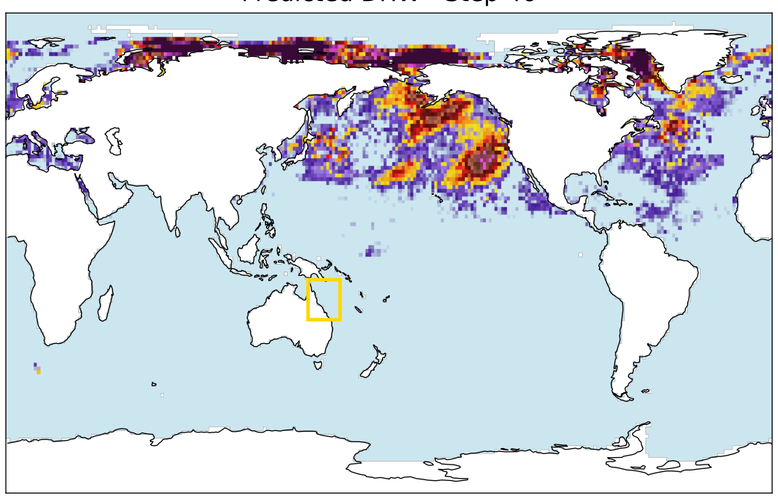
Error (Pred - Actual) - Step 1 [scale ±0.074]



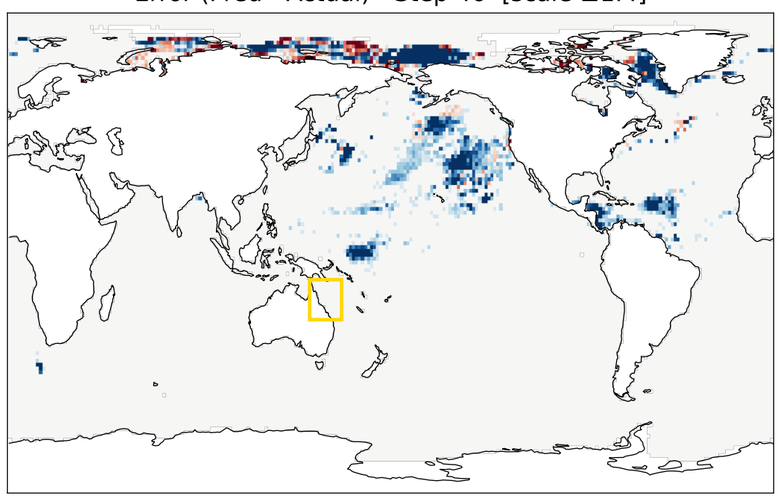
Actual DHW - Step 46



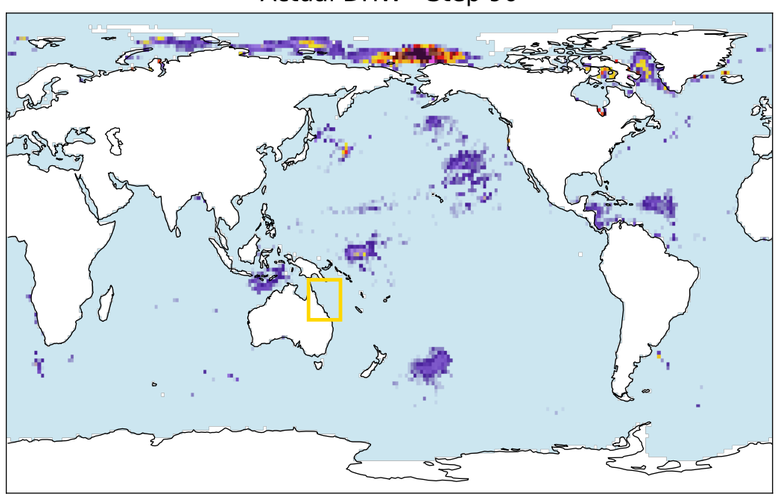
Predicted DHW - Step 46



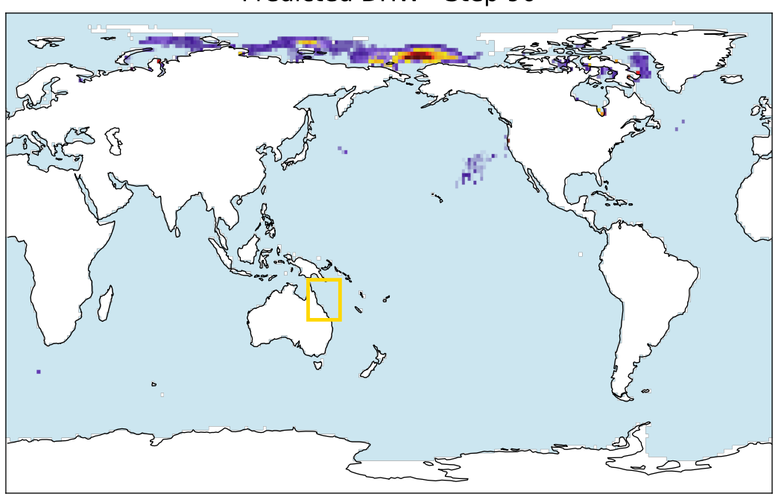
Error (Pred - Actual) - Step 46 [scale ±1.4]



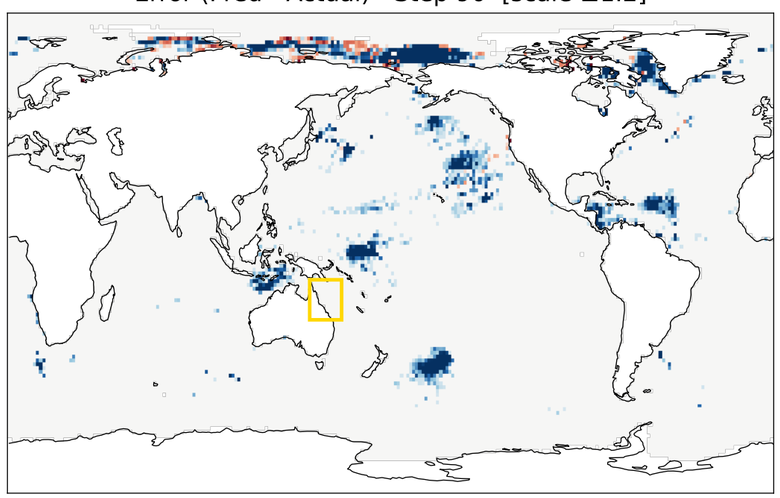
Actual DHW - Step 90

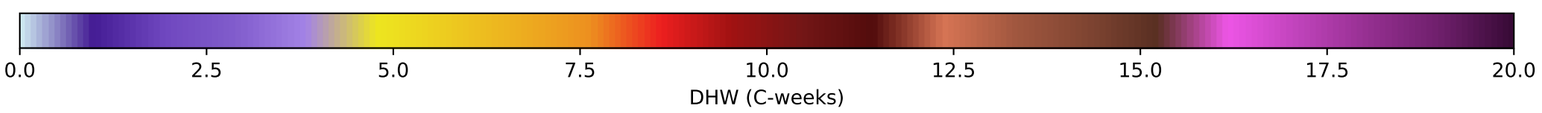


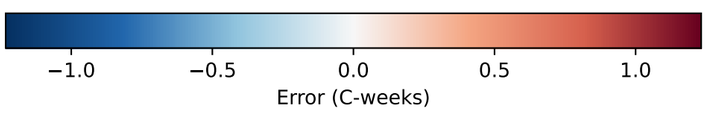
Predicted DHW - Step 90



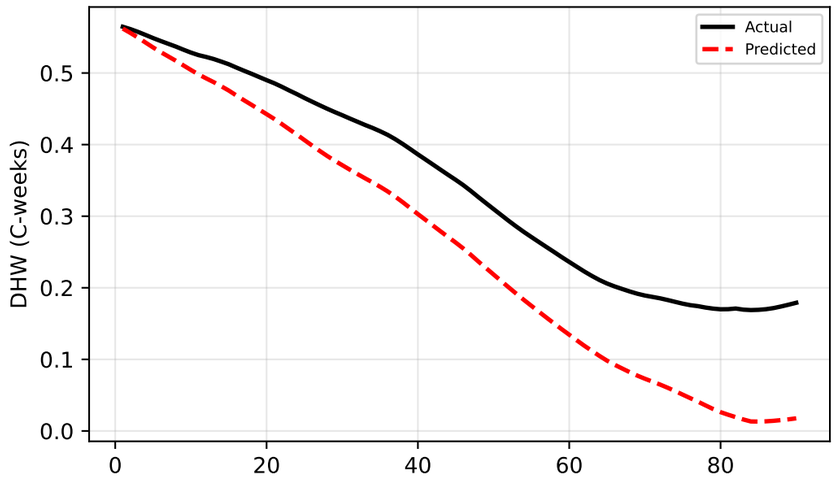
Error (Pred - Actual) - Step 90 [scale ±1.2]

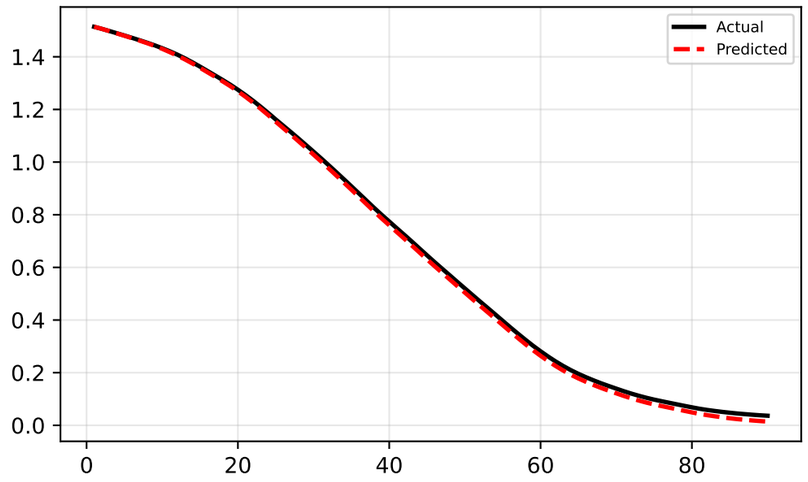


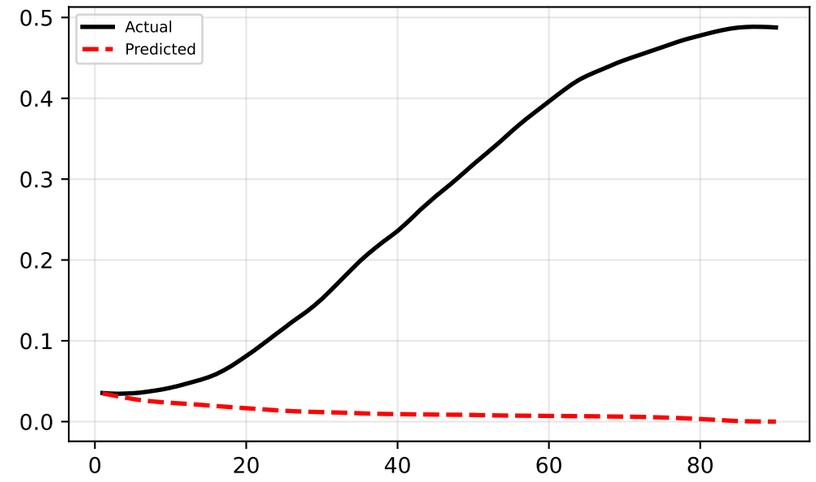


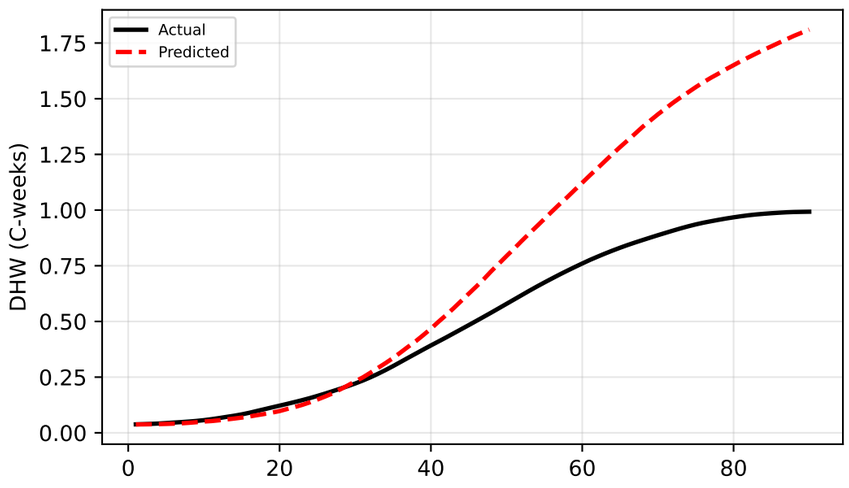


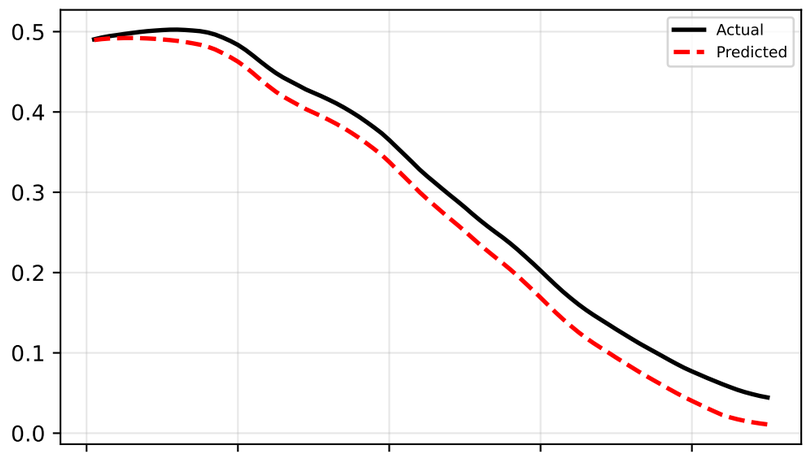
Global-Mean DHW - Val (C-weeks)

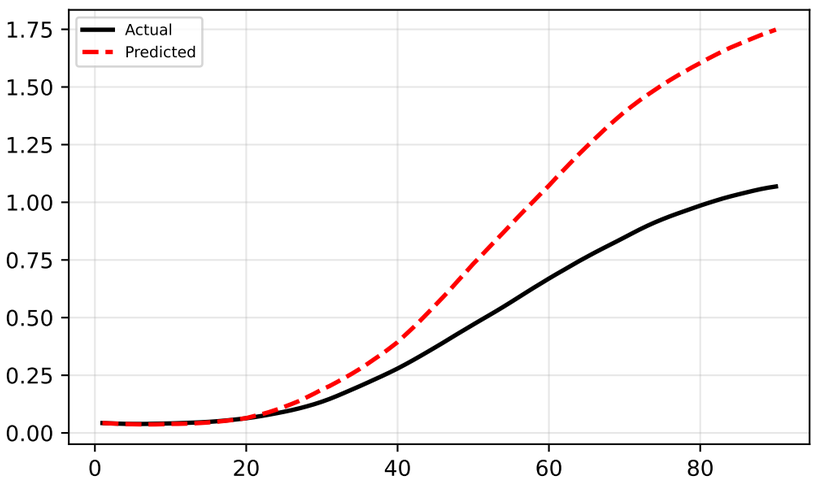


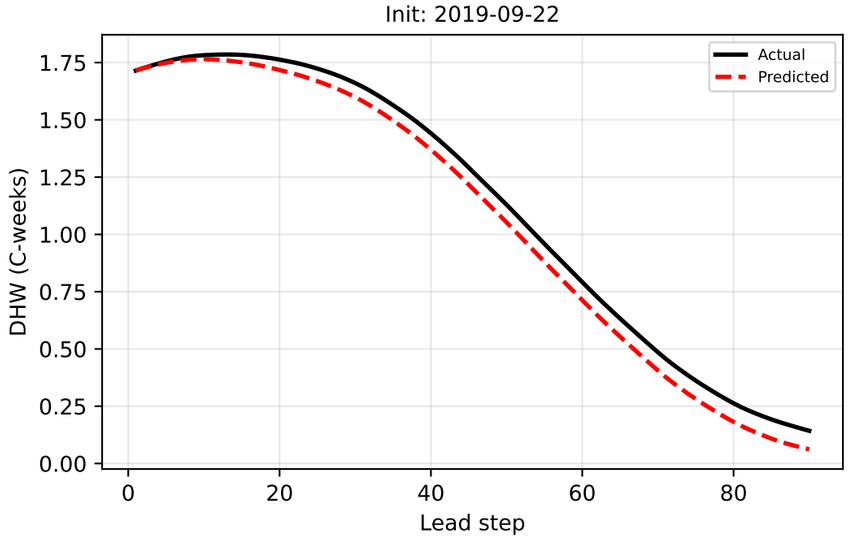




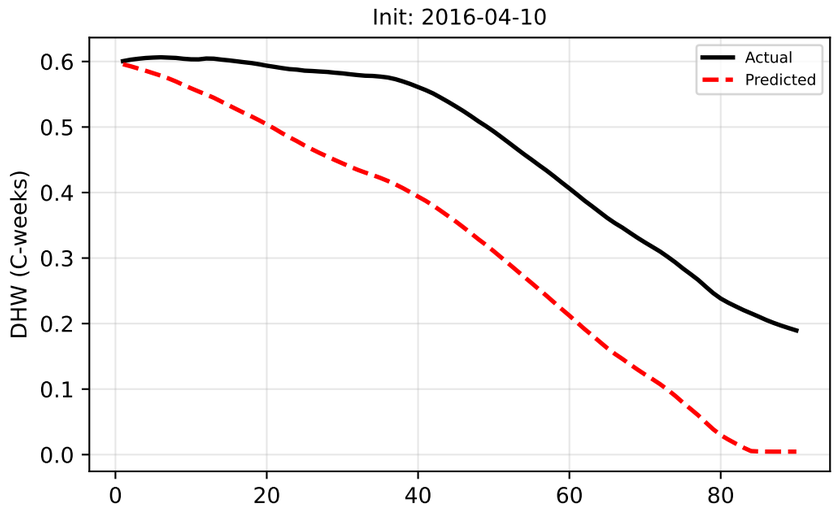


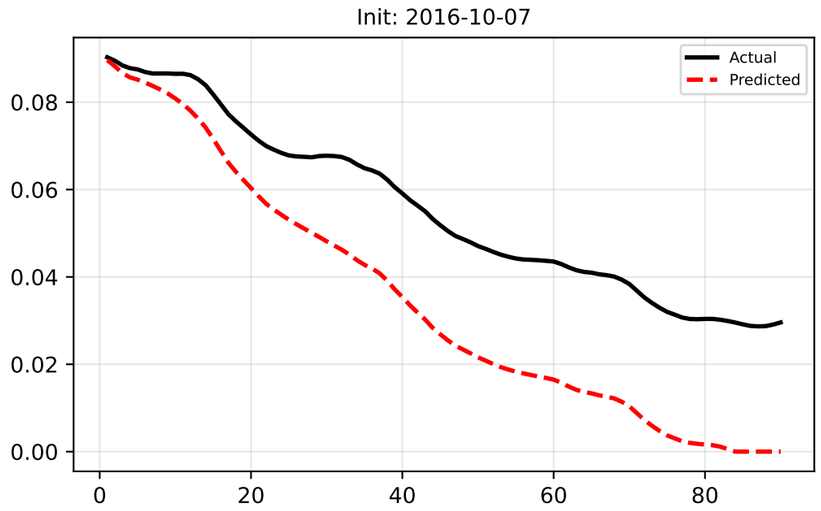


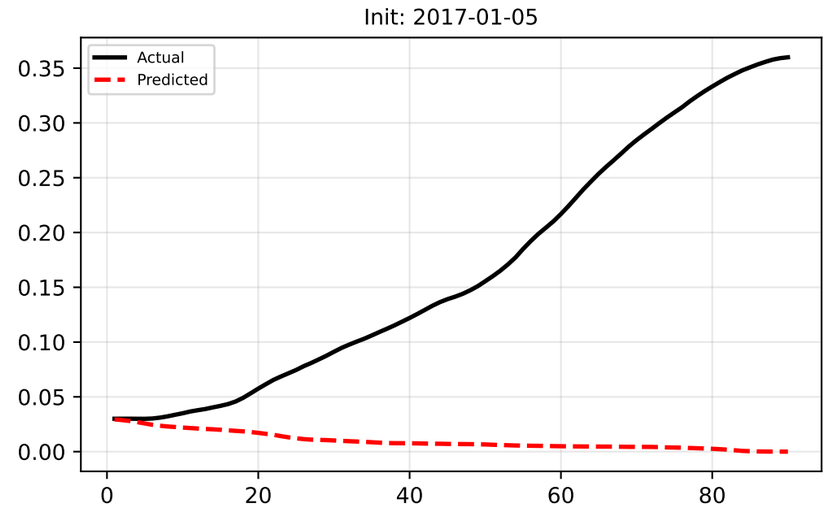


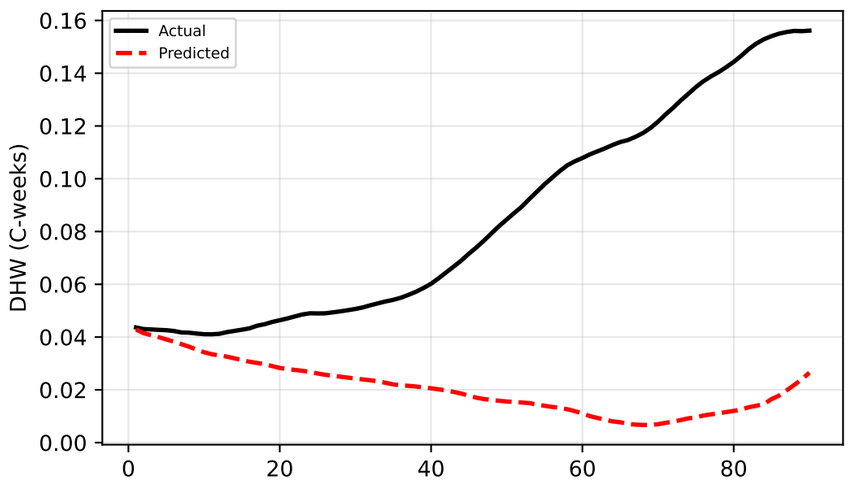


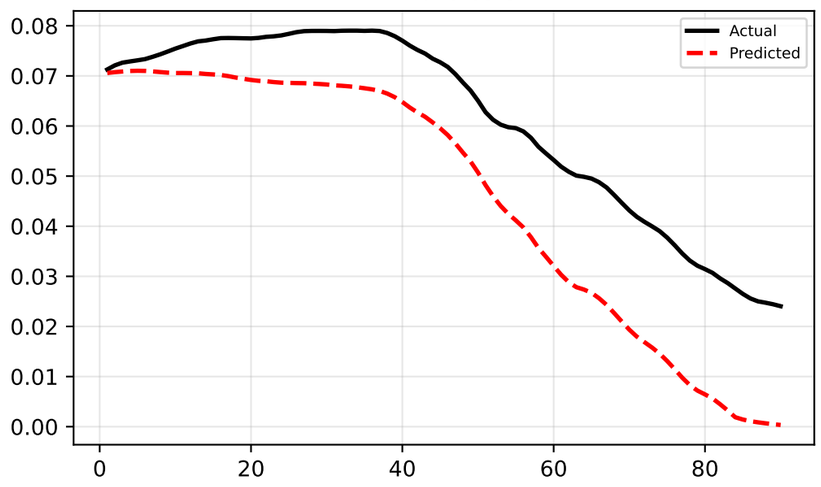
Tropical-Mean DHW (|lat|<=30) - Val (C-weeks)

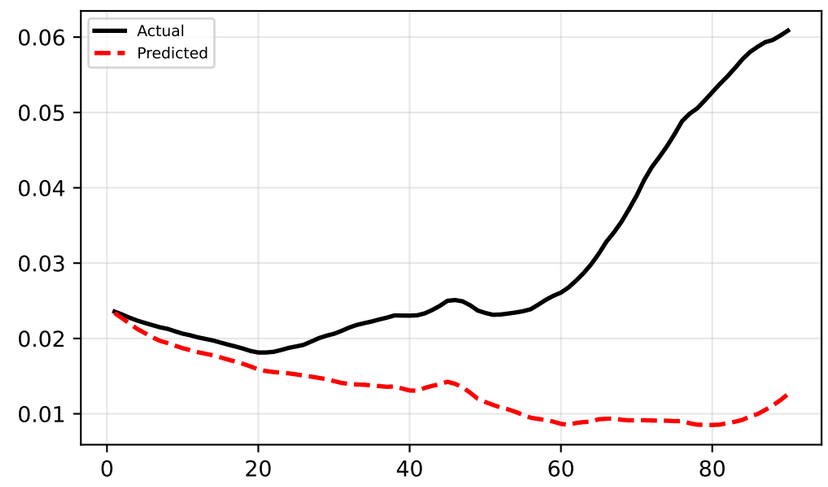


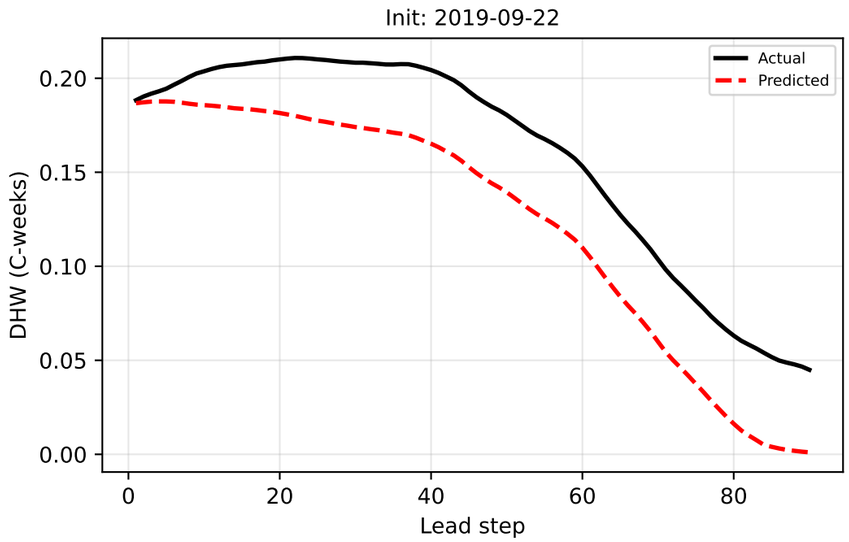




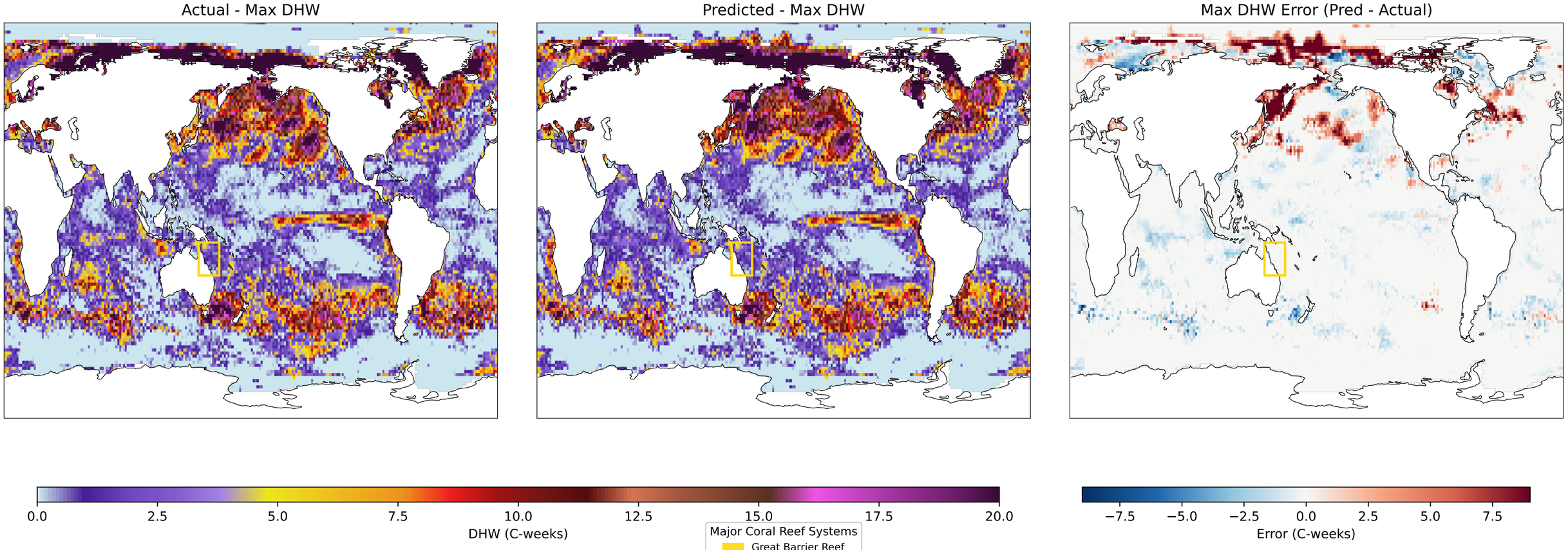


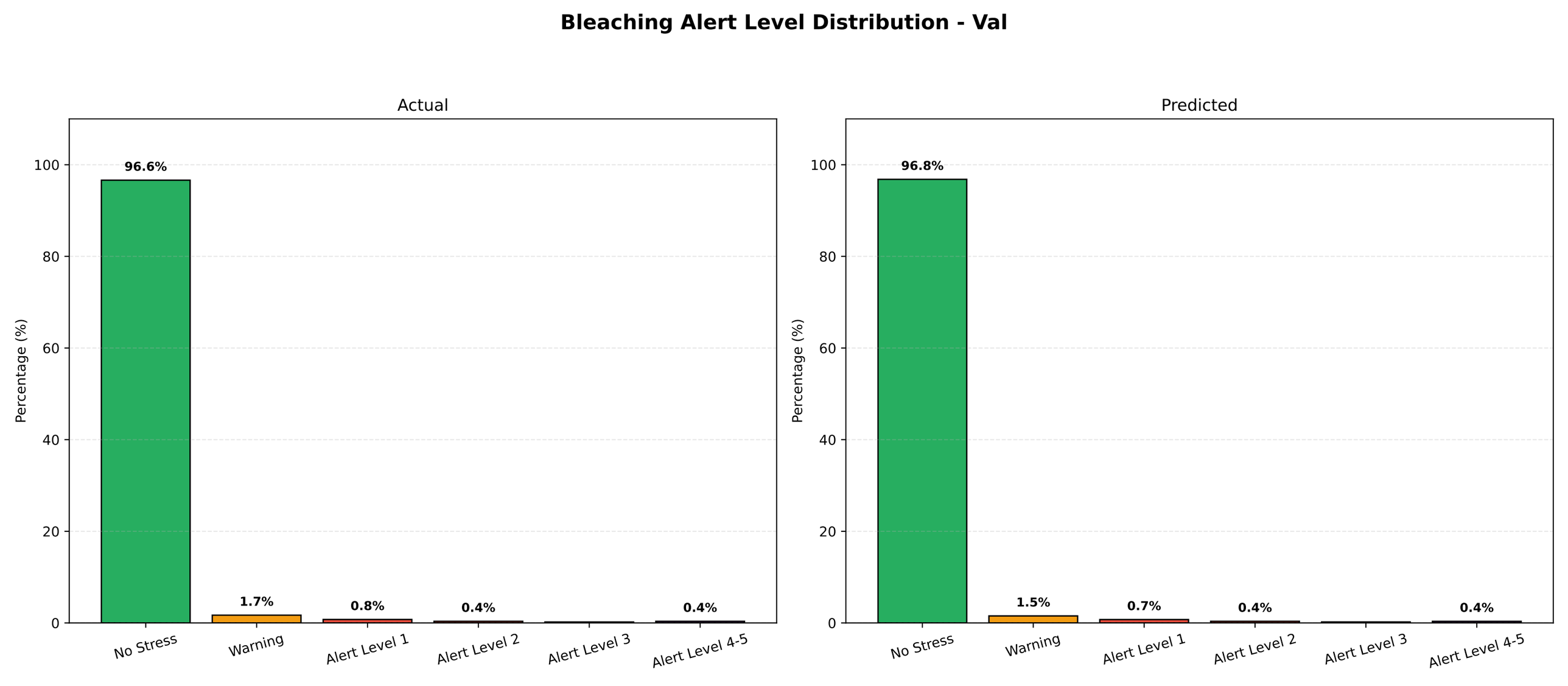






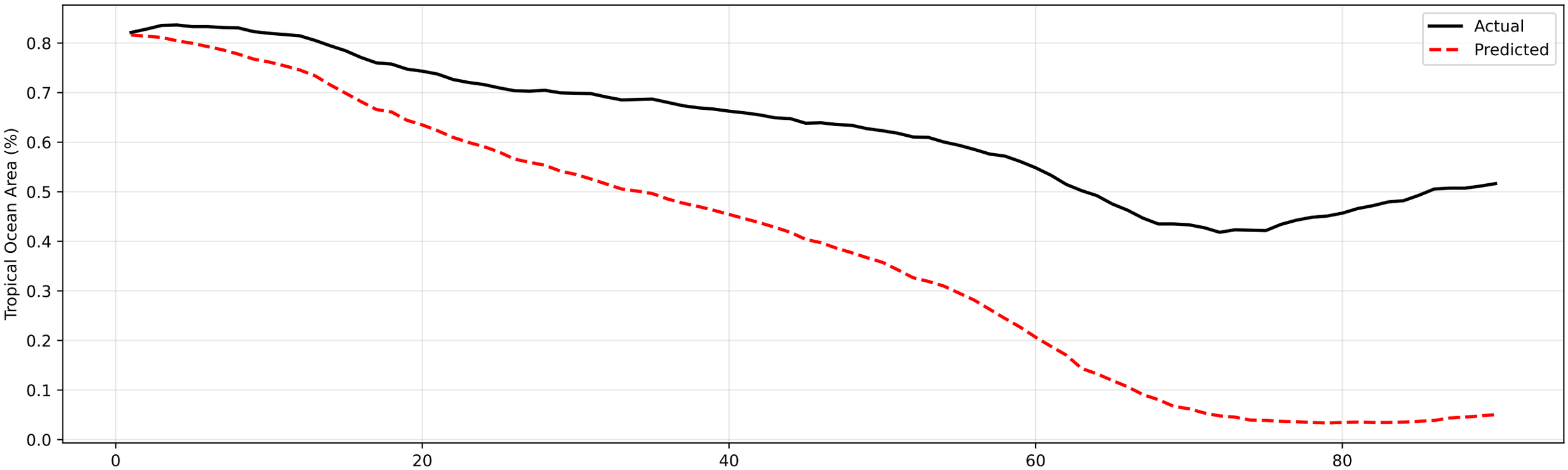
Peak DHW (max over all chains & leads) - Val (C-weeks)



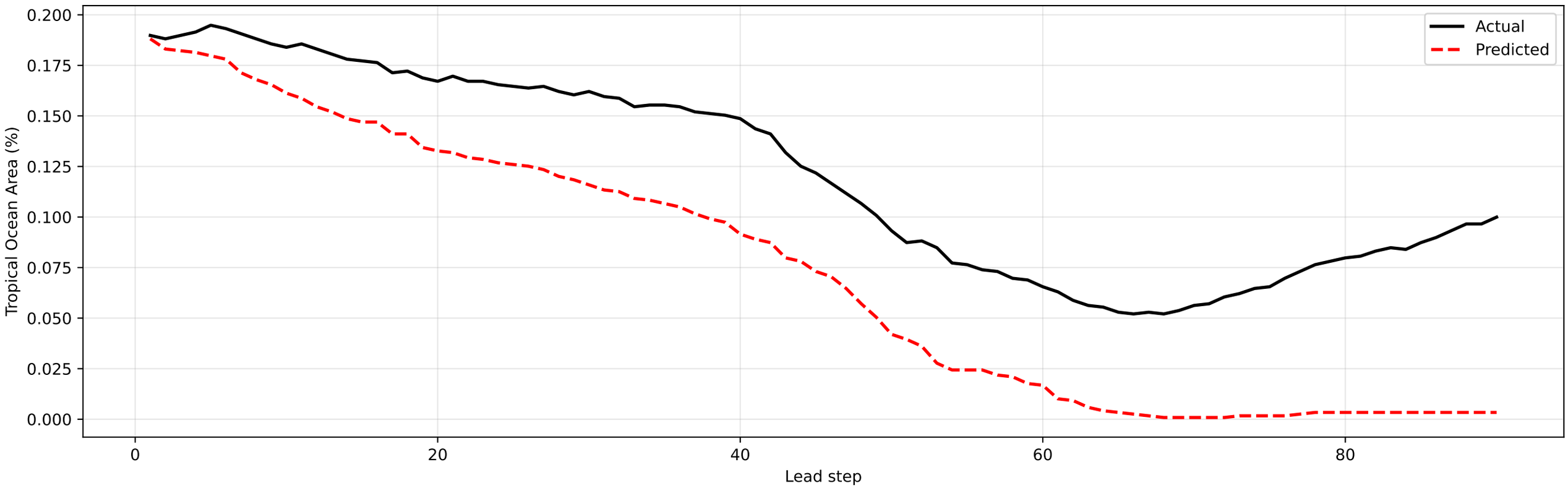

Bleaching Alert Level Distribution - Val
Actual
Predicted
Percentage (%)
0
20
40
60
80
100
96.6%
1.7%
0.8%
0.4%
0.4%
96.8%
1.5%
0.7%
0.4%
0.4%
No Stress
Warning
Alert Level 1
Alert Level 2
Alert Level 3
Alert Level 4-5

Tropical Ocean Area Under Bleaching Stress - Val

DHW >= 4 C-weeks (Alert Level 1 DHW threshold)



DHW >= 8 C-weeks (Alert Level 2 DHW threshold)



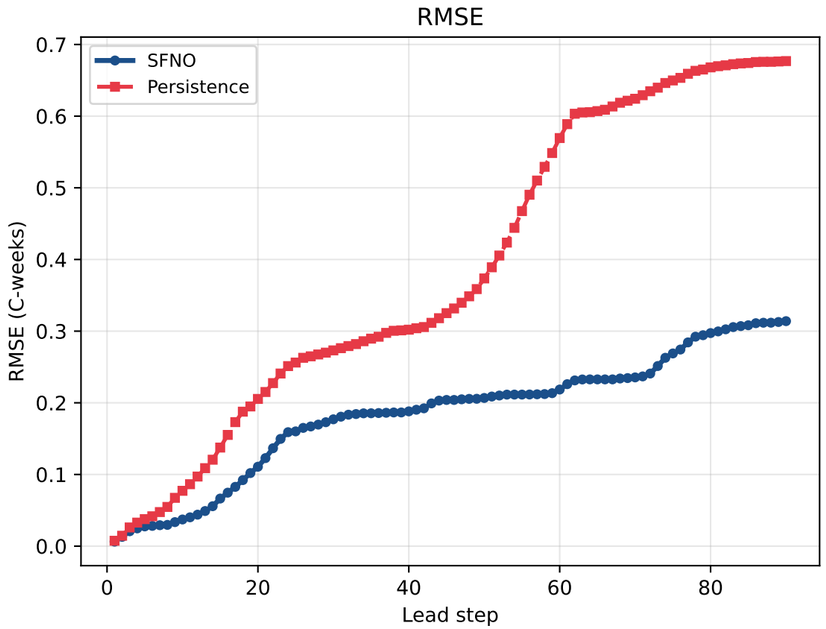
Great Barrier Reef Analysis - Val

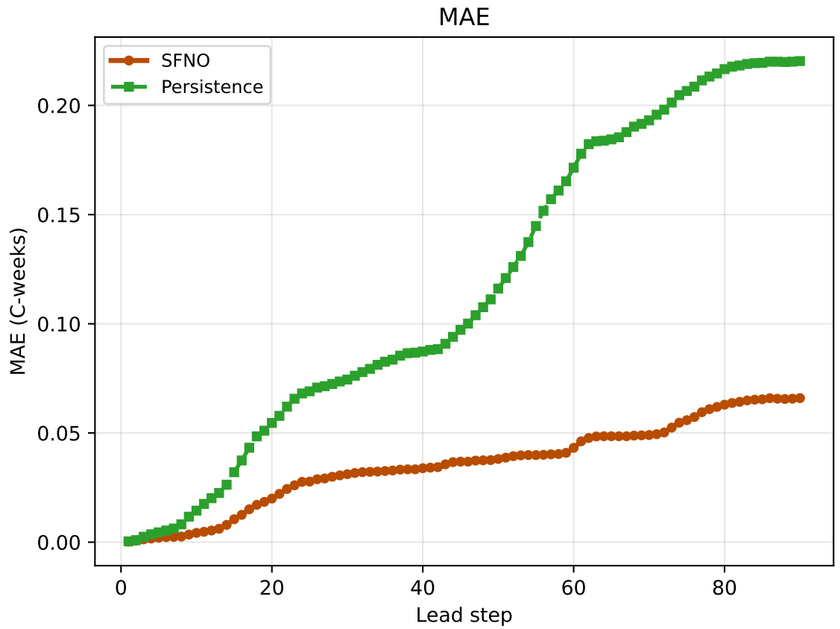
```
                    GREAT BARRIER REEF - VAL
Lat -25.0 deg to -10.0 deg  |  Lon 142.0 deg to 157.0 deg  (0-360)
                    Grid: 11 lat x 11 lon points
```

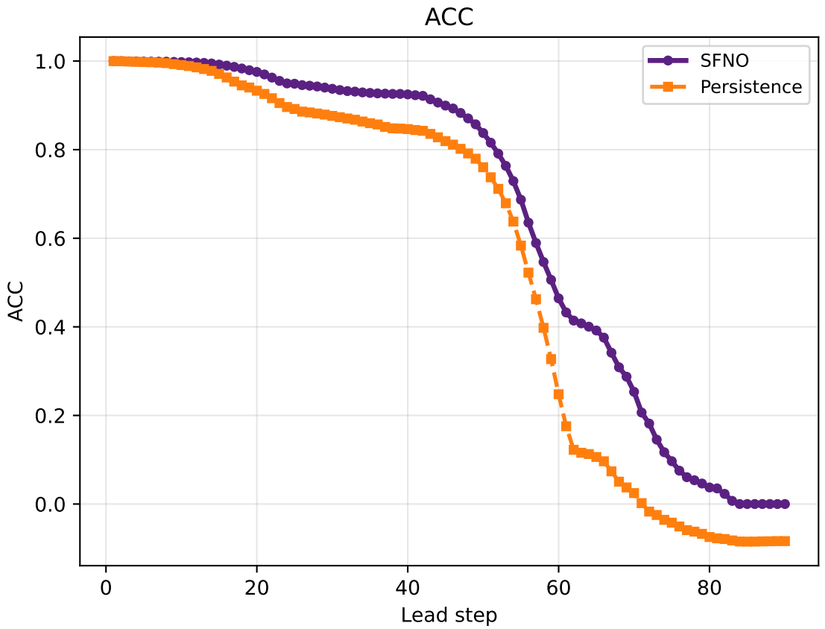
DHW Metrics by Lead Step - Val - GBR (15 chains scored at every lead)

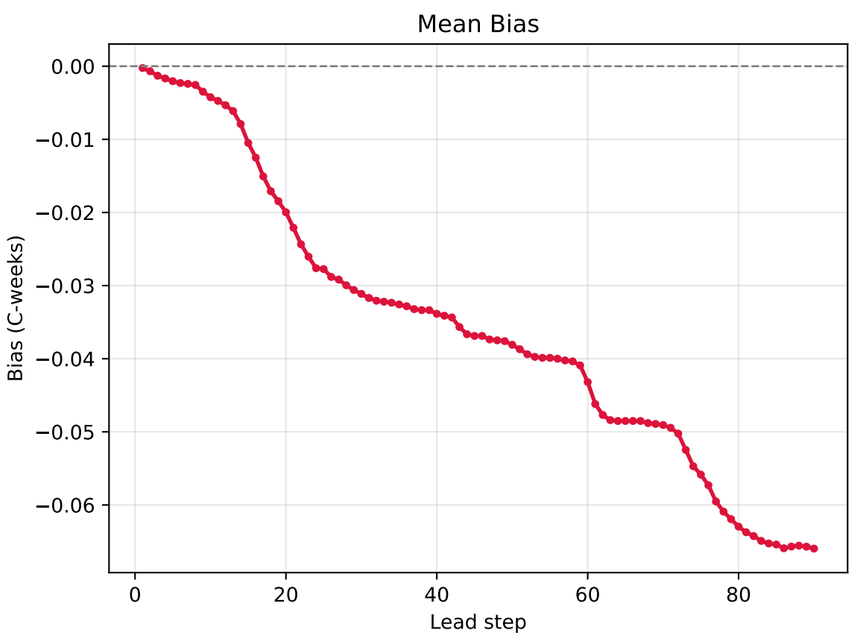
Lead Step	SFNO RMSE	Persist RMSE	Skill %	SFNO MAE	Persist MAE	SFNO Bias	SFNO ACC	Persist ACC	SFNO R²
1	0.0063	0.0075	16.0%	0.0002	0.0004	-0.0002	0.9999	0.9999	0.9999
2	0.0130	0.0145	10.7%	0.0007	0.0009	-0.0007	0.9997	0.9997	0.9995
3	0.0208	0.0261	20.3%	0.0013	0.0025	-0.0013	0.9993	0.9989	0.9987
4	0.0248	0.0328	24.6%	0.0017	0.0036	-0.0017	0.9990	0.9983	0.9982
5	0.0275	0.0377	27.0%	0.0020	0.0045	-0.0020	0.9988	0.9978	0.9977
6	0.0283	0.0417	32.2%	0.0023	0.0053	-0.0023	0.9987	0.9973	0.9976
7	0.0292	0.0476	38.6%	0.0024	0.0062	-0.0024	0.9986	0.9965	0.9974
8	0.0297	0.0547	45.8%	0.0026	0.0082	-0.0026	0.9986	0.9954	0.9973
9	0.0335	0.0674	50.3%	0.0035	0.0117	-0.0035	0.9981	0.9931	0.9965
10	0.0372	0.0772	51.8%	0.0042	0.0144	-0.0042	0.9977	0.9910	0.9956
11	0.0402	0.0864	53.5%	0.0047	0.0175	-0.0047	0.9973	0.9887	0.9948
12	0.0439	0.0971	54.8%	0.0053	0.0201	-0.0053	0.9967	0.9857	0.9937
13	0.0490	0.1088	55.0%	0.0061	0.0225	-0.0061	0.9958	0.9820	0.9920
14	0.0557	0.1206	53.8%	0.0079	0.0263	-0.0079	0.9945	0.9778	0.9896
15	0.0664	0.1375	51.7%	0.0105	0.0320	-0.0105	0.9920	0.9710	0.9849
16	0.0746	0.1552	51.9%	0.0125	0.0374	-0.0125	0.9896	0.9629	0.9803
17	0.0827	0.1730	52.2%	0.0151	0.0432	-0.0151	0.9868	0.9536	0.9751
18	0.0922	0.1876	50.9%	0.0171	0.0484	-0.0171	0.9832	0.9451	0.9684
19	0.1020	0.1949	47.7%	0.0185	0.0510	-0.0185	0.9794	0.9403	0.9612
20	0.1108	0.2054	46.0%	0.0200	0.0546	-0.0200	0.9755	0.9330	0.9539
21	0.1229	0.2151	42.9%	0.0221	0.0578	-0.0221	0.9698	0.9258	0.9435
22	0.1366	0.2275	39.9%	0.0244	0.0621	-0.0244	0.9627	0.9161	0.9304
23	0.1496	0.2409	37.9%	0.0261	0.0657	-0.0261	0.9552	0.9051	0.9169
24	0.1590	0.2512	36.7%	0.0276	0.0681	-0.0276	0.9495	0.8962	0.9067
25	0.1601	0.2562	37.5%	0.0277	0.0691	-0.0277	0.9488	0.8919	0.9053
26	0.1650	0.2628	37.2%	0.0288	0.0708	-0.0288	0.9457	0.8859	0.8998
27	0.1670	0.2649	36.9%	0.0292	0.0714	-0.0292	0.9443	0.8840	0.8974
28	0.1696	0.2675	36.6%	0.0300	0.0725	-0.0300	0.9426	0.8816	0.8943
29	0.1729	0.2700	36.0%	0.0306	0.0736	-0.0306	0.9401	0.8792	0.8900
30	0.1770	0.2732	35.2%	0.0311	0.0745	-0.0311	0.9374	0.8762	0.8852
31	0.1806	0.2762	34.6%	0.0317	0.0762	-0.0317	0.9345	0.8732	0.8802
32	0.1834	0.2792	34.3%	0.0321	0.0779	-0.0321	0.9322	0.8703	0.8763
33	0.1843	0.2819	34.6%	0.0322	0.0794	-0.0322	0.9311	0.8676	0.8743
34	0.1853	0.2859	35.2%	0.0323	0.0812	-0.0323	0.9292	0.8635	0.8711
35	0.1854	0.2895	36.0%	0.0326	0.0826	-0.0326	0.9281	0.8597	0.8690
36	0.1857	0.2922	36.4%	0.0328	0.0836	-0.0328	0.9271	0.8569	0.8672
37	0.1861	0.2976	37.5%	0.0332	0.0854	-0.0332	0.9264	0.8512	0.8662
38	0.1865	0.3005	37.9%	0.0334	0.0866	-0.0334	0.9259	0.8481	0.8652
39	0.1865	0.3011	38.0%	0.0334	0.0868	-0.0334	0.9258	0.8475	0.8651
40	0.1881	0.3020	37.7%	0.0339	0.0873	-0.0339	0.9247	0.8465	0.8632
41	0.1903	0.3041	37.4%	0.0341	0.0880	-0.0341	0.9229	0.8443	0.8601
42	0.1923	0.3057	37.1%	0.0344	0.0884	-0.0344	0.9214	0.8425	0.8575
43	0.1992	0.3116	36.1%	0.0357	0.0909	-0.0357	0.9136	0.8357	0.8438
44	0.2031	0.3180	36.1%	0.0367	0.0940	-0.0367	0.9063	0.8279	0.8312
45	0.2040	0.3251	37.3%	0.0369	0.0972	-0.0369	0.8995	0.8193	0.8196
46	0.2040	0.3317	38.5%	0.0369	0.1001	-0.0369	0.8930	0.8114	0.8085
47	0.2049	0.3397	39.7%	0.0374	0.1039	-0.0374	0.8828	0.8019	0.7914
48	0.2055	0.3486	41.0%	0.0375	0.1076	-0.0375	0.8706	0.7915	0.7710
49	0.2056	0.3587	42.7%	0.0376	0.1112	-0.0376	0.8572	0.7796	0.7489
50	0.2068	0.3735	44.7%	0.0381	0.1161	-0.0381	0.8375	0.7603	0.7172
51	0.2087	0.3891	46.4%	0.0387	0.1209	-0.0387	0.8157	0.7376	0.6828
52	0.2102	0.4055	48.2%	0.0394	0.1260	-0.0394	0.7910	0.7115	0.6451
53	0.2114	0.4235	50.1%	0.0397	0.1311	-0.0397	0.7634	0.6790	0.6043
54	0.2115	0.4441	52.4%	0.0399	0.1374	-0.0399	0.7290	0.6376	0.5564
55	0.2115	0.4675	54.8%	0.0399	0.1448	-0.0399	0.6870	0.5834	0.5008
56	0.2116	0.4904	56.8%	0.0400	0.1517	-0.0400	0.6353	0.5223	0.4359
57	0.2119	0.5101	58.5%	0.0402	0.1570	-0.0402	0.5892	0.4623	0.3801
58	0.2122	0.5292	59.9%	0.0404	0.1610	-0.0404	0.5464	0.3978	0.3309
59	0.2135	0.5485	61.1%	0.0409	0.1653	-0.0409	0.5061	0.3272	0.2869
60	0.2185	0.5694	61.6%	0.0432	0.1715	-0.0432	0.4643	0.2479	0.2427
61	0.2260	0.5888	61.6%	0.0462	0.1779	-0.0462	0.4325	0.1755	0.2101
62	0.2313	0.6033	61.7%	0.0477	0.1822	-0.0477	0.4141	0.1224	0.1930
63	0.2326	0.6051	61.6%	0.0484	0.1836	-0.0484	0.4078	0.1157	0.1877
64	0.2327	0.6055	61.6%	0.0485	0.1838	-0.0485	0.4004	0.1129	0.1811
65	0.2327	0.6070	61.7%	0.0485	0.1845	-0.0485	0.3918	0.1062	0.1731
66	0.2327	0.6089	61.8%	0.0485	0.1854	-0.0485	0.3756	0.0965	0.1592
67	0.2327	0.6135	62.1%	0.0485	0.1878	-0.0485	0.3416	0.0739	0.1321
68	0.2339	0.6187	62.2%	0.0488	0.1903	-0.0488	0.3088	0.0504	0.1073
69	0.2345	0.6215	62.3%	0.0489	0.1916	-0.0489	0.2875	0.0374	0.0921
70	0.2355	0.6242	62.3%	0.0491	0.1932	-0.0491	0.2533	0.0247	0.0697
71	0.2368	0.6292	62.4%	0.0494	0.1958	-0.0494	0.2065	0.0020	0.0418
72	0.2408	0.6347	62.1%	0.0502	0.1981	-0.0502	0.1816	-0.0171	0.0290
73	0.2513	0.6397	60.7%	0.0525	0.2013	-0.0525	0.1454	-0.0248	0.0124
74	0.2628	0.6462	59.3%	0.0547	0.2047	-0.0547	0.1169	-0.0357	0.0008
75	0.2689	0.6498	58.6%	0.0559	0.2066	-0.0559	0.0967	-0.0421	-0.0072
76	0.2743	0.6534	58.0%	0.0573	0.2086	-0.0573	0.0753	-0.0506	-0.0161
77	0.2843	0.6591	56.9%	0.0595	0.2115	-0.0595	0.0606	-0.0589	-0.0216
78	0.2923	0.6632	55.9%	0.0609	0.2132	-0.0609	0.0537	-0.0623	-0.0237
79	0.2943	0.6651	55.7%	0.0619	0.2146	-0.0619	0.0463	-0.0673	-0.0271
80	0.2973	0.6680	55.5%	0.0630	0.2166	-0.0630	0.0374	-0.0744	-0.0304
81	0.2996	0.6697	55.3%	0.0637	0.2178	-0.0637	0.0353	-0.0774	-0.0312
82	0.3025	0.6709	54.9%	0.0643	0.2183	-0.0643	0.0230	-0.0790	-0.0366
83	0.3057	0.6725	54.6%	0.0649	0.2190	-0.0649	0.0071	-0.0822	-0.0439
84	0.3071	0.6736	54.4%	0.0653	0.2194	-0.0653	0.0000	-0.0846	-0.0473
85	0.3082	0.6741	54.3%	0.0654	0.2195	-0.0654	-0.0000	-0.0849	-0.0471
86	0.3111	0.6755	53.9%	0.0659	0.2200	-0.0659	0.0000	-0.0848	-0.0470
87	0.3117	0.6758	53.9%	0.0657	0.2200	-0.0657	0.0000	-0.0843	-0.0465
88	0.3118	0.6758	53.9%	0.0656	0.2199	-0.0656	0.0000	-0.0841	-0.0463
89	0.3128	0.6763	53.8%	0.0657	0.2200	-0.0657	-0.0000	-0.0838	-0.0462
90	0.3139	0.6768	53.6%	0.0660	0.2203	-0.0660	0.0000	-0.0838	-0.0462

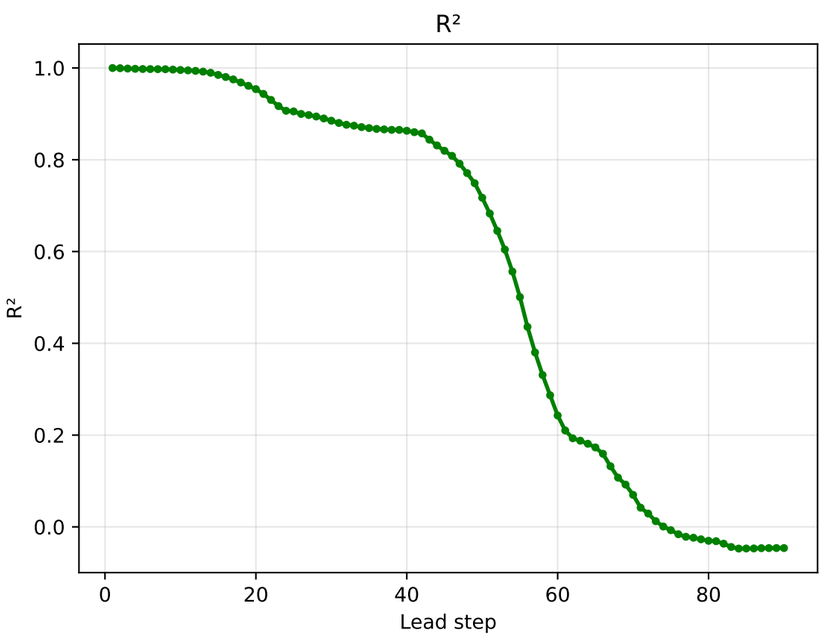
DHW Forecast Skill vs Lead Step - Val - GBR

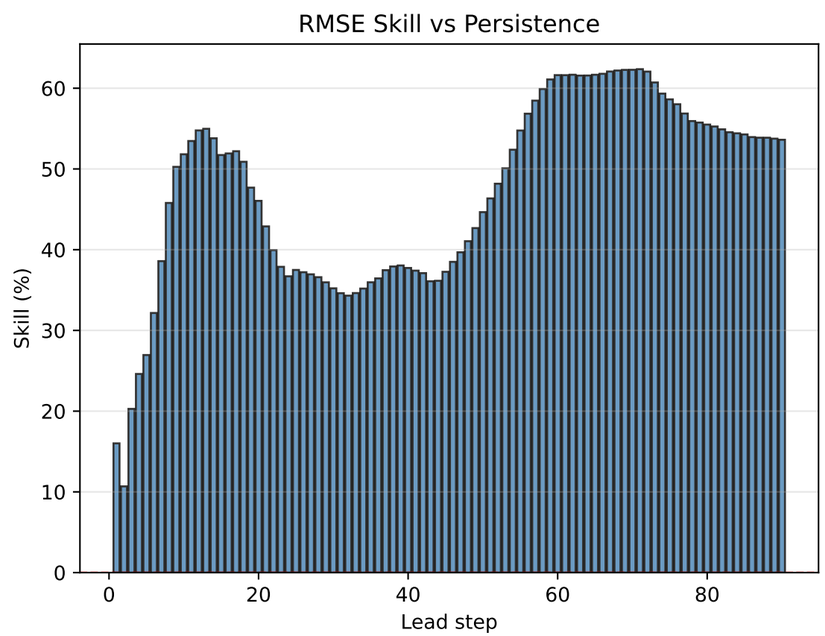


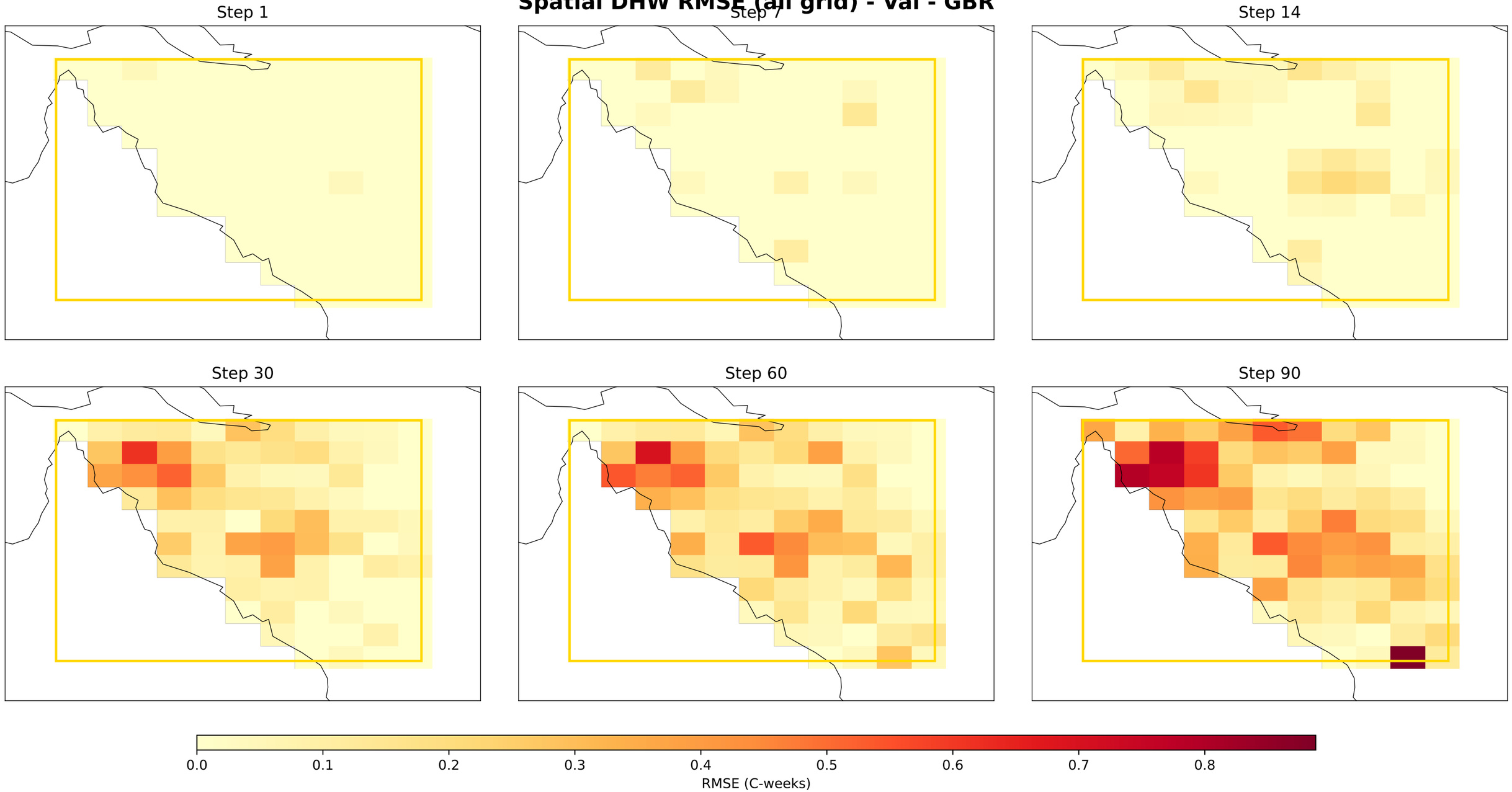


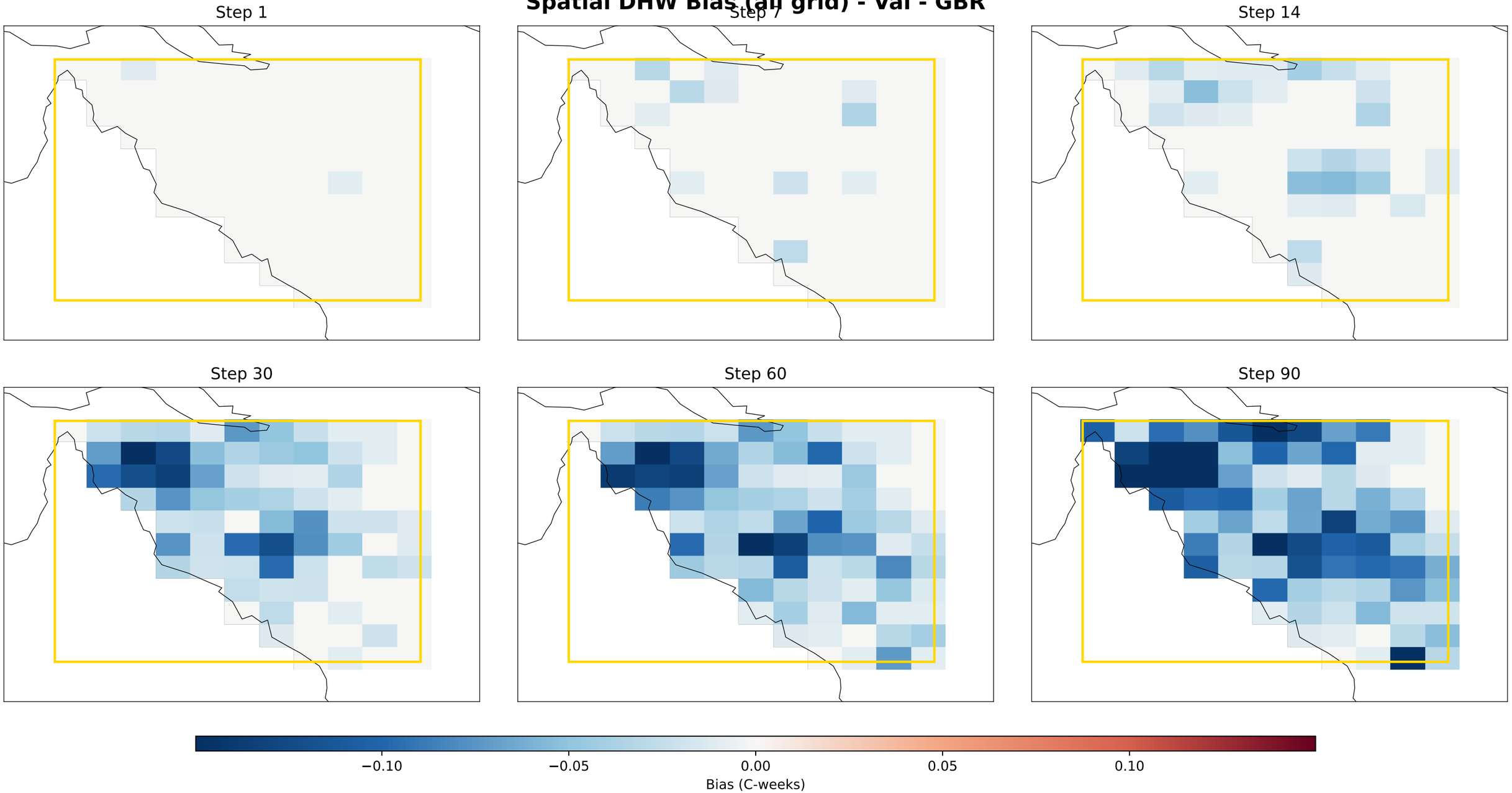


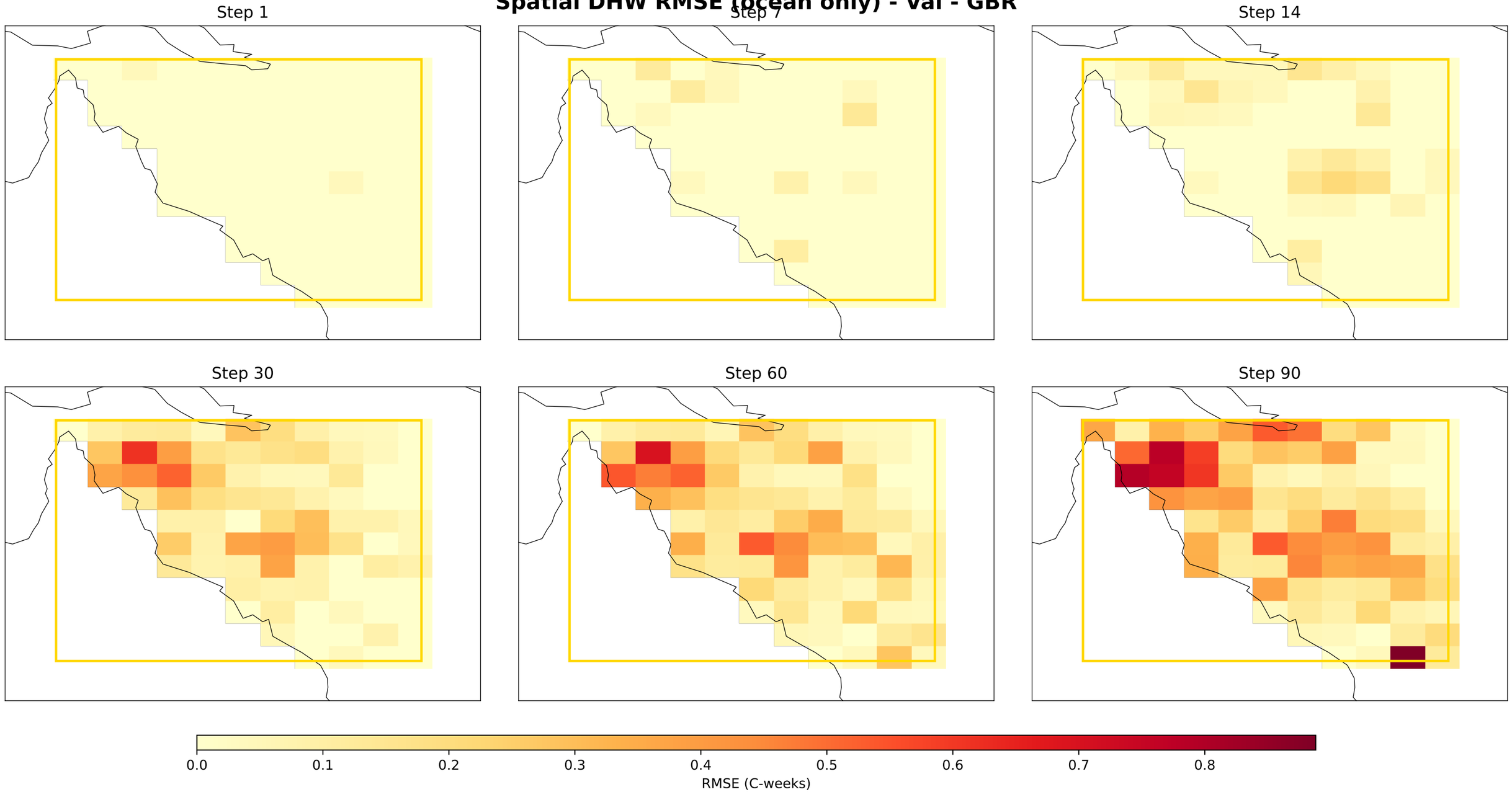


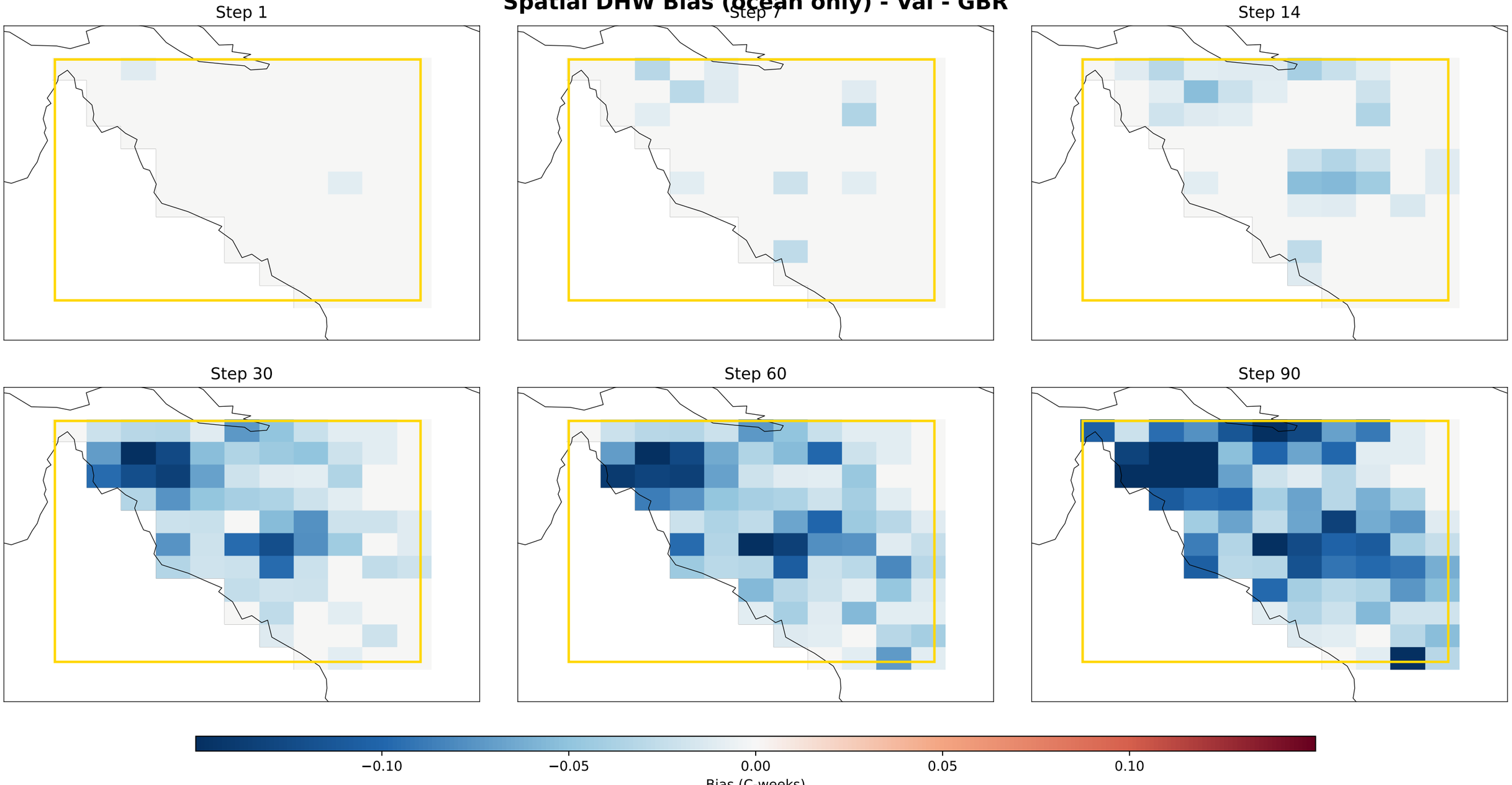





Spatial DHW RMSE (all grid) - Val - GBR
Step 1
Step 7
Step 14
Step 30
Step 60
Step 90
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
RMSE (C-weeks)

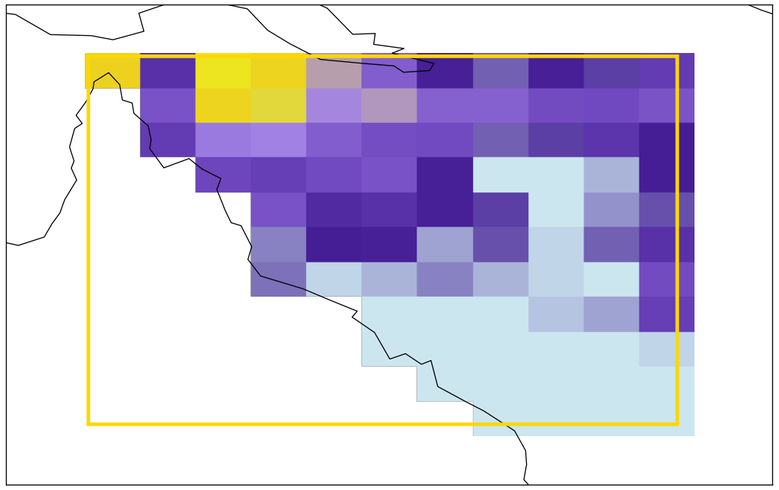

Spatial DHW Bias (all grid) - Val - GBR
Step 1
Step 7
Step 14
Step 30
Step 60
Step 90
−0.10
−0.05
0.00
0.05
0.10
Bias (C-weeks)


Spatial DHW RMSE (ocean only) - Val - GBR
Step 1
Step 7
Step 14
Step 30
Step 60
Step 90
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
RMSE (C-weeks)

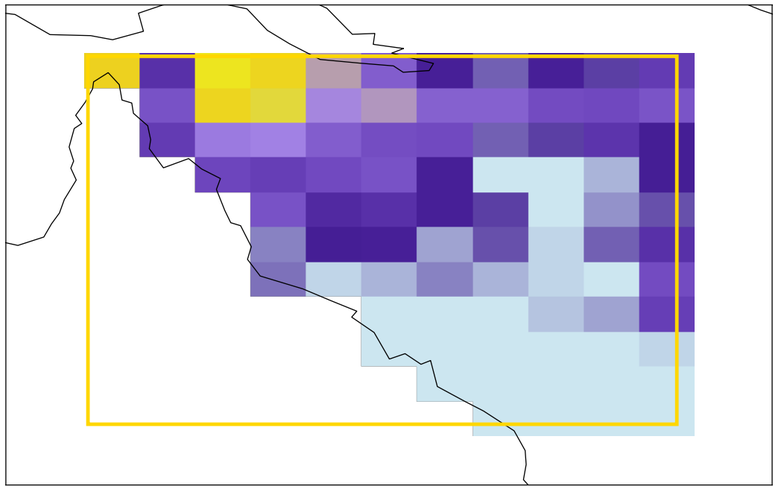

Spatial DHW Bias (ocean only) - Val - GBR
Step 1
Step 7
Step 14
Step 30
Step 60
Step 90
−0.10
−0.05
0.00
0.05
0.10
Bias (C-weeks)

DHW Chain init: 2016-04-10 - Val - GBR (C-weeks)

Actual DHW - Step 1



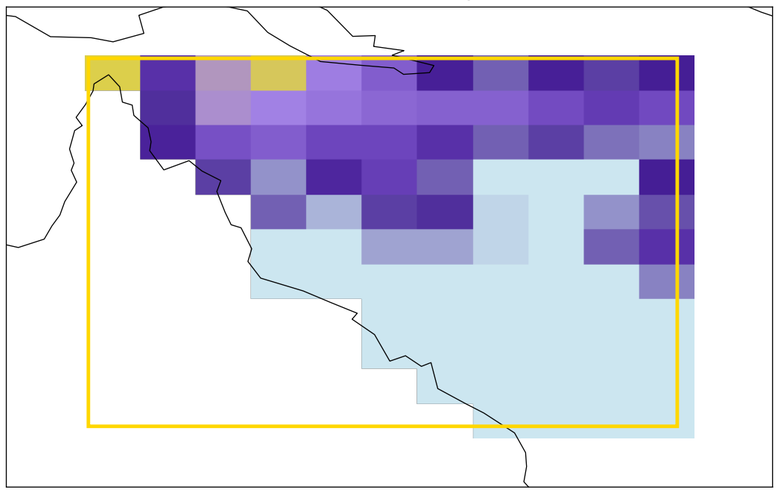
Predicted DHW - Step 1



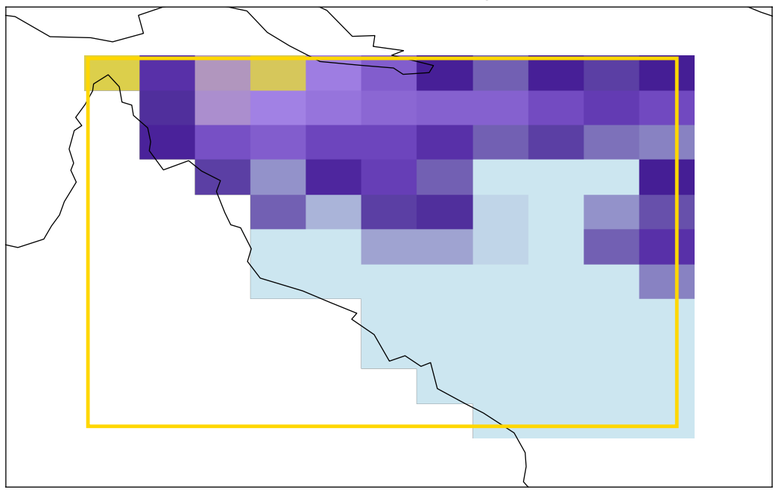
Error (Pred - Actual) - Step 1 [scale ±8]



Actual DHW - Step 46



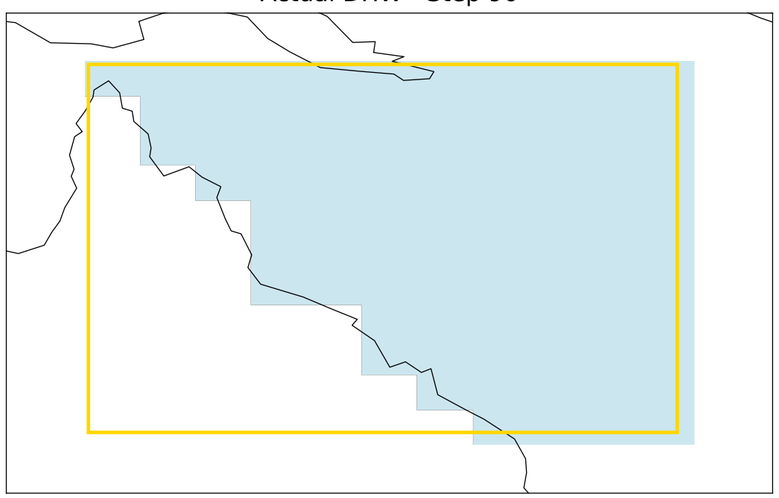
Predicted DHW - Step 46



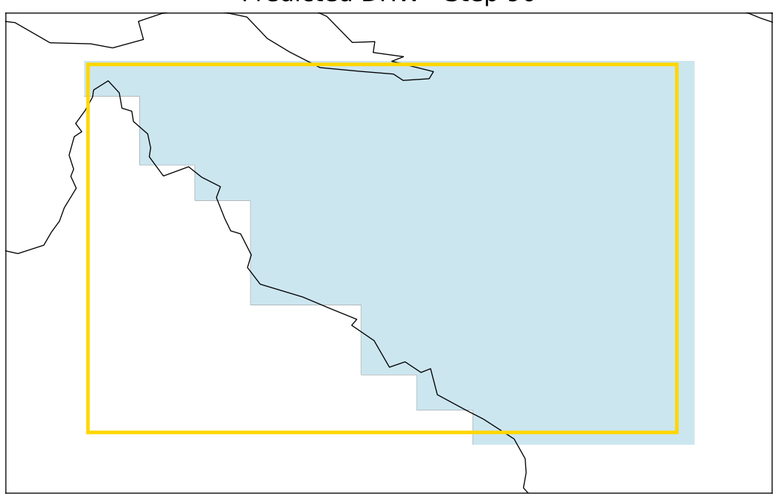
Error (Pred - Actual) - Step 46 [scale ±8]



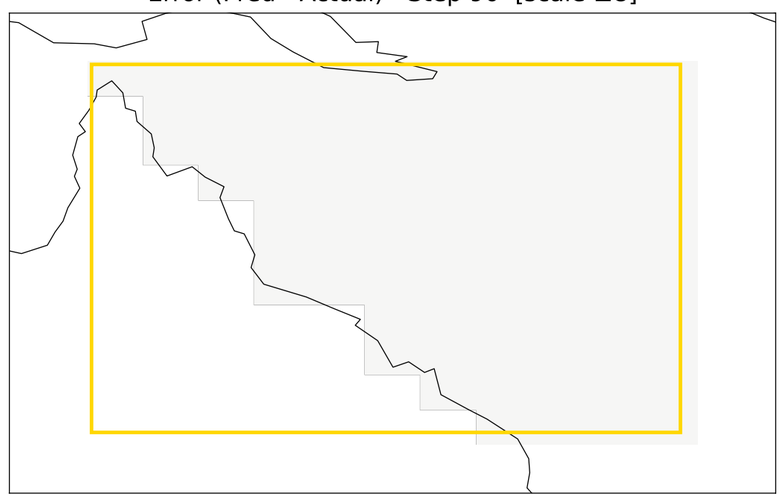
Actual DHW - Step 90

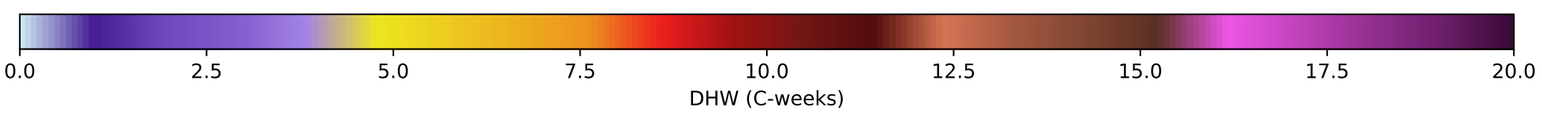


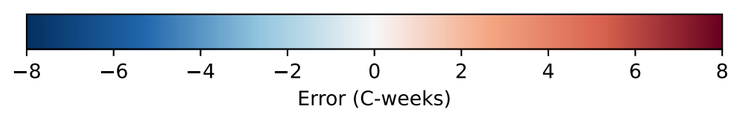
Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±8]

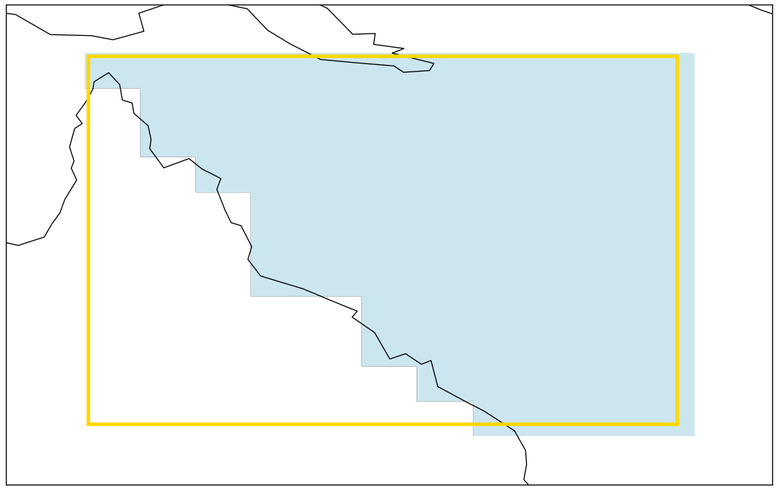




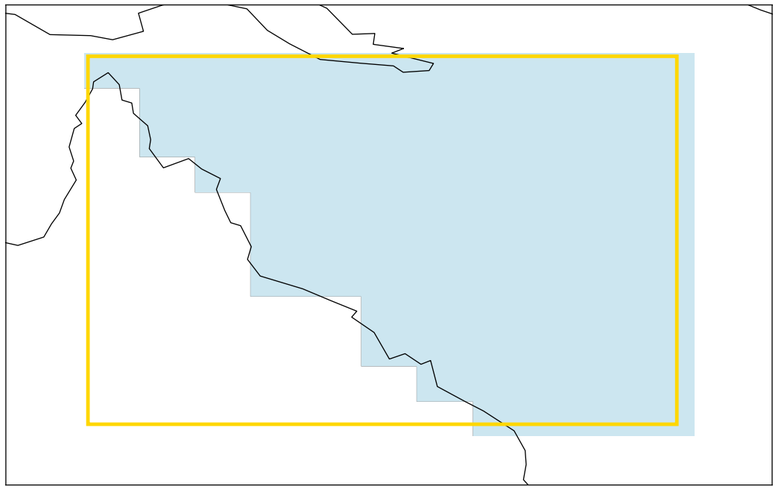


DHW Chain init: 2016-10-07 - Val - GBR (C-weeks)

Actual DHW - Step 1



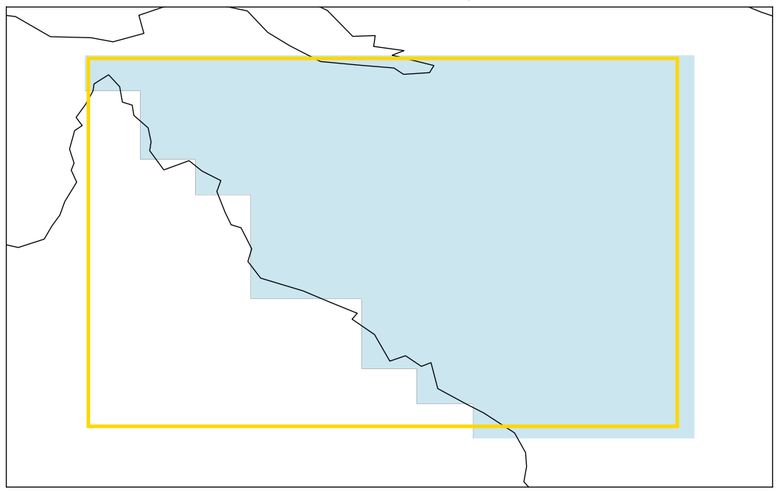
Predicted DHW - Step 1



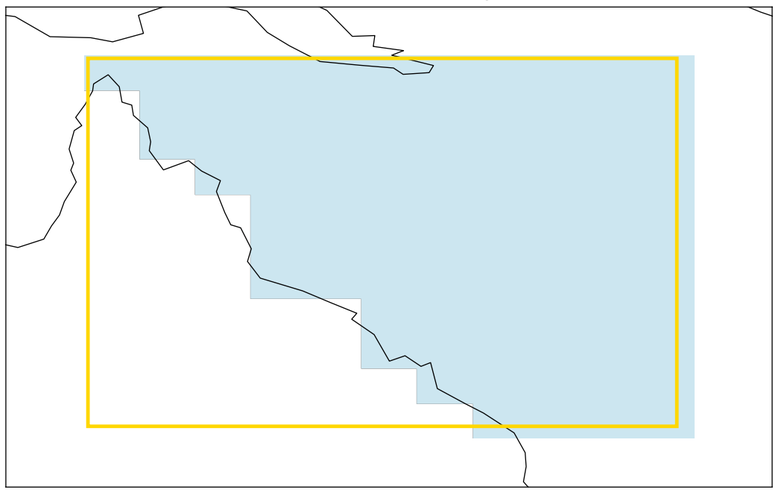
Error (Pred - Actual) - Step 1 [scale ±8]



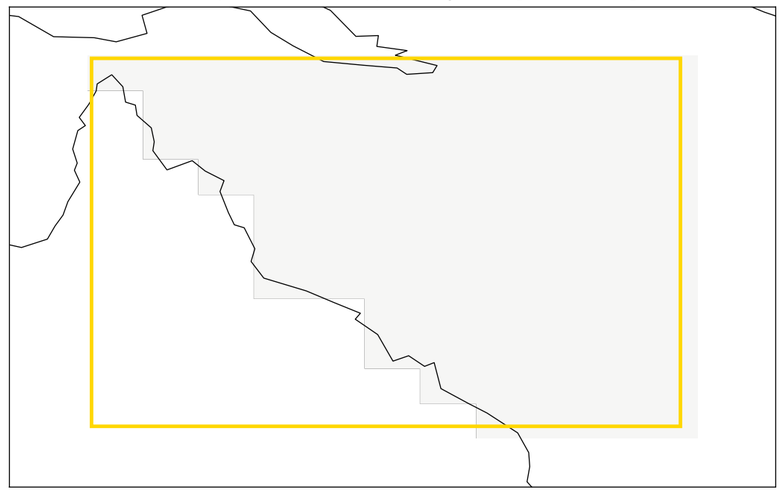
Actual DHW - Step 46



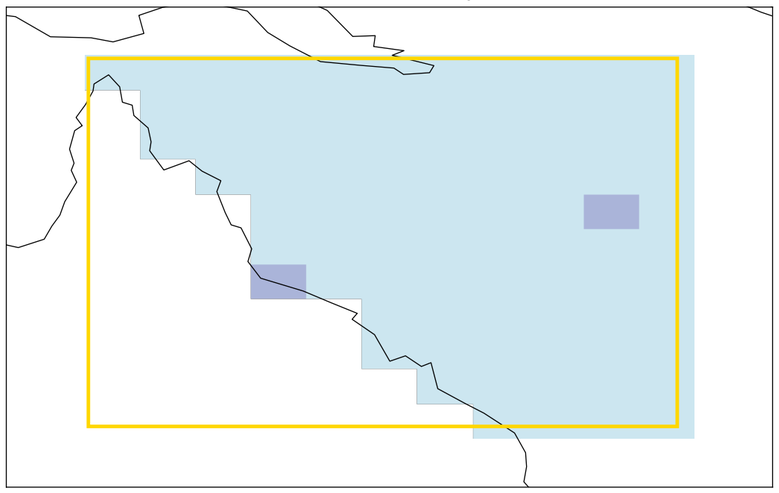
Predicted DHW - Step 46



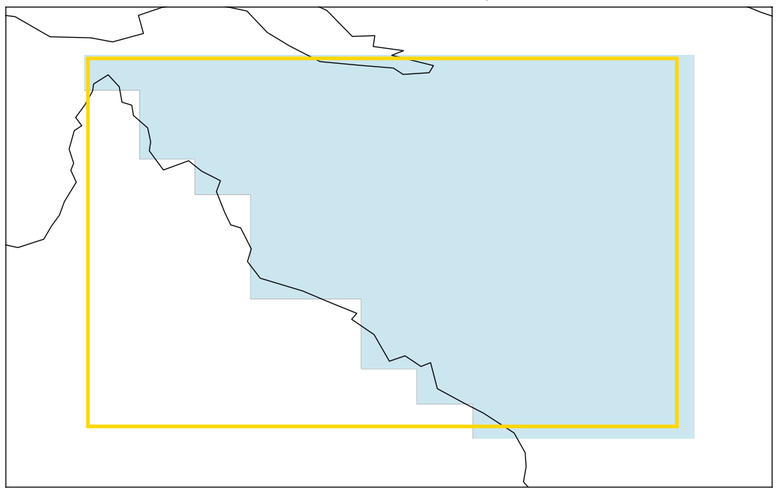
Error (Pred - Actual) - Step 46 [scale ±8]



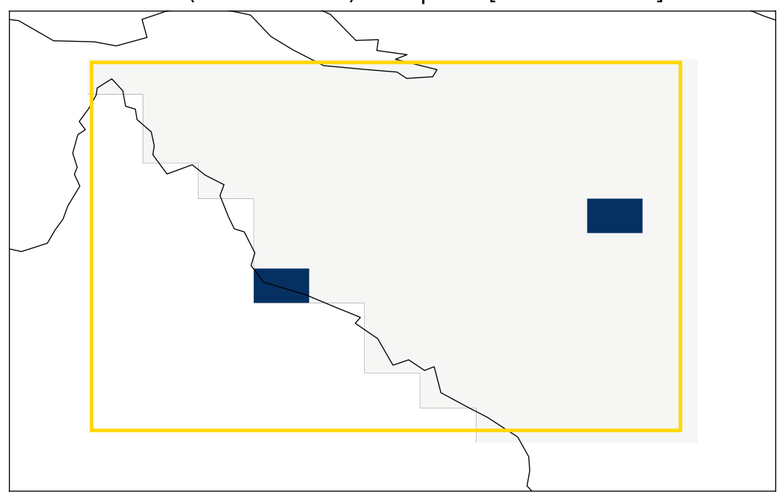
Actual DHW - Step 90

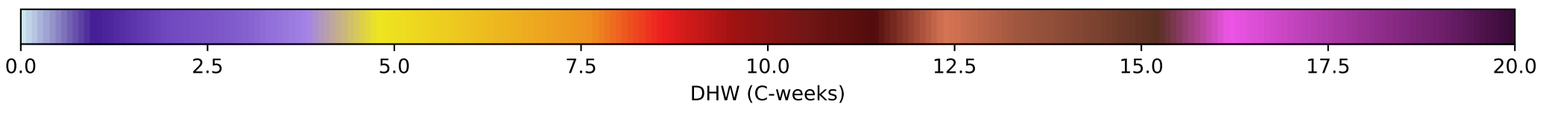


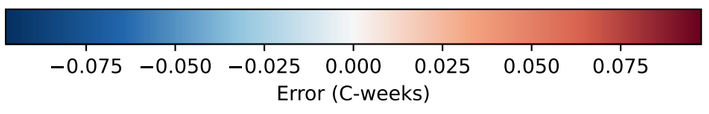
Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±0.098]

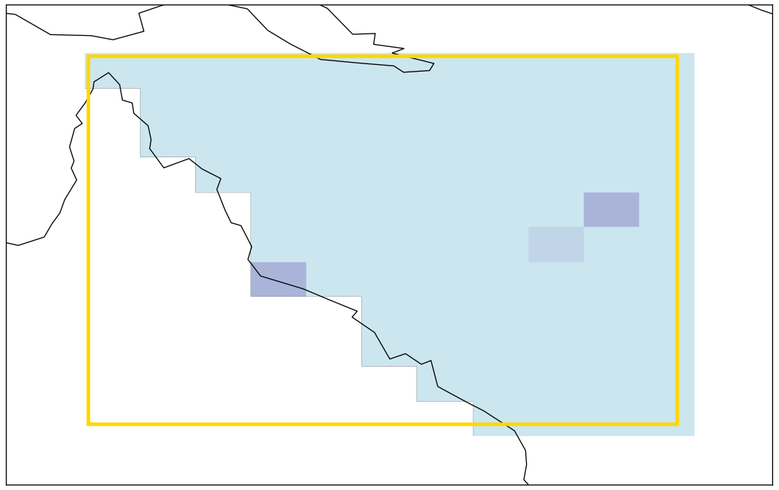




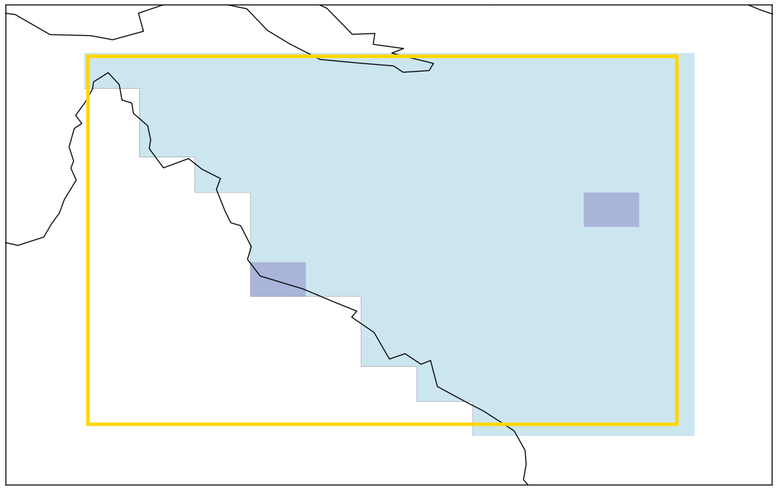


DHW Chain init: 2017-01-05 - Val - GBR (C-weeks)

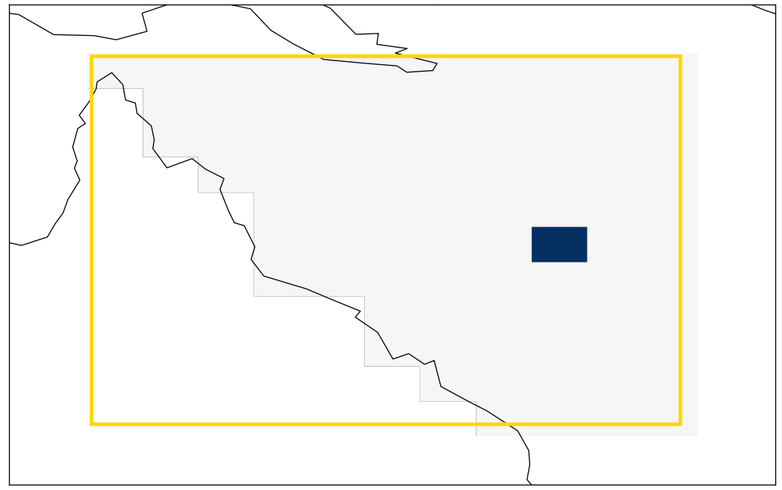
Actual DHW - Step 1



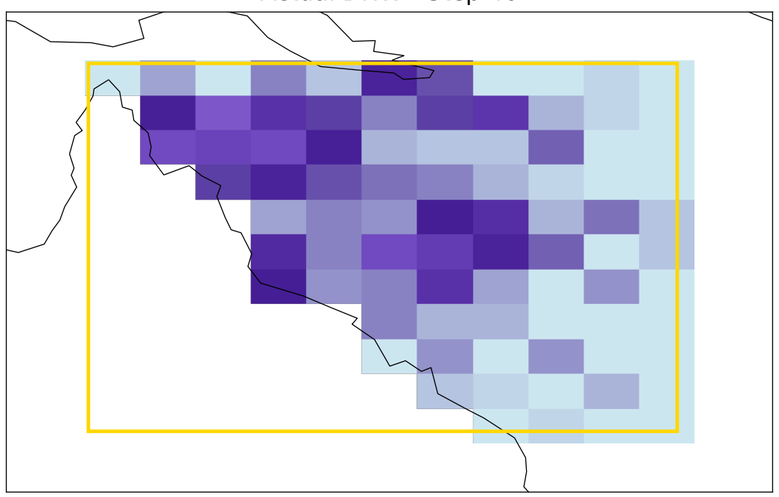
Predicted DHW - Step 1



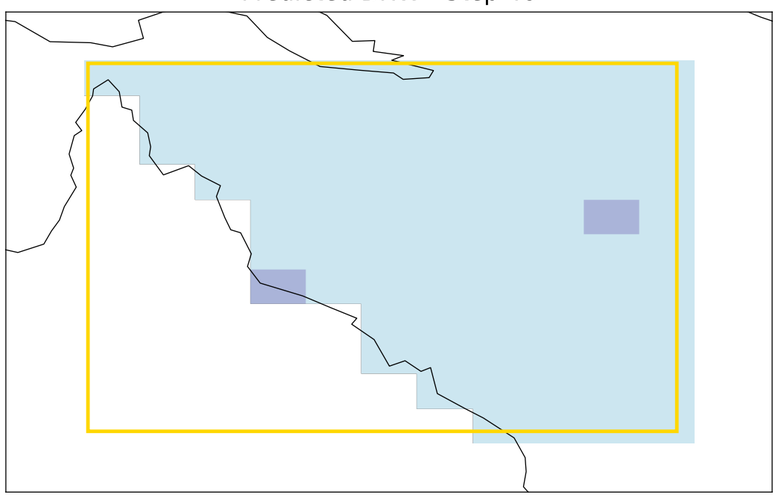
Error (Pred - Actual) - Step 1 [scale ±0.15]



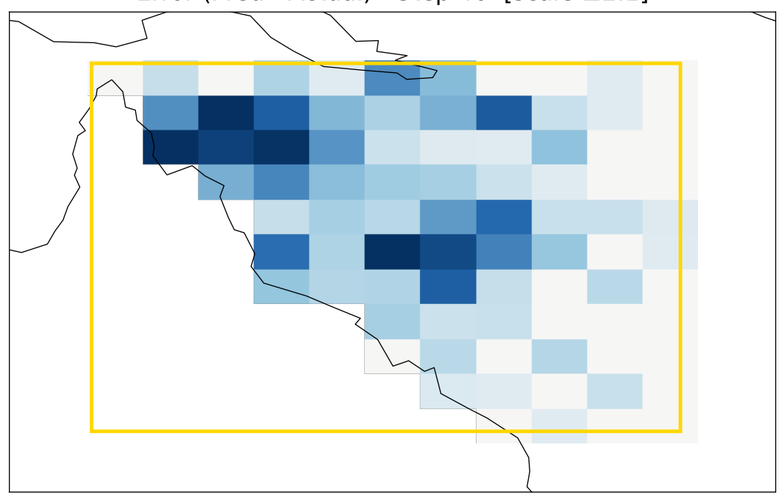
Actual DHW - Step 46



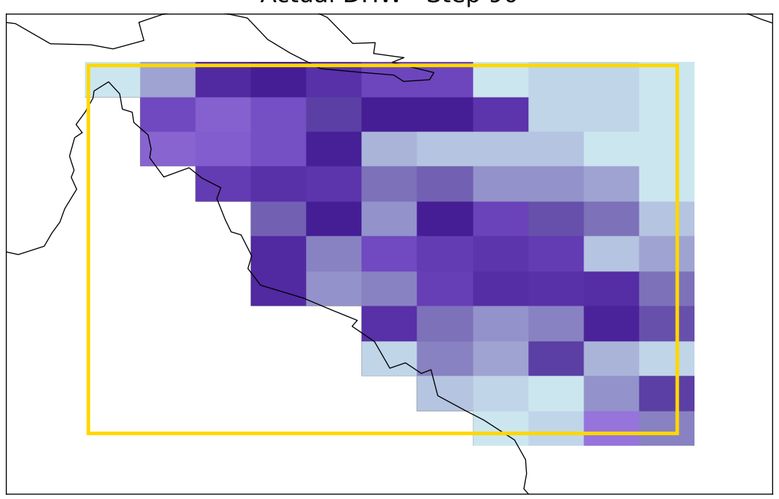
Predicted DHW - Step 46



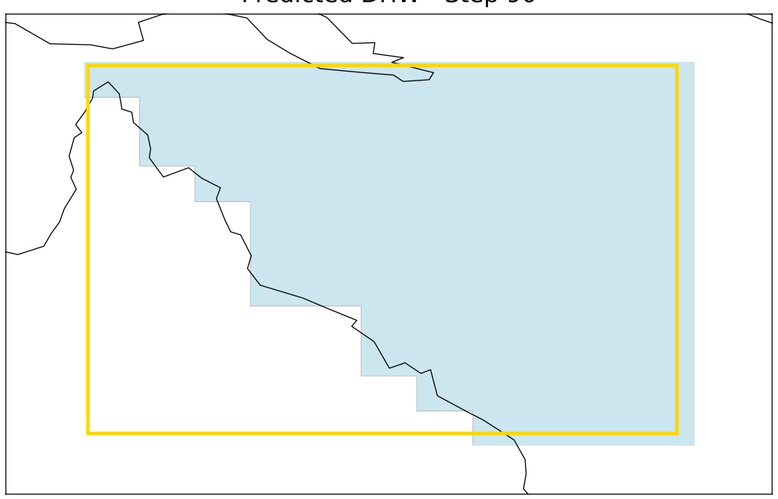
Error (Pred - Actual) - Step 46 [scale ±2.1]



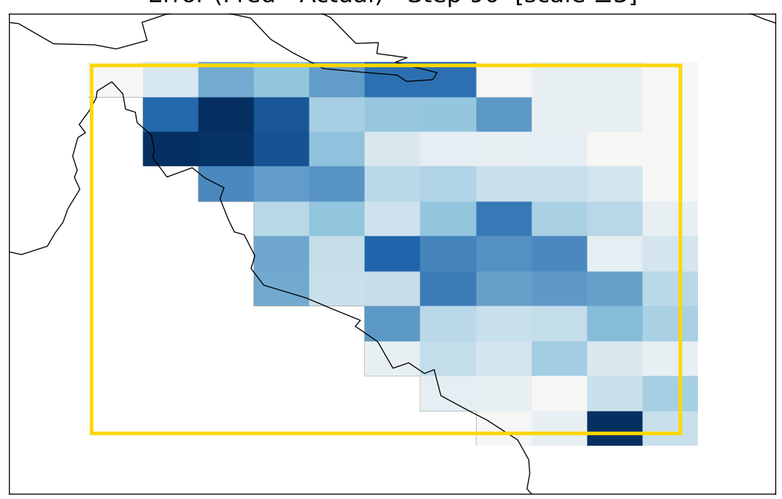
Actual DHW - Step 90

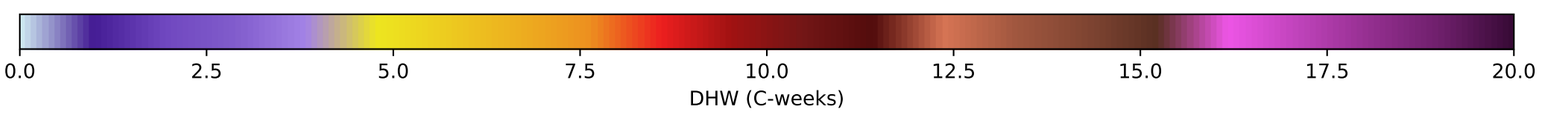


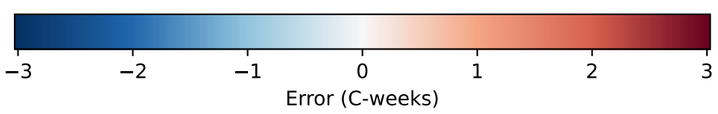
Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±3]

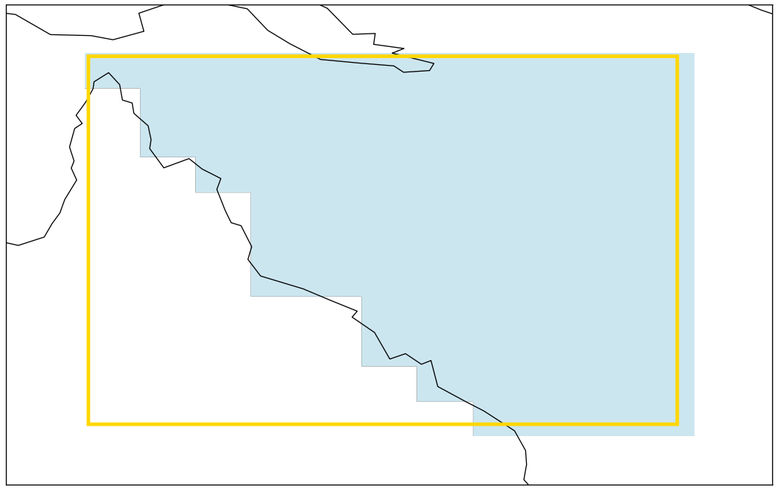




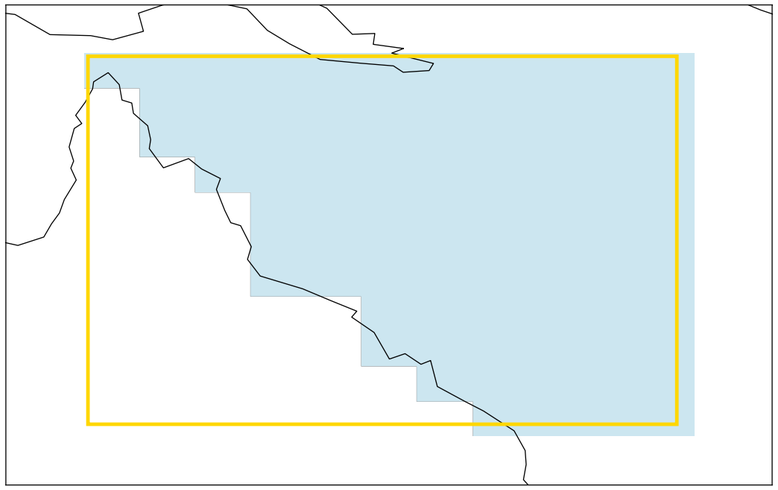


DHW Chain init: 2017-07-04 - Val - GBR (C-weeks)

Actual DHW - Step 1



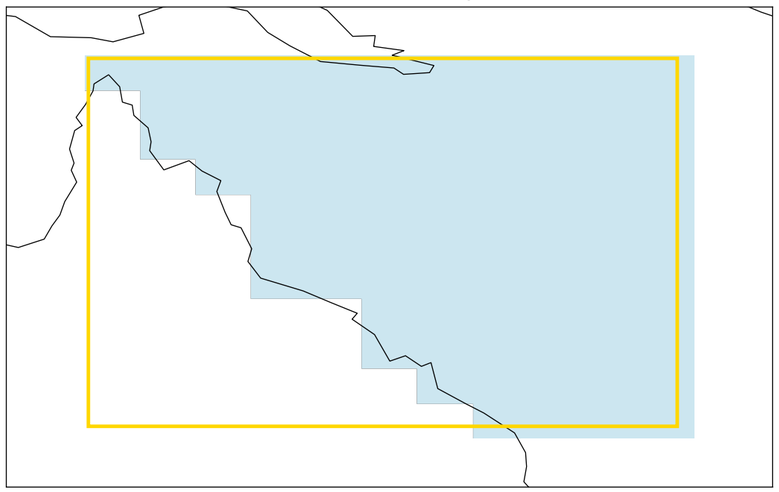
Predicted DHW - Step 1



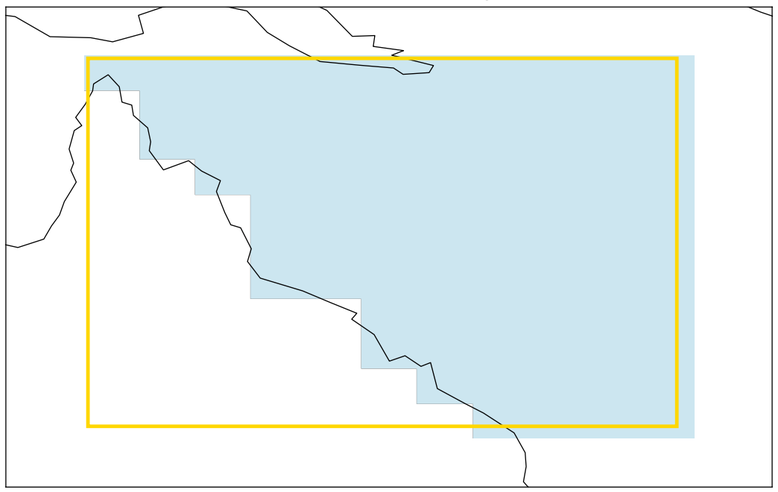
Error (Pred - Actual) - Step 1 [scale ±8]



Actual DHW - Step 46



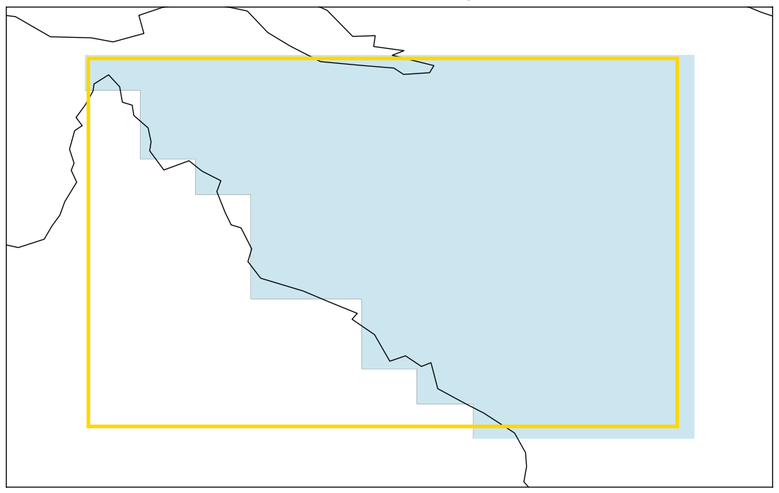
Predicted DHW - Step 46



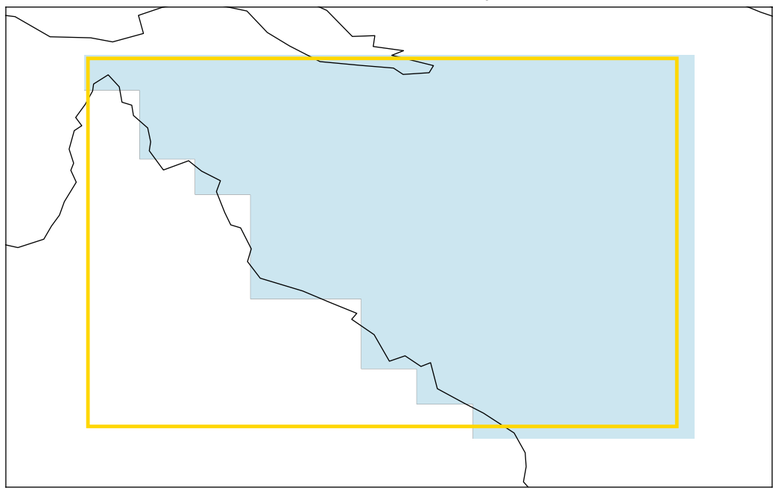
Error (Pred - Actual) - Step 46 [scale ±8]



Actual DHW - Step 90

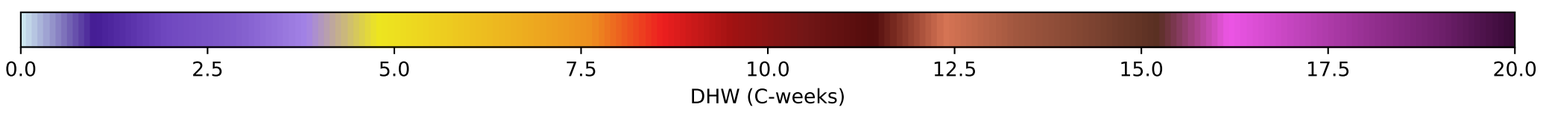


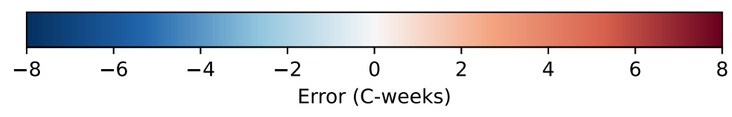
Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±8]

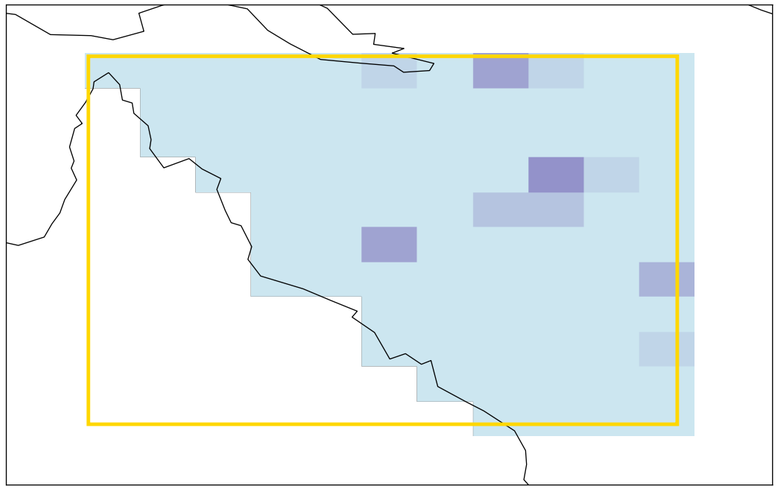




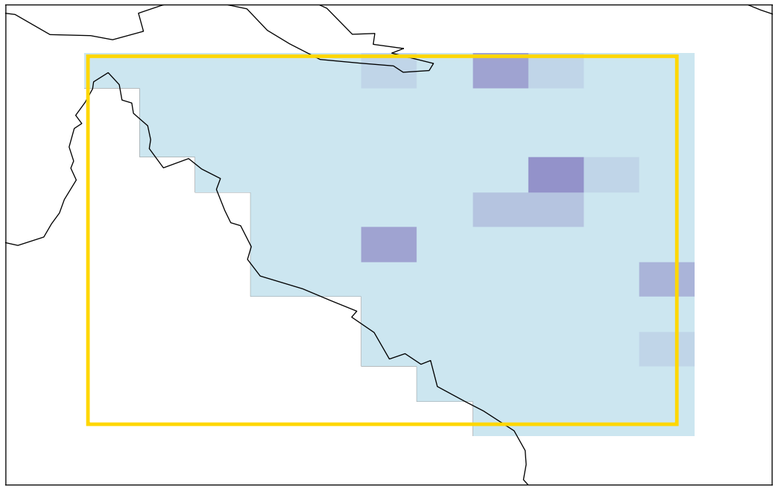


DHW Chain init: 2018-03-31 - Val - GBR (C-weeks)

Actual DHW - Step 1



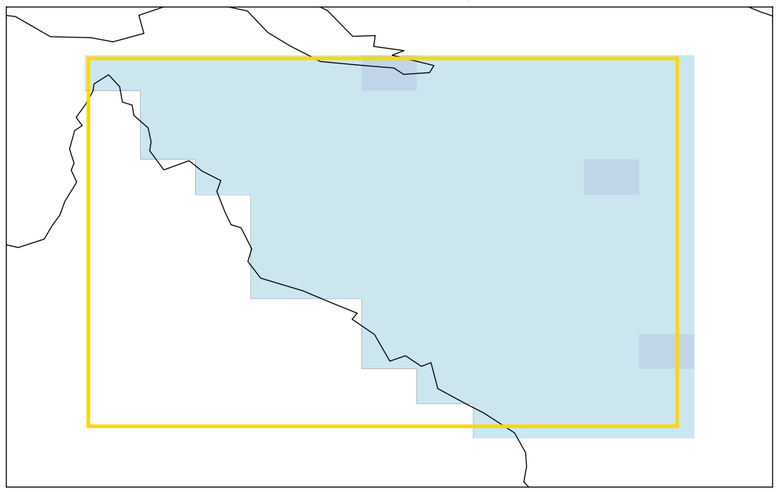
Predicted DHW - Step 1



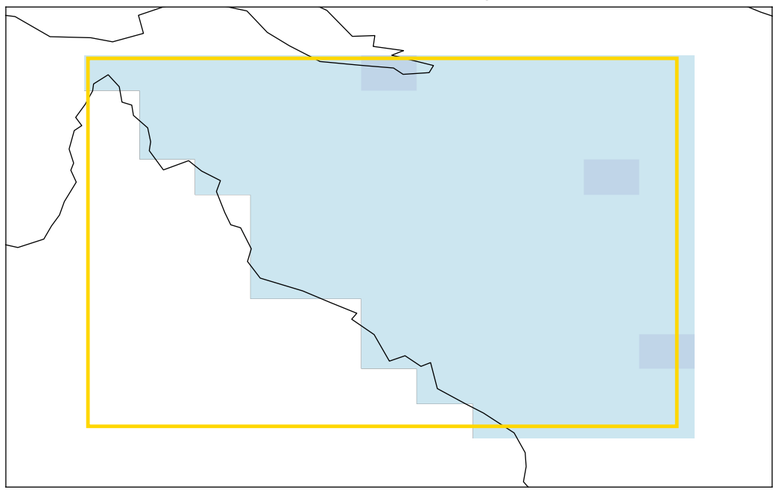
Error (Pred - Actual) - Step 1 [scale ±8]



Actual DHW - Step 46



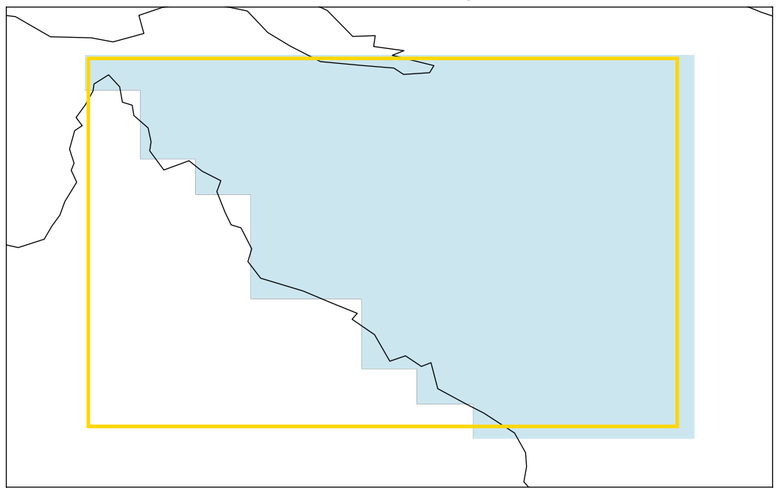
Predicted DHW - Step 46



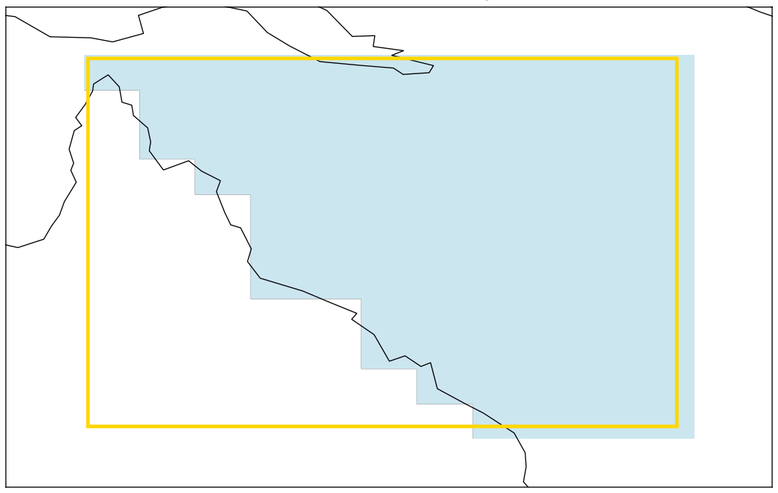
Error (Pred - Actual) - Step 46 [scale ±8]



Actual DHW - Step 90

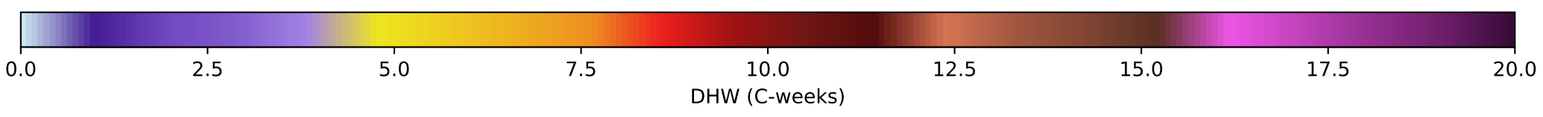


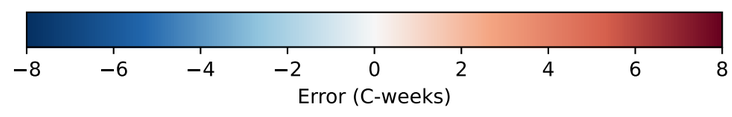
Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±8]

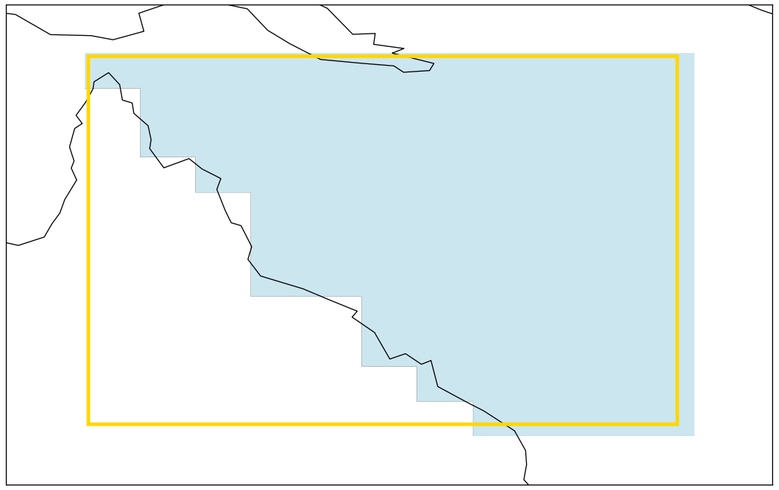




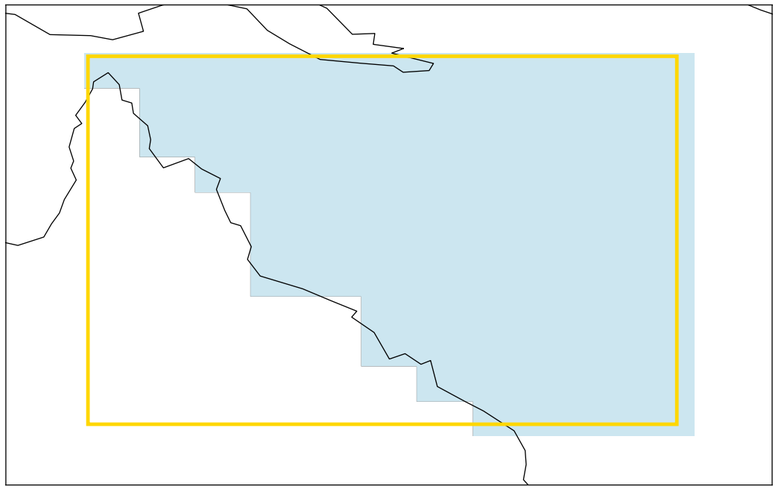


DHW Chain init: 2018-06-29 - Val - GBR (C-weeks)

Actual DHW - Step 1



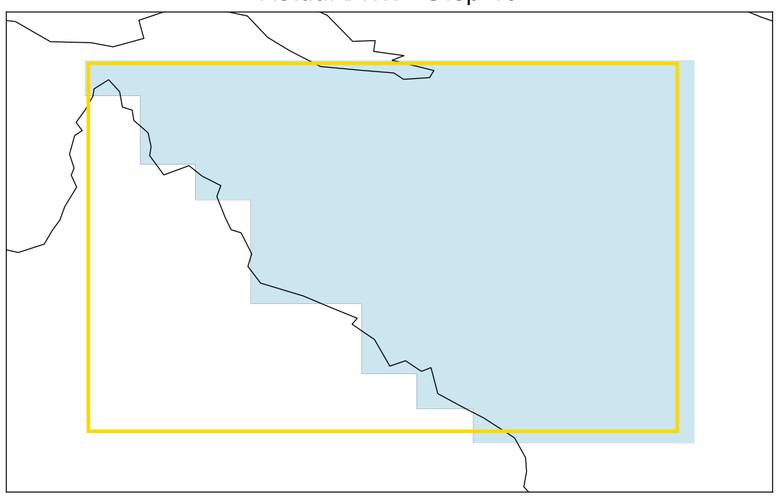
Predicted DHW - Step 1



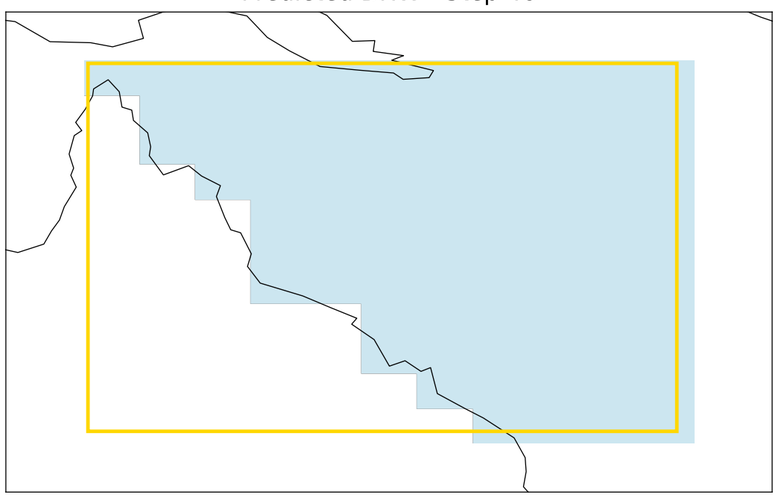
Error (Pred - Actual) - Step 1 [scale ±8]



Actual DHW - Step 46



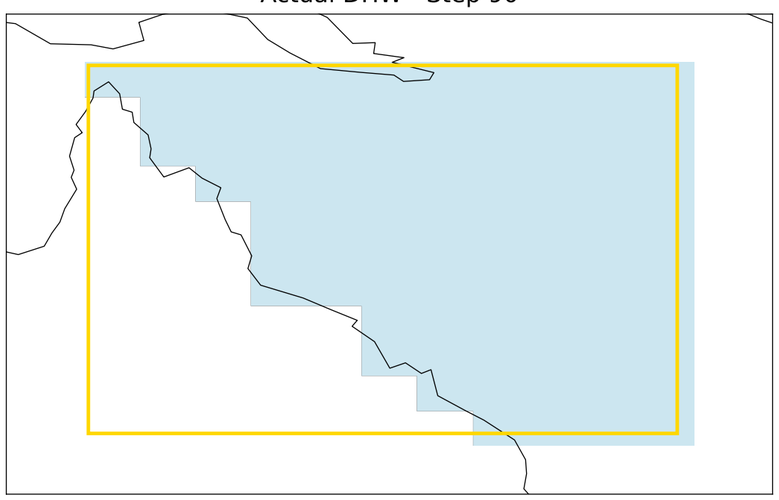
Predicted DHW - Step 46



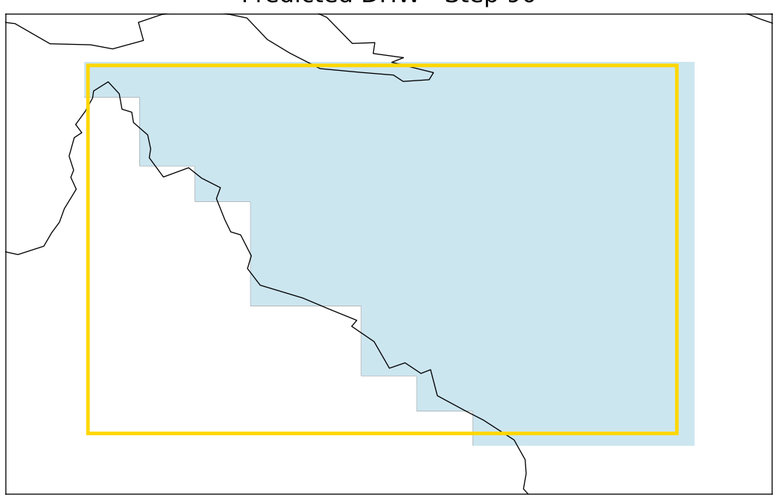
Error (Pred - Actual) - Step 46 [scale ±8]



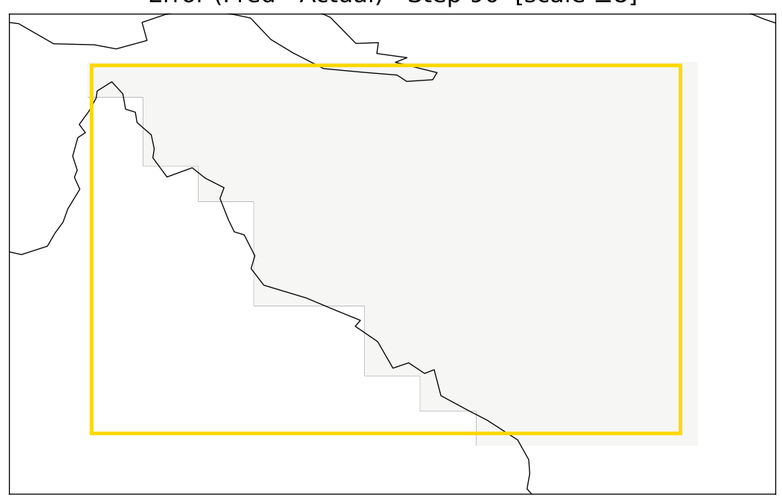
Actual DHW - Step 90

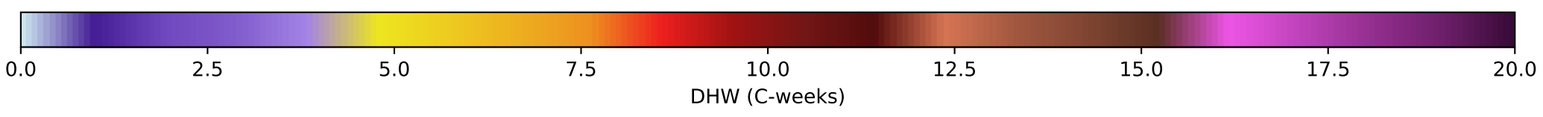


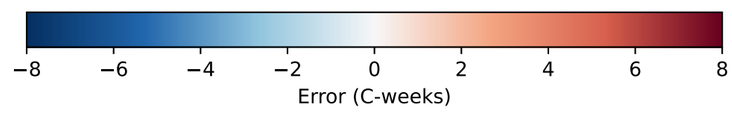
Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±8]

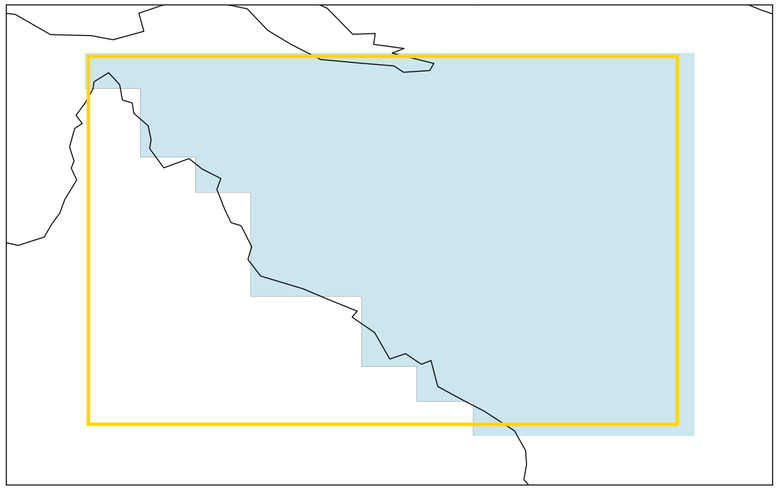




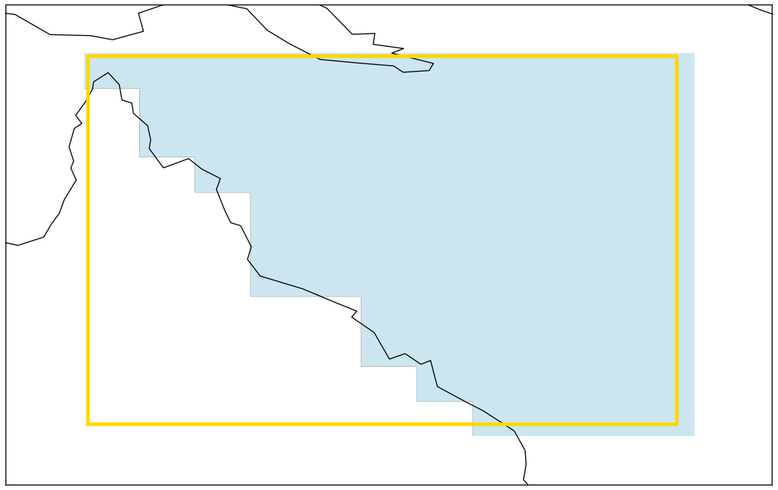


DHW Chain init: 2019-09-22 - Val - GBR (C-weeks)

Actual DHW - Step 1



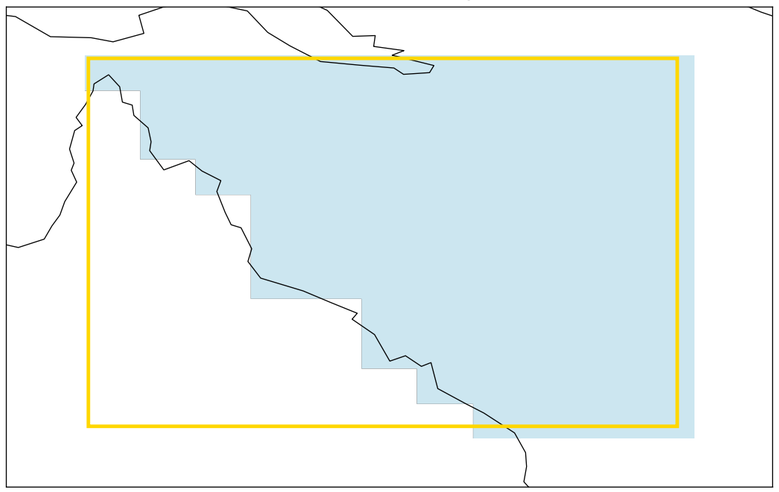
Predicted DHW - Step 1



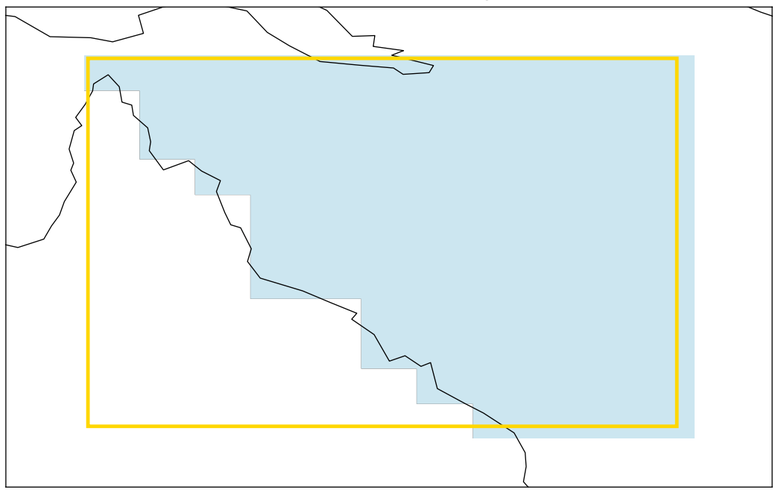
Error (Pred - Actual) - Step 1 [scale ±8]



Actual DHW - Step 46



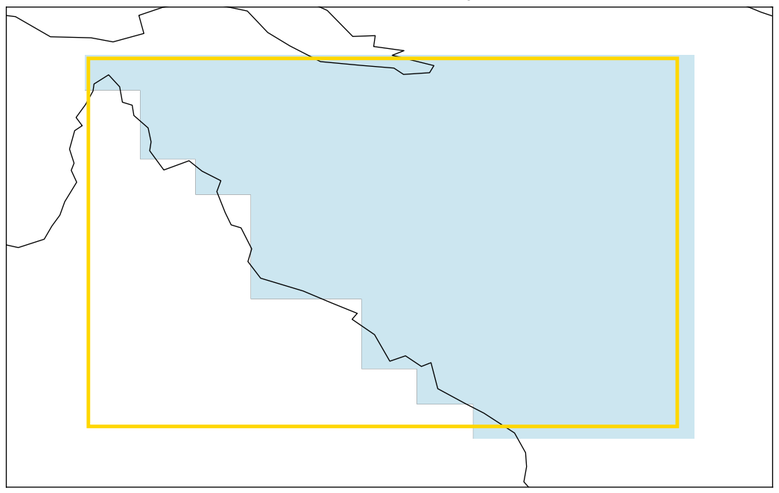
Predicted DHW - Step 46



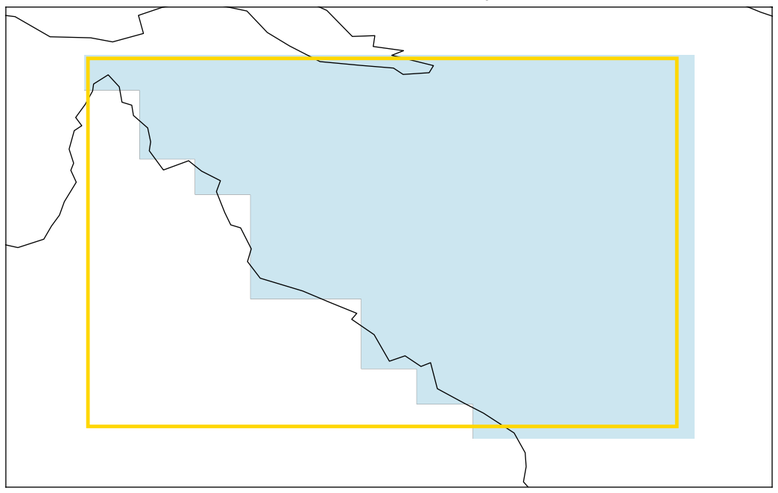
Error (Pred - Actual) - Step 46 [scale ±8]



Actual DHW - Step 90

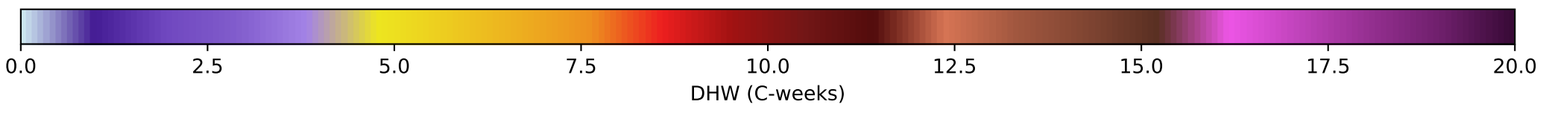


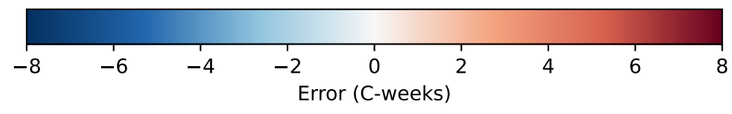
Predicted DHW - Step 90



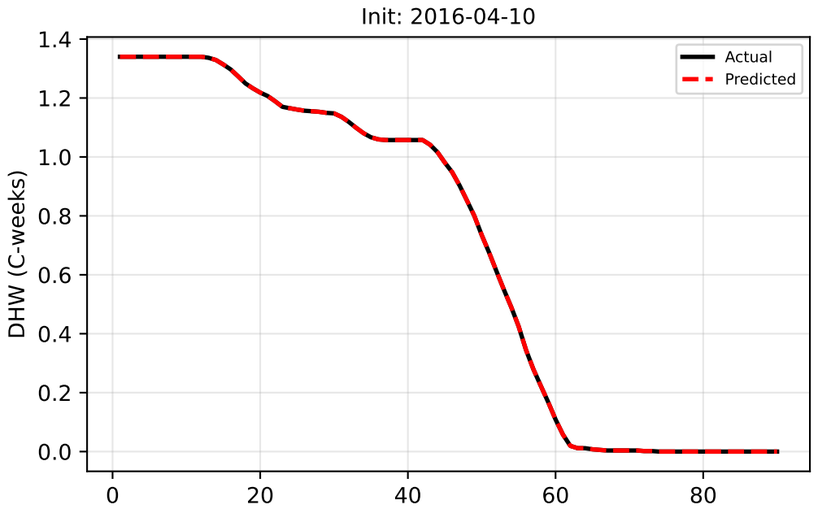
Error (Pred - Actual) - Step 90 [scale ±8]

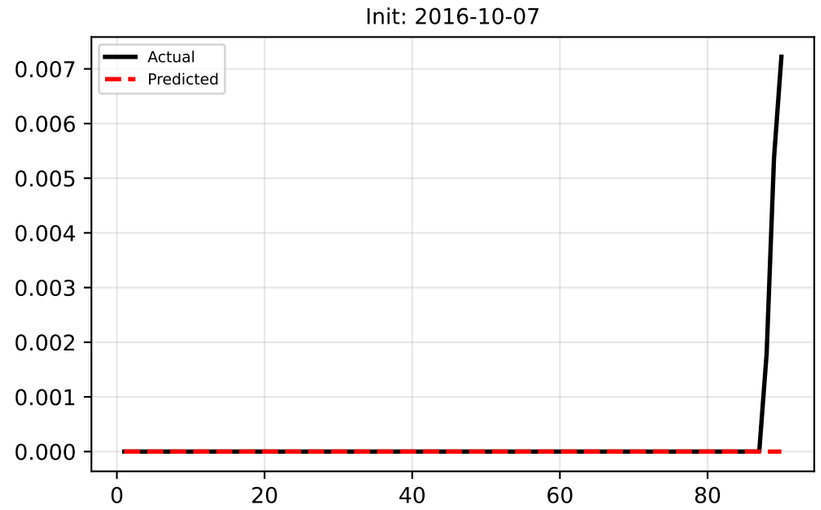


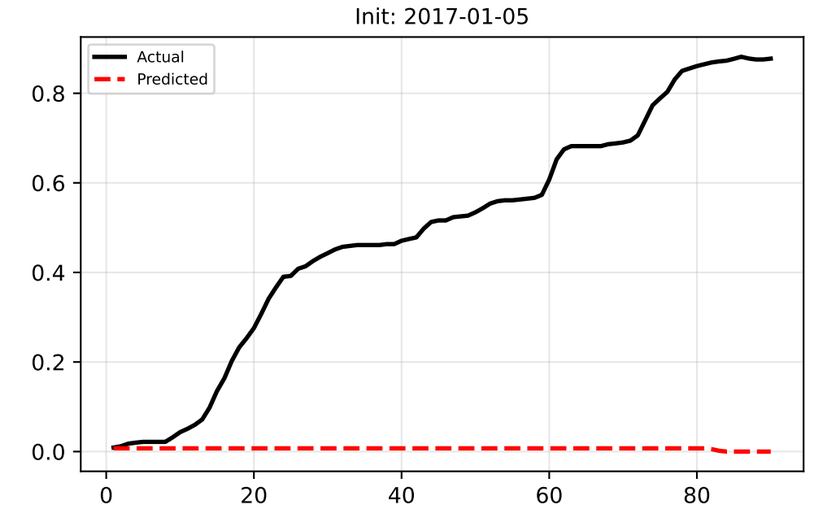


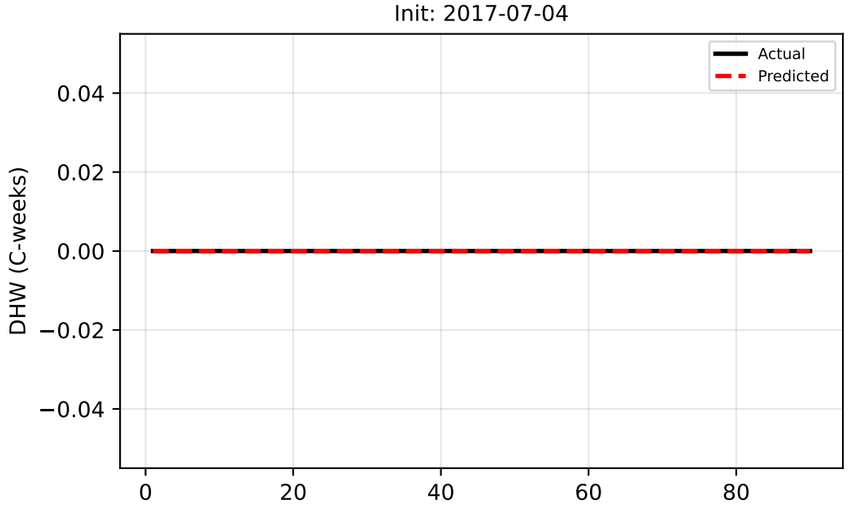


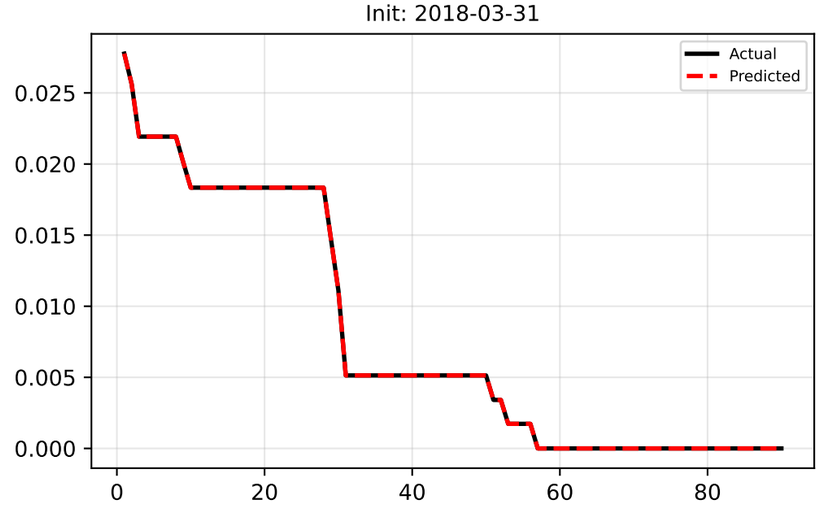
Global-Mean DHW - Val - GBR (C-weeks)

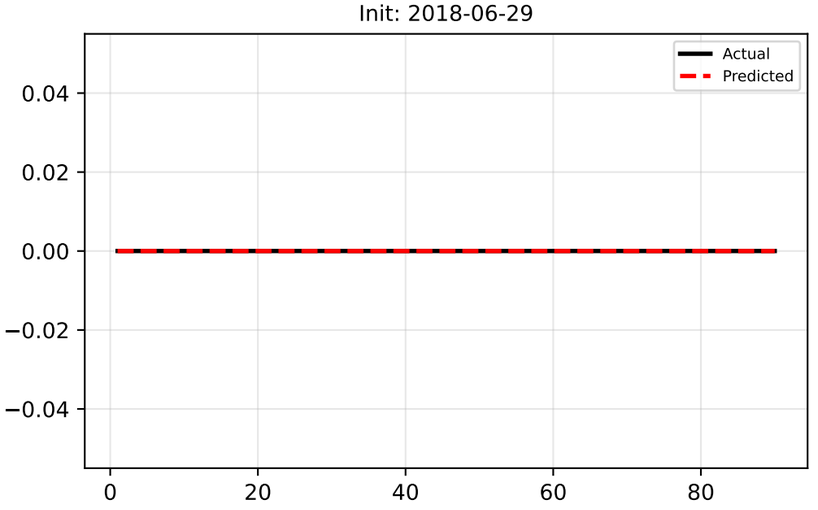


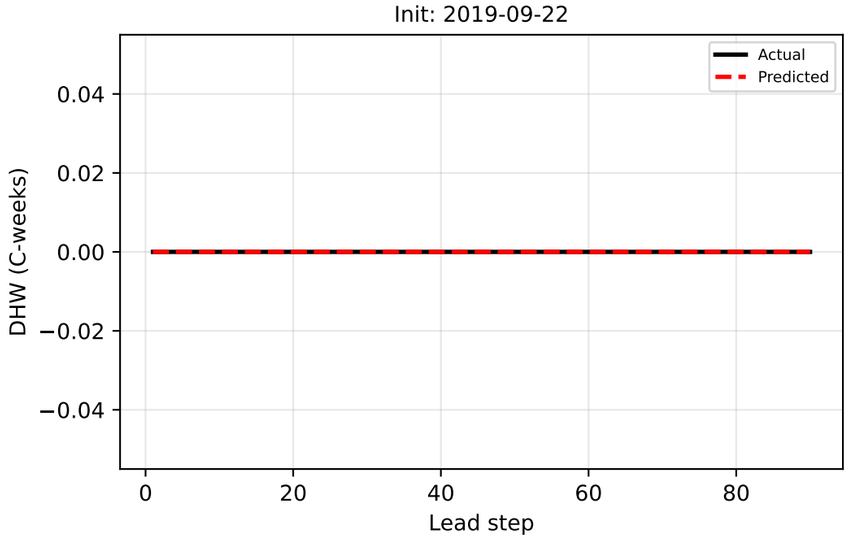


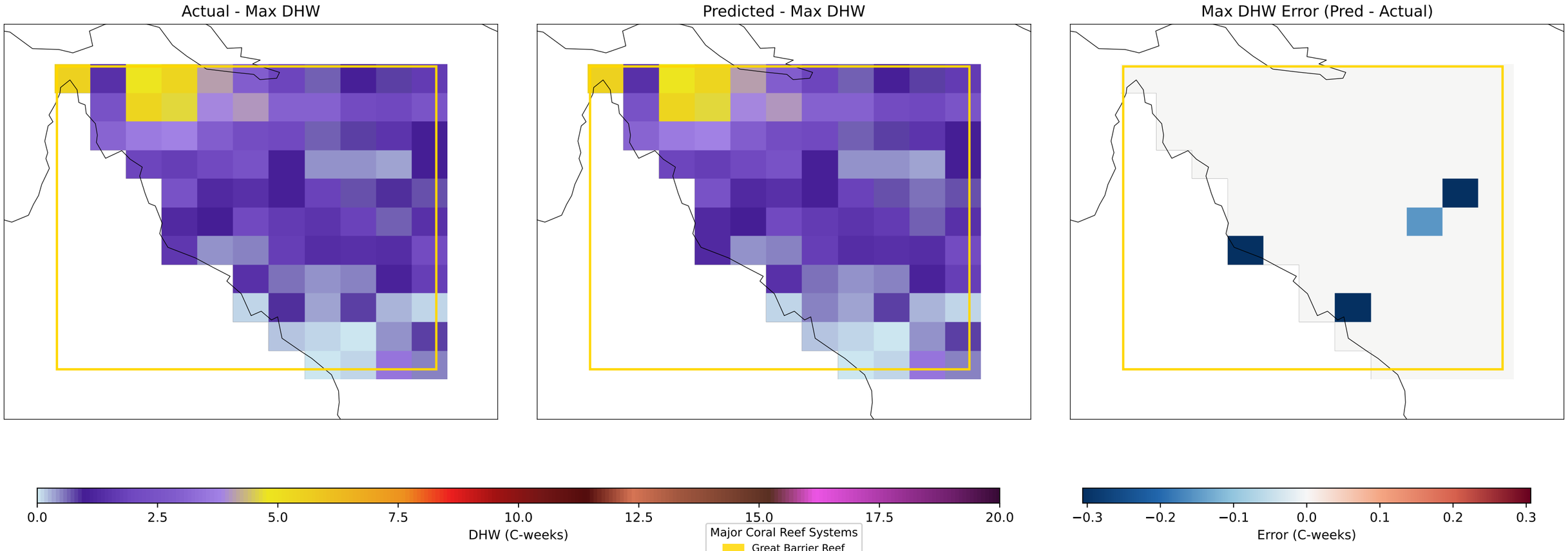


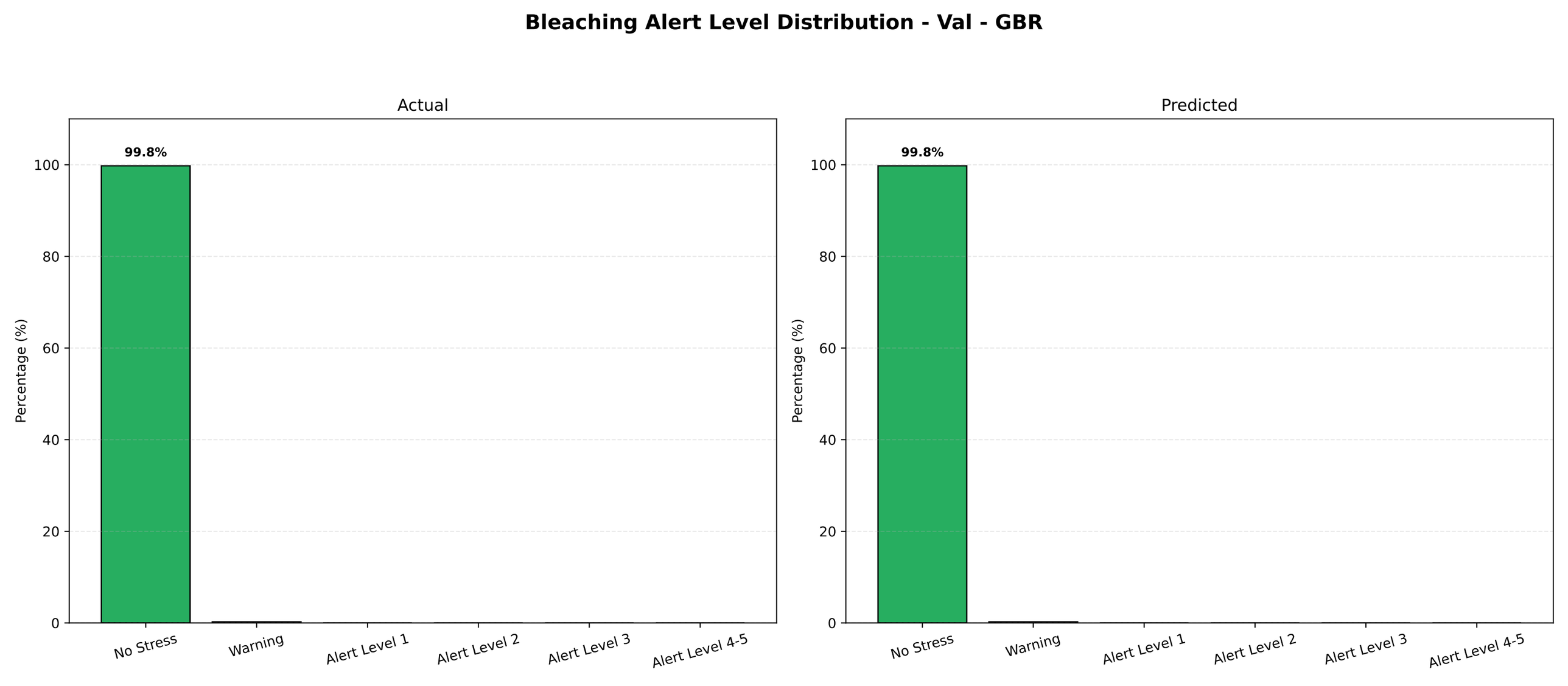






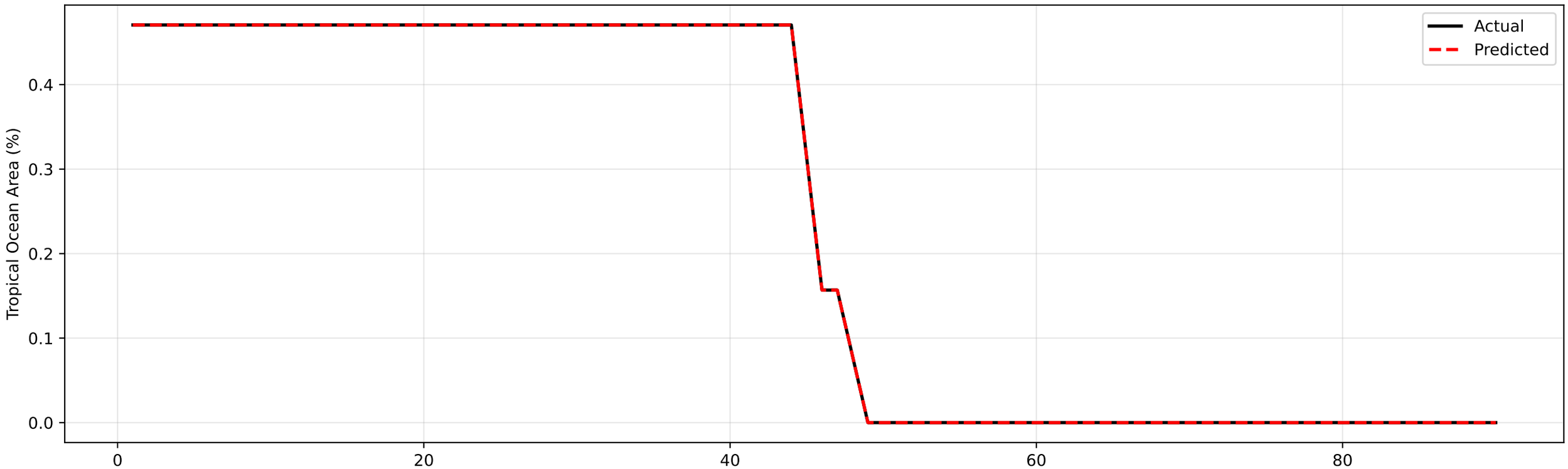



Peak DHW (max over all chains & leads) - Val - GBR (C-weeks)
Actual - Max DHW
Predicted - Max DHW
Max DHW Error (Pred - Actual)
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
DHW (C-weeks)
Major Coral Reef Systems
Great Barrier Reef
−0.3
−0.2
−0.1
0.0
0.1
0.2
0.3
Error (C-weeks)


Bleaching Alert Level Distribution - Val - GBR
Actual
Predicted
Percentage (%)
0
20
40
60
80
100
99.8%
No Stress
Warning
Alert Level 1
Alert Level 2
Alert Level 3
Alert Level 4-5

Tropical Ocean Area Under Bleaching Stress - Val - GBR

DHW >= 4 C-weeks (Alert Level 1 DHW threshold)



DHW >= 8 C-weeks (Alert Level 2 DHW threshold)

